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In [1]: # This Python 3 environment comes with many helpful analytics libraries i
# It is defined by the kaggle/python Docker image: https://github.com/kag
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/" directory
# For example, running this (by clicking run or pressing Shift+Enter) wil

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/) th
# You can also write temporary files to /kaggle/temp/, but they won't be
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In [2]: import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import DataLoader
from torchvision import datasets, transforms
import matplotlib.pyplot as plt
import numpy as np
from dataclasses import dataclass

# =====
# Color Palette
# =====
COLORS = {
    'primary': '#6366f1',      # Indigo
    'secondary': '#8b5cf6',    # Purple
    'accent': '#ec4899',      # Pink
    'success': '#10b981',     # Emerald
    'warning': '#f59e0b',      # Amber
    'error': '#ef4444',        # Red
    'info': '#3b82f6',         # Blue
    'teal': '#14b8a6',         # Teal
    'cyan': '#06b6d4',         # Cyan
    'background': '#f8fafc',   # Slate 50
    'text': '#1e293b',         # Slate 800
    'muted': '#94a3b8',        # Slate 400
}

def create_colormap(n_colors=512):
    """Create a soft, vibrant colormap for UIs"""
    from matplotlib.colors import LinearSegmentedColormap, to_rgb

    colors_list = [
        '#6366f1', '#8b5cf6', '#ec4899', '#f43f5e',
        '#f97316', '#f59e0b', '#eab308', '#84cc16',
        '#22c55e', '#10b981', '#14b8a6', '#06b6d4',
        '#0ea5e9', '#3b82f6',
    ]

    colors_rgb = [to_rgb(c) for c in colors_list]
    positions = np.linspace(0, 1, len(colors_rgb))
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cdict = {'red': [], 'green': [], 'blue': []}
for i, pos in enumerate(positions):
    r, g, b = colors_rgb[i]
    cdict['red'].append((pos, r, r))
    cdict['green'].append((pos, g, g))
    cdict['blue'].append((pos, b, b))

    return LinearSegmentedColormap('', cdict, N=n_colors)

# =====
# Configuration
# =====
@dataclass
class VQVAEConfig:
    batch_size: int = 64
    num_workers: int = 4
    in_channels: int = 3
    hidden_channels: int = 256
    latent_dim: int = 64
    num_embeddings: int = 512
    commitment_cost: float = 0.25
    num_epochs: int = 500
    learning_rate: float = 1e-4
    early_stopping: bool = True
    patience: int = 20
    min_delta: float = 1e-4
    use_multi_gpu: bool = True
    num_vis_samples: int = 5
    inner_radius: float = 0.8 # For ring plots

config = VQVAEConfig()
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
num_gpus = torch.cuda.device_count()
print(f"Using device: {device}")
print(f"Number of GPUs available: {num_gpus}")

# =====
# Data Loading
# =====
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5))
])

train_dataset = datasets.CIFAR10(root='./data', train=True, download=True)
test_dataset = datasets.CIFAR10(root='./data', train=False, download=True)

train_loader = DataLoader(train_dataset, batch_size=config.batch_size, sh
                           num_workers=config.num_workers, pin_memory=True)
test_loader = DataLoader(test_dataset, batch_size=config.batch_size, shuf
                          num_workers=config.num_workers, pin_memory=True)

# =====
# Model Components
# =====
class Encoder(nn.Module):
    def __init__(self, in_channels=3, hidden_channels=128, latent_dim=64):
        super().__init__()
        self.conv1 = nn.Conv2d(in_channels, hidden_channels, 4, stride=2,

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self.conv2 = nn.Conv2d(hidden_channels, hidden_channels, 4, stride=2)
self.conv3 = nn.Conv2d(hidden_channels, latent_dim, 1)
self.relu = nn.ReLU(inplace=True)

def forward(self, x):
    x = self.relu(self.conv1(x))
    x = self.relu(self.conv2(x))
    x = self.conv3(x)
    return x

class Decoder(nn.Module):
    def __init__(self, latent_dim=64, hidden_channels=128, out_channels=3):
        super().__init__()
        self.conv1 = nn.Conv2d(latent_dim, hidden_channels, 3, padding=1)
        self.conv2 = nn.ConvTranspose2d(hidden_channels, hidden_channels, 2, padding=1)
        self.conv3 = nn.ConvTranspose2d(hidden_channels, out_channels, 4, padding=1)
        self.relu = nn.ReLU(inplace=True)

    def forward(self, x):
        x = self.relu(self.conv1(x))
        x = self.relu(self.conv2(x))
        x = torch.tanh(self.conv3(x))
        return x

class VectorQuantizer(nn.Module):
    def __init__(self, num_embeddings, embedding_dim, commitment_cost=0.2):
        super().__init__()
        self.embedding_dim = embedding_dim
        self.num_embeddings = num_embeddings
        self.commitment_cost = commitment_cost
        self.embeddings = nn.Embedding(num_embeddings, embedding_dim)
        self.embeddings.weight.data.uniform_(-1/num_embeddings, 1/num_embeddings)

    def forward(self, x):
        B, D, H, W = x.shape
        flat_x = x.permute(0, 2, 3, 1).contiguous().view(-1, self.embedding_dim)
        distances = (flat_x.pow(2).sum(1, keepdim=True)
                     + 2 * flat_x @ self.embeddings.weight.t()
                     + self.embeddings.weight.pow(2).sum(1, keepdim=True)).view(B, H, W)
        encoding_indices = torch.argmin(distances, dim=-1)
        quantized = self.embeddings(encoding_indices).view(B, H, W, D)
        quantized = quantized.permute(0, 3, 1, 2).contiguous()
        e_latent_loss = F.mse_loss(quantized.detach(), x, reduction='mean')
        q_latent_loss = F.mse_loss(quantized, x.detach(), reduction='mean')
        loss = q_latent_loss + self.commitment_cost * e_latent_loss
        quantized = x + (quantized - x).detach()
        avg_probs = torch.mean(F.one_hot(encoding_indices, self.num_embeddings), dim=0)
        perplexity = torch.exp(-torch.sum(avg_probs * torch.log(avg_probs + 1e-5)))
        return quantized, loss, perplexity

class VQVAE(nn.Module):
    def __init__(self, config):
        super().__init__()
        self.encoder = Encoder(config.in_channels, config.hidden_channels)
        self.quantizer = VectorQuantizer(config.num_embeddings, config.embedding_dim)
        self.decoder = Decoder(config.latent_dim, config.hidden_channels, config.out_channels)

    def forward(self, x):
        z_e = self.encoder(x)
        z_q, vq_loss, perplexity = self.quantizer(z_e)

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        x_recon = self.decoder(z_q)
        return {'recon': x_recon, 'vq_loss': vq_loss, 'perplexity': perpl

    def encode(self, x):
        with torch.no_grad():
            z_e = self.encoder(x)
            B, D, H, W = z_e.shape
            flat_x = z_e.permute(0, 2, 3, 1).contiguous().view(-1, D)
            distances = (flat_x.pow(2).sum(1, keepdim=True)
                        - 2 * flat_x @ self.quantizer.embeddings.weight.t
                        + self.quantizer.embeddings.weight.pow(2).sum(1,
                        encoding_indices = torch.argmin(distances, dim=1)
            return encoding_indices.view(B, H, W)

    def decode(self, indices):
        with torch.no_grad():
            B, H, W = indices.shape
            quantized = self.quantizer.embeddings(indices).view(B, H, W,
            quantized = quantized.permute(0, 3, 1, 2).contiguous()
            return self.decoder(quantized)

# =====
# Training Loop
# =====
model = VQVAE(config)
if config.use_multi_gpu and num_gpus > 1:
    print(f"Using DataParallel with {num_gpus} GPUs")
    model = nn.DataParallel(model)
model = model.to(device)

optimizer = torch.optim.Adam(model.parameters(), lr=config.learning_rate)
best_loss = float('inf')
patience_counter = 0
best_epoch = 0

train_losses = []
val_losses = []
train_recon_losses = []
val_recon_losses = []
train_vq_losses = []
val_vq_losses = []
train_perplexities = []
val_perplexities = []

for epoch in range(config.num_epochs):
    model.train()
    train_loss = 0
    train_recon_loss = 0
    train_vq_loss = 0
    train_perplexity = 0

    for batch_idx, (imgs, _) in enumerate(train_loader):
        imgs = imgs.to(device)
        optimizer.zero_grad()
        outputs = model(imgs)

        if isinstance(outputs, dict):
            recon = outputs['recon']
            vq_loss = outputs['vq_loss'].mean()
            perplexity = outputs['perplexity'].mean()

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else:
    recon, vq_loss, perplexity = outputs

    recon_loss = F.mse_loss(recon, imgs, reduction='mean')
    loss = recon_loss + vq_loss
    loss.backward()
    optimizer.step()

    train_loss += loss.item()
    train_recon_loss += recon_loss.item()
    train_vq_loss += vq_loss.item()
    train_perplexity += perplexity.item()

    if (batch_idx + 1) % 100 == 0:
        print(f'Epoch {epoch+1}/{config.num_epochs}, Batch {batch_idx}
              f'Loss: {loss.item():.4f}, Recon: {recon_loss.item():.4f}
              f'VQ: {vq_loss.item():.4f}, Perplexity: {perplexity.item():.4f}')

n_batches = len(train_loader)
avg_train_loss = train_loss / n_batches
avg_train_recon_loss = train_recon_loss / n_batches
avg_train_vq_loss = train_vq_loss / n_batches
avg_train_perplexity = train_perplexity / n_batches

model.eval()
val_loss = 0
val_recon_loss = 0
val_vq_loss = 0
val_perplexity = 0

with torch.no_grad():
    for imgs, _ in test_loader:
        imgs = imgs.to(device)
        outputs = model(imgs)

        if isinstance(outputs, dict):
            recon = outputs['recon']
            vq_loss_batch = outputs['vq_loss'].mean()
            perplexity_batch = outputs['perplexity'].mean()
        else:
            recon, vq_loss_batch, perplexity_batch = outputs

        recon_loss_batch = F.mse_loss(recon, imgs, reduction='mean')
        loss_batch = recon_loss_batch + vq_loss_batch

        val_loss += loss_batch.item()
        val_recon_loss += recon_loss_batch.item()
        val_vq_loss += vq_loss_batch.item()
        val_perplexity += perplexity_batch.item()

avg_val_loss = val_loss / len(test_loader)
avg_val_recon_loss = val_recon_loss / len(test_loader)
avg_val_vq_loss = val_vq_loss / len(test_loader)
avg_val_perplexity = val_perplexity / len(test_loader)

train_losses.append(avg_train_loss)
val_losses.append(avg_val_loss)
train_recon_losses.append(avg_train_recon_loss)
val_recon_losses.append(avg_val_recon_loss)
train_vq_losses.append(avg_train_vq_loss)

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val_vq_losses.append(avg_val_vq_loss)
train_perplexities.append(avg_train_perplexity)
val_perplexities.append(avg_val_perplexity)

print(f'\n=== Epoch {epoch+1}/{config.num_epochs} Summary ===')
print(f'Train Loss: {avg_train_loss:.4f} | Val Loss: {avg_val_loss:.4f}')
print(f'Train Recon: {avg_train_recon_loss:.4f} | Val Recon: {avg_val_recon_loss:.4f}')
print(f'Train VQ: {avg_train_vq_loss:.4f} | Val VQ: {avg_val_vq_loss:.4f}')
print(f'Train Perplexity: {avg_train_perplexity:.2f} | Val Perplexity: {avg_val_perplexity:.2f}')

if avg_val_loss < best_loss - config.min_delta:
    best_loss = avg_val_loss
    best_epoch = epoch + 1
    patience_counter = 0

    save_path = 'vqvae_cifar10_best.pt'
    if isinstance(model, nn.DataParallel):
        torch.save({'epoch': epoch + 1, 'model_state_dict': model.module.state_dict(), 'optimizer_state_dict': optimizer.state_dict(), 'train_loss': avg_train_loss, 'val_loss': avg_val_loss, 'train_recon_loss': avg_train_recon_loss, 'val_recon_loss': avg_val_recon_loss, 'train_vq_loss': avg_train_vq_loss, 'val_vq_loss': avg_val_vq_loss, 'train_perplexity': avg_train_perplexity, 'val_perplexity': avg_val_perplexity}, save_path)
    else:
        torch.save({'epoch': epoch + 1, 'model_state_dict': model.state_dict(), 'optimizer_state_dict': optimizer.state_dict(), 'train_loss': avg_train_loss, 'val_loss': avg_val_loss, 'train_recon_loss': avg_train_recon_loss, 'val_recon_loss': avg_val_recon_loss, 'train_vq_loss': avg_train_vq_loss, 'val_vq_loss': avg_val_vq_loss, 'train_perplexity': avg_train_perplexity, 'val_perplexity': avg_val_perplexity}, save_path)
    print(f'✓ Best model saved! (Val Loss: {avg_val_loss:.4f})')
else:
    patience_counter += 1
    print(f'No improvement. Patience: {patience_counter}/{config.patience}')

print(f'Best Loss: {best_loss:.4f} (Epoch {best_epoch})\n')

if config.early_stopping and patience_counter >= config.patience:
    print(f'\n⚠ Early stopping triggered after {epoch+1} epochs')
    print(f'Best model was at epoch {best_epoch} with loss {best_loss:.4f}')
    break

# =====
# Training History
# =====
print("\nPlotting training history...")
epochs_range = range(1, len(train_losses) + 1)

plt.style.use('seaborn-v0_8-darkgrid')
fig, axes = plt.subplots(2, 2, figsize=(14, 10), facecolor='white')

for ax in axes.flat:
    ax.set_facecolor('#fafafa')
    ax.grid(True, alpha=0.2, linestyle='--', linewidth=0.5)
    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)

axes[0, 0].plot(epochs_range, train_losses, color=COLORS['primary'], line
axes[0, 0].plot(epochs_range, val_losses, color=COLORS['accent'], line
axes[0, 0].set_xlabel('Epoch', fontsize=11, color=COLORS['text'])

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axes[0, 0].set_ylabel('Total Loss', fontsize=11, color=_COLORS['text'])
axes[0, 0].set_title('Total Loss (Recon + VQ)', fontsize=12, fontweight='bold')
axes[0, 0].legend(fontsize=10, framealpha=0.9)
axes[0, 0].axvline(best_epoch, color=_COLORS['success'], linestyle='--',

axes[0, 1].plot(epochs_range, train_recon_losses, color=_COLORS['primary'])
axes[0, 1].plot(epochs_range, val_recon_losses, color=_COLORS['accent'],
axes[0, 1].set_xlabel('Epoch', fontsize=11, color=_COLORS['text'])
axes[0, 1].set_ylabel('Reconstruction Loss', fontsize=11, color=_COLORS['text'])
axes[0, 1].set_title('Reconstruction Loss (MSE)', fontsize=12, fontweight='bold')
axes[0, 1].legend(fontsize=10, framealpha=0.9)
axes[0, 1].axvline(best_epoch, color=_COLORS['success'], linestyle='--',

axes[1, 0].plot(epochs_range, train_vq_losses, color=_COLORS['primary'],
axes[1, 0].plot(epochs_range, val_vq_losses, color=_COLORS['accent'], lin
axes[1, 0].set_xlabel('Epoch', fontsize=11, color=_COLORS['text'])
axes[1, 0].set_ylabel('VQ Loss', fontsize=11, color=_COLORS['text'])
axes[1, 0].set_title('Vector Quantization Loss', fontsize=12, fontweight='bold')
axes[1, 0].legend(fontsize=10, framealpha=0.9)
axes[1, 0].axvline(best_epoch, color=_COLORS['success'], linestyle='--',

axes[1, 1].plot(epochs_range, train_perplexities, color=_COLORS['primary'])
axes[1, 1].plot(epochs_range, val_perplexities, color=_COLORS['accent'],
axes[1, 1].set_xlabel('Epoch', fontsize=11, color=_COLORS['text'])
axes[1, 1].set_ylabel('Perplexity', fontsize=11, color=_COLORS['text'])
axes[1, 1].set_title(f'Codebook Utilization (Max: {config.num_embeddings})',
axes[1, 1].legend(fontsize=10, framealpha=0.9)
axes[1, 1].axvline(best_epoch, color=_COLORS['success'], linestyle='--',
axes[1, 1].axhline(config.num_embeddings, color=_COLORS['muted'], linesty

plt.suptitle(f'Training History - Best Model at Epoch {best_epoch}', font
plt.tight_layout()
plt.savefig('vqvae_training_history.png', dpi=150, bbox_inches='tight', f
plt.show()

# =====
# Load Best Model
# =====
print(f'\nLoading best model from epoch {best_epoch}...')
checkpoint = torch.load('vqvae_cifar10_best.pt')
if isinstance(model, nn.DataParallel):
    model.module.load_state_dict(checkpoint['model_state_dict'])
else:
    model.load_state_dict(checkpoint['model_state_dict'])

model.eval()
imgs, _ = next(iter(test_loader))
imgs = imgs.to(device)

with torch.no_grad():
    outputs = model(imgs)
    if isinstance(outputs, dict):
        recon = outputs['recon']
    else:
        recon, _, _ = outputs

def denormalize(x):
    return ((x + 1) / 2).clamp(0, 1)

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# Visualizations
# =====
print("\n=== Generating Visualizations ===")

_cmap = create_colormap(config.num_embeddings)
colors_array = _cmap(np.linspace(0, 1, config.num_embeddings))

# 1. DNA Sequences
print("1. Creating DNA-style sequences...")
fig = plt.figure(figsize=(18, 2*config.num_vis_samples), facecolor='white')
gs = fig.add_gridspec(config.num_vis_samples, 20, hspace=0.4, wspace=0.1)

for i in range(config.num_vis_samples):
    with torch.no_grad():
        if isinstance(model, nn.DataParallel):
            indices = model.module.encode(imgs[i:i+1])
        else:
            indices = model.encode(imgs[i:i+1])
        indices = indices[0].cpu().numpy()

    H, W = indices.shape
    sequence = indices.flatten()

    ax_img = fig.add_subplot(gs[i, :2])
    img_orig = denormalize(imgs[i]).cpu().permute(1, 2, 0).numpy()
    ax_img.imshow(img_orig)
    ax_img.axis('off')
    ax_img.set_title(f'Image {i+1}', fontsize=9, pad=5, color=_COLORS['teal'])

    ax_seq = fig.add_subplot(gs[i, 2:])
    ax_seq.set_facecolor('white')
    x_positions = np.arange(len(sequence))
    colors_seq = [colors_array[idx] for idx in sequence]

    ax_seq.bar(x_positions, np.ones(len(sequence)), color=colors_seq,
              width=1.0, edgecolor='white', linewidth=0.2, alpha=0.95)
    ax_seq.set_xlim(-0.5, len(sequence)-0.5)
    ax_seq.set_ylim(0, 1)
    ax_seq.set_yticks([])
    ax_seq.set_xticks([])

    for spine in ax_seq.spines.values():
        spine.set_visible(False)

    unique_codes = len(np.unique(sequence))
    ax_seq.set_xlabel(f'{len(sequence)} codes | {unique_codes} unique',
                    fontsize=9, style='italic', color=_COLORS['muted'])

plt.suptitle('DNA-Style Codeword Sequences', fontsize=14, y=0.98,
            fontweight='bold', color=_COLORS['text'])
plt.tight_layout()
plt.savefig('vqvae_dna_sequences.png', dpi=150, bbox_inches='tight', facecolor='white')
plt.show()

# 2. 2D Genome Map (NO COLORBAR)
print("2. Creating 2D genome map...")
fig, axes = plt.subplots(config.num_vis_samples, 2, figsize=(16, 2.5*config.num_vis_samples),
                        facecolor='white', gridspec_kw={'width_ratios': [1, 1]})
if config.num_vis_samples == 1:
    axes = axes.reshape(1, -1)

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for i in range(config.num_vis_samples):
    with torch.no_grad():
        if isinstance(model, nn.DataParallel):
            indices = model.module.encode(imgs[i:i+1])
        else:
            indices = model.encode(imgs[i:i+1])
        indices = indices[0].cpu().numpy()

    img_orig = denormalize(imgs[i]).cpu().permute(1, 2, 0).numpy()
    axes[i, 0].imshow(img_orig)
    axes[i, 0].axis('off')
    axes[i, 0].set_title(f'Image {i+1}', fontsize=10, color=_COLORS['text'])

    # Create the grid visualization using pcolormesh for exact cell bound
    axes[i, 1].set_facecolor('#fafafa')
    H, W = indices.shape

    # Use pcolormesh for precise grid control
    x_edges = np.arange(W + 1)
    y_edges = np.arange(H + 1)

    mesh = axes[i, 1].pcolormesh(x_edges, y_edges, indices, cmap=_cmap,
                                  vmin=0, vmax=config.num_embeddings-1,
                                  edgecolors='white', linewidth=0.8, alph

    axes[i, 1].set_title(f'Codebook Sequence ({H}x{W} grid)',
                        fontsize=10, color=_COLORS['text'])

    # Set exact ticks at cell centers
    axes[i, 1].set_xticks(np.arange(W) + 0.5)
    axes[i, 1].set_yticks(np.arange(H) + 0.5)
    axes[i, 1].set_xticklabels(range(W))
    axes[i, 1].set_yticklabels(range(H))

    axes[i, 1].set_xlim(0, W)
    axes[i, 1].set_ylim(0, H)
    axes[i, 1].invert_yaxis() # Match image coordinates

    axes[i, 1].tick_params(labelsize=8, colors=_COLORS['muted'])
    axes[i, 1].set_xlabel('Width Position', fontsize=9, color=_COLORS['te
    axes[i, 1].set_ylabel('Height Position', fontsize=9, color=_COLORS['t

    # Remove spines
    for spine in axes[i, 1].spines.values():
        spine.set_visible(False)

plt.suptitle('2D Codebook Genome Map', fontsize=14, y=0.995,
            fontweight='bold', color=_COLORS['text'])
plt.tight_layout()
plt.savefig('vqvae_genome_map.png', dpi=150, bbox_inches='tight', facecol
plt.show()

# 3. Ring Plot
print("3. Creating ring plot...")
n_cols = 2
n_rows = (config.num_vis_samples + n_cols - 1) // n_cols
fig = plt.figure(figsize=(7*n_cols, 6*n_rows), facecolor='white')

for i in range(config.num_vis_samples):

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with torch.no_grad():
    if isinstance(model, nn.DataParallel):
        indices = model.module.encode(imgs[i:i+1])
    else:
        indices = model.encode(imgs[i:i+1])
    indices = indices[0].cpu().numpy().flatten()

ax = fig.add_subplot(n_rows, n_cols, i+1, projection='polar')
ax.set_facecolor('white')

n_codes = len(indices)
theta = np.linspace(0, 2*np.pi, n_codes, endpoint=False)
width = 2*np.pi / n_codes

outer_radius = 1.0
inner_radius = config.inner_radius
ring_height = outer_radius - inner_radius

for j, (angle, code_idx) in enumerate(zip(theta, indices)):
    ax.bar(angle, ring_height, width=width, bottom=inner_radius,
           color=colors_array[code_idx], edgecolor='white', linewidth=1)

ax.set_ylim(0, outer_radius * 1.1)
ax.set_yticks([])
ax.set_xticks([])
ax.spines['polar'].set_visible(False)

center_circle = plt.Circle((0, 0), inner_radius, color='#fafafa',
                           transform=ax.transData._b, zorder=10)
ax.add_artist(center_circle)

unique_codes = len(np.unique(indices))
ax.set_title(f'Image {i+1} - Circular Genome\n{n_codes} codes | {unique_codes} unique codes',
             fontsize=11, pad=20, color=_COLORS['text'], fontweight='bold')

plt.suptitle('Circular Codebook Genome (Ring View)', fontsize=14,
             fontweight='bold', color=_COLORS['text'], y=0.98)
plt.tight_layout()
plt.savefig('vqvae_circular_genome.png', dpi=150, bbox_inches='tight', facecolor='white')
plt.show()

# 4. Latent Codes
print("4. Visualizing latent codes...")
fig, axes = plt.subplots(2, config.num_vis_samples,
                        figsize=(2*config.num_vis_samples, 5), facecolor='white')

for i in range(config.num_vis_samples):
    img_orig = denormalize(imgs[i]).cpu().permute(1, 2, 0).numpy()
    axes[0, i].imshow(img_orig)
    axes[0, i].axis('off')
    if i == 0:
        axes[0, i].set_title('Original', fontsize=10, pad=10, color=_COLORS['text'])

    with torch.no_grad():
        if isinstance(model, nn.DataParallel):
            indices = model.module.encode(imgs[i:i+1])
        else:
            indices = model.encode(imgs[i:i+1])
        indices = indices[0].cpu().numpy()

```

```

        im = axes[1, i].imshow(indices, cmap=_cmap, interpolation='nearest',
                                vmin=0, vmax=config.num_embeddings-1)
        axes[1, i].axis('off')
        if i == 0:
            axes[1, i].set_title('Latent Codes', fontsize=10, pad=10, color=_

plt.tight_layout()
plt.savefig('vqvae_latent_codes.png', dpi=150, bbox_inches='tight', facec
plt.show()

# 5. Codebook Usage
print("5. Analyzing codebook usage...")
all_indices = []
with torch.no_grad():
    for batch_imgs, _ in test_loader:
        batch_imgs = batch_imgs.to(device)
        if isinstance(model, nn.DataParallel):
            indices = model.module.encode(batch_imgs)
        else:
            indices = model.encode(batch_imgs)
        all_indices.append(indices.cpu().flatten())

all_indices = torch.cat(all_indices).numpy()
code_counts = np.bincount(all_indices, minlength=config.num_embeddings)
code_usage = (code_counts > 0).sum()
usage_percent = 100 * code_usage / config.num_embeddings

fig, axes = plt.subplots(1, 2, figsize=(14, 5), facecolor='white')

axes[0].set_facecolor('#fafafa')
axes[0].bar(range(config.num_embeddings), code_counts, color=_COLORS['pri
            alpha=0.8, edgecolor='white', linewidth=0.5)
axes[0].set_xlabel('Codebook Index', fontsize=11, color=_COLORS['text'])
axes[0].set_ylabel('Usage Count', fontsize=11, color=_COLORS['text'])
axes[0].set_title(f'Codebook Usage Distribution\n{code_usage}/{config.num
            fontsize=12, pad=10, color=_COLORS['text'], fontweight=
axes[0].grid(axis='y', alpha=0.2, linestyle='-', linewidth=0.5)
axes[0].spines['top'].set_visible(False)
axes[0].spines['right'].set_visible(False)

axes[1].set_facecolor('#fafafa')
sorted_counts = np.sort(code_counts)[::-1]
axes[1].plot(sorted_counts, color=_COLORS['accent'], linewidth=2.5, alpha
axes[1].set_xlabel('Code Rank (sorted by usage)', fontsize=11, color=_COL
axes[1].set_ylabel('Usage Count', fontsize=11, color=_COLORS['text'])
axes[1].set_title('Code Usage (Sorted)', fontsize=12, pad=10,
            color=_COLORS['text'], fontweight='500')
axes[1].grid(alpha=0.2, linestyle='-', linewidth=0.5)
axes[1].set_yscale('log')
axes[1].spines['top'].set_visible(False)
axes[1].spines['right'].set_visible(False)

plt.tight_layout()
plt.savefig('vqvae_codebook_usage.png', dpi=150, bbox_inches='tight', fac
plt.show()

print(f"    - Active codes: {code_usage}/{config.num_embeddings} ({usage_p
print(f"    - Most used code: {code_counts.max()} times")
print(f"    - Least used active code: {code_counts[code_counts > 0].min()}

```

```

# 6. Codebook Structure (PCA)
print("6. Visualizing codebook structure...")
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

if isinstance(model, nn.DataParallel):
    embeddings = model.module.quantizer.embeddings.weight.data.cpu().numpy()
else:
    embeddings = model.quantizer.embeddings.weight.data.cpu().numpy()

scaler = StandardScaler()
embeddings_scaled = scaler.fit_transform(embeddings)
pca = PCA(n_components=2)
embeddings_2d = pca.fit_transform(embeddings_scaled)

fig, axes = plt.subplots(1, 2, figsize=(14, 6), facecolor='white')

axes[0].set_facecolor('#fafafa')
scatter = axes[0].scatter(embeddings_2d[:, 0], embeddings_2d[:, 1],
                          c=code_counts, cmap='viridis', s=60, alpha=0.8,
                          edgecolors='white', linewidth=0.5)
axes[0].set_xlabel(f'PC1 ({pca.explained_variance_ratio_[0]*100:.1f}% var',
                  fontsize=11, color=_COLORS['text'])
axes[0].set_ylabel(f'PC2 ({pca.explained_variance_ratio_[1]*100:.1f}% var',
                  fontsize=11, color=_COLORS['text'])
axes[0].set_title('Codebook Structure (colored by usage)', fontsize=12, color=_COLORS['text'], fontweight='500')
axes[0].grid(alpha=0.2, linestyle='--', linewidth=0.5)
axes[0].spines['top'].set_visible(False)
axes[0].spines['right'].set_visible(False)
cbar = plt.colorbar(scatter, ax=axes[0])
cbar.set_label('Usage Count', fontsize=10, color=_COLORS['text'])

axes[1].set_facecolor('#fafafa')
used_mask = code_counts > 0
axes[1].scatter(embeddings_2d[~used_mask, 0], embeddings_2d[~used_mask, 1],
                c=_COLORS['error'], s=60, alpha=0.7,
                label=f'Unused ({(~used_mask).sum()})',
                edgecolors='white', linewidth=0.5)
axes[1].scatter(embeddings_2d[used_mask, 0], embeddings_2d[used_mask, 1],
                c=_COLORS['primary'], s=60, alpha=0.7,
                label=f'Used ({used_mask.sum()})',
                edgecolors='white', linewidth=0.5)
axes[1].set_xlabel(f'PC1 ({pca.explained_variance_ratio_[0]*100:.1f}% var',
                  fontsize=11, color=_COLORS['text'])
axes[1].set_ylabel(f'PC2 ({pca.explained_variance_ratio_[1]*100:.1f}% var',
                  fontsize=11, color=_COLORS['text'])
axes[1].set_title('Codebook Structure (used vs unused)', fontsize=12, pad=10, color=_COLORS['text'], fontweight='500')
axes[1].grid(alpha=0.2, linestyle='--', linewidth=0.5)
axes[1].spines['top'].set_visible(False)
axes[1].spines['right'].set_visible(False)
axes[1].legend(fontsize=10, framealpha=0.9)

plt.tight_layout()
plt.savefig('vqvae_codebook_structure.png', dpi=150, bbox_inches='tight',
plt.show()

# 7. Top Used Codes
print("7. Visualizing individual codebook entries...")

```

```

top_k = 16
top_indices = np.argsort(code_counts)[::-1][:top_k]

fig, axes = plt.subplots(4, 4, figsize=(10, 10), facecolor='white')
axes = axes.flatten()

for idx, code_idx in enumerate(top_indices):
    H, W = 8, 8
    single_code_grid = torch.full((1, H, W), code_idx, dtype=torch.long).

    with torch.no_grad():
        if isinstance(model, nn.DataParallel):
            decoded = model.module.decode(single_code_grid)
        else:
            decoded = model.decode(single_code_grid)

    img = denormalize(decoded[0]).cpu().permute(1, 2, 0).numpy()
    axes[idx].imshow(img)
    axes[idx].set_title(f'Code {code_idx}\n({code_counts[code_idx]} uses)
                        fontsize=9, color=_COLORS['text'])
    axes[idx].axis('off')

plt.suptitle('Top 16 Most Used Codebook Entries', fontsize=14, y=0.995,
             fontweight='bold', color=_COLORS['text'])
plt.tight_layout()
plt.savefig('vqvae_top_codes.png', dpi=150, bbox_inches='tight', facecolor='white')
plt.show()

# 8. Codebook Interpolation
print("8. Visualizing codebook interpolation...")
code_a_idx = top_indices[0]
code_b_idx = top_indices[1]

if isinstance(model, nn.DataParallel):
    code_a = model.module.quantizer.embeddings.weight[code_a_idx].unsqueeze(0)
    code_b = model.module.quantizer.embeddings.weight[code_b_idx].unsqueeze(0)
else:
    code_a = model.quantizer.embeddings.weight[code_a_idx].unsqueeze(0)
    code_b = model.quantizer.embeddings.weight[code_b_idx].unsqueeze(0)

n_interp = 8
fig, axes = plt.subplots(1, n_interp, figsize=(2*n_interp, 2.5), facecolor='white')

for i, alpha in enumerate(np.linspace(0, 1, n_interp)):
    interpolated = (1 - alpha) * code_a + alpha * code_b

    H, W = 8, 8
    interp_grid = interpolated.unsqueeze(1).unsqueeze(2).repeat(1, H, W, 1)
    interp_grid = interp_grid.permute(0, 3, 1, 2).contiguous()

    with torch.no_grad():
        if isinstance(model, nn.DataParallel):
            decoded = model.module.decoder(interp_grid)
        else:
            decoded = model.decoder(interp_grid)

    img = denormalize(decoded[0]).cpu().permute(1, 2, 0).numpy()
    axes[i].imshow(img)
    axes[i].set_title(f'α={alpha:.2f}', fontsize=9, color=_COLORS['text'])
    axes[i].axis('off')

```

```

plt.suptitle(f'Interpolation: Code {code_a_idx} → Code {code_b_idx}',
             fontsize=12, y=0.98, fontweight='500', color=_COLORS['text'])
plt.tight_layout()
plt.savefig('vqvae_interpolation.png', dpi=150, bbox_inches='tight', face
plt.show()

# 9. Basic Reconstruction
print("9. Creating basic reconstruction comparison...")
fig, axes = plt.subplots(2, config.num_vis_samples,
                        figsize=(2*config.num_vis_samples, 4), facecolor

for i in range(config.num_vis_samples):
    img_orig = denormalize(imgs[i]).cpu().permute(1, 2, 0).numpy()
    axes[0, i].imshow(img_orig)
    axes[0, i].axis('off')
    if i == 0:
        axes[0, i].set_title('Original', fontsize=10, color=_COLORS['text

    img_recon = denormalize(recon[i]).cpu().permute(1, 2, 0).numpy()
    axes[1, i].imshow(img_recon)
    axes[1, i].axis('off')
    if i == 0:
        axes[1, i].set_title('Reconstructed', fontsize=10, color=_COLORS[

plt.tight_layout()
plt.savefig('vqvae_reconstruction.png', dpi=150, bbox_inches='tight', fac
plt.show()

print("\n=== All Visualizations Complete ===")
print("Generated files:")
print(" 0. vqvae_training_history.png - training curves")
print(" 1. vqvae_reconstruction.png - Original vs Reconstructed")
print(" 2. vqvae_dna_sequences.png - DNA-style linear sequences")
print(" 3. vqvae_genome_map.png - 2D genome map (no colorbar)")
print(" 4. vqvae_circular_genome.png - ring plot")
print(" 5. vqvae_latent_codes.png - Discrete latent representations")
print(" 6. vqvae_codebook_usage.png - Usage statistics")
print(" 7. vqvae_codebook_structure.png - PCA visualization")
print(" 8. vqvae_top_codes.png - Most frequently used codes")
print(" 9. vqvae_interpolation.png - Smooth interpolation")
print("\nAll visualizations use , soft color palette!")
print(f"Ring plot inner radius: {config.inner_radius} (adjustable in conf

```

Using device: cuda

Number of GPUs available: 2

100%|██████████| 170M/170M [00:03<00:00, 53.6MB/s]

Using DataParallel with 2 GPUs

```

/usr/local/lib/python3.11/dist-packages/torch/nn/parallel/_functions.py:7
0: UserWarning: Was asked to gather along dimension 0, but all input tenso
rs were scalars; will instead unsqueeze and return a vector.
  warnings.warn(

```

Epoch 1/500, Batch 100/782, Loss: 34.0943, Recon: 0.1465, VQ: 33.9478, Perplexity: 2.20
Epoch 1/500, Batch 200/782, Loss: 31.0026, Recon: 0.1091, VQ: 30.8934, Perplexity: 2.06
Epoch 1/500, Batch 300/782, Loss: 17.0053, Recon: 0.1054, VQ: 16.8999, Perplexity: 2.39
Epoch 1/500, Batch 400/782, Loss: 12.5347, Recon: 0.0882, VQ: 12.4465, Perplexity: 2.95
Epoch 1/500, Batch 500/782, Loss: 9.4022, Recon: 0.0781, VQ: 9.3241, Perplexity: 4.48
Epoch 1/500, Batch 600/782, Loss: 7.1431, Recon: 0.0658, VQ: 7.0773, Perplexity: 5.44
Epoch 1/500, Batch 700/782, Loss: 9.1607, Recon: 0.0704, VQ: 9.0903, Perplexity: 6.13

=== Epoch 1/500 Summary ===

Train Loss: 14.8735 | Val Loss: 6.0853
Train Recon: 0.1055 | Val Recon: 0.0701
Train VQ: 14.7680 | Val VQ: 6.0152
Train Perplexity: 3.79 | Val Perplexity: 6.70
✓ Best model saved! (Val Loss: 6.0853)
Best Loss: 6.0853 (Epoch 1)

Epoch 2/500, Batch 100/782, Loss: 4.1057, Recon: 0.0672, VQ: 4.0384, Perplexity: 6.72
Epoch 2/500, Batch 200/782, Loss: 4.1914, Recon: 0.0695, VQ: 4.1219, Perplexity: 6.39
Epoch 2/500, Batch 300/782, Loss: 3.5182, Recon: 0.0618, VQ: 3.4564, Perplexity: 5.86
Epoch 2/500, Batch 400/782, Loss: 3.5813, Recon: 0.0628, VQ: 3.5185, Perplexity: 5.59
Epoch 2/500, Batch 500/782, Loss: 4.0813, Recon: 0.0622, VQ: 4.0191, Perplexity: 5.67
Epoch 2/500, Batch 600/782, Loss: 3.1895, Recon: 0.0640, VQ: 3.1255, Perplexity: 5.68
Epoch 2/500, Batch 700/782, Loss: 3.2435, Recon: 0.0635, VQ: 3.1800, Perplexity: 5.70

=== Epoch 2/500 Summary ===

Train Loss: 3.7811 | Val Loss: 3.0443
Train Recon: 0.0661 | Val Recon: 0.0635
Train VQ: 3.7150 | Val VQ: 2.9807
Train Perplexity: 6.01 | Val Perplexity: 5.90
✓ Best model saved! (Val Loss: 3.0443)
Best Loss: 3.0443 (Epoch 2)

Epoch 3/500, Batch 100/782, Loss: 2.5890, Recon: 0.0563, VQ: 2.5327, Perplexity: 5.91
Epoch 3/500, Batch 200/782, Loss: 2.0559, Recon: 0.0597, VQ: 1.9962, Perplexity: 5.85
Epoch 3/500, Batch 300/782, Loss: 1.7076, Recon: 0.0553, VQ: 1.6523, Perplexity: 5.78
Epoch 3/500, Batch 400/782, Loss: 1.6932, Recon: 0.0597, VQ: 1.6335, Perplexity: 5.93
Epoch 3/500, Batch 500/782, Loss: 1.3813, Recon: 0.0572, VQ: 1.3242, Perplexity: 5.88
Epoch 3/500, Batch 600/782, Loss: 1.1271, Recon: 0.0554, VQ: 1.0717, Perplexity: 5.88
Epoch 3/500, Batch 700/782, Loss: 1.0926, Recon: 0.0562, VQ: 1.0363, Perplexity: 6.34

=== Epoch 3/500 Summary ===

Train Loss: 1.7852 | Val Loss: 1.1351
Train Recon: 0.0595 | Val Recon: 0.0560
Train VQ: 1.7257 | Val VQ: 1.0792
Train Perplexity: 5.95 | Val Perplexity: 6.42
✓ Best model saved! (Val Loss: 1.1351)
Best Loss: 1.1351 (Epoch 3)

Epoch 4/500, Batch 100/782, Loss: 1.3169, Recon: 0.0609, VQ: 1.2560, Perplexity: 6.55
Epoch 4/500, Batch 200/782, Loss: 0.8254, Recon: 0.0506, VQ: 0.7749, Perplexity: 6.70
Epoch 4/500, Batch 300/782, Loss: 1.0552, Recon: 0.0550, VQ: 1.0002, Perplexity: 6.82
Epoch 4/500, Batch 400/782, Loss: 0.9241, Recon: 0.0549, VQ: 0.8692, Perplexity: 6.81
Epoch 4/500, Batch 500/782, Loss: 0.9045, Recon: 0.0522, VQ: 0.8524, Perplexity: 7.07
Epoch 4/500, Batch 600/782, Loss: 0.6768, Recon: 0.0525, VQ: 0.6244, Perplexity: 7.10
Epoch 4/500, Batch 700/782, Loss: 0.6999, Recon: 0.0553, VQ: 0.6446, Perplexity: 7.20

=== Epoch 4/500 Summary ===

Train Loss: 0.8530 | Val Loss: 0.6706
Train Recon: 0.0535 | Val Recon: 0.0519
Train VQ: 0.7994 | Val VQ: 0.6187
Train Perplexity: 6.91 | Val Perplexity: 7.26
✓ Best model saved! (Val Loss: 0.6706)
Best Loss: 0.6706 (Epoch 4)

Epoch 5/500, Batch 100/782, Loss: 0.7311, Recon: 0.0531, VQ: 0.6780, Perplexity: 7.56
Epoch 5/500, Batch 200/782, Loss: 0.5068, Recon: 0.0490, VQ: 0.4578, Perplexity: 7.66
Epoch 5/500, Batch 300/782, Loss: 0.6309, Recon: 0.0502, VQ: 0.5807, Perplexity: 7.70
Epoch 5/500, Batch 400/782, Loss: 0.5779, Recon: 0.0488, VQ: 0.5291, Perplexity: 7.93
Epoch 5/500, Batch 500/782, Loss: 0.5222, Recon: 0.0489, VQ: 0.4733, Perplexity: 8.01
Epoch 5/500, Batch 600/782, Loss: 0.6068, Recon: 0.0513, VQ: 0.5554, Perplexity: 7.98
Epoch 5/500, Batch 700/782, Loss: 0.5355, Recon: 0.0475, VQ: 0.4880, Perplexity: 8.31

=== Epoch 5/500 Summary ===

Train Loss: 0.5929 | Val Loss: 0.5244
Train Recon: 0.0503 | Val Recon: 0.0486
Train VQ: 0.5425 | Val VQ: 0.4758
Train Perplexity: 7.82 | Val Perplexity: 8.44
✓ Best model saved! (Val Loss: 0.5244)
Best Loss: 0.5244 (Epoch 5)

Epoch 6/500, Batch 100/782, Loss: 0.4627, Recon: 0.0478, VQ: 0.4149, Perplexity: 8.69
Epoch 6/500, Batch 200/782, Loss: 0.6255, Recon: 0.0479, VQ: 0.5775, Perplexity: 9.04
Epoch 6/500, Batch 300/782, Loss: 0.4845, Recon: 0.0470, VQ: 0.4375, Perplexity:

exity: 9.35
Epoch 6/500, Batch 400/782, Loss: 0.4613, Recon: 0.0465, VQ: 0.4149, Perplexity: 9.66
Epoch 6/500, Batch 500/782, Loss: 0.4227, Recon: 0.0460, VQ: 0.3767, Perplexity: 9.59
Epoch 6/500, Batch 600/782, Loss: 0.4570, Recon: 0.0469, VQ: 0.4102, Perplexity: 10.18
Epoch 6/500, Batch 700/782, Loss: 0.3683, Recon: 0.0455, VQ: 0.3227, Perplexity: 10.52

=== Epoch 6/500 Summary ===
Train Loss: 0.4525 | Val Loss: 0.3787
Train Recon: 0.0468 | Val Recon: 0.0450
Train VQ: 0.4058 | Val VQ: 0.3337
Train Perplexity: 9.63 | Val Perplexity: 10.80
✓ Best model saved! (Val Loss: 0.3787)
Best Loss: 0.3787 (Epoch 6)

Epoch 7/500, Batch 100/782, Loss: 0.3957, Recon: 0.0470, VQ: 0.3487, Perplexity: 10.92
Epoch 7/500, Batch 200/782, Loss: 0.3542, Recon: 0.0417, VQ: 0.3125, Perplexity: 11.47
Epoch 7/500, Batch 300/782, Loss: 0.3716, Recon: 0.0463, VQ: 0.3253, Perplexity: 11.67
Epoch 7/500, Batch 400/782, Loss: 0.3153, Recon: 0.0430, VQ: 0.2722, Perplexity: 11.51
Epoch 7/500, Batch 500/782, Loss: 0.3428, Recon: 0.0438, VQ: 0.2990, Perplexity: 12.44
Epoch 7/500, Batch 600/782, Loss: 0.2983, Recon: 0.0380, VQ: 0.2603, Perplexity: 12.64
Epoch 7/500, Batch 700/782, Loss: 0.3608, Recon: 0.0464, VQ: 0.3144, Perplexity: 12.98

=== Epoch 7/500 Summary ===
Train Loss: 0.3440 | Val Loss: 0.3223
Train Recon: 0.0432 | Val Recon: 0.0423
Train VQ: 0.3008 | Val VQ: 0.2800
Train Perplexity: 11.98 | Val Perplexity: 13.33
✓ Best model saved! (Val Loss: 0.3223)
Best Loss: 0.3223 (Epoch 7)

Epoch 8/500, Batch 100/782, Loss: 0.2889, Recon: 0.0392, VQ: 0.2497, Perplexity: 13.63
Epoch 8/500, Batch 200/782, Loss: 0.2548, Recon: 0.0361, VQ: 0.2187, Perplexity: 13.71
Epoch 8/500, Batch 300/782, Loss: 0.2652, Recon: 0.0380, VQ: 0.2272, Perplexity: 13.93
Epoch 8/500, Batch 400/782, Loss: 0.2828, Recon: 0.0391, VQ: 0.2437, Perplexity: 13.85
Epoch 8/500, Batch 500/782, Loss: 0.2616, Recon: 0.0386, VQ: 0.2230, Perplexity: 14.49
Epoch 8/500, Batch 600/782, Loss: 0.2282, Recon: 0.0370, VQ: 0.1911, Perplexity: 14.17
Epoch 8/500, Batch 700/782, Loss: 0.2774, Recon: 0.0395, VQ: 0.2379, Perplexity: 14.41

=== Epoch 8/500 Summary ===
Train Loss: 0.2867 | Val Loss: 0.2600
Train Recon: 0.0402 | Val Recon: 0.0389
Train VQ: 0.2465 | Val VQ: 0.2212

Train Perplexity: 14.17 | Val Perplexity: 15.09

✓ Best model saved! (Val Loss: 0.2600)

Best Loss: 0.2600 (Epoch 8)

Epoch 9/500, Batch 100/782, Loss: 0.2304, Recon: 0.0373, VQ: 0.1932, Perplexity: 15.64

Epoch 9/500, Batch 200/782, Loss: 0.2633, Recon: 0.0378, VQ: 0.2255, Perplexity: 15.70

Epoch 9/500, Batch 300/782, Loss: 0.2685, Recon: 0.0411, VQ: 0.2274, Perplexity: 16.01

Epoch 9/500, Batch 400/782, Loss: 0.2548, Recon: 0.0379, VQ: 0.2169, Perplexity: 15.90

Epoch 9/500, Batch 500/782, Loss: 0.2274, Recon: 0.0356, VQ: 0.1918, Perplexity: 16.43

Epoch 9/500, Batch 600/782, Loss: 0.2413, Recon: 0.0367, VQ: 0.2046, Perplexity: 16.81

Epoch 9/500, Batch 700/782, Loss: 0.2644, Recon: 0.0371, VQ: 0.2274, Perplexity: 17.35

=== Epoch 9/500 Summary ===

Train Loss: 0.2451 | Val Loss: 0.2314

Train Recon: 0.0376 | Val Recon: 0.0365

Train VQ: 0.2075 | Val VQ: 0.1949

Train Perplexity: 15.90 | Val Perplexity: 16.66

✓ Best model saved! (Val Loss: 0.2314)

Best Loss: 0.2314 (Epoch 9)

Epoch 10/500, Batch 100/782, Loss: 0.2316, Recon: 0.0375, VQ: 0.1941, Perplexity: 16.88

Epoch 10/500, Batch 200/782, Loss: 0.2237, Recon: 0.0373, VQ: 0.1864, Perplexity: 16.82

Epoch 10/500, Batch 300/782, Loss: 0.1931, Recon: 0.0333, VQ: 0.1598, Perplexity: 17.47

Epoch 10/500, Batch 400/782, Loss: 0.1800, Recon: 0.0305, VQ: 0.1495, Perplexity: 16.84

Epoch 10/500, Batch 500/782, Loss: 0.2418, Recon: 0.0377, VQ: 0.2042, Perplexity: 17.90

Epoch 10/500, Batch 600/782, Loss: 0.1998, Recon: 0.0336, VQ: 0.1661, Perplexity: 17.05

Epoch 10/500, Batch 700/782, Loss: 0.2327, Recon: 0.0373, VQ: 0.1954, Perplexity: 17.94

=== Epoch 10/500 Summary ===

Train Loss: 0.2198 | Val Loss: 0.2091

Train Recon: 0.0356 | Val Recon: 0.0348

Train VQ: 0.1842 | Val VQ: 0.1743

Train Perplexity: 17.44 | Val Perplexity: 17.92

✓ Best model saved! (Val Loss: 0.2091)

Best Loss: 0.2091 (Epoch 10)

Epoch 11/500, Batch 100/782, Loss: 0.2277, Recon: 0.0360, VQ: 0.1918, Perplexity: 18.52

Epoch 11/500, Batch 200/782, Loss: 0.2015, Recon: 0.0336, VQ: 0.1679, Perplexity: 17.83

Epoch 11/500, Batch 300/782, Loss: 0.2022, Recon: 0.0340, VQ: 0.1682, Perplexity: 18.57

Epoch 11/500, Batch 400/782, Loss: 0.1894, Recon: 0.0323, VQ: 0.1571, Perplexity: 17.53

Epoch 11/500, Batch 500/782, Loss: 0.2103, Recon: 0.0343, VQ: 0.1761, Perplexity: 18.00

Epoch 11/500, Batch 600/782, Loss: 0.2505, Recon: 0.0373, VQ: 0.2132, Perplexity: 18.29
Epoch 11/500, Batch 700/782, Loss: 0.1840, Recon: 0.0325, VQ: 0.1516, Perplexity: 18.11

=== Epoch 11/500 Summary ===

Train Loss: 0.2011 | Val Loss: 0.1933
Train Recon: 0.0340 | Val Recon: 0.0335
Train VQ: 0.1671 | Val VQ: 0.1599
Train Perplexity: 18.07 | Val Perplexity: 18.29
✓ Best model saved! (Val Loss: 0.1933)
Best Loss: 0.1933 (Epoch 11)

Epoch 12/500, Batch 100/782, Loss: 0.2128, Recon: 0.0359, VQ: 0.1769, Perplexity: 18.94
Epoch 12/500, Batch 200/782, Loss: 0.1907, Recon: 0.0327, VQ: 0.1580, Perplexity: 18.28
Epoch 12/500, Batch 300/782, Loss: 0.2129, Recon: 0.0340, VQ: 0.1790, Perplexity: 18.46
Epoch 12/500, Batch 400/782, Loss: 0.2011, Recon: 0.0347, VQ: 0.1664, Perplexity: 19.69
Epoch 12/500, Batch 500/782, Loss: 0.1889, Recon: 0.0327, VQ: 0.1562, Perplexity: 19.80
Epoch 12/500, Batch 600/782, Loss: 0.1978, Recon: 0.0351, VQ: 0.1626, Perplexity: 19.18
Epoch 12/500, Batch 700/782, Loss: 0.1597, Recon: 0.0312, VQ: 0.1284, Perplexity: 20.03

=== Epoch 12/500 Summary ===

Train Loss: 0.1883 | Val Loss: 0.1819
Train Recon: 0.0329 | Val Recon: 0.0323
Train VQ: 0.1553 | Val VQ: 0.1496
Train Perplexity: 18.92 | Val Perplexity: 19.55
✓ Best model saved! (Val Loss: 0.1819)
Best Loss: 0.1819 (Epoch 12)

Epoch 13/500, Batch 100/782, Loss: 0.1916, Recon: 0.0323, VQ: 0.1593, Perplexity: 20.67
Epoch 13/500, Batch 200/782, Loss: 0.1808, Recon: 0.0324, VQ: 0.1484, Perplexity: 21.28
Epoch 13/500, Batch 300/782, Loss: 0.1758, Recon: 0.0338, VQ: 0.1420, Perplexity: 20.28
Epoch 13/500, Batch 400/782, Loss: 0.1717, Recon: 0.0292, VQ: 0.1425, Perplexity: 20.74
Epoch 13/500, Batch 500/782, Loss: 0.1648, Recon: 0.0302, VQ: 0.1346, Perplexity: 19.47
Epoch 13/500, Batch 600/782, Loss: 0.1641, Recon: 0.0313, VQ: 0.1329, Perplexity: 20.98
Epoch 13/500, Batch 700/782, Loss: 0.1681, Recon: 0.0313, VQ: 0.1368, Perplexity: 21.09

=== Epoch 13/500 Summary ===

Train Loss: 0.1770 | Val Loss: 0.1679
Train Recon: 0.0317 | Val Recon: 0.0316
Train VQ: 0.1453 | Val VQ: 0.1363
Train Perplexity: 20.51 | Val Perplexity: 21.20
✓ Best model saved! (Val Loss: 0.1679)
Best Loss: 0.1679 (Epoch 13)

Epoch 14/500, Batch 100/782, Loss: 0.2358, Recon: 0.0376, VQ: 0.1982, Perplexity:

lexity: 20.99
Epoch 14/500, Batch 200/782, Loss: 0.1663, Recon: 0.0299, VQ: 0.1365, Perp
lexity: 21.06
Epoch 14/500, Batch 300/782, Loss: 0.1805, Recon: 0.0323, VQ: 0.1482, Perp
lexity: 23.17
Epoch 14/500, Batch 400/782, Loss: 0.1742, Recon: 0.0320, VQ: 0.1422, Perp
lexity: 22.73
Epoch 14/500, Batch 500/782, Loss: 0.1674, Recon: 0.0305, VQ: 0.1369, Perp
lexity: 21.87
Epoch 14/500, Batch 600/782, Loss: 0.1590, Recon: 0.0290, VQ: 0.1300, Perp
lexity: 22.05
Epoch 14/500, Batch 700/782, Loss: 0.1543, Recon: 0.0273, VQ: 0.1270, Perp
lexity: 22.74

=== Epoch 14/500 Summary ===

Train Loss: 0.1696 | Val Loss: 0.1668
Train Recon: 0.0307 | Val Recon: 0.0304
Train VQ: 0.1388 | Val VQ: 0.1363
Train Perplexity: 21.87 | Val Perplexity: 22.38
✓ Best model saved! (Val Loss: 0.1668)
Best Loss: 0.1668 (Epoch 14)

Epoch 15/500, Batch 100/782, Loss: 0.1510, Recon: 0.0287, VQ: 0.1223, Perp
lexity: 21.97
Epoch 15/500, Batch 200/782, Loss: 0.1463, Recon: 0.0279, VQ: 0.1183, Perp
lexity: 22.31
Epoch 15/500, Batch 300/782, Loss: 0.1684, Recon: 0.0307, VQ: 0.1377, Perp
lexity: 22.67
Epoch 15/500, Batch 400/782, Loss: 0.1600, Recon: 0.0313, VQ: 0.1287, Perp
lexity: 23.60
Epoch 15/500, Batch 500/782, Loss: 0.1633, Recon: 0.0296, VQ: 0.1337, Perp
lexity: 23.35
Epoch 15/500, Batch 600/782, Loss: 0.1277, Recon: 0.0256, VQ: 0.1022, Perp
lexity: 22.20
Epoch 15/500, Batch 700/782, Loss: 0.1725, Recon: 0.0306, VQ: 0.1418, Perp
lexity: 23.91

=== Epoch 15/500 Summary ===

Train Loss: 0.1621 | Val Loss: 0.1594
Train Recon: 0.0298 | Val Recon: 0.0293
Train VQ: 0.1323 | Val VQ: 0.1300
Train Perplexity: 23.12 | Val Perplexity: 23.86
✓ Best model saved! (Val Loss: 0.1594)
Best Loss: 0.1594 (Epoch 15)

Epoch 16/500, Batch 100/782, Loss: 0.1585, Recon: 0.0285, VQ: 0.1300, Perp
lexity: 24.33
Epoch 16/500, Batch 200/782, Loss: 0.1470, Recon: 0.0283, VQ: 0.1187, Perp
lexity: 24.29
Epoch 16/500, Batch 300/782, Loss: 0.1312, Recon: 0.0258, VQ: 0.1054, Perp
lexity: 23.67
Epoch 16/500, Batch 400/782, Loss: 0.1489, Recon: 0.0294, VQ: 0.1196, Perp
lexity: 24.24
Epoch 16/500, Batch 500/782, Loss: 0.1480, Recon: 0.0301, VQ: 0.1179, Perp
lexity: 25.58
Epoch 16/500, Batch 600/782, Loss: 0.1699, Recon: 0.0297, VQ: 0.1402, Perp
lexity: 25.37
Epoch 16/500, Batch 700/782, Loss: 0.1753, Recon: 0.0313, VQ: 0.1441, Perp
lexity: 25.53

=== Epoch 16/500 Summary ===

Train Loss: 0.1552 | Val Loss: 0.1521

Train Recon: 0.0290 | Val Recon: 0.0287

Train VQ: 0.1262 | Val VQ: 0.1233

Train Perplexity: 24.59 | Val Perplexity: 25.26

✓ Best model saved! (Val Loss: 0.1521)

Best Loss: 0.1521 (Epoch 16)

Epoch 17/500, Batch 100/782, Loss: 0.1502, Recon: 0.0288, VQ: 0.1214, Perplexity: 25.59

Epoch 17/500, Batch 200/782, Loss: 0.1376, Recon: 0.0280, VQ: 0.1095, Perplexity: 25.02

Epoch 17/500, Batch 300/782, Loss: 0.1447, Recon: 0.0279, VQ: 0.1168, Perplexity: 25.73

Epoch 17/500, Batch 400/782, Loss: 0.1518, Recon: 0.0298, VQ: 0.1220, Perplexity: 25.70

Epoch 17/500, Batch 500/782, Loss: 0.1569, Recon: 0.0283, VQ: 0.1286, Perplexity: 25.80

Epoch 17/500, Batch 600/782, Loss: 0.1363, Recon: 0.0255, VQ: 0.1108, Perplexity: 25.13

Epoch 17/500, Batch 700/782, Loss: 0.1767, Recon: 0.0294, VQ: 0.1473, Perplexity: 26.35

=== Epoch 17/500 Summary ===

Train Loss: 0.1495 | Val Loss: 0.1462

Train Recon: 0.0283 | Val Recon: 0.0282

Train VQ: 0.1212 | Val VQ: 0.1180

Train Perplexity: 25.63 | Val Perplexity: 25.94

✓ Best model saved! (Val Loss: 0.1462)

Best Loss: 0.1462 (Epoch 17)

Epoch 18/500, Batch 100/782, Loss: 0.1660, Recon: 0.0311, VQ: 0.1349, Perplexity: 25.97

Epoch 18/500, Batch 200/782, Loss: 0.1580, Recon: 0.0286, VQ: 0.1294, Perplexity: 25.93

Epoch 18/500, Batch 300/782, Loss: 0.1525, Recon: 0.0299, VQ: 0.1226, Perplexity: 27.08

Epoch 18/500, Batch 400/782, Loss: 0.1773, Recon: 0.0322, VQ: 0.1451, Perplexity: 27.60

Epoch 18/500, Batch 500/782, Loss: 0.1282, Recon: 0.0262, VQ: 0.1021, Perplexity: 26.61

Epoch 18/500, Batch 600/782, Loss: 0.1548, Recon: 0.0285, VQ: 0.1263, Perplexity: 26.42

Epoch 18/500, Batch 700/782, Loss: 0.1279, Recon: 0.0258, VQ: 0.1021, Perplexity: 26.37

=== Epoch 18/500 Summary ===

Train Loss: 0.1457 | Val Loss: 0.1456

Train Recon: 0.0278 | Val Recon: 0.0276

Train VQ: 0.1179 | Val VQ: 0.1180

Train Perplexity: 26.44 | Val Perplexity: 26.88

✓ Best model saved! (Val Loss: 0.1456)

Best Loss: 0.1456 (Epoch 18)

Epoch 19/500, Batch 100/782, Loss: 0.1209, Recon: 0.0244, VQ: 0.0966, Perplexity: 26.52

Epoch 19/500, Batch 200/782, Loss: 0.1463, Recon: 0.0292, VQ: 0.1171, Perplexity: 26.97

Epoch 19/500, Batch 300/782, Loss: 0.1342, Recon: 0.0269, VQ: 0.1072, Perplexity: 26.95

Epoch 19/500, Batch 400/782, Loss: 0.1366, Recon: 0.0282, VQ: 0.1084, Perplexity: 26.70
Epoch 19/500, Batch 500/782, Loss: 0.1497, Recon: 0.0290, VQ: 0.1207, Perplexity: 26.90
Epoch 19/500, Batch 600/782, Loss: 0.1440, Recon: 0.0273, VQ: 0.1167, Perplexity: 27.57
Epoch 19/500, Batch 700/782, Loss: 0.1268, Recon: 0.0258, VQ: 0.1010, Perplexity: 26.77

=== Epoch 19/500 Summary ===

Train Loss: 0.1430 | Val Loss: 0.1428
Train Recon: 0.0273 | Val Recon: 0.0273
Train VQ: 0.1156 | Val VQ: 0.1155
Train Perplexity: 26.87 | Val Perplexity: 27.06
✓ Best model saved! (Val Loss: 0.1428)
Best Loss: 0.1428 (Epoch 19)

Epoch 20/500, Batch 100/782, Loss: 0.1349, Recon: 0.0268, VQ: 0.1081, Perplexity: 27.13
Epoch 20/500, Batch 200/782, Loss: 0.1477, Recon: 0.0287, VQ: 0.1190, Perplexity: 27.87
Epoch 20/500, Batch 300/782, Loss: 0.1317, Recon: 0.0256, VQ: 0.1061, Perplexity: 26.73
Epoch 20/500, Batch 400/782, Loss: 0.1358, Recon: 0.0275, VQ: 0.1083, Perplexity: 26.45
Epoch 20/500, Batch 500/782, Loss: 0.1498, Recon: 0.0298, VQ: 0.1200, Perplexity: 28.06
Epoch 20/500, Batch 600/782, Loss: 0.1453, Recon: 0.0275, VQ: 0.1178, Perplexity: 27.79
Epoch 20/500, Batch 700/782, Loss: 0.1480, Recon: 0.0294, VQ: 0.1186, Perplexity: 28.02

=== Epoch 20/500 Summary ===

Train Loss: 0.1408 | Val Loss: 0.1404
Train Recon: 0.0269 | Val Recon: 0.0271
Train VQ: 0.1139 | Val VQ: 0.1133
Train Perplexity: 27.31 | Val Perplexity: 27.43
✓ Best model saved! (Val Loss: 0.1404)
Best Loss: 0.1404 (Epoch 20)

Epoch 21/500, Batch 100/782, Loss: 0.1374, Recon: 0.0273, VQ: 0.1101, Perplexity: 28.64
Epoch 21/500, Batch 200/782, Loss: 0.1402, Recon: 0.0272, VQ: 0.1130, Perplexity: 28.24
Epoch 21/500, Batch 300/782, Loss: 0.1279, Recon: 0.0241, VQ: 0.1038, Perplexity: 27.38
Epoch 21/500, Batch 400/782, Loss: 0.1359, Recon: 0.0270, VQ: 0.1089, Perplexity: 26.21
Epoch 21/500, Batch 500/782, Loss: 0.1380, Recon: 0.0273, VQ: 0.1107, Perplexity: 28.52
Epoch 21/500, Batch 600/782, Loss: 0.1897, Recon: 0.0273, VQ: 0.1624, Perplexity: 27.76
Epoch 21/500, Batch 700/782, Loss: 0.1446, Recon: 0.0263, VQ: 0.1182, Perplexity: 27.99

=== Epoch 21/500 Summary ===

Train Loss: 0.1390 | Val Loss: 0.1383
Train Recon: 0.0266 | Val Recon: 0.0266
Train VQ: 0.1124 | Val VQ: 0.1117
Train Perplexity: 27.64 | Val Perplexity: 27.83

✓ Best model saved! (Val Loss: 0.1383)
Best Loss: 0.1383 (Epoch 21)

Epoch 22/500, Batch 100/782, Loss: 0.1545, Recon: 0.0277, VQ: 0.1268, Perplexity: 28.76
Epoch 22/500, Batch 200/782, Loss: 0.1282, Recon: 0.0264, VQ: 0.1018, Perplexity: 27.55
Epoch 22/500, Batch 300/782, Loss: 0.1464, Recon: 0.0292, VQ: 0.1172, Perplexity: 29.44
Epoch 22/500, Batch 400/782, Loss: 0.1261, Recon: 0.0242, VQ: 0.1019, Perplexity: 28.08
Epoch 22/500, Batch 500/782, Loss: 0.1300, Recon: 0.0246, VQ: 0.1054, Perplexity: 28.11
Epoch 22/500, Batch 600/782, Loss: 0.1420, Recon: 0.0261, VQ: 0.1159, Perplexity: 29.06
Epoch 22/500, Batch 700/782, Loss: 0.1574, Recon: 0.0272, VQ: 0.1302, Perplexity: 29.20

=== Epoch 22/500 Summary ===
Train Loss: 0.1375 | Val Loss: 0.1370
Train Recon: 0.0262 | Val Recon: 0.0262
Train VQ: 0.1112 | Val VQ: 0.1108
Train Perplexity: 28.43 | Val Perplexity: 28.72
✓ Best model saved! (Val Loss: 0.1370)
Best Loss: 0.1370 (Epoch 22)

Epoch 23/500, Batch 100/782, Loss: 0.1353, Recon: 0.0259, VQ: 0.1094, Perplexity: 28.48
Epoch 23/500, Batch 200/782, Loss: 0.1350, Recon: 0.0262, VQ: 0.1088, Perplexity: 28.67
Epoch 23/500, Batch 300/782, Loss: 0.1386, Recon: 0.0258, VQ: 0.1128, Perplexity: 28.15
Epoch 23/500, Batch 400/782, Loss: 0.1293, Recon: 0.0267, VQ: 0.1026, Perplexity: 28.88
Epoch 23/500, Batch 500/782, Loss: 0.1870, Recon: 0.0284, VQ: 0.1587, Perplexity: 29.76
Epoch 23/500, Batch 600/782, Loss: 0.1265, Recon: 0.0236, VQ: 0.1029, Perplexity: 28.93
Epoch 23/500, Batch 700/782, Loss: 0.1369, Recon: 0.0251, VQ: 0.1118, Perplexity: 29.79

=== Epoch 23/500 Summary ===
Train Loss: 0.1362 | Val Loss: 0.1363
Train Recon: 0.0259 | Val Recon: 0.0260
Train VQ: 0.1103 | Val VQ: 0.1103
Train Perplexity: 28.96 | Val Perplexity: 29.37
✓ Best model saved! (Val Loss: 0.1363)
Best Loss: 0.1363 (Epoch 23)

Epoch 24/500, Batch 100/782, Loss: 0.1205, Recon: 0.0225, VQ: 0.0980, Perplexity: 28.39
Epoch 24/500, Batch 200/782, Loss: 0.1316, Recon: 0.0260, VQ: 0.1056, Perplexity: 29.58
Epoch 24/500, Batch 300/782, Loss: 0.1300, Recon: 0.0247, VQ: 0.1053, Perplexity: 29.53
Epoch 24/500, Batch 400/782, Loss: 0.1360, Recon: 0.0269, VQ: 0.1091, Perplexity: 29.99
Epoch 24/500, Batch 500/782, Loss: 0.1385, Recon: 0.0276, VQ: 0.1109, Perplexity: 30.49
Epoch 24/500, Batch 600/782, Loss: 0.1281, Recon: 0.0260, VQ: 0.1021, Perplexity: 29.20

lexity: 30.57

Epoch 24/500, Batch 700/782, Loss: 0.1214, Recon: 0.0247, VQ: 0.0967, Perp
lexity: 30.20

=== Epoch 24/500 Summary ===

Train Loss: 0.1351 | Val Loss: 0.1346

Train Recon: 0.0256 | Val Recon: 0.0256

Train VQ: 0.1095 | Val VQ: 0.1089

Train Perplexity: 29.76 | Val Perplexity: 30.27

✓ Best model saved! (Val Loss: 0.1346)

Best Loss: 0.1346 (Epoch 24)

Epoch 25/500, Batch 100/782, Loss: 0.1458, Recon: 0.0265, VQ: 0.1193, Perp
lexity: 30.08

Epoch 25/500, Batch 200/782, Loss: 0.1179, Recon: 0.0230, VQ: 0.0949, Perp
lexity: 29.74

Epoch 25/500, Batch 300/782, Loss: 0.1332, Recon: 0.0257, VQ: 0.1074, Perp
lexity: 30.35

Epoch 25/500, Batch 400/782, Loss: 0.1423, Recon: 0.0275, VQ: 0.1148, Perp
lexity: 31.38

Epoch 25/500, Batch 500/782, Loss: 0.2290, Recon: 0.0244, VQ: 0.2046, Perp
lexity: 30.81

Epoch 25/500, Batch 600/782, Loss: 0.1422, Recon: 0.0276, VQ: 0.1145, Perp
lexity: 31.17

Epoch 25/500, Batch 700/782, Loss: 0.1346, Recon: 0.0262, VQ: 0.1083, Perp
lexity: 30.69

=== Epoch 25/500 Summary ===

Train Loss: 0.1339 | Val Loss: 0.1333

Train Recon: 0.0253 | Val Recon: 0.0254

Train VQ: 0.1086 | Val VQ: 0.1079

Train Perplexity: 30.41 | Val Perplexity: 30.65

✓ Best model saved! (Val Loss: 0.1333)

Best Loss: 0.1333 (Epoch 25)

Epoch 26/500, Batch 100/782, Loss: 0.1270, Recon: 0.0259, VQ: 0.1011, Perp
lexity: 30.86

Epoch 26/500, Batch 200/782, Loss: 0.1401, Recon: 0.0261, VQ: 0.1140, Perp
lexity: 32.25

Epoch 26/500, Batch 300/782, Loss: 0.1159, Recon: 0.0236, VQ: 0.0923, Perp
lexity: 31.31

Epoch 26/500, Batch 400/782, Loss: 0.1862, Recon: 0.0259, VQ: 0.1603, Perp
lexity: 32.03

Epoch 26/500, Batch 500/782, Loss: 0.1342, Recon: 0.0249, VQ: 0.1093, Perp
lexity: 31.17

Epoch 26/500, Batch 600/782, Loss: 0.1306, Recon: 0.0249, VQ: 0.1057, Perp
lexity: 32.18

Epoch 26/500, Batch 700/782, Loss: 0.1201, Recon: 0.0238, VQ: 0.0963, Perp
lexity: 31.38

=== Epoch 26/500 Summary ===

Train Loss: 0.1327 | Val Loss: 0.1326

Train Recon: 0.0250 | Val Recon: 0.0250

Train VQ: 0.1077 | Val VQ: 0.1076

Train Perplexity: 31.40 | Val Perplexity: 31.87

✓ Best model saved! (Val Loss: 0.1326)

Best Loss: 0.1326 (Epoch 26)

Epoch 27/500, Batch 100/782, Loss: 0.1346, Recon: 0.0257, VQ: 0.1089, Perp
lexity: 32.59

Epoch 27/500, Batch 200/782, Loss: 0.1307, Recon: 0.0252, VQ: 0.1055, Perplexity: 32.46
Epoch 27/500, Batch 300/782, Loss: 0.1548, Recon: 0.0254, VQ: 0.1294, Perplexity: 32.08
Epoch 27/500, Batch 400/782, Loss: 0.1319, Recon: 0.0249, VQ: 0.1070, Perplexity: 31.63
Epoch 27/500, Batch 500/782, Loss: 0.1285, Recon: 0.0242, VQ: 0.1044, Perplexity: 32.93
Epoch 27/500, Batch 600/782, Loss: 0.1364, Recon: 0.0265, VQ: 0.1099, Perplexity: 33.23
Epoch 27/500, Batch 700/782, Loss: 0.1160, Recon: 0.0227, VQ: 0.0933, Perplexity: 31.82

=== Epoch 27/500 Summary ===

Train Loss: 0.1311 | Val Loss: 0.1318
Train Recon: 0.0247 | Val Recon: 0.0247
Train VQ: 0.1063 | Val VQ: 0.1071
Train Perplexity: 32.37 | Val Perplexity: 32.98
✓ Best model saved! (Val Loss: 0.1318)
Best Loss: 0.1318 (Epoch 27)

Epoch 28/500, Batch 100/782, Loss: 0.1281, Recon: 0.0248, VQ: 0.1033, Perplexity: 33.00
Epoch 28/500, Batch 200/782, Loss: 0.1455, Recon: 0.0267, VQ: 0.1188, Perplexity: 33.67
Epoch 28/500, Batch 300/782, Loss: 0.1339, Recon: 0.0243, VQ: 0.1096, Perplexity: 32.72
Epoch 28/500, Batch 400/782, Loss: 0.1511, Recon: 0.0245, VQ: 0.1266, Perplexity: 33.53
Epoch 28/500, Batch 500/782, Loss: 0.1287, Recon: 0.0241, VQ: 0.1046, Perplexity: 31.79
Epoch 28/500, Batch 600/782, Loss: 0.1161, Recon: 0.0225, VQ: 0.0936, Perplexity: 31.77
Epoch 28/500, Batch 700/782, Loss: 0.1280, Recon: 0.0242, VQ: 0.1037, Perplexity: 33.10

=== Epoch 28/500 Summary ===

Train Loss: 0.1301 | Val Loss: 0.1299
Train Recon: 0.0245 | Val Recon: 0.0246
Train VQ: 0.1056 | Val VQ: 0.1053
Train Perplexity: 33.00 | Val Perplexity: 33.21
✓ Best model saved! (Val Loss: 0.1299)
Best Loss: 0.1299 (Epoch 28)

Epoch 29/500, Batch 100/782, Loss: 0.1507, Recon: 0.0254, VQ: 0.1253, Perplexity: 34.15
Epoch 29/500, Batch 200/782, Loss: 0.1104, Recon: 0.0227, VQ: 0.0877, Perplexity: 32.50
Epoch 29/500, Batch 300/782, Loss: 0.1225, Recon: 0.0234, VQ: 0.0991, Perplexity: 33.33
Epoch 29/500, Batch 400/782, Loss: 0.1355, Recon: 0.0257, VQ: 0.1098, Perplexity: 34.42
Epoch 29/500, Batch 500/782, Loss: 0.1210, Recon: 0.0240, VQ: 0.0970, Perplexity: 32.99
Epoch 29/500, Batch 600/782, Loss: 0.1403, Recon: 0.0266, VQ: 0.1137, Perplexity: 34.54
Epoch 29/500, Batch 700/782, Loss: 0.1424, Recon: 0.0260, VQ: 0.1164, Perplexity: 34.37

=== Epoch 29/500 Summary ===

Train Loss: 0.1295 | Val Loss: 0.1305
Train Recon: 0.0243 | Val Recon: 0.0245
Train VQ: 0.1052 | Val VQ: 0.1060
Train Perplexity: 33.65 | Val Perplexity: 33.92
No improvement. Patience: 1/20
Best Loss: 0.1299 (Epoch 28)

Epoch 30/500, Batch 100/782, Loss: 0.1243, Recon: 0.0235, VQ: 0.1008, Perplexity: 34.46
Epoch 30/500, Batch 200/782, Loss: 0.1324, Recon: 0.0256, VQ: 0.1068, Perplexity: 35.34
Epoch 30/500, Batch 300/782, Loss: 0.1154, Recon: 0.0232, VQ: 0.0922, Perplexity: 34.21
Epoch 30/500, Batch 400/782, Loss: 0.1267, Recon: 0.0238, VQ: 0.1030, Perplexity: 33.86
Epoch 30/500, Batch 500/782, Loss: 0.1405, Recon: 0.0238, VQ: 0.1167, Perplexity: 33.90
Epoch 30/500, Batch 600/782, Loss: 0.1327, Recon: 0.0234, VQ: 0.1093, Perplexity: 33.25
Epoch 30/500, Batch 700/782, Loss: 0.1452, Recon: 0.0264, VQ: 0.1188, Perplexity: 35.63

=== Epoch 30/500 Summary ===
Train Loss: 0.1292 | Val Loss: 0.1301
Train Recon: 0.0241 | Val Recon: 0.0242
Train VQ: 0.1051 | Val VQ: 0.1059
Train Perplexity: 34.12 | Val Perplexity: 34.40
No improvement. Patience: 2/20
Best Loss: 0.1299 (Epoch 28)

Epoch 31/500, Batch 100/782, Loss: 0.1347, Recon: 0.0250, VQ: 0.1097, Perplexity: 34.82
Epoch 31/500, Batch 200/782, Loss: 0.1302, Recon: 0.0254, VQ: 0.1048, Perplexity: 35.01
Epoch 31/500, Batch 300/782, Loss: 0.1196, Recon: 0.0223, VQ: 0.0973, Perplexity: 34.05
Epoch 31/500, Batch 400/782, Loss: 0.1286, Recon: 0.0245, VQ: 0.1041, Perplexity: 35.63
Epoch 31/500, Batch 500/782, Loss: 0.1306, Recon: 0.0255, VQ: 0.1052, Perplexity: 32.97
Epoch 31/500, Batch 600/782, Loss: 0.1275, Recon: 0.0233, VQ: 0.1042, Perplexity: 34.61
Epoch 31/500, Batch 700/782, Loss: 0.1152, Recon: 0.0223, VQ: 0.0929, Perplexity: 33.51

=== Epoch 31/500 Summary ===
Train Loss: 0.1291 | Val Loss: 0.1290
Train Recon: 0.0240 | Val Recon: 0.0242
Train VQ: 0.1051 | Val VQ: 0.1048
Train Perplexity: 34.41 | Val Perplexity: 34.38
✓ Best model saved! (Val Loss: 0.1290)
Best Loss: 0.1290 (Epoch 31)

Epoch 32/500, Batch 100/782, Loss: 0.1140, Recon: 0.0221, VQ: 0.0918, Perplexity: 34.94
Epoch 32/500, Batch 200/782, Loss: 0.1256, Recon: 0.0235, VQ: 0.1020, Perplexity: 33.70
Epoch 32/500, Batch 300/782, Loss: 0.1222, Recon: 0.0235, VQ: 0.0987, Perplexity: 33.99
Epoch 32/500, Batch 400/782, Loss: 0.1275, Recon: 0.0246, VQ: 0.1029, Perplexity:

lexity: 35.47
Epoch 32/500, Batch 500/782, Loss: 0.1226, Recon: 0.0235, VQ: 0.0992, Perp
lexity: 34.24
Epoch 32/500, Batch 600/782, Loss: 0.1216, Recon: 0.0222, VQ: 0.0994, Perp
lexity: 32.72
Epoch 32/500, Batch 700/782, Loss: 0.1314, Recon: 0.0263, VQ: 0.1052, Perp
lexity: 35.17

=== Epoch 32/500 Summary ===

Train Loss: 0.1284 | Val Loss: 0.1291
Train Recon: 0.0238 | Val Recon: 0.0241
Train VQ: 0.1046 | Val VQ: 0.1050
Train Perplexity: 34.68 | Val Perplexity: 34.45
No improvement. Patience: 1/20
Best Loss: 0.1290 (Epoch 31)

Epoch 33/500, Batch 100/782, Loss: 0.1261, Recon: 0.0235, VQ: 0.1026, Perp
lexity: 33.94
Epoch 33/500, Batch 200/782, Loss: 0.1262, Recon: 0.0224, VQ: 0.1038, Perp
lexity: 34.61
Epoch 33/500, Batch 300/782, Loss: 0.1316, Recon: 0.0255, VQ: 0.1061, Perp
lexity: 35.69
Epoch 33/500, Batch 400/782, Loss: 0.1646, Recon: 0.0273, VQ: 0.1373, Perp
lexity: 37.81
Epoch 33/500, Batch 500/782, Loss: 0.1418, Recon: 0.0239, VQ: 0.1179, Perp
lexity: 35.16
Epoch 33/500, Batch 600/782, Loss: 0.1258, Recon: 0.0243, VQ: 0.1014, Perp
lexity: 36.21
Epoch 33/500, Batch 700/782, Loss: 0.1169, Recon: 0.0232, VQ: 0.0937, Perp
lexity: 34.82

=== Epoch 33/500 Summary ===

Train Loss: 0.1281 | Val Loss: 0.1289
Train Recon: 0.0237 | Val Recon: 0.0238
Train VQ: 0.1044 | Val VQ: 0.1051
Train Perplexity: 34.93 | Val Perplexity: 35.23
✓ Best model saved! (Val Loss: 0.1289)
Best Loss: 0.1289 (Epoch 33)

Epoch 34/500, Batch 100/782, Loss: 0.1241, Recon: 0.0238, VQ: 0.1003, Perp
lexity: 34.58
Epoch 34/500, Batch 200/782, Loss: 0.1316, Recon: 0.0241, VQ: 0.1075, Perp
lexity: 36.42
Epoch 34/500, Batch 300/782, Loss: 0.1360, Recon: 0.0249, VQ: 0.1112, Perp
lexity: 35.37
Epoch 34/500, Batch 400/782, Loss: 0.1174, Recon: 0.0224, VQ: 0.0950, Perp
lexity: 35.28
Epoch 34/500, Batch 500/782, Loss: 0.1265, Recon: 0.0230, VQ: 0.1035, Perp
lexity: 33.98
Epoch 34/500, Batch 600/782, Loss: 0.1257, Recon: 0.0242, VQ: 0.1015, Perp
lexity: 37.31
Epoch 34/500, Batch 700/782, Loss: 0.1178, Recon: 0.0230, VQ: 0.0948, Perp
lexity: 34.56

=== Epoch 34/500 Summary ===

Train Loss: 0.1276 | Val Loss: 0.1297
Train Recon: 0.0236 | Val Recon: 0.0237
Train VQ: 0.1040 | Val VQ: 0.1060
Train Perplexity: 35.28 | Val Perplexity: 35.71
No improvement. Patience: 1/20

Best Loss: 0.1289 (Epoch 33)

Epoch 35/500, Batch 100/782, Loss: 0.1247, Recon: 0.0238, VQ: 0.1009, Perplexity: 36.58
Epoch 35/500, Batch 200/782, Loss: 0.1128, Recon: 0.0223, VQ: 0.0905, Perplexity: 35.62
Epoch 35/500, Batch 300/782, Loss: 0.1251, Recon: 0.0235, VQ: 0.1016, Perplexity: 36.05
Epoch 35/500, Batch 400/782, Loss: 0.1148, Recon: 0.0225, VQ: 0.0923, Perplexity: 35.34
Epoch 35/500, Batch 500/782, Loss: 0.1308, Recon: 0.0243, VQ: 0.1065, Perplexity: 35.64
Epoch 35/500, Batch 600/782, Loss: 0.1312, Recon: 0.0233, VQ: 0.1079, Perplexity: 35.25
Epoch 35/500, Batch 700/782, Loss: 0.1202, Recon: 0.0239, VQ: 0.0963, Perplexity: 34.96

=== Epoch 35/500 Summary ===

Train Loss: 0.1273 | Val Loss: 0.1264
Train Recon: 0.0234 | Val Recon: 0.0236
Train VQ: 0.1039 | Val VQ: 0.1028
Train Perplexity: 35.71 | Val Perplexity: 35.72
✓ Best model saved! (Val Loss: 0.1264)
Best Loss: 0.1264 (Epoch 35)

Epoch 36/500, Batch 100/782, Loss: 0.1260, Recon: 0.0236, VQ: 0.1024, Perplexity: 35.67
Epoch 36/500, Batch 200/782, Loss: 0.1206, Recon: 0.0225, VQ: 0.0980, Perplexity: 34.83
Epoch 36/500, Batch 300/782, Loss: 0.1195, Recon: 0.0229, VQ: 0.0965, Perplexity: 36.68
Epoch 36/500, Batch 400/782, Loss: 0.1374, Recon: 0.0238, VQ: 0.1136, Perplexity: 36.60
Epoch 36/500, Batch 500/782, Loss: 0.1182, Recon: 0.0229, VQ: 0.0953, Perplexity: 35.48
Epoch 36/500, Batch 600/782, Loss: 0.2223, Recon: 0.0230, VQ: 0.1993, Perplexity: 36.35
Epoch 36/500, Batch 700/782, Loss: 0.1130, Recon: 0.0218, VQ: 0.0913, Perplexity: 34.33

=== Epoch 36/500 Summary ===

Train Loss: 0.1269 | Val Loss: 0.1267
Train Recon: 0.0233 | Val Recon: 0.0235
Train VQ: 0.1036 | Val VQ: 0.1033
Train Perplexity: 35.86 | Val Perplexity: 36.10
No improvement. Patience: 1/20
Best Loss: 0.1264 (Epoch 35)

Epoch 37/500, Batch 100/782, Loss: 0.1089, Recon: 0.0213, VQ: 0.0875, Perplexity: 35.86
Epoch 37/500, Batch 200/782, Loss: 0.1213, Recon: 0.0228, VQ: 0.0985, Perplexity: 35.96
Epoch 37/500, Batch 300/782, Loss: 0.1366, Recon: 0.0232, VQ: 0.1134, Perplexity: 35.49
Epoch 37/500, Batch 400/782, Loss: 0.1222, Recon: 0.0216, VQ: 0.1006, Perplexity: 35.95
Epoch 37/500, Batch 500/782, Loss: 0.1317, Recon: 0.0234, VQ: 0.1083, Perplexity: 35.74
Epoch 37/500, Batch 600/782, Loss: 0.1102, Recon: 0.0222, VQ: 0.0879, Perplexity: 35.35

Epoch 37/500, Batch 700/782, Loss: 0.1190, Recon: 0.0222, VQ: 0.0968, Perplexity: 35.33

=== Epoch 37/500 Summary ===

Train Loss: 0.1268 | Val Loss: 0.1263

Train Recon: 0.0232 | Val Recon: 0.0235

Train VQ: 0.1036 | Val VQ: 0.1028

Train Perplexity: 36.06 | Val Perplexity: 35.94

✓ Best model saved! (Val Loss: 0.1263)

Best Loss: 0.1263 (Epoch 37)

Epoch 38/500, Batch 100/782, Loss: 0.1206, Recon: 0.0228, VQ: 0.0979, Perplexity: 35.95

Epoch 38/500, Batch 200/782, Loss: 0.1262, Recon: 0.0238, VQ: 0.1025, Perplexity: 37.03

Epoch 38/500, Batch 300/782, Loss: 0.1182, Recon: 0.0229, VQ: 0.0952, Perplexity: 36.43

Epoch 38/500, Batch 400/782, Loss: 0.1282, Recon: 0.0240, VQ: 0.1041, Perplexity: 37.37

Epoch 38/500, Batch 500/782, Loss: 0.1376, Recon: 0.0238, VQ: 0.1139, Perplexity: 36.94

Epoch 38/500, Batch 600/782, Loss: 0.1134, Recon: 0.0218, VQ: 0.0916, Perplexity: 35.71

Epoch 38/500, Batch 700/782, Loss: 0.1381, Recon: 0.0244, VQ: 0.1137, Perplexity: 38.02

=== Epoch 38/500 Summary ===

Train Loss: 0.1269 | Val Loss: 0.1265

Train Recon: 0.0231 | Val Recon: 0.0234

Train VQ: 0.1038 | Val VQ: 0.1032

Train Perplexity: 36.44 | Val Perplexity: 36.57

No improvement. Patience: 1/20

Best Loss: 0.1263 (Epoch 37)

Epoch 39/500, Batch 100/782, Loss: 0.1287, Recon: 0.0224, VQ: 0.1063, Perplexity: 35.64

Epoch 39/500, Batch 200/782, Loss: 0.1236, Recon: 0.0223, VQ: 0.1013, Perplexity: 36.46

Epoch 39/500, Batch 300/782, Loss: 0.1314, Recon: 0.0236, VQ: 0.1078, Perplexity: 37.34

Epoch 39/500, Batch 400/782, Loss: 0.1293, Recon: 0.0226, VQ: 0.1067, Perplexity: 37.17

Epoch 39/500, Batch 500/782, Loss: 0.1170, Recon: 0.0221, VQ: 0.0949, Perplexity: 36.13

Epoch 39/500, Batch 600/782, Loss: 0.1152, Recon: 0.0224, VQ: 0.0928, Perplexity: 36.39

Epoch 39/500, Batch 700/782, Loss: 0.1008, Recon: 0.0205, VQ: 0.0804, Perplexity: 33.63

=== Epoch 39/500 Summary ===

Train Loss: 0.1268 | Val Loss: 0.1265

Train Recon: 0.0230 | Val Recon: 0.0233

Train VQ: 0.1038 | Val VQ: 0.1032

Train Perplexity: 36.74 | Val Perplexity: 37.02

No improvement. Patience: 2/20

Best Loss: 0.1263 (Epoch 37)

Epoch 40/500, Batch 100/782, Loss: 0.1353, Recon: 0.0228, VQ: 0.1124, Perplexity: 36.57

Epoch 40/500, Batch 200/782, Loss: 0.1339, Recon: 0.0236, VQ: 0.1103, Perplexity: 36.57

lexity: 37.85
Epoch 40/500, Batch 300/782, Loss: 0.1076, Recon: 0.0209, VQ: 0.0868, Perp
lexity: 35.50
Epoch 40/500, Batch 400/782, Loss: 0.1203, Recon: 0.0229, VQ: 0.0973, Perp
lexity: 36.63
Epoch 40/500, Batch 500/782, Loss: 0.1056, Recon: 0.0209, VQ: 0.0847, Perp
lexity: 35.93
Epoch 40/500, Batch 600/782, Loss: 0.1376, Recon: 0.0265, VQ: 0.1110, Perp
lexity: 39.22
Epoch 40/500, Batch 700/782, Loss: 0.1350, Recon: 0.0229, VQ: 0.1121, Perp
lexity: 38.81

=== Epoch 40/500 Summary ===
Train Loss: 0.1269 | Val Loss: 0.1269
Train Recon: 0.0229 | Val Recon: 0.0234
Train VQ: 0.1040 | Val VQ: 0.1035
Train Perplexity: 37.13 | Val Perplexity: 37.27
No improvement. Patience: 3/20
Best Loss: 0.1263 (Epoch 37)

Epoch 41/500, Batch 100/782, Loss: 0.1142, Recon: 0.0212, VQ: 0.0929, Perp
lexity: 36.64
Epoch 41/500, Batch 200/782, Loss: 0.1250, Recon: 0.0246, VQ: 0.1004, Perp
lexity: 38.81
Epoch 41/500, Batch 300/782, Loss: 0.1129, Recon: 0.0217, VQ: 0.0912, Perp
lexity: 37.37
Epoch 41/500, Batch 400/782, Loss: 0.1269, Recon: 0.0236, VQ: 0.1033, Perp
lexity: 38.01
Epoch 41/500, Batch 500/782, Loss: 0.1340, Recon: 0.0241, VQ: 0.1099, Perp
lexity: 38.40
Epoch 41/500, Batch 600/782, Loss: 0.1260, Recon: 0.0224, VQ: 0.1037, Perp
lexity: 38.53
Epoch 41/500, Batch 700/782, Loss: 0.1202, Recon: 0.0210, VQ: 0.0992, Perp
lexity: 36.73

=== Epoch 41/500 Summary ===
Train Loss: 0.1269 | Val Loss: 0.1267
Train Recon: 0.0228 | Val Recon: 0.0231
Train VQ: 0.1041 | Val VQ: 0.1036
Train Perplexity: 37.51 | Val Perplexity: 37.68
No improvement. Patience: 4/20
Best Loss: 0.1263 (Epoch 37)

Epoch 42/500, Batch 100/782, Loss: 0.1175, Recon: 0.0197, VQ: 0.0977, Perp
lexity: 36.56
Epoch 42/500, Batch 200/782, Loss: 0.1346, Recon: 0.0232, VQ: 0.1115, Perp
lexity: 38.26
Epoch 42/500, Batch 300/782, Loss: 0.1352, Recon: 0.0233, VQ: 0.1119, Perp
lexity: 37.87
Epoch 42/500, Batch 400/782, Loss: 0.1096, Recon: 0.0210, VQ: 0.0886, Perp
lexity: 36.11
Epoch 42/500, Batch 500/782, Loss: 0.1356, Recon: 0.0215, VQ: 0.1141, Perp
lexity: 37.43
Epoch 42/500, Batch 600/782, Loss: 0.1257, Recon: 0.0219, VQ: 0.1038, Perp
lexity: 37.67
Epoch 42/500, Batch 700/782, Loss: 0.1157, Recon: 0.0225, VQ: 0.0932, Perp
lexity: 37.42

=== Epoch 42/500 Summary ===
Train Loss: 0.1265 | Val Loss: 0.1266

Train Recon: 0.0227 | Val Recon: 0.0230
Train VQ: 0.1038 | Val VQ: 0.1036
Train Perplexity: 37.74 | Val Perplexity: 37.55
No improvement. Patience: 5/20
Best Loss: 0.1263 (Epoch 37)

Epoch 43/500, Batch 100/782, Loss: 0.1162, Recon: 0.0219, VQ: 0.0943, Perplexity: 36.00
Epoch 43/500, Batch 200/782, Loss: 0.1201, Recon: 0.0211, VQ: 0.0991, Perplexity: 37.50
Epoch 43/500, Batch 300/782, Loss: 0.1210, Recon: 0.0210, VQ: 0.1000, Perplexity: 37.89
Epoch 43/500, Batch 400/782, Loss: 0.1255, Recon: 0.0230, VQ: 0.1025, Perplexity: 37.57
Epoch 43/500, Batch 500/782, Loss: 0.1193, Recon: 0.0230, VQ: 0.0963, Perplexity: 37.27
Epoch 43/500, Batch 600/782, Loss: 0.1040, Recon: 0.0203, VQ: 0.0837, Perplexity: 35.76
Epoch 43/500, Batch 700/782, Loss: 0.1338, Recon: 0.0227, VQ: 0.1111, Perplexity: 38.84

=== Epoch 43/500 Summary ===
Train Loss: 0.1267 | Val Loss: 0.1276
Train Recon: 0.0227 | Val Recon: 0.0230
Train VQ: 0.1040 | Val VQ: 0.1046
Train Perplexity: 37.99 | Val Perplexity: 38.00
No improvement. Patience: 6/20
Best Loss: 0.1263 (Epoch 37)

Epoch 44/500, Batch 100/782, Loss: 0.1162, Recon: 0.0223, VQ: 0.0939, Perplexity: 38.76
Epoch 44/500, Batch 200/782, Loss: 0.1237, Recon: 0.0230, VQ: 0.1007, Perplexity: 38.08
Epoch 44/500, Batch 300/782, Loss: 0.1331, Recon: 0.0238, VQ: 0.1093, Perplexity: 39.30
Epoch 44/500, Batch 400/782, Loss: 0.1254, Recon: 0.0226, VQ: 0.1029, Perplexity: 36.88
Epoch 44/500, Batch 500/782, Loss: 0.1237, Recon: 0.0236, VQ: 0.1001, Perplexity: 38.97
Epoch 44/500, Batch 600/782, Loss: 0.1299, Recon: 0.0236, VQ: 0.1063, Perplexity: 39.73
Epoch 44/500, Batch 700/782, Loss: 0.1198, Recon: 0.0227, VQ: 0.0971, Perplexity: 38.17

=== Epoch 44/500 Summary ===
Train Loss: 0.1265 | Val Loss: 0.1265
Train Recon: 0.0226 | Val Recon: 0.0229
Train VQ: 0.1039 | Val VQ: 0.1035
Train Perplexity: 38.23 | Val Perplexity: 38.40
No improvement. Patience: 7/20
Best Loss: 0.1263 (Epoch 37)

Epoch 45/500, Batch 100/782, Loss: 0.1079, Recon: 0.0204, VQ: 0.0876, Perplexity: 36.29
Epoch 45/500, Batch 200/782, Loss: 0.1121, Recon: 0.0211, VQ: 0.0910, Perplexity: 37.10
Epoch 45/500, Batch 300/782, Loss: 0.1391, Recon: 0.0243, VQ: 0.1148, Perplexity: 39.36
Epoch 45/500, Batch 400/782, Loss: 0.1265, Recon: 0.0225, VQ: 0.1040, Perplexity: 38.51

Epoch 45/500, Batch 500/782, Loss: 0.1225, Recon: 0.0220, VQ: 0.1005, Perplexity: 37.50
Epoch 45/500, Batch 600/782, Loss: 0.1254, Recon: 0.0229, VQ: 0.1025, Perplexity: 39.92
Epoch 45/500, Batch 700/782, Loss: 0.2120, Recon: 0.0259, VQ: 0.1860, Perplexity: 39.65

=== Epoch 45/500 Summary ===

Train Loss: 0.1265 | Val Loss: 0.1263
Train Recon: 0.0225 | Val Recon: 0.0229
Train VQ: 0.1040 | Val VQ: 0.1034
Train Perplexity: 38.43 | Val Perplexity: 38.72
No improvement. Patience: 8/20
Best Loss: 0.1263 (Epoch 37)

Epoch 46/500, Batch 100/782, Loss: 0.1318, Recon: 0.0228, VQ: 0.1090, Perplexity: 38.94
Epoch 46/500, Batch 200/782, Loss: 0.1221, Recon: 0.0230, VQ: 0.0991, Perplexity: 38.89
Epoch 46/500, Batch 300/782, Loss: 0.1220, Recon: 0.0221, VQ: 0.0999, Perplexity: 38.21
Epoch 46/500, Batch 400/782, Loss: 0.1084, Recon: 0.0198, VQ: 0.0886, Perplexity: 37.69
Epoch 46/500, Batch 500/782, Loss: 0.1236, Recon: 0.0226, VQ: 0.1010, Perplexity: 38.99
Epoch 46/500, Batch 600/782, Loss: 0.1330, Recon: 0.0233, VQ: 0.1097, Perplexity: 39.33
Epoch 46/500, Batch 700/782, Loss: 0.1222, Recon: 0.0227, VQ: 0.0995, Perplexity: 39.39

=== Epoch 46/500 Summary ===

Train Loss: 0.1266 | Val Loss: 0.1258
Train Recon: 0.0225 | Val Recon: 0.0228
Train VQ: 0.1041 | Val VQ: 0.1030
Train Perplexity: 38.62 | Val Perplexity: 38.93
✓ Best model saved! (Val Loss: 0.1258)
Best Loss: 0.1258 (Epoch 46)

Epoch 47/500, Batch 100/782, Loss: 0.1324, Recon: 0.0240, VQ: 0.1083, Perplexity: 39.60
Epoch 47/500, Batch 200/782, Loss: 0.1176, Recon: 0.0222, VQ: 0.0953, Perplexity: 39.25
Epoch 47/500, Batch 300/782, Loss: 0.1206, Recon: 0.0213, VQ: 0.0993, Perplexity: 38.67
Epoch 47/500, Batch 400/782, Loss: 0.1285, Recon: 0.0218, VQ: 0.1067, Perplexity: 37.95
Epoch 47/500, Batch 500/782, Loss: 0.1134, Recon: 0.0214, VQ: 0.0919, Perplexity: 38.45
Epoch 47/500, Batch 600/782, Loss: 0.1125, Recon: 0.0215, VQ: 0.0911, Perplexity: 38.42
Epoch 47/500, Batch 700/782, Loss: 0.1081, Recon: 0.0199, VQ: 0.0882, Perplexity: 38.88

=== Epoch 47/500 Summary ===

Train Loss: 0.1264 | Val Loss: 0.1262
Train Recon: 0.0224 | Val Recon: 0.0228
Train VQ: 0.1040 | Val VQ: 0.1034
Train Perplexity: 38.89 | Val Perplexity: 39.01
No improvement. Patience: 1/20
Best Loss: 0.1258 (Epoch 46)

Epoch 48/500, Batch 100/782, Loss: 0.1312, Recon: 0.0208, VQ: 0.1103, Perplexity: 38.03
Epoch 48/500, Batch 200/782, Loss: 0.1246, Recon: 0.0228, VQ: 0.1019, Perplexity: 39.92
Epoch 48/500, Batch 300/782, Loss: 0.1108, Recon: 0.0200, VQ: 0.0908, Perplexity: 38.57
Epoch 48/500, Batch 400/782, Loss: 0.1188, Recon: 0.0221, VQ: 0.0967, Perplexity: 40.30
Epoch 48/500, Batch 500/782, Loss: 0.1316, Recon: 0.0240, VQ: 0.1077, Perplexity: 39.80
Epoch 48/500, Batch 600/782, Loss: 0.1482, Recon: 0.0242, VQ: 0.1240, Perplexity: 41.03
Epoch 48/500, Batch 700/782, Loss: 0.1400, Recon: 0.0239, VQ: 0.1161, Perplexity: 40.44

=== Epoch 48/500 Summary ===

Train Loss: 0.1261 | Val Loss: 0.1255
Train Recon: 0.0223 | Val Recon: 0.0226
Train VQ: 0.1038 | Val VQ: 0.1029
Train Perplexity: 39.51 | Val Perplexity: 39.58
✓ Best model saved! (Val Loss: 0.1255)
Best Loss: 0.1255 (Epoch 48)

Epoch 49/500, Batch 100/782, Loss: 0.1397, Recon: 0.0238, VQ: 0.1159, Perplexity: 41.69
Epoch 49/500, Batch 200/782, Loss: 0.1180, Recon: 0.0204, VQ: 0.0976, Perplexity: 37.93
Epoch 49/500, Batch 300/782, Loss: 0.1316, Recon: 0.0222, VQ: 0.1094, Perplexity: 39.68
Epoch 49/500, Batch 400/782, Loss: 0.1121, Recon: 0.0207, VQ: 0.0914, Perplexity: 37.59
Epoch 49/500, Batch 500/782, Loss: 0.1085, Recon: 0.0201, VQ: 0.0884, Perplexity: 39.33
Epoch 49/500, Batch 600/782, Loss: 0.1128, Recon: 0.0206, VQ: 0.0922, Perplexity: 39.41
Epoch 49/500, Batch 700/782, Loss: 0.1157, Recon: 0.0210, VQ: 0.0947, Perplexity: 38.63

=== Epoch 49/500 Summary ===

Train Loss: 0.1254 | Val Loss: 0.1242
Train Recon: 0.0222 | Val Recon: 0.0226
Train VQ: 0.1032 | Val VQ: 0.1017
Train Perplexity: 39.84 | Val Perplexity: 40.01
✓ Best model saved! (Val Loss: 0.1242)
Best Loss: 0.1242 (Epoch 49)

Epoch 50/500, Batch 100/782, Loss: 0.1174, Recon: 0.0226, VQ: 0.0948, Perplexity: 39.28
Epoch 50/500, Batch 200/782, Loss: 0.1212, Recon: 0.0219, VQ: 0.0993, Perplexity: 40.11
Epoch 50/500, Batch 300/782, Loss: 0.1242, Recon: 0.0223, VQ: 0.1019, Perplexity: 39.38
Epoch 50/500, Batch 400/782, Loss: 0.1066, Recon: 0.0198, VQ: 0.0868, Perplexity: 37.76
Epoch 50/500, Batch 500/782, Loss: 0.1072, Recon: 0.0201, VQ: 0.0871, Perplexity: 38.89
Epoch 50/500, Batch 600/782, Loss: 0.1341, Recon: 0.0241, VQ: 0.1100, Perplexity: 40.55
Epoch 50/500, Batch 700/782, Loss: 0.1534, Recon: 0.0242, VQ: 0.1292, Perplexity: 40.55

lexity: 40.39

=== Epoch 50/500 Summary ===

Train Loss: 0.1250 | Val Loss: 0.1255
Train Recon: 0.0222 | Val Recon: 0.0225
Train VQ: 0.1028 | Val VQ: 0.1030
Train Perplexity: 40.06 | Val Perplexity: 40.50
No improvement. Patience: 1/20
Best Loss: 0.1242 (Epoch 49)

Epoch 51/500, Batch 100/782, Loss: 0.1483, Recon: 0.0231, VQ: 0.1253, Perplexity: 40.54
Epoch 51/500, Batch 200/782, Loss: 0.1212, Recon: 0.0216, VQ: 0.0997, Perplexity: 39.18
Epoch 51/500, Batch 300/782, Loss: 0.1450, Recon: 0.0232, VQ: 0.1217, Perplexity: 40.95
Epoch 51/500, Batch 400/782, Loss: 0.1318, Recon: 0.0222, VQ: 0.1096, Perplexity: 41.29
Epoch 51/500, Batch 500/782, Loss: 0.1194, Recon: 0.0210, VQ: 0.0984, Perplexity: 39.72
Epoch 51/500, Batch 600/782, Loss: 0.1671, Recon: 0.0234, VQ: 0.1437, Perplexity: 41.91
Epoch 51/500, Batch 700/782, Loss: 0.1167, Recon: 0.0210, VQ: 0.0957, Perplexity: 39.16

=== Epoch 51/500 Summary ===

Train Loss: 0.1249 | Val Loss: 0.1240
Train Recon: 0.0221 | Val Recon: 0.0225
Train VQ: 0.1027 | Val VQ: 0.1015
Train Perplexity: 40.25 | Val Perplexity: 40.24
✓ Best model saved! (Val Loss: 0.1240)
Best Loss: 0.1240 (Epoch 51)

Epoch 52/500, Batch 100/782, Loss: 0.1208, Recon: 0.0226, VQ: 0.0983, Perplexity: 39.96
Epoch 52/500, Batch 200/782, Loss: 0.1143, Recon: 0.0220, VQ: 0.0923, Perplexity: 40.27
Epoch 52/500, Batch 300/782, Loss: 0.1312, Recon: 0.0248, VQ: 0.1063, Perplexity: 41.52
Epoch 52/500, Batch 400/782, Loss: 0.1287, Recon: 0.0236, VQ: 0.1051, Perplexity: 40.96
Epoch 52/500, Batch 500/782, Loss: 0.1222, Recon: 0.0212, VQ: 0.1010, Perplexity: 40.69
Epoch 52/500, Batch 600/782, Loss: 0.1107, Recon: 0.0203, VQ: 0.0904, Perplexity: 39.85
Epoch 52/500, Batch 700/782, Loss: 0.1195, Recon: 0.0226, VQ: 0.0970, Perplexity: 41.95

=== Epoch 52/500 Summary ===

Train Loss: 0.1241 | Val Loss: 0.1231
Train Recon: 0.0221 | Val Recon: 0.0225
Train VQ: 0.1020 | Val VQ: 0.1006
Train Perplexity: 40.43 | Val Perplexity: 40.76
✓ Best model saved! (Val Loss: 0.1231)
Best Loss: 0.1231 (Epoch 52)

Epoch 53/500, Batch 100/782, Loss: 0.1147, Recon: 0.0203, VQ: 0.0944, Perplexity: 40.14
Epoch 53/500, Batch 200/782, Loss: 0.1191, Recon: 0.0225, VQ: 0.0966, Perplexity: 40.72

Epoch 53/500, Batch 300/782, Loss: 0.1280, Recon: 0.0233, VQ: 0.1046, Perplexity: 41.54
Epoch 53/500, Batch 400/782, Loss: 0.1183, Recon: 0.0222, VQ: 0.0961, Perplexity: 40.95
Epoch 53/500, Batch 500/782, Loss: 0.1087, Recon: 0.0205, VQ: 0.0882, Perplexity: 40.75
Epoch 53/500, Batch 600/782, Loss: 0.1120, Recon: 0.0221, VQ: 0.0899, Perplexity: 39.24
Epoch 53/500, Batch 700/782, Loss: 0.1074, Recon: 0.0203, VQ: 0.0871, Perplexity: 40.09

=== Epoch 53/500 Summary ===

Train Loss: 0.1241 | Val Loss: 0.1242
Train Recon: 0.0220 | Val Recon: 0.0224
Train VQ: 0.1020 | Val VQ: 0.1018
Train Perplexity: 40.79 | Val Perplexity: 40.84
No improvement. Patience: 1/20
Best Loss: 0.1231 (Epoch 52)

Epoch 54/500, Batch 100/782, Loss: 0.1198, Recon: 0.0219, VQ: 0.0979, Perplexity: 41.15
Epoch 54/500, Batch 200/782, Loss: 0.1392, Recon: 0.0235, VQ: 0.1158, Perplexity: 42.35
Epoch 54/500, Batch 300/782, Loss: 0.1084, Recon: 0.0205, VQ: 0.0880, Perplexity: 39.90
Epoch 54/500, Batch 400/782, Loss: 0.1177, Recon: 0.0217, VQ: 0.0959, Perplexity: 41.18
Epoch 54/500, Batch 500/782, Loss: 0.1250, Recon: 0.0211, VQ: 0.1039, Perplexity: 41.35
Epoch 54/500, Batch 600/782, Loss: 0.1082, Recon: 0.0207, VQ: 0.0875, Perplexity: 41.17
Epoch 54/500, Batch 700/782, Loss: 0.1654, Recon: 0.0230, VQ: 0.1425, Perplexity: 40.47

=== Epoch 54/500 Summary ===

Train Loss: 0.1234 | Val Loss: 0.1224
Train Recon: 0.0220 | Val Recon: 0.0223
Train VQ: 0.1014 | Val VQ: 0.1001
Train Perplexity: 41.13 | Val Perplexity: 41.16
✓ Best model saved! (Val Loss: 0.1224)
Best Loss: 0.1224 (Epoch 54)

Epoch 55/500, Batch 100/782, Loss: 0.1180, Recon: 0.0210, VQ: 0.0970, Perplexity: 40.22
Epoch 55/500, Batch 200/782, Loss: 0.1176, Recon: 0.0207, VQ: 0.0969, Perplexity: 41.30
Epoch 55/500, Batch 300/782, Loss: 0.1235, Recon: 0.0239, VQ: 0.0996, Perplexity: 42.27
Epoch 55/500, Batch 400/782, Loss: 0.1135, Recon: 0.0211, VQ: 0.0924, Perplexity: 41.38
Epoch 55/500, Batch 500/782, Loss: 0.1152, Recon: 0.0207, VQ: 0.0945, Perplexity: 39.98
Epoch 55/500, Batch 600/782, Loss: 0.1329, Recon: 0.0227, VQ: 0.1102, Perplexity: 42.58
Epoch 55/500, Batch 700/782, Loss: 0.1194, Recon: 0.0229, VQ: 0.0965, Perplexity: 43.39

=== Epoch 55/500 Summary ===

Train Loss: 0.1230 | Val Loss: 0.1236
Train Recon: 0.0219 | Val Recon: 0.0223

Train VQ: 0.1011 | Val VQ: 0.1013
Train Perplexity: 41.55 | Val Perplexity: 41.92
No improvement. Patience: 1/20
Best Loss: 0.1224 (Epoch 54)

Epoch 56/500, Batch 100/782, Loss: 0.1091, Recon: 0.0201, VQ: 0.0891, Perplexity: 41.00
Epoch 56/500, Batch 200/782, Loss: 0.1028, Recon: 0.0197, VQ: 0.0831, Perplexity: 40.65
Epoch 56/500, Batch 300/782, Loss: 0.1180, Recon: 0.0214, VQ: 0.0966, Perplexity: 40.48
Epoch 56/500, Batch 400/782, Loss: 0.1179, Recon: 0.0215, VQ: 0.0963, Perplexity: 42.25
Epoch 56/500, Batch 500/782, Loss: 0.1063, Recon: 0.0204, VQ: 0.0859, Perplexity: 41.41
Epoch 56/500, Batch 600/782, Loss: 0.1084, Recon: 0.0197, VQ: 0.0887, Perplexity: 41.00
Epoch 56/500, Batch 700/782, Loss: 0.1120, Recon: 0.0216, VQ: 0.0904, Perplexity: 39.32

=== Epoch 56/500 Summary ===
Train Loss: 0.1224 | Val Loss: 0.1224
Train Recon: 0.0218 | Val Recon: 0.0222
Train VQ: 0.1006 | Val VQ: 0.1002
Train Perplexity: 41.97 | Val Perplexity: 42.00
No improvement. Patience: 2/20
Best Loss: 0.1224 (Epoch 54)

Epoch 57/500, Batch 100/782, Loss: 0.1170, Recon: 0.0204, VQ: 0.0966, Perplexity: 41.59
Epoch 57/500, Batch 200/782, Loss: 0.1205, Recon: 0.0225, VQ: 0.0980, Perplexity: 43.31
Epoch 57/500, Batch 300/782, Loss: 0.1240, Recon: 0.0209, VQ: 0.1031, Perplexity: 41.47
Epoch 57/500, Batch 400/782, Loss: 0.1260, Recon: 0.0217, VQ: 0.1042, Perplexity: 42.73
Epoch 57/500, Batch 500/782, Loss: 0.1138, Recon: 0.0204, VQ: 0.0934, Perplexity: 42.24
Epoch 57/500, Batch 600/782, Loss: 0.1124, Recon: 0.0217, VQ: 0.0907, Perplexity: 41.04
Epoch 57/500, Batch 700/782, Loss: 0.1162, Recon: 0.0217, VQ: 0.0945, Perplexity: 41.78

=== Epoch 57/500 Summary ===
Train Loss: 0.1222 | Val Loss: 0.1203
Train Recon: 0.0218 | Val Recon: 0.0221
Train VQ: 0.1005 | Val VQ: 0.0982
Train Perplexity: 42.09 | Val Perplexity: 42.05
✓ Best model saved! (Val Loss: 0.1203)
Best Loss: 0.1203 (Epoch 57)

Epoch 58/500, Batch 100/782, Loss: 0.1180, Recon: 0.0215, VQ: 0.0965, Perplexity: 42.35
Epoch 58/500, Batch 200/782, Loss: 0.1211, Recon: 0.0214, VQ: 0.0996, Perplexity: 42.69
Epoch 58/500, Batch 300/782, Loss: 0.1150, Recon: 0.0217, VQ: 0.0933, Perplexity: 43.24
Epoch 58/500, Batch 400/782, Loss: 0.1095, Recon: 0.0208, VQ: 0.0887, Perplexity: 41.01
Epoch 58/500, Batch 500/782, Loss: 0.1193, Recon: 0.0220, VQ: 0.0973, Perplexity:

lexity: 42.82
Epoch 58/500, Batch 600/782, Loss: 0.1152, Recon: 0.0213, VQ: 0.0939, Perp
lexity: 42.54
Epoch 58/500, Batch 700/782, Loss: 0.1230, Recon: 0.0229, VQ: 0.1002, Perp
lexity: 43.32

=== Epoch 58/500 Summary ===

Train Loss: 0.1219 | Val Loss: 0.1226
Train Recon: 0.0217 | Val Recon: 0.0222
Train VQ: 0.1002 | Val VQ: 0.1005
Train Perplexity: 42.11 | Val Perplexity: 42.11
No improvement. Patience: 1/20
Best Loss: 0.1203 (Epoch 57)

Epoch 59/500, Batch 100/782, Loss: 0.1094, Recon: 0.0208, VQ: 0.0887, Perp
lexity: 40.78
Epoch 59/500, Batch 200/782, Loss: 0.1015, Recon: 0.0193, VQ: 0.0822, Perp
lexity: 40.79
Epoch 59/500, Batch 300/782, Loss: 0.1183, Recon: 0.0210, VQ: 0.0973, Perp
lexity: 41.50
Epoch 59/500, Batch 400/782, Loss: 0.1188, Recon: 0.0219, VQ: 0.0969, Perp
lexity: 41.63
Epoch 59/500, Batch 500/782, Loss: 0.1171, Recon: 0.0211, VQ: 0.0960, Perp
lexity: 41.34
Epoch 59/500, Batch 600/782, Loss: 0.1194, Recon: 0.0201, VQ: 0.0992, Perp
lexity: 41.56
Epoch 59/500, Batch 700/782, Loss: 0.1167, Recon: 0.0212, VQ: 0.0955, Perp
lexity: 40.61

=== Epoch 59/500 Summary ===

Train Loss: 0.1218 | Val Loss: 0.1223
Train Recon: 0.0217 | Val Recon: 0.0221
Train VQ: 0.1001 | Val VQ: 0.1002
Train Perplexity: 42.13 | Val Perplexity: 42.15
No improvement. Patience: 2/20
Best Loss: 0.1203 (Epoch 57)

Epoch 60/500, Batch 100/782, Loss: 0.1127, Recon: 0.0212, VQ: 0.0915, Perp
lexity: 41.61
Epoch 60/500, Batch 200/782, Loss: 0.1307, Recon: 0.0215, VQ: 0.1092, Perp
lexity: 42.12
Epoch 60/500, Batch 300/782, Loss: 0.0998, Recon: 0.0197, VQ: 0.0800, Perp
lexity: 41.68
Epoch 60/500, Batch 400/782, Loss: 0.1204, Recon: 0.0234, VQ: 0.0970, Perp
lexity: 42.51
Epoch 60/500, Batch 500/782, Loss: 0.1219, Recon: 0.0226, VQ: 0.0993, Perp
lexity: 43.97
Epoch 60/500, Batch 600/782, Loss: 0.1160, Recon: 0.0210, VQ: 0.0950, Perp
lexity: 41.28
Epoch 60/500, Batch 700/782, Loss: 0.1329, Recon: 0.0219, VQ: 0.1110, Perp
lexity: 42.37

=== Epoch 60/500 Summary ===

Train Loss: 0.1217 | Val Loss: 0.1219
Train Recon: 0.0216 | Val Recon: 0.0220
Train VQ: 0.1001 | Val VQ: 0.0998
Train Perplexity: 42.25 | Val Perplexity: 42.52
No improvement. Patience: 3/20
Best Loss: 0.1203 (Epoch 57)

Epoch 61/500, Batch 100/782, Loss: 0.1230, Recon: 0.0190, VQ: 0.1040, Perplexity: 40.01
Epoch 61/500, Batch 200/782, Loss: 0.1123, Recon: 0.0195, VQ: 0.0928, Perplexity: 40.03
Epoch 61/500, Batch 300/782, Loss: 0.1269, Recon: 0.0219, VQ: 0.1051, Perplexity: 42.42
Epoch 61/500, Batch 400/782, Loss: 0.1182, Recon: 0.0223, VQ: 0.0958, Perplexity: 43.45
Epoch 61/500, Batch 500/782, Loss: 0.1138, Recon: 0.0210, VQ: 0.0928, Perplexity: 42.50
Epoch 61/500, Batch 600/782, Loss: 0.1220, Recon: 0.0225, VQ: 0.0996, Perplexity: 43.29
Epoch 61/500, Batch 700/782, Loss: 0.1056, Recon: 0.0203, VQ: 0.0853, Perplexity: 41.71

=== Epoch 61/500 Summary ===

Train Loss: 0.1215 | Val Loss: 0.1221
Train Recon: 0.0216 | Val Recon: 0.0220
Train VQ: 0.0999 | Val VQ: 0.1001
Train Perplexity: 42.39 | Val Perplexity: 42.31
No improvement. Patience: 4/20
Best Loss: 0.1203 (Epoch 57)

Epoch 62/500, Batch 100/782, Loss: 0.1128, Recon: 0.0207, VQ: 0.0922, Perplexity: 42.21
Epoch 62/500, Batch 200/782, Loss: 0.1149, Recon: 0.0204, VQ: 0.0944, Perplexity: 40.93
Epoch 62/500, Batch 300/782, Loss: 0.1278, Recon: 0.0227, VQ: 0.1051, Perplexity: 44.85
Epoch 62/500, Batch 400/782, Loss: 0.1135, Recon: 0.0203, VQ: 0.0932, Perplexity: 42.02
Epoch 62/500, Batch 500/782, Loss: 0.1183, Recon: 0.0224, VQ: 0.0959, Perplexity: 43.18
Epoch 62/500, Batch 600/782, Loss: 0.1164, Recon: 0.0224, VQ: 0.0940, Perplexity: 43.81
Epoch 62/500, Batch 700/782, Loss: 0.1085, Recon: 0.0200, VQ: 0.0884, Perplexity: 41.29

=== Epoch 62/500 Summary ===

Train Loss: 0.1218 | Val Loss: 0.1210
Train Recon: 0.0215 | Val Recon: 0.0220
Train VQ: 0.1003 | Val VQ: 0.0990
Train Perplexity: 42.58 | Val Perplexity: 42.74
No improvement. Patience: 5/20
Best Loss: 0.1203 (Epoch 57)

Epoch 63/500, Batch 100/782, Loss: 0.1210, Recon: 0.0216, VQ: 0.0995, Perplexity: 43.08
Epoch 63/500, Batch 200/782, Loss: 0.1130, Recon: 0.0202, VQ: 0.0928, Perplexity: 43.27
Epoch 63/500, Batch 300/782, Loss: 0.1296, Recon: 0.0222, VQ: 0.1074, Perplexity: 42.34
Epoch 63/500, Batch 400/782, Loss: 0.1318, Recon: 0.0239, VQ: 0.1079, Perplexity: 44.64
Epoch 63/500, Batch 500/782, Loss: 0.1212, Recon: 0.0202, VQ: 0.1010, Perplexity: 42.09
Epoch 63/500, Batch 600/782, Loss: 0.1054, Recon: 0.0189, VQ: 0.0865, Perplexity: 42.13
Epoch 63/500, Batch 700/782, Loss: 0.1099, Recon: 0.0200, VQ: 0.0899, Perplexity: 42.46

=== Epoch 63/500 Summary ===

Train Loss: 0.1215 | Val Loss: 0.1225

Train Recon: 0.0215 | Val Recon: 0.0220

Train VQ: 0.1001 | Val VQ: 0.1005

Train Perplexity: 42.88 | Val Perplexity: 43.15

No improvement. Patience: 6/20

Best Loss: 0.1203 (Epoch 57)

Epoch 64/500, Batch 100/782, Loss: 0.1276, Recon: 0.0214, VQ: 0.1062, Perplexity: 44.26

Epoch 64/500, Batch 200/782, Loss: 0.1216, Recon: 0.0233, VQ: 0.0983, Perplexity: 44.54

Epoch 64/500, Batch 300/782, Loss: 0.1370, Recon: 0.0210, VQ: 0.1159, Perplexity: 42.02

Epoch 64/500, Batch 400/782, Loss: 0.1127, Recon: 0.0212, VQ: 0.0915, Perplexity: 42.77

Epoch 64/500, Batch 500/782, Loss: 0.1383, Recon: 0.0254, VQ: 0.1129, Perplexity: 45.18

Epoch 64/500, Batch 600/782, Loss: 0.1143, Recon: 0.0211, VQ: 0.0933, Perplexity: 40.93

Epoch 64/500, Batch 700/782, Loss: 0.1086, Recon: 0.0201, VQ: 0.0885, Perplexity: 41.62

=== Epoch 64/500 Summary ===

Train Loss: 0.1216 | Val Loss: 0.1220

Train Recon: 0.0214 | Val Recon: 0.0219

Train VQ: 0.1001 | Val VQ: 0.1000

Train Perplexity: 43.08 | Val Perplexity: 43.07

No improvement. Patience: 7/20

Best Loss: 0.1203 (Epoch 57)

Epoch 65/500, Batch 100/782, Loss: 0.1236, Recon: 0.0214, VQ: 0.1022, Perplexity: 42.04

Epoch 65/500, Batch 200/782, Loss: 0.1228, Recon: 0.0233, VQ: 0.0995, Perplexity: 44.28

Epoch 65/500, Batch 300/782, Loss: 0.1133, Recon: 0.0205, VQ: 0.0929, Perplexity: 41.96

Epoch 65/500, Batch 400/782, Loss: 0.1177, Recon: 0.0210, VQ: 0.0967, Perplexity: 44.24

Epoch 65/500, Batch 500/782, Loss: 0.1334, Recon: 0.0223, VQ: 0.1112, Perplexity: 43.61

Epoch 65/500, Batch 600/782, Loss: 0.1170, Recon: 0.0218, VQ: 0.0952, Perplexity: 43.24

Epoch 65/500, Batch 700/782, Loss: 0.1167, Recon: 0.0210, VQ: 0.0958, Perplexity: 43.51

=== Epoch 65/500 Summary ===

Train Loss: 0.1212 | Val Loss: 0.1228

Train Recon: 0.0214 | Val Recon: 0.0219

Train VQ: 0.0998 | Val VQ: 0.1009

Train Perplexity: 43.25 | Val Perplexity: 43.51

No improvement. Patience: 8/20

Best Loss: 0.1203 (Epoch 57)

Epoch 66/500, Batch 100/782, Loss: 0.1119, Recon: 0.0208, VQ: 0.0912, Perplexity: 42.67

Epoch 66/500, Batch 200/782, Loss: 0.1291, Recon: 0.0234, VQ: 0.1057, Perplexity: 44.71

Epoch 66/500, Batch 300/782, Loss: 0.1378, Recon: 0.0237, VQ: 0.1141, Perplexity:

lexity: 44.18
Epoch 66/500, Batch 400/782, Loss: 0.1129, Recon: 0.0198, VQ: 0.0931, Perp
lexity: 42.55
Epoch 66/500, Batch 500/782, Loss: 0.1282, Recon: 0.0222, VQ: 0.1060, Perp
lexity: 43.32
Epoch 66/500, Batch 600/782, Loss: 0.1266, Recon: 0.0226, VQ: 0.1040, Perp
lexity: 45.76
Epoch 66/500, Batch 700/782, Loss: 0.1216, Recon: 0.0215, VQ: 0.1002, Perp
lexity: 43.76

=== Epoch 66/500 Summary ===

Train Loss: 0.1211 | Val Loss: 0.1214
Train Recon: 0.0214 | Val Recon: 0.0219
Train VQ: 0.0997 | Val VQ: 0.0995
Train Perplexity: 43.34 | Val Perplexity: 43.49
No improvement. Patience: 9/20
Best Loss: 0.1203 (Epoch 57)

Epoch 67/500, Batch 100/782, Loss: 0.1267, Recon: 0.0217, VQ: 0.1050, Perp
lexity: 44.08
Epoch 67/500, Batch 200/782, Loss: 0.1062, Recon: 0.0197, VQ: 0.0865, Perp
lexity: 41.89
Epoch 67/500, Batch 300/782, Loss: 0.1191, Recon: 0.0215, VQ: 0.0976, Perp
lexity: 44.62
Epoch 67/500, Batch 400/782, Loss: 0.1151, Recon: 0.0195, VQ: 0.0956, Perp
lexity: 42.77
Epoch 67/500, Batch 500/782, Loss: 0.1043, Recon: 0.0203, VQ: 0.0840, Perp
lexity: 41.49
Epoch 67/500, Batch 600/782, Loss: 0.1151, Recon: 0.0213, VQ: 0.0938, Perp
lexity: 41.76
Epoch 67/500, Batch 700/782, Loss: 0.1161, Recon: 0.0214, VQ: 0.0947, Perp
lexity: 43.18

=== Epoch 67/500 Summary ===

Train Loss: 0.1206 | Val Loss: 0.1210
Train Recon: 0.0213 | Val Recon: 0.0218
Train VQ: 0.0992 | Val VQ: 0.0992
Train Perplexity: 43.40 | Val Perplexity: 43.59
No improvement. Patience: 10/20
Best Loss: 0.1203 (Epoch 57)

Epoch 68/500, Batch 100/782, Loss: 0.1201, Recon: 0.0227, VQ: 0.0974, Perp
lexity: 43.56
Epoch 68/500, Batch 200/782, Loss: 0.1079, Recon: 0.0198, VQ: 0.0880, Perp
lexity: 43.17
Epoch 68/500, Batch 300/782, Loss: 0.1268, Recon: 0.0224, VQ: 0.1044, Perp
lexity: 43.21
Epoch 68/500, Batch 400/782, Loss: 0.1136, Recon: 0.0201, VQ: 0.0935, Perp
lexity: 42.87
Epoch 68/500, Batch 500/782, Loss: 0.1142, Recon: 0.0201, VQ: 0.0940, Perp
lexity: 43.20
Epoch 68/500, Batch 600/782, Loss: 0.1141, Recon: 0.0205, VQ: 0.0935, Perp
lexity: 43.69
Epoch 68/500, Batch 700/782, Loss: 0.1199, Recon: 0.0227, VQ: 0.0971, Perp
lexity: 42.31

=== Epoch 68/500 Summary ===

Train Loss: 0.1205 | Val Loss: 0.1205
Train Recon: 0.0213 | Val Recon: 0.0218
Train VQ: 0.0993 | Val VQ: 0.0987

Train Perplexity: 43.51 | Val Perplexity: 43.33

No improvement. Patience: 11/20

Best Loss: 0.1203 (Epoch 57)

Epoch 69/500, Batch 100/782, Loss: 0.1292, Recon: 0.0230, VQ: 0.1062, Perplexity: 43.70

Epoch 69/500, Batch 200/782, Loss: 0.1279, Recon: 0.0222, VQ: 0.1056, Perplexity: 43.24

Epoch 69/500, Batch 300/782, Loss: 0.1101, Recon: 0.0197, VQ: 0.0904, Perplexity: 41.73

Epoch 69/500, Batch 400/782, Loss: 0.1250, Recon: 0.0212, VQ: 0.1038, Perplexity: 44.62

Epoch 69/500, Batch 500/782, Loss: 0.1141, Recon: 0.0209, VQ: 0.0931, Perplexity: 42.31

Epoch 69/500, Batch 600/782, Loss: 0.1111, Recon: 0.0201, VQ: 0.0910, Perplexity: 42.30

Epoch 69/500, Batch 700/782, Loss: 0.1119, Recon: 0.0205, VQ: 0.0914, Perplexity: 42.63

=== Epoch 69/500 Summary ===

Train Loss: 0.1205 | Val Loss: 0.1202

Train Recon: 0.0212 | Val Recon: 0.0217

Train VQ: 0.0992 | Val VQ: 0.0985

Train Perplexity: 43.64 | Val Perplexity: 43.73

No improvement. Patience: 12/20

Best Loss: 0.1203 (Epoch 57)

Epoch 70/500, Batch 100/782, Loss: 0.1103, Recon: 0.0201, VQ: 0.0902, Perplexity: 43.72

Epoch 70/500, Batch 200/782, Loss: 0.1991, Recon: 0.0234, VQ: 0.1757, Perplexity: 45.71

Epoch 70/500, Batch 300/782, Loss: 0.1234, Recon: 0.0225, VQ: 0.1009, Perplexity: 44.14

Epoch 70/500, Batch 400/782, Loss: 0.1086, Recon: 0.0206, VQ: 0.0879, Perplexity: 42.52

Epoch 70/500, Batch 500/782, Loss: 0.1196, Recon: 0.0224, VQ: 0.0972, Perplexity: 44.58

Epoch 70/500, Batch 600/782, Loss: 0.1289, Recon: 0.0216, VQ: 0.1072, Perplexity: 44.90

Epoch 70/500, Batch 700/782, Loss: 0.1271, Recon: 0.0228, VQ: 0.1043, Perplexity: 45.40

=== Epoch 70/500 Summary ===

Train Loss: 0.1202 | Val Loss: 0.1229

Train Recon: 0.0212 | Val Recon: 0.0218

Train VQ: 0.0990 | Val VQ: 0.1011

Train Perplexity: 43.85 | Val Perplexity: 44.50

No improvement. Patience: 13/20

Best Loss: 0.1203 (Epoch 57)

Epoch 71/500, Batch 100/782, Loss: 0.1136, Recon: 0.0223, VQ: 0.0913, Perplexity: 44.08

Epoch 71/500, Batch 200/782, Loss: 0.1351, Recon: 0.0215, VQ: 0.1136, Perplexity: 44.58

Epoch 71/500, Batch 300/782, Loss: 0.1097, Recon: 0.0202, VQ: 0.0895, Perplexity: 43.04

Epoch 71/500, Batch 400/782, Loss: 0.1210, Recon: 0.0209, VQ: 0.1000, Perplexity: 44.97

Epoch 71/500, Batch 500/782, Loss: 0.1269, Recon: 0.0212, VQ: 0.1058, Perplexity: 43.88

Epoch 71/500, Batch 600/782, Loss: 0.1273, Recon: 0.0218, VQ: 0.1054, Perplexity: 45.19
Epoch 71/500, Batch 700/782, Loss: 0.1169, Recon: 0.0207, VQ: 0.0962, Perplexity: 43.55

=== Epoch 71/500 Summary ===

Train Loss: 0.1200 | Val Loss: 0.1193
Train Recon: 0.0212 | Val Recon: 0.0218
Train VQ: 0.0989 | Val VQ: 0.0976
Train Perplexity: 43.96 | Val Perplexity: 44.09
✓ Best model saved! (Val Loss: 0.1193)
Best Loss: 0.1193 (Epoch 71)

Epoch 72/500, Batch 100/782, Loss: 0.1179, Recon: 0.0191, VQ: 0.0988, Perplexity: 43.47
Epoch 72/500, Batch 200/782, Loss: 0.1064, Recon: 0.0199, VQ: 0.0865, Perplexity: 42.32
Epoch 72/500, Batch 300/782, Loss: 0.1304, Recon: 0.0207, VQ: 0.1097, Perplexity: 44.06
Epoch 72/500, Batch 400/782, Loss: 0.1151, Recon: 0.0210, VQ: 0.0941, Perplexity: 44.74
Epoch 72/500, Batch 500/782, Loss: 0.1216, Recon: 0.0207, VQ: 0.1008, Perplexity: 45.55
Epoch 72/500, Batch 600/782, Loss: 0.1286, Recon: 0.0211, VQ: 0.1076, Perplexity: 44.78
Epoch 72/500, Batch 700/782, Loss: 0.1345, Recon: 0.0224, VQ: 0.1121, Perplexity: 46.08

=== Epoch 72/500 Summary ===

Train Loss: 0.1199 | Val Loss: 0.1204
Train Recon: 0.0211 | Val Recon: 0.0216
Train VQ: 0.0988 | Val VQ: 0.0988
Train Perplexity: 44.25 | Val Perplexity: 44.64
No improvement. Patience: 1/20
Best Loss: 0.1193 (Epoch 71)

Epoch 73/500, Batch 100/782, Loss: 0.1109, Recon: 0.0207, VQ: 0.0902, Perplexity: 44.53
Epoch 73/500, Batch 200/782, Loss: 0.1204, Recon: 0.0213, VQ: 0.0991, Perplexity: 45.65
Epoch 73/500, Batch 300/782, Loss: 0.1177, Recon: 0.0204, VQ: 0.0973, Perplexity: 44.55
Epoch 73/500, Batch 400/782, Loss: 0.1271, Recon: 0.0220, VQ: 0.1051, Perplexity: 44.93
Epoch 73/500, Batch 500/782, Loss: 0.1069, Recon: 0.0197, VQ: 0.0872, Perplexity: 43.38
Epoch 73/500, Batch 600/782, Loss: 0.1137, Recon: 0.0197, VQ: 0.0940, Perplexity: 44.26
Epoch 73/500, Batch 700/782, Loss: 0.1421, Recon: 0.0225, VQ: 0.1196, Perplexity: 44.59

=== Epoch 73/500 Summary ===

Train Loss: 0.1198 | Val Loss: 0.1181
Train Recon: 0.0211 | Val Recon: 0.0216
Train VQ: 0.0987 | Val VQ: 0.0965
Train Perplexity: 44.49 | Val Perplexity: 44.28
✓ Best model saved! (Val Loss: 0.1181)
Best Loss: 0.1181 (Epoch 73)

Epoch 74/500, Batch 100/782, Loss: 0.1132, Recon: 0.0197, VQ: 0.0935, Perplexity: 44.59

lexity: 43.92
Epoch 74/500, Batch 200/782, Loss: 0.1341, Recon: 0.0212, VQ: 0.1129, Perp
lexity: 42.95
Epoch 74/500, Batch 300/782, Loss: 0.1098, Recon: 0.0203, VQ: 0.0895, Perp
lexity: 43.54
Epoch 74/500, Batch 400/782, Loss: 0.1111, Recon: 0.0196, VQ: 0.0916, Perp
lexity: 44.26
Epoch 74/500, Batch 500/782, Loss: 0.1265, Recon: 0.0224, VQ: 0.1041, Perp
lexity: 45.56
Epoch 74/500, Batch 600/782, Loss: 0.1088, Recon: 0.0197, VQ: 0.0891, Perp
lexity: 42.98
Epoch 74/500, Batch 700/782, Loss: 0.1341, Recon: 0.0223, VQ: 0.1118, Perp
lexity: 46.21

=== Epoch 74/500 Summary ===

Train Loss: 0.1196 | Val Loss: 0.1183
Train Recon: 0.0210 | Val Recon: 0.0216
Train VQ: 0.0985 | Val VQ: 0.0967
Train Perplexity: 44.62 | Val Perplexity: 44.37
No improvement. Patience: 1/20
Best Loss: 0.1181 (Epoch 73)

Epoch 75/500, Batch 100/782, Loss: 0.1171, Recon: 0.0213, VQ: 0.0958, Perp
lexity: 45.27
Epoch 75/500, Batch 200/782, Loss: 0.1188, Recon: 0.0215, VQ: 0.0972, Perp
lexity: 46.24
Epoch 75/500, Batch 300/782, Loss: 0.1205, Recon: 0.0213, VQ: 0.0992, Perp
lexity: 44.68
Epoch 75/500, Batch 400/782, Loss: 0.1159, Recon: 0.0203, VQ: 0.0956, Perp
lexity: 45.30
Epoch 75/500, Batch 500/782, Loss: 0.1182, Recon: 0.0216, VQ: 0.0966, Perp
lexity: 45.60
Epoch 75/500, Batch 600/782, Loss: 0.1313, Recon: 0.0218, VQ: 0.1095, Perp
lexity: 46.05
Epoch 75/500, Batch 700/782, Loss: 0.1252, Recon: 0.0213, VQ: 0.1039, Perp
lexity: 46.26

=== Epoch 75/500 Summary ===

Train Loss: 0.1191 | Val Loss: 0.1195
Train Recon: 0.0210 | Val Recon: 0.0217
Train VQ: 0.0981 | Val VQ: 0.0978
Train Perplexity: 44.77 | Val Perplexity: 44.86
No improvement. Patience: 2/20
Best Loss: 0.1181 (Epoch 73)

Epoch 76/500, Batch 100/782, Loss: 0.1234, Recon: 0.0225, VQ: 0.1009, Perp
lexity: 45.63
Epoch 76/500, Batch 200/782, Loss: 0.1041, Recon: 0.0198, VQ: 0.0844, Perp
lexity: 43.51
Epoch 76/500, Batch 300/782, Loss: 0.1308, Recon: 0.0232, VQ: 0.1076, Perp
lexity: 46.18
Epoch 76/500, Batch 400/782, Loss: 0.1046, Recon: 0.0191, VQ: 0.0855, Perp
lexity: 43.71
Epoch 76/500, Batch 500/782, Loss: 0.1169, Recon: 0.0202, VQ: 0.0968, Perp
lexity: 44.97
Epoch 76/500, Batch 600/782, Loss: 0.1048, Recon: 0.0195, VQ: 0.0854, Perp
lexity: 44.23
Epoch 76/500, Batch 700/782, Loss: 0.1087, Recon: 0.0201, VQ: 0.0886, Perp
lexity: 44.34

=== Epoch 76/500 Summary ===

Train Loss: 0.1191 | Val Loss: 0.1172
Train Recon: 0.0210 | Val Recon: 0.0216
Train VQ: 0.0981 | Val VQ: 0.0957
Train Perplexity: 44.88 | Val Perplexity: 44.99
✓ Best model saved! (Val Loss: 0.1172)
Best Loss: 0.1172 (Epoch 76)

Epoch 77/500, Batch 100/782, Loss: 0.1291, Recon: 0.0221, VQ: 0.1069, Perplexity: 45.98
Epoch 77/500, Batch 200/782, Loss: 0.1062, Recon: 0.0190, VQ: 0.0871, Perplexity: 44.22
Epoch 77/500, Batch 300/782, Loss: 0.1073, Recon: 0.0198, VQ: 0.0875, Perplexity: 44.38
Epoch 77/500, Batch 400/782, Loss: 0.1149, Recon: 0.0202, VQ: 0.0946, Perplexity: 44.50
Epoch 77/500, Batch 500/782, Loss: 0.1245, Recon: 0.0217, VQ: 0.1028, Perplexity: 47.03
Epoch 77/500, Batch 600/782, Loss: 0.1183, Recon: 0.0216, VQ: 0.0967, Perplexity: 46.89
Epoch 77/500, Batch 700/782, Loss: 0.1162, Recon: 0.0220, VQ: 0.0942, Perplexity: 45.92

=== Epoch 77/500 Summary ===

Train Loss: 0.1187 | Val Loss: 0.1177
Train Recon: 0.0210 | Val Recon: 0.0215
Train VQ: 0.0978 | Val VQ: 0.0962
Train Perplexity: 45.03 | Val Perplexity: 44.80
No improvement. Patience: 1/20
Best Loss: 0.1172 (Epoch 76)

Epoch 78/500, Batch 100/782, Loss: 0.1214, Recon: 0.0218, VQ: 0.0996, Perplexity: 45.89
Epoch 78/500, Batch 200/782, Loss: 0.1247, Recon: 0.0236, VQ: 0.1012, Perplexity: 47.23
Epoch 78/500, Batch 300/782, Loss: 0.1206, Recon: 0.0217, VQ: 0.0989, Perplexity: 46.28
Epoch 78/500, Batch 400/782, Loss: 0.1172, Recon: 0.0203, VQ: 0.0968, Perplexity: 44.45
Epoch 78/500, Batch 500/782, Loss: 0.1171, Recon: 0.0214, VQ: 0.0957, Perplexity: 45.60
Epoch 78/500, Batch 600/782, Loss: 0.1259, Recon: 0.0221, VQ: 0.1038, Perplexity: 47.34
Epoch 78/500, Batch 700/782, Loss: 0.1224, Recon: 0.0226, VQ: 0.0998, Perplexity: 45.44

=== Epoch 78/500 Summary ===

Train Loss: 0.1187 | Val Loss: 0.1188
Train Recon: 0.0209 | Val Recon: 0.0215
Train VQ: 0.0977 | Val VQ: 0.0973
Train Perplexity: 45.19 | Val Perplexity: 45.59
No improvement. Patience: 2/20
Best Loss: 0.1172 (Epoch 76)

Epoch 79/500, Batch 100/782, Loss: 0.1118, Recon: 0.0199, VQ: 0.0918, Perplexity: 42.40
Epoch 79/500, Batch 200/782, Loss: 0.1161, Recon: 0.0205, VQ: 0.0957, Perplexity: 44.60
Epoch 79/500, Batch 300/782, Loss: 0.1002, Recon: 0.0195, VQ: 0.0807, Perplexity: 43.55

Epoch 79/500, Batch 400/782, Loss: 0.1201, Recon: 0.0215, VQ: 0.0987, Perplexity: 46.35
Epoch 79/500, Batch 500/782, Loss: 0.1238, Recon: 0.0218, VQ: 0.1020, Perplexity: 46.29
Epoch 79/500, Batch 600/782, Loss: 0.1025, Recon: 0.0184, VQ: 0.0841, Perplexity: 43.76
Epoch 79/500, Batch 700/782, Loss: 0.1401, Recon: 0.0237, VQ: 0.1164, Perplexity: 47.40

=== Epoch 79/500 Summary ===

Train Loss: 0.1187 | Val Loss: 0.1181
Train Recon: 0.0209 | Val Recon: 0.0215
Train VQ: 0.0978 | Val VQ: 0.0966
Train Perplexity: 45.49 | Val Perplexity: 45.69
No improvement. Patience: 3/20
Best Loss: 0.1172 (Epoch 76)

Epoch 80/500, Batch 100/782, Loss: 0.1067, Recon: 0.0189, VQ: 0.0878, Perplexity: 45.20
Epoch 80/500, Batch 200/782, Loss: 0.1159, Recon: 0.0201, VQ: 0.0959, Perplexity: 46.64
Epoch 80/500, Batch 300/782, Loss: 0.1107, Recon: 0.0214, VQ: 0.0892, Perplexity: 44.60
Epoch 80/500, Batch 400/782, Loss: 0.1038, Recon: 0.0197, VQ: 0.0841, Perplexity: 43.87
Epoch 80/500, Batch 500/782, Loss: 0.1101, Recon: 0.0202, VQ: 0.0899, Perplexity: 45.39
Epoch 80/500, Batch 600/782, Loss: 0.1334, Recon: 0.0230, VQ: 0.1103, Perplexity: 48.82
Epoch 80/500, Batch 700/782, Loss: 0.1326, Recon: 0.0216, VQ: 0.1111, Perplexity: 45.04

=== Epoch 80/500 Summary ===

Train Loss: 0.1185 | Val Loss: 0.1194
Train Recon: 0.0208 | Val Recon: 0.0214
Train VQ: 0.0977 | Val VQ: 0.0980
Train Perplexity: 45.79 | Val Perplexity: 46.11
No improvement. Patience: 4/20
Best Loss: 0.1172 (Epoch 76)

Epoch 81/500, Batch 100/782, Loss: 0.1128, Recon: 0.0209, VQ: 0.0918, Perplexity: 45.22
Epoch 81/500, Batch 200/782, Loss: 0.1250, Recon: 0.0226, VQ: 0.1024, Perplexity: 46.91
Epoch 81/500, Batch 300/782, Loss: 0.1325, Recon: 0.0215, VQ: 0.1109, Perplexity: 47.22
Epoch 81/500, Batch 400/782, Loss: 0.1120, Recon: 0.0200, VQ: 0.0920, Perplexity: 46.87
Epoch 81/500, Batch 500/782, Loss: 0.1077, Recon: 0.0204, VQ: 0.0873, Perplexity: 45.67
Epoch 81/500, Batch 600/782, Loss: 0.1116, Recon: 0.0203, VQ: 0.0913, Perplexity: 46.12
Epoch 81/500, Batch 700/782, Loss: 0.1342, Recon: 0.0243, VQ: 0.1099, Perplexity: 47.63

=== Epoch 81/500 Summary ===

Train Loss: 0.1179 | Val Loss: 0.1175
Train Recon: 0.0208 | Val Recon: 0.0214
Train VQ: 0.0971 | Val VQ: 0.0961
Train Perplexity: 45.95 | Val Perplexity: 45.85

No improvement. Patience: 5/20
Best Loss: 0.1172 (Epoch 76)

Epoch 82/500, Batch 100/782, Loss: 0.1100, Recon: 0.0209, VQ: 0.0891, Perplexity: 45.56
Epoch 82/500, Batch 200/782, Loss: 0.1330, Recon: 0.0197, VQ: 0.1133, Perplexity: 45.09
Epoch 82/500, Batch 300/782, Loss: 0.1057, Recon: 0.0201, VQ: 0.0856, Perplexity: 45.15
Epoch 82/500, Batch 400/782, Loss: 0.1531, Recon: 0.0223, VQ: 0.1308, Perplexity: 46.90
Epoch 82/500, Batch 500/782, Loss: 0.1155, Recon: 0.0206, VQ: 0.0949, Perplexity: 45.18
Epoch 82/500, Batch 600/782, Loss: 0.0974, Recon: 0.0183, VQ: 0.0791, Perplexity: 43.80
Epoch 82/500, Batch 700/782, Loss: 0.1128, Recon: 0.0192, VQ: 0.0936, Perplexity: 45.74

=== Epoch 82/500 Summary ===
Train Loss: 0.1176 | Val Loss: 0.1173
Train Recon: 0.0208 | Val Recon: 0.0214
Train VQ: 0.0968 | Val VQ: 0.0959
Train Perplexity: 46.04 | Val Perplexity: 46.24
No improvement. Patience: 6/20
Best Loss: 0.1172 (Epoch 76)

Epoch 83/500, Batch 100/782, Loss: 0.1092, Recon: 0.0202, VQ: 0.0890, Perplexity: 45.75
Epoch 83/500, Batch 200/782, Loss: 0.0987, Recon: 0.0178, VQ: 0.0809, Perplexity: 42.86
Epoch 83/500, Batch 300/782, Loss: 0.1063, Recon: 0.0200, VQ: 0.0863, Perplexity: 46.52
Epoch 83/500, Batch 400/782, Loss: 0.1080, Recon: 0.0198, VQ: 0.0882, Perplexity: 45.34
Epoch 83/500, Batch 500/782, Loss: 0.1265, Recon: 0.0209, VQ: 0.1056, Perplexity: 47.27
Epoch 83/500, Batch 600/782, Loss: 0.1195, Recon: 0.0207, VQ: 0.0988, Perplexity: 45.64
Epoch 83/500, Batch 700/782, Loss: 0.1119, Recon: 0.0213, VQ: 0.0905, Perplexity: 45.37

=== Epoch 83/500 Summary ===
Train Loss: 0.1173 | Val Loss: 0.1172
Train Recon: 0.0208 | Val Recon: 0.0214
Train VQ: 0.0965 | Val VQ: 0.0958
Train Perplexity: 46.12 | Val Perplexity: 46.00
No improvement. Patience: 7/20
Best Loss: 0.1172 (Epoch 76)

Epoch 84/500, Batch 100/782, Loss: 0.1120, Recon: 0.0213, VQ: 0.0907, Perplexity: 47.51
Epoch 84/500, Batch 200/782, Loss: 0.1096, Recon: 0.0210, VQ: 0.0885, Perplexity: 46.33
Epoch 84/500, Batch 300/782, Loss: 0.1153, Recon: 0.0205, VQ: 0.0948, Perplexity: 46.30
Epoch 84/500, Batch 400/782, Loss: 0.1083, Recon: 0.0204, VQ: 0.0879, Perplexity: 46.71
Epoch 84/500, Batch 500/782, Loss: 0.0996, Recon: 0.0187, VQ: 0.0808, Perplexity: 43.63
Epoch 84/500, Batch 600/782, Loss: 0.1066, Recon: 0.0199, VQ: 0.0867, Perplexity: 46.33

lexity: 45.37

Epoch 84/500, Batch 700/782, Loss: 0.1107, Recon: 0.0195, VQ: 0.0911, Perp
lexity: 44.58

=== Epoch 84/500 Summary ===

Train Loss: 0.1170 | Val Loss: 0.1175

Train Recon: 0.0207 | Val Recon: 0.0213

Train VQ: 0.0962 | Val VQ: 0.0961

Train Perplexity: 46.20 | Val Perplexity: 46.29

No improvement. Patience: 8/20

Best Loss: 0.1172 (Epoch 76)

Epoch 85/500, Batch 100/782, Loss: 0.1240, Recon: 0.0226, VQ: 0.1014, Perp
lexity: 48.29

Epoch 85/500, Batch 200/782, Loss: 0.1074, Recon: 0.0196, VQ: 0.0878, Perp
lexity: 45.07

Epoch 85/500, Batch 300/782, Loss: 0.1194, Recon: 0.0214, VQ: 0.0980, Perp
lexity: 47.69

Epoch 85/500, Batch 400/782, Loss: 0.1135, Recon: 0.0194, VQ: 0.0941, Perp
lexity: 45.89

Epoch 85/500, Batch 500/782, Loss: 0.1066, Recon: 0.0195, VQ: 0.0871, Perp
lexity: 45.23

Epoch 85/500, Batch 600/782, Loss: 0.1167, Recon: 0.0206, VQ: 0.0961, Perp
lexity: 46.98

Epoch 85/500, Batch 700/782, Loss: 0.1032, Recon: 0.0190, VQ: 0.0842, Perp
lexity: 45.13

=== Epoch 85/500 Summary ===

Train Loss: 0.1167 | Val Loss: 0.1162

Train Recon: 0.0207 | Val Recon: 0.0213

Train VQ: 0.0960 | Val VQ: 0.0949

Train Perplexity: 46.47 | Val Perplexity: 46.64

✓ Best model saved! (Val Loss: 0.1162)

Best Loss: 0.1162 (Epoch 85)

Epoch 86/500, Batch 100/782, Loss: 0.0940, Recon: 0.0185, VQ: 0.0754, Perp
lexity: 44.22

Epoch 86/500, Batch 200/782, Loss: 0.1145, Recon: 0.0203, VQ: 0.0942, Perp
lexity: 47.13

Epoch 86/500, Batch 300/782, Loss: 0.1055, Recon: 0.0199, VQ: 0.0856, Perp
lexity: 44.84

Epoch 86/500, Batch 400/782, Loss: 0.1171, Recon: 0.0214, VQ: 0.0957, Perp
lexity: 46.94

Epoch 86/500, Batch 500/782, Loss: 0.1285, Recon: 0.0219, VQ: 0.1066, Perp
lexity: 46.72

Epoch 86/500, Batch 600/782, Loss: 0.1069, Recon: 0.0199, VQ: 0.0869, Perp
lexity: 45.23

Epoch 86/500, Batch 700/782, Loss: 0.0982, Recon: 0.0187, VQ: 0.0795, Perp
lexity: 44.26

=== Epoch 86/500 Summary ===

Train Loss: 0.1163 | Val Loss: 0.1174

Train Recon: 0.0207 | Val Recon: 0.0213

Train VQ: 0.0957 | Val VQ: 0.0961

Train Perplexity: 46.62 | Val Perplexity: 47.01

No improvement. Patience: 1/20

Best Loss: 0.1162 (Epoch 85)

Epoch 87/500, Batch 100/782, Loss: 0.1228, Recon: 0.0209, VQ: 0.1019, Perp
lexity: 46.00

Epoch 87/500, Batch 200/782, Loss: 0.1175, Recon: 0.0210, VQ: 0.0965, Perplexity: 47.15
Epoch 87/500, Batch 300/782, Loss: 0.1001, Recon: 0.0183, VQ: 0.0818, Perplexity: 43.52
Epoch 87/500, Batch 400/782, Loss: 0.1153, Recon: 0.0217, VQ: 0.0937, Perplexity: 47.62
Epoch 87/500, Batch 500/782, Loss: 0.1143, Recon: 0.0208, VQ: 0.0935, Perplexity: 45.66
Epoch 87/500, Batch 600/782, Loss: 0.1203, Recon: 0.0207, VQ: 0.0996, Perplexity: 46.01
Epoch 87/500, Batch 700/782, Loss: 0.1197, Recon: 0.0206, VQ: 0.0991, Perplexity: 47.29

=== Epoch 87/500 Summary ===

Train Loss: 0.1163 | Val Loss: 0.1158
Train Recon: 0.0206 | Val Recon: 0.0212
Train VQ: 0.0957 | Val VQ: 0.0945
Train Perplexity: 46.78 | Val Perplexity: 46.94
✓ Best model saved! (Val Loss: 0.1158)
Best Loss: 0.1158 (Epoch 87)

Epoch 88/500, Batch 100/782, Loss: 0.1228, Recon: 0.0206, VQ: 0.1022, Perplexity: 45.72
Epoch 88/500, Batch 200/782, Loss: 0.1136, Recon: 0.0208, VQ: 0.0928, Perplexity: 46.27
Epoch 88/500, Batch 300/782, Loss: 0.1049, Recon: 0.0196, VQ: 0.0852, Perplexity: 44.90
Epoch 88/500, Batch 400/782, Loss: 0.1189, Recon: 0.0221, VQ: 0.0968, Perplexity: 48.43
Epoch 88/500, Batch 500/782, Loss: 0.1131, Recon: 0.0205, VQ: 0.0926, Perplexity: 46.44
Epoch 88/500, Batch 600/782, Loss: 0.1027, Recon: 0.0195, VQ: 0.0832, Perplexity: 45.68
Epoch 88/500, Batch 700/782, Loss: 0.1075, Recon: 0.0211, VQ: 0.0864, Perplexity: 47.64

=== Epoch 88/500 Summary ===

Train Loss: 0.1160 | Val Loss: 0.1151
Train Recon: 0.0206 | Val Recon: 0.0212
Train VQ: 0.0953 | Val VQ: 0.0939
Train Perplexity: 46.89 | Val Perplexity: 46.93
✓ Best model saved! (Val Loss: 0.1151)
Best Loss: 0.1151 (Epoch 88)

Epoch 89/500, Batch 100/782, Loss: 0.1041, Recon: 0.0187, VQ: 0.0854, Perplexity: 45.33
Epoch 89/500, Batch 200/782, Loss: 0.1154, Recon: 0.0210, VQ: 0.0944, Perplexity: 46.55
Epoch 89/500, Batch 300/782, Loss: 0.1209, Recon: 0.0209, VQ: 0.1000, Perplexity: 47.51
Epoch 89/500, Batch 400/782, Loss: 0.1110, Recon: 0.0197, VQ: 0.0913, Perplexity: 45.70
Epoch 89/500, Batch 500/782, Loss: 0.1029, Recon: 0.0186, VQ: 0.0843, Perplexity: 46.52
Epoch 89/500, Batch 600/782, Loss: 0.1033, Recon: 0.0187, VQ: 0.0846, Perplexity: 47.45
Epoch 89/500, Batch 700/782, Loss: 0.1061, Recon: 0.0202, VQ: 0.0859, Perplexity: 45.15

=== Epoch 89/500 Summary ===

Train Loss: 0.1163 | Val Loss: 0.1168
Train Recon: 0.0206 | Val Recon: 0.0212
Train VQ: 0.0957 | Val VQ: 0.0955
Train Perplexity: 47.00 | Val Perplexity: 47.20
No improvement. Patience: 1/20
Best Loss: 0.1151 (Epoch 88)

Epoch 90/500, Batch 100/782, Loss: 0.1140, Recon: 0.0201, VQ: 0.0939, Perplexity: 45.73
Epoch 90/500, Batch 200/782, Loss: 0.1196, Recon: 0.0219, VQ: 0.0978, Perplexity: 48.45
Epoch 90/500, Batch 300/782, Loss: 0.1263, Recon: 0.0213, VQ: 0.1051, Perplexity: 48.64
Epoch 90/500, Batch 400/782, Loss: 0.1200, Recon: 0.0209, VQ: 0.0990, Perplexity: 46.71
Epoch 90/500, Batch 500/782, Loss: 0.1131, Recon: 0.0214, VQ: 0.0917, Perplexity: 48.23
Epoch 90/500, Batch 600/782, Loss: 0.1134, Recon: 0.0215, VQ: 0.0919, Perplexity: 47.90
Epoch 90/500, Batch 700/782, Loss: 0.1123, Recon: 0.0197, VQ: 0.0926, Perplexity: 46.35

=== Epoch 90/500 Summary ===
Train Loss: 0.1163 | Val Loss: 0.1162
Train Recon: 0.0206 | Val Recon: 0.0212
Train VQ: 0.0957 | Val VQ: 0.0950
Train Perplexity: 47.18 | Val Perplexity: 47.54
No improvement. Patience: 2/20
Best Loss: 0.1151 (Epoch 88)

Epoch 91/500, Batch 100/782, Loss: 0.1370, Recon: 0.0204, VQ: 0.1167, Perplexity: 48.73
Epoch 91/500, Batch 200/782, Loss: 0.1143, Recon: 0.0200, VQ: 0.0943, Perplexity: 46.31
Epoch 91/500, Batch 300/782, Loss: 0.1079, Recon: 0.0195, VQ: 0.0884, Perplexity: 45.86
Epoch 91/500, Batch 400/782, Loss: 0.1218, Recon: 0.0216, VQ: 0.1001, Perplexity: 48.23
Epoch 91/500, Batch 500/782, Loss: 0.1147, Recon: 0.0218, VQ: 0.0929, Perplexity: 48.77
Epoch 91/500, Batch 600/782, Loss: 0.1313, Recon: 0.0222, VQ: 0.1091, Perplexity: 48.48
Epoch 91/500, Batch 700/782, Loss: 0.1093, Recon: 0.0203, VQ: 0.0891, Perplexity: 47.75

=== Epoch 91/500 Summary ===
Train Loss: 0.1162 | Val Loss: 0.1147
Train Recon: 0.0205 | Val Recon: 0.0212
Train VQ: 0.0957 | Val VQ: 0.0935
Train Perplexity: 47.41 | Val Perplexity: 47.27
✓ Best model saved! (Val Loss: 0.1147)
Best Loss: 0.1147 (Epoch 91)

Epoch 92/500, Batch 100/782, Loss: 0.1103, Recon: 0.0203, VQ: 0.0900, Perplexity: 47.42
Epoch 92/500, Batch 200/782, Loss: 0.1136, Recon: 0.0212, VQ: 0.0924, Perplexity: 47.58
Epoch 92/500, Batch 300/782, Loss: 0.1117, Recon: 0.0198, VQ: 0.0919, Perplexity: 46.95
Epoch 92/500, Batch 400/782, Loss: 0.1071, Recon: 0.0190, VQ: 0.0882, Perplexity: 46.35

lexity: 46.48
Epoch 92/500, Batch 500/782, Loss: 0.1222, Recon: 0.0209, VQ: 0.1014, Perp
lexity: 49.91
Epoch 92/500, Batch 600/782, Loss: 0.1158, Recon: 0.0208, VQ: 0.0950, Perp
lexity: 48.22
Epoch 92/500, Batch 700/782, Loss: 0.1059, Recon: 0.0193, VQ: 0.0866, Perp
lexity: 46.77

=== Epoch 92/500 Summary ===

Train Loss: 0.1159 | Val Loss: 0.1147
Train Recon: 0.0205 | Val Recon: 0.0212
Train VQ: 0.0955 | Val VQ: 0.0936
Train Perplexity: 47.63 | Val Perplexity: 47.80
No improvement. Patience: 1/20
Best Loss: 0.1147 (Epoch 91)

Epoch 93/500, Batch 100/782, Loss: 0.1154, Recon: 0.0199, VQ: 0.0954, Perp
lexity: 46.18
Epoch 93/500, Batch 200/782, Loss: 0.1053, Recon: 0.0189, VQ: 0.0864, Perp
lexity: 45.43
Epoch 93/500, Batch 300/782, Loss: 0.1208, Recon: 0.0206, VQ: 0.1003, Perp
lexity: 48.41
Epoch 93/500, Batch 400/782, Loss: 0.1271, Recon: 0.0223, VQ: 0.1048, Perp
lexity: 48.85
Epoch 93/500, Batch 500/782, Loss: 0.1191, Recon: 0.0206, VQ: 0.0986, Perp
lexity: 46.93
Epoch 93/500, Batch 600/782, Loss: 0.0993, Recon: 0.0186, VQ: 0.0806, Perp
lexity: 45.03
Epoch 93/500, Batch 700/782, Loss: 0.1177, Recon: 0.0220, VQ: 0.0957, Perp
lexity: 50.42

=== Epoch 93/500 Summary ===

Train Loss: 0.1156 | Val Loss: 0.1155
Train Recon: 0.0205 | Val Recon: 0.0211
Train VQ: 0.0951 | Val VQ: 0.0944
Train Perplexity: 47.81 | Val Perplexity: 47.89
No improvement. Patience: 2/20
Best Loss: 0.1147 (Epoch 91)

Epoch 94/500, Batch 100/782, Loss: 0.1029, Recon: 0.0193, VQ: 0.0836, Perp
lexity: 46.93
Epoch 94/500, Batch 200/782, Loss: 0.1082, Recon: 0.0191, VQ: 0.0890, Perp
lexity: 47.39
Epoch 94/500, Batch 300/782, Loss: 0.1155, Recon: 0.0206, VQ: 0.0949, Perp
lexity: 46.64
Epoch 94/500, Batch 400/782, Loss: 0.1111, Recon: 0.0200, VQ: 0.0911, Perp
lexity: 47.86
Epoch 94/500, Batch 500/782, Loss: 0.1042, Recon: 0.0203, VQ: 0.0839, Perp
lexity: 46.18
Epoch 94/500, Batch 600/782, Loss: 0.1100, Recon: 0.0205, VQ: 0.0895, Perp
lexity: 48.22
Epoch 94/500, Batch 700/782, Loss: 0.1240, Recon: 0.0212, VQ: 0.1028, Perp
lexity: 48.32

=== Epoch 94/500 Summary ===

Train Loss: 0.1155 | Val Loss: 0.1150
Train Recon: 0.0204 | Val Recon: 0.0211
Train VQ: 0.0950 | Val VQ: 0.0940
Train Perplexity: 47.86 | Val Perplexity: 47.99
No improvement. Patience: 3/20

Best Loss: 0.1147 (Epoch 91)

Epoch 95/500, Batch 100/782, Loss: 0.1194, Recon: 0.0206, VQ: 0.0988, Perplexity: 49.69
Epoch 95/500, Batch 200/782, Loss: 0.1058, Recon: 0.0196, VQ: 0.0862, Perplexity: 48.55
Epoch 95/500, Batch 300/782, Loss: 0.1087, Recon: 0.0201, VQ: 0.0885, Perplexity: 47.47
Epoch 95/500, Batch 400/782, Loss: 0.1126, Recon: 0.0212, VQ: 0.0915, Perplexity: 49.29
Epoch 95/500, Batch 500/782, Loss: 0.1011, Recon: 0.0185, VQ: 0.0826, Perplexity: 45.79
Epoch 95/500, Batch 600/782, Loss: 0.1231, Recon: 0.0211, VQ: 0.1020, Perplexity: 48.92
Epoch 95/500, Batch 700/782, Loss: 0.1018, Recon: 0.0187, VQ: 0.0831, Perplexity: 46.54

=== Epoch 95/500 Summary ===

Train Loss: 0.1153 | Val Loss: 0.1172
Train Recon: 0.0204 | Val Recon: 0.0211
Train VQ: 0.0949 | Val VQ: 0.0961
Train Perplexity: 47.99 | Val Perplexity: 48.10
No improvement. Patience: 4/20
Best Loss: 0.1147 (Epoch 91)

Epoch 96/500, Batch 100/782, Loss: 0.0956, Recon: 0.0180, VQ: 0.0776, Perplexity: 45.42
Epoch 96/500, Batch 200/782, Loss: 0.1157, Recon: 0.0202, VQ: 0.0956, Perplexity: 49.18
Epoch 96/500, Batch 300/782, Loss: 0.1132, Recon: 0.0212, VQ: 0.0920, Perplexity: 48.04
Epoch 96/500, Batch 400/782, Loss: 0.1302, Recon: 0.0224, VQ: 0.1077, Perplexity: 50.87
Epoch 96/500, Batch 500/782, Loss: 0.1070, Recon: 0.0208, VQ: 0.0862, Perplexity: 49.28
Epoch 96/500, Batch 600/782, Loss: 0.1204, Recon: 0.0218, VQ: 0.0985, Perplexity: 50.03
Epoch 96/500, Batch 700/782, Loss: 0.1070, Recon: 0.0202, VQ: 0.0868, Perplexity: 48.15

=== Epoch 96/500 Summary ===

Train Loss: 0.1152 | Val Loss: 0.1152
Train Recon: 0.0204 | Val Recon: 0.0211
Train VQ: 0.0948 | Val VQ: 0.0941
Train Perplexity: 48.04 | Val Perplexity: 48.00
No improvement. Patience: 5/20
Best Loss: 0.1147 (Epoch 91)

Epoch 97/500, Batch 100/782, Loss: 0.1126, Recon: 0.0207, VQ: 0.0920, Perplexity: 49.87
Epoch 97/500, Batch 200/782, Loss: 0.1199, Recon: 0.0194, VQ: 0.1005, Perplexity: 48.57
Epoch 97/500, Batch 300/782, Loss: 0.1062, Recon: 0.0186, VQ: 0.0876, Perplexity: 46.59
Epoch 97/500, Batch 400/782, Loss: 0.1081, Recon: 0.0206, VQ: 0.0875, Perplexity: 48.06
Epoch 97/500, Batch 500/782, Loss: 0.1087, Recon: 0.0205, VQ: 0.0882, Perplexity: 48.12
Epoch 97/500, Batch 600/782, Loss: 0.1172, Recon: 0.0217, VQ: 0.0956, Perplexity: 50.25

Epoch 97/500, Batch 700/782, Loss: 0.1132, Recon: 0.0195, VQ: 0.0937, Perplexity: 48.09

=== Epoch 97/500 Summary ===

Train Loss: 0.1150 | Val Loss: 0.1160
Train Recon: 0.0204 | Val Recon: 0.0210
Train VQ: 0.0946 | Val VQ: 0.0949
Train Perplexity: 48.34 | Val Perplexity: 48.43
No improvement. Patience: 6/20
Best Loss: 0.1147 (Epoch 91)

Epoch 98/500, Batch 100/782, Loss: 0.1178, Recon: 0.0204, VQ: 0.0974, Perplexity: 48.34
Epoch 98/500, Batch 200/782, Loss: 0.1248, Recon: 0.0217, VQ: 0.1031, Perplexity: 49.06
Epoch 98/500, Batch 300/782, Loss: 0.1222, Recon: 0.0207, VQ: 0.1015, Perplexity: 50.26
Epoch 98/500, Batch 400/782, Loss: 0.1201, Recon: 0.0205, VQ: 0.0997, Perplexity: 49.17
Epoch 98/500, Batch 500/782, Loss: 0.1207, Recon: 0.0211, VQ: 0.0996, Perplexity: 48.73
Epoch 98/500, Batch 600/782, Loss: 0.1206, Recon: 0.0210, VQ: 0.0996, Perplexity: 49.42
Epoch 98/500, Batch 700/782, Loss: 0.1277, Recon: 0.0217, VQ: 0.1059, Perplexity: 50.64

=== Epoch 98/500 Summary ===

Train Loss: 0.1151 | Val Loss: 0.1146
Train Recon: 0.0203 | Val Recon: 0.0210
Train VQ: 0.0948 | Val VQ: 0.0935
Train Perplexity: 48.68 | Val Perplexity: 48.91
✓ Best model saved! (Val Loss: 0.1146)
Best Loss: 0.1146 (Epoch 98)

Epoch 99/500, Batch 100/782, Loss: 0.1064, Recon: 0.0187, VQ: 0.0877, Perplexity: 47.06
Epoch 99/500, Batch 200/782, Loss: 0.1091, Recon: 0.0195, VQ: 0.0896, Perplexity: 49.89
Epoch 99/500, Batch 300/782, Loss: 0.1059, Recon: 0.0185, VQ: 0.0874, Perplexity: 48.37
Epoch 99/500, Batch 400/782, Loss: 0.1083, Recon: 0.0192, VQ: 0.0891, Perplexity: 49.11
Epoch 99/500, Batch 500/782, Loss: 0.1188, Recon: 0.0201, VQ: 0.0987, Perplexity: 49.22
Epoch 99/500, Batch 600/782, Loss: 0.1093, Recon: 0.0204, VQ: 0.0890, Perplexity: 48.89
Epoch 99/500, Batch 700/782, Loss: 0.1114, Recon: 0.0202, VQ: 0.0912, Perplexity: 48.30

=== Epoch 99/500 Summary ===

Train Loss: 0.1147 | Val Loss: 0.1142
Train Recon: 0.0203 | Val Recon: 0.0210
Train VQ: 0.0944 | Val VQ: 0.0932
Train Perplexity: 48.81 | Val Perplexity: 48.89
✓ Best model saved! (Val Loss: 0.1142)
Best Loss: 0.1142 (Epoch 99)

Epoch 100/500, Batch 100/782, Loss: 0.1066, Recon: 0.0196, VQ: 0.0869, Perplexity: 49.15
Epoch 100/500, Batch 200/782, Loss: 0.1125, Recon: 0.0207, VQ: 0.0918, Perplexity: 49.15

plexity: 50.10
Epoch 100/500, Batch 300/782, Loss: 0.1065, Recon: 0.0198, VQ: 0.0867, Per
plexity: 49.26
Epoch 100/500, Batch 400/782, Loss: 0.1085, Recon: 0.0197, VQ: 0.0888, Per
plexity: 48.07
Epoch 100/500, Batch 500/782, Loss: 0.1145, Recon: 0.0206, VQ: 0.0940, Per
plexity: 49.42
Epoch 100/500, Batch 600/782, Loss: 0.1086, Recon: 0.0196, VQ: 0.0890, Per
plexity: 48.23
Epoch 100/500, Batch 700/782, Loss: 0.0984, Recon: 0.0193, VQ: 0.0791, Per
plexity: 47.62

=== Epoch 100/500 Summary ===
Train Loss: 0.1148 | Val Loss: 0.1143
Train Recon: 0.0203 | Val Recon: 0.0210
Train VQ: 0.0945 | Val VQ: 0.0934
Train Perplexity: 48.93 | Val Perplexity: 49.05
No improvement. Patience: 1/20
Best Loss: 0.1142 (Epoch 99)

Epoch 101/500, Batch 100/782, Loss: 0.1028, Recon: 0.0190, VQ: 0.0838, Per
plexity: 47.52
Epoch 101/500, Batch 200/782, Loss: 0.1169, Recon: 0.0209, VQ: 0.0959, Per
plexity: 48.62
Epoch 101/500, Batch 300/782, Loss: 0.1187, Recon: 0.0193, VQ: 0.0994, Per
plexity: 47.01
Epoch 101/500, Batch 400/782, Loss: 0.1137, Recon: 0.0206, VQ: 0.0931, Per
plexity: 48.67
Epoch 101/500, Batch 500/782, Loss: 0.1118, Recon: 0.0200, VQ: 0.0919, Per
plexity: 49.88
Epoch 101/500, Batch 600/782, Loss: 0.1302, Recon: 0.0224, VQ: 0.1077, Per
plexity: 51.25
Epoch 101/500, Batch 700/782, Loss: 0.1176, Recon: 0.0202, VQ: 0.0974, Per
plexity: 49.59

=== Epoch 101/500 Summary ===
Train Loss: 0.1144 | Val Loss: 0.1143
Train Recon: 0.0202 | Val Recon: 0.0210
Train VQ: 0.0942 | Val VQ: 0.0934
Train Perplexity: 48.97 | Val Perplexity: 49.13
No improvement. Patience: 2/20
Best Loss: 0.1142 (Epoch 99)

Epoch 102/500, Batch 100/782, Loss: 0.1110, Recon: 0.0204, VQ: 0.0906, Per
plexity: 48.97
Epoch 102/500, Batch 200/782, Loss: 0.1219, Recon: 0.0226, VQ: 0.0993, Per
plexity: 51.47
Epoch 102/500, Batch 300/782, Loss: 0.1091, Recon: 0.0199, VQ: 0.0892, Per
plexity: 49.81
Epoch 102/500, Batch 400/782, Loss: 0.1254, Recon: 0.0218, VQ: 0.1036, Per
plexity: 49.57
Epoch 102/500, Batch 500/782, Loss: 0.1016, Recon: 0.0179, VQ: 0.0837, Per
plexity: 45.97
Epoch 102/500, Batch 600/782, Loss: 0.1133, Recon: 0.0202, VQ: 0.0931, Per
plexity: 49.57
Epoch 102/500, Batch 700/782, Loss: 0.1007, Recon: 0.0175, VQ: 0.0831, Per
plexity: 47.30

=== Epoch 102/500 Summary ===
Train Loss: 0.1144 | Val Loss: 0.1138

Train Recon: 0.0202 | Val Recon: 0.0209
Train VQ: 0.0941 | Val VQ: 0.0929
Train Perplexity: 49.05 | Val Perplexity: 49.14
✓ Best model saved! (Val Loss: 0.1138)
Best Loss: 0.1138 (Epoch 102)

Epoch 103/500, Batch 100/782, Loss: 0.1251, Recon: 0.0219, VQ: 0.1031, Perplexity: 51.14
Epoch 103/500, Batch 200/782, Loss: 0.1127, Recon: 0.0199, VQ: 0.0928, Perplexity: 49.03
Epoch 103/500, Batch 300/782, Loss: 0.0990, Recon: 0.0186, VQ: 0.0804, Perplexity: 48.17
Epoch 103/500, Batch 400/782, Loss: 0.1109, Recon: 0.0199, VQ: 0.0910, Perplexity: 49.05
Epoch 103/500, Batch 500/782, Loss: 0.1178, Recon: 0.0194, VQ: 0.0984, Perplexity: 48.56
Epoch 103/500, Batch 600/782, Loss: 0.1162, Recon: 0.0207, VQ: 0.0955, Perplexity: 48.59
Epoch 103/500, Batch 700/782, Loss: 0.1151, Recon: 0.0207, VQ: 0.0944, Perplexity: 48.06

=== Epoch 103/500 Summary ===
Train Loss: 0.1141 | Val Loss: 0.1143
Train Recon: 0.0202 | Val Recon: 0.0210
Train VQ: 0.0939 | Val VQ: 0.0934
Train Perplexity: 49.06 | Val Perplexity: 49.36
No improvement. Patience: 1/20
Best Loss: 0.1138 (Epoch 102)

Epoch 104/500, Batch 100/782, Loss: 0.1252, Recon: 0.0213, VQ: 0.1038, Perplexity: 51.18
Epoch 104/500, Batch 200/782, Loss: 0.1021, Recon: 0.0189, VQ: 0.0832, Perplexity: 48.05
Epoch 104/500, Batch 300/782, Loss: 0.1112, Recon: 0.0212, VQ: 0.0899, Perplexity: 48.87
Epoch 104/500, Batch 400/782, Loss: 0.1007, Recon: 0.0181, VQ: 0.0826, Perplexity: 47.73
Epoch 104/500, Batch 500/782, Loss: 0.1019, Recon: 0.0192, VQ: 0.0827, Perplexity: 47.53
Epoch 104/500, Batch 600/782, Loss: 0.1139, Recon: 0.0197, VQ: 0.0943, Perplexity: 49.07
Epoch 104/500, Batch 700/782, Loss: 0.1284, Recon: 0.0216, VQ: 0.1068, Perplexity: 51.59

=== Epoch 104/500 Summary ===
Train Loss: 0.1142 | Val Loss: 0.1142
Train Recon: 0.0202 | Val Recon: 0.0209
Train VQ: 0.0940 | Val VQ: 0.0933
Train Perplexity: 49.11 | Val Perplexity: 49.18
No improvement. Patience: 2/20
Best Loss: 0.1138 (Epoch 102)

Epoch 105/500, Batch 100/782, Loss: 0.0967, Recon: 0.0186, VQ: 0.0781, Perplexity: 48.29
Epoch 105/500, Batch 200/782, Loss: 0.1067, Recon: 0.0190, VQ: 0.0877, Perplexity: 49.46
Epoch 105/500, Batch 300/782, Loss: 0.1104, Recon: 0.0204, VQ: 0.0900, Perplexity: 48.82
Epoch 105/500, Batch 400/782, Loss: 0.1109, Recon: 0.0195, VQ: 0.0914, Perplexity: 48.77

Epoch 105/500, Batch 500/782, Loss: 0.1091, Recon: 0.0193, VQ: 0.0898, Perplexity: 50.65
Epoch 105/500, Batch 600/782, Loss: 0.1174, Recon: 0.0218, VQ: 0.0956, Perplexity: 50.20
Epoch 105/500, Batch 700/782, Loss: 0.1071, Recon: 0.0199, VQ: 0.0872, Perplexity: 49.24

=== Epoch 105/500 Summary ===

Train Loss: 0.1141 | Val Loss: 0.1157
Train Recon: 0.0202 | Val Recon: 0.0210
Train VQ: 0.0939 | Val VQ: 0.0948
Train Perplexity: 49.14 | Val Perplexity: 49.25
No improvement. Patience: 3/20
Best Loss: 0.1138 (Epoch 102)

Epoch 106/500, Batch 100/782, Loss: 0.1270, Recon: 0.0221, VQ: 0.1049, Perplexity: 52.43
Epoch 106/500, Batch 200/782, Loss: 0.1070, Recon: 0.0197, VQ: 0.0873, Perplexity: 49.59
Epoch 106/500, Batch 300/782, Loss: 0.1070, Recon: 0.0203, VQ: 0.0867, Perplexity: 47.56
Epoch 106/500, Batch 400/782, Loss: 0.1230, Recon: 0.0225, VQ: 0.1005, Perplexity: 51.40
Epoch 106/500, Batch 500/782, Loss: 0.1105, Recon: 0.0192, VQ: 0.0913, Perplexity: 48.54
Epoch 106/500, Batch 600/782, Loss: 0.1171, Recon: 0.0210, VQ: 0.0960, Perplexity: 49.63
Epoch 106/500, Batch 700/782, Loss: 0.1198, Recon: 0.0197, VQ: 0.1000, Perplexity: 50.08

=== Epoch 106/500 Summary ===

Train Loss: 0.1139 | Val Loss: 0.1139
Train Recon: 0.0202 | Val Recon: 0.0211
Train VQ: 0.0938 | Val VQ: 0.0928
Train Perplexity: 49.19 | Val Perplexity: 49.24
No improvement. Patience: 4/20
Best Loss: 0.1138 (Epoch 102)

Epoch 107/500, Batch 100/782, Loss: 0.1205, Recon: 0.0219, VQ: 0.0986, Perplexity: 50.74
Epoch 107/500, Batch 200/782, Loss: 0.1181, Recon: 0.0227, VQ: 0.0955, Perplexity: 51.85
Epoch 107/500, Batch 300/782, Loss: 0.1087, Recon: 0.0189, VQ: 0.0898, Perplexity: 48.91
Epoch 107/500, Batch 400/782, Loss: 0.1128, Recon: 0.0200, VQ: 0.0928, Perplexity: 50.06
Epoch 107/500, Batch 500/782, Loss: 0.1172, Recon: 0.0209, VQ: 0.0963, Perplexity: 50.82
Epoch 107/500, Batch 600/782, Loss: 0.1154, Recon: 0.0208, VQ: 0.0947, Perplexity: 50.66
Epoch 107/500, Batch 700/782, Loss: 0.1095, Recon: 0.0196, VQ: 0.0899, Perplexity: 47.14

=== Epoch 107/500 Summary ===

Train Loss: 0.1140 | Val Loss: 0.1141
Train Recon: 0.0201 | Val Recon: 0.0210
Train VQ: 0.0938 | Val VQ: 0.0931
Train Perplexity: 49.26 | Val Perplexity: 49.53
No improvement. Patience: 5/20
Best Loss: 0.1138 (Epoch 102)

Epoch 108/500, Batch 100/782, Loss: 0.1264, Recon: 0.0208, VQ: 0.1056, Perplexity: 49.21
Epoch 108/500, Batch 200/782, Loss: 0.1172, Recon: 0.0207, VQ: 0.0966, Perplexity: 51.97
Epoch 108/500, Batch 300/782, Loss: 0.1181, Recon: 0.0204, VQ: 0.0977, Perplexity: 49.06
Epoch 108/500, Batch 400/782, Loss: 0.1023, Recon: 0.0192, VQ: 0.0832, Perplexity: 48.18
Epoch 108/500, Batch 500/782, Loss: 0.1076, Recon: 0.0195, VQ: 0.0881, Perplexity: 48.46
Epoch 108/500, Batch 600/782, Loss: 0.1093, Recon: 0.0210, VQ: 0.0883, Perplexity: 47.86
Epoch 108/500, Batch 700/782, Loss: 0.0970, Recon: 0.0179, VQ: 0.0791, Perplexity: 46.53

=== Epoch 108/500 Summary ===

Train Loss: 0.1141 | Val Loss: 0.1137
Train Recon: 0.0201 | Val Recon: 0.0209
Train VQ: 0.0940 | Val VQ: 0.0929
Train Perplexity: 49.31 | Val Perplexity: 49.22
✓ Best model saved! (Val Loss: 0.1137)
Best Loss: 0.1137 (Epoch 108)

Epoch 109/500, Batch 100/782, Loss: 0.1054, Recon: 0.0196, VQ: 0.0857, Perplexity: 48.62
Epoch 109/500, Batch 200/782, Loss: 0.1328, Recon: 0.0228, VQ: 0.1100, Perplexity: 52.62
Epoch 109/500, Batch 300/782, Loss: 0.1131, Recon: 0.0201, VQ: 0.0930, Perplexity: 50.29
Epoch 109/500, Batch 400/782, Loss: 0.1032, Recon: 0.0183, VQ: 0.0849, Perplexity: 48.73
Epoch 109/500, Batch 500/782, Loss: 0.1153, Recon: 0.0210, VQ: 0.0943, Perplexity: 48.52
Epoch 109/500, Batch 600/782, Loss: 0.1108, Recon: 0.0198, VQ: 0.0909, Perplexity: 50.42
Epoch 109/500, Batch 700/782, Loss: 0.1128, Recon: 0.0194, VQ: 0.0934, Perplexity: 48.38

=== Epoch 109/500 Summary ===

Train Loss: 0.1141 | Val Loss: 0.1134
Train Recon: 0.0201 | Val Recon: 0.0209
Train VQ: 0.0940 | Val VQ: 0.0925
Train Perplexity: 49.36 | Val Perplexity: 49.21
✓ Best model saved! (Val Loss: 0.1134)
Best Loss: 0.1134 (Epoch 109)

Epoch 110/500, Batch 100/782, Loss: 0.1119, Recon: 0.0200, VQ: 0.0919, Perplexity: 49.85
Epoch 110/500, Batch 200/782, Loss: 0.1144, Recon: 0.0201, VQ: 0.0943, Perplexity: 49.80
Epoch 110/500, Batch 300/782, Loss: 0.1263, Recon: 0.0215, VQ: 0.1049, Perplexity: 50.74
Epoch 110/500, Batch 400/782, Loss: 0.1069, Recon: 0.0184, VQ: 0.0885, Perplexity: 47.41
Epoch 110/500, Batch 500/782, Loss: 0.1065, Recon: 0.0185, VQ: 0.0879, Perplexity: 48.28
Epoch 110/500, Batch 600/782, Loss: 0.1179, Recon: 0.0214, VQ: 0.0965, Perplexity: 50.50
Epoch 110/500, Batch 700/782, Loss: 0.1084, Recon: 0.0204, VQ: 0.0880, Per

plexity: 49.27

=== Epoch 110/500 Summary ===

Train Loss: 0.1141 | Val Loss: 0.1150

Train Recon: 0.0201 | Val Recon: 0.0209

Train VQ: 0.0940 | Val VQ: 0.0941

Train Perplexity: 49.41 | Val Perplexity: 49.68

No improvement. Patience: 1/20

Best Loss: 0.1134 (Epoch 109)

Epoch 111/500, Batch 100/782, Loss: 0.1116, Recon: 0.0199, VQ: 0.0917, Perplexity: 50.78

Epoch 111/500, Batch 200/782, Loss: 0.1090, Recon: 0.0200, VQ: 0.0890, Perplexity: 49.79

Epoch 111/500, Batch 300/782, Loss: 0.1250, Recon: 0.0218, VQ: 0.1032, Perplexity: 52.64

Epoch 111/500, Batch 400/782, Loss: 0.1045, Recon: 0.0193, VQ: 0.0852, Perplexity: 48.85

Epoch 111/500, Batch 500/782, Loss: 0.1091, Recon: 0.0198, VQ: 0.0893, Perplexity: 47.06

Epoch 111/500, Batch 600/782, Loss: 0.1185, Recon: 0.0201, VQ: 0.0985, Perplexity: 50.06

Epoch 111/500, Batch 700/782, Loss: 0.1148, Recon: 0.0211, VQ: 0.0937, Perplexity: 51.29

=== Epoch 111/500 Summary ===

Train Loss: 0.1143 | Val Loss: 0.1142

Train Recon: 0.0201 | Val Recon: 0.0208

Train VQ: 0.0942 | Val VQ: 0.0934

Train Perplexity: 49.47 | Val Perplexity: 49.52

No improvement. Patience: 2/20

Best Loss: 0.1134 (Epoch 109)

Epoch 112/500, Batch 100/782, Loss: 0.1110, Recon: 0.0195, VQ: 0.0915, Perplexity: 50.08

Epoch 112/500, Batch 200/782, Loss: 0.1084, Recon: 0.0195, VQ: 0.0889, Perplexity: 49.40

Epoch 112/500, Batch 300/782, Loss: 0.1234, Recon: 0.0215, VQ: 0.1019, Perplexity: 50.84

Epoch 112/500, Batch 400/782, Loss: 0.1136, Recon: 0.0214, VQ: 0.0921, Perplexity: 50.42

Epoch 112/500, Batch 500/782, Loss: 0.1135, Recon: 0.0200, VQ: 0.0935, Perplexity: 50.98

Epoch 112/500, Batch 600/782, Loss: 0.1042, Recon: 0.0193, VQ: 0.0849, Perplexity: 48.49

Epoch 112/500, Batch 700/782, Loss: 0.1165, Recon: 0.0210, VQ: 0.0955, Perplexity: 49.39

=== Epoch 112/500 Summary ===

Train Loss: 0.1141 | Val Loss: 0.1117

Train Recon: 0.0201 | Val Recon: 0.0210

Train VQ: 0.0940 | Val VQ: 0.0907

Train Perplexity: 49.51 | Val Perplexity: 49.61

✓ Best model saved! (Val Loss: 0.1117)

Best Loss: 0.1117 (Epoch 112)

Epoch 113/500, Batch 100/782, Loss: 0.1170, Recon: 0.0215, VQ: 0.0955, Perplexity: 52.48

Epoch 113/500, Batch 200/782, Loss: 0.1156, Recon: 0.0201, VQ: 0.0954, Perplexity: 51.10

Epoch 113/500, Batch 300/782, Loss: 0.1026, Recon: 0.0180, VQ: 0.0846, Perplexity: 48.03
Epoch 113/500, Batch 400/782, Loss: 0.1026, Recon: 0.0194, VQ: 0.0832, Perplexity: 49.09
Epoch 113/500, Batch 500/782, Loss: 0.1095, Recon: 0.0197, VQ: 0.0898, Perplexity: 49.41
Epoch 113/500, Batch 600/782, Loss: 0.1183, Recon: 0.0195, VQ: 0.0988, Perplexity: 50.24
Epoch 113/500, Batch 700/782, Loss: 0.1061, Recon: 0.0194, VQ: 0.0866, Perplexity: 48.29

=== Epoch 113/500 Summary ===

Train Loss: 0.1138 | Val Loss: 0.1138
Train Recon: 0.0200 | Val Recon: 0.0209
Train VQ: 0.0938 | Val VQ: 0.0929
Train Perplexity: 49.66 | Val Perplexity: 49.74
No improvement. Patience: 1/20
Best Loss: 0.1117 (Epoch 112)

Epoch 114/500, Batch 100/782, Loss: 0.1340, Recon: 0.0217, VQ: 0.1123, Perplexity: 51.79
Epoch 114/500, Batch 200/782, Loss: 0.1061, Recon: 0.0210, VQ: 0.0851, Perplexity: 50.47
Epoch 114/500, Batch 300/782, Loss: 0.1138, Recon: 0.0202, VQ: 0.0936, Perplexity: 50.64
Epoch 114/500, Batch 400/782, Loss: 0.1019, Recon: 0.0186, VQ: 0.0832, Perplexity: 49.44
Epoch 114/500, Batch 500/782, Loss: 0.0998, Recon: 0.0188, VQ: 0.0810, Perplexity: 49.09
Epoch 114/500, Batch 600/782, Loss: 0.1185, Recon: 0.0222, VQ: 0.0963, Perplexity: 51.91
Epoch 114/500, Batch 700/782, Loss: 0.1179, Recon: 0.0218, VQ: 0.0961, Perplexity: 51.86

=== Epoch 114/500 Summary ===

Train Loss: 0.1138 | Val Loss: 0.1126
Train Recon: 0.0200 | Val Recon: 0.0208
Train VQ: 0.0938 | Val VQ: 0.0918
Train Perplexity: 49.69 | Val Perplexity: 49.81
No improvement. Patience: 2/20
Best Loss: 0.1117 (Epoch 112)

Epoch 115/500, Batch 100/782, Loss: 0.1073, Recon: 0.0204, VQ: 0.0869, Perplexity: 49.13
Epoch 115/500, Batch 200/782, Loss: 0.1103, Recon: 0.0204, VQ: 0.0899, Perplexity: 50.61
Epoch 115/500, Batch 300/782, Loss: 0.1102, Recon: 0.0199, VQ: 0.0903, Perplexity: 50.71
Epoch 115/500, Batch 400/782, Loss: 0.1089, Recon: 0.0196, VQ: 0.0893, Perplexity: 50.37
Epoch 115/500, Batch 500/782, Loss: 0.1172, Recon: 0.0217, VQ: 0.0955, Perplexity: 50.81
Epoch 115/500, Batch 600/782, Loss: 0.1143, Recon: 0.0194, VQ: 0.0949, Perplexity: 50.79
Epoch 115/500, Batch 700/782, Loss: 0.1127, Recon: 0.0206, VQ: 0.0921, Perplexity: 50.23

=== Epoch 115/500 Summary ===

Train Loss: 0.1135 | Val Loss: 0.1128
Train Recon: 0.0200 | Val Recon: 0.0208

Train VQ: 0.0935 | Val VQ: 0.0920
Train Perplexity: 49.77 | Val Perplexity: 49.88
No improvement. Patience: 3/20
Best Loss: 0.1117 (Epoch 112)

Epoch 116/500, Batch 100/782, Loss: 0.1016, Recon: 0.0177, VQ: 0.0839, Perplexity: 48.68
Epoch 116/500, Batch 200/782, Loss: 0.1209, Recon: 0.0191, VQ: 0.1018, Perplexity: 50.36
Epoch 116/500, Batch 300/782, Loss: 0.1057, Recon: 0.0183, VQ: 0.0874, Perplexity: 47.13
Epoch 116/500, Batch 400/782, Loss: 0.1067, Recon: 0.0190, VQ: 0.0877, Perplexity: 51.07
Epoch 116/500, Batch 500/782, Loss: 0.1034, Recon: 0.0200, VQ: 0.0834, Perplexity: 48.26
Epoch 116/500, Batch 600/782, Loss: 0.1129, Recon: 0.0206, VQ: 0.0922, Perplexity: 49.89
Epoch 116/500, Batch 700/782, Loss: 0.1054, Recon: 0.0199, VQ: 0.0854, Perplexity: 49.89

=== Epoch 116/500 Summary ===
Train Loss: 0.1134 | Val Loss: 0.1131
Train Recon: 0.0200 | Val Recon: 0.0209
Train VQ: 0.0934 | Val VQ: 0.0922
Train Perplexity: 49.94 | Val Perplexity: 49.80
No improvement. Patience: 4/20
Best Loss: 0.1117 (Epoch 112)

Epoch 117/500, Batch 100/782, Loss: 0.1046, Recon: 0.0192, VQ: 0.0854, Perplexity: 48.34
Epoch 117/500, Batch 200/782, Loss: 0.1193, Recon: 0.0212, VQ: 0.0981, Perplexity: 51.43
Epoch 117/500, Batch 300/782, Loss: 0.1225, Recon: 0.0202, VQ: 0.1023, Perplexity: 49.91
Epoch 117/500, Batch 400/782, Loss: 0.1025, Recon: 0.0184, VQ: 0.0841, Perplexity: 50.16
Epoch 117/500, Batch 500/782, Loss: 0.1201, Recon: 0.0217, VQ: 0.0983, Perplexity: 51.78
Epoch 117/500, Batch 600/782, Loss: 0.1095, Recon: 0.0199, VQ: 0.0896, Perplexity: 51.46
Epoch 117/500, Batch 700/782, Loss: 0.3977, Recon: 0.0228, VQ: 0.3749, Perplexity: 50.71

=== Epoch 117/500 Summary ===
Train Loss: 0.1132 | Val Loss: 0.1127
Train Recon: 0.0199 | Val Recon: 0.0208
Train VQ: 0.0933 | Val VQ: 0.0919
Train Perplexity: 50.32 | Val Perplexity: 50.94
No improvement. Patience: 5/20
Best Loss: 0.1117 (Epoch 112)

Epoch 118/500, Batch 100/782, Loss: 0.1024, Recon: 0.0184, VQ: 0.0840, Perplexity: 48.63
Epoch 118/500, Batch 200/782, Loss: 0.1125, Recon: 0.0197, VQ: 0.0928, Perplexity: 50.78
Epoch 118/500, Batch 300/782, Loss: 0.1092, Recon: 0.0200, VQ: 0.0892, Perplexity: 48.50
Epoch 118/500, Batch 400/782, Loss: 0.0975, Recon: 0.0179, VQ: 0.0796, Perplexity: 49.42
Epoch 118/500, Batch 500/782, Loss: 0.1117, Recon: 0.0191, VQ: 0.0926, Per

plexity: 49.66
Epoch 118/500, Batch 600/782, Loss: 0.1145, Recon: 0.0202, VQ: 0.0943, Per
plexity: 51.76
Epoch 118/500, Batch 700/782, Loss: 0.1103, Recon: 0.0197, VQ: 0.0906, Per
plexity: 50.35

=== Epoch 118/500 Summary ===
Train Loss: 0.1131 | Val Loss: 0.1137
Train Recon: 0.0199 | Val Recon: 0.0208
Train VQ: 0.0931 | Val VQ: 0.0929
Train Perplexity: 50.62 | Val Perplexity: 50.63
No improvement. Patience: 6/20
Best Loss: 0.1117 (Epoch 112)

Epoch 119/500, Batch 100/782, Loss: 0.1040, Recon: 0.0195, VQ: 0.0845, Per
plexity: 49.35
Epoch 119/500, Batch 200/782, Loss: 0.1148, Recon: 0.0199, VQ: 0.0949, Per
plexity: 51.63
Epoch 119/500, Batch 300/782, Loss: 0.1102, Recon: 0.0205, VQ: 0.0898, Per
plexity: 51.92
Epoch 119/500, Batch 400/782, Loss: 0.1238, Recon: 0.0217, VQ: 0.1021, Per
plexity: 52.08
Epoch 119/500, Batch 500/782, Loss: 0.1057, Recon: 0.0190, VQ: 0.0867, Per
plexity: 52.07
Epoch 119/500, Batch 600/782, Loss: 0.1079, Recon: 0.0200, VQ: 0.0879, Per
plexity: 51.93
Epoch 119/500, Batch 700/782, Loss: 0.1062, Recon: 0.0186, VQ: 0.0876, Per
plexity: 50.91

=== Epoch 119/500 Summary ===
Train Loss: 0.1132 | Val Loss: 0.1128
Train Recon: 0.0199 | Val Recon: 0.0207
Train VQ: 0.0933 | Val VQ: 0.0921
Train Perplexity: 50.74 | Val Perplexity: 51.17
No improvement. Patience: 7/20
Best Loss: 0.1117 (Epoch 112)

Epoch 120/500, Batch 100/782, Loss: 0.1101, Recon: 0.0203, VQ: 0.0899, Per
plexity: 51.35
Epoch 120/500, Batch 200/782, Loss: 0.1116, Recon: 0.0186, VQ: 0.0930, Per
plexity: 50.34
Epoch 120/500, Batch 300/782, Loss: 0.1127, Recon: 0.0198, VQ: 0.0929, Per
plexity: 51.07
Epoch 120/500, Batch 400/782, Loss: 0.0985, Recon: 0.0184, VQ: 0.0801, Per
plexity: 49.57
Epoch 120/500, Batch 500/782, Loss: 0.1049, Recon: 0.0185, VQ: 0.0864, Per
plexity: 49.72
Epoch 120/500, Batch 600/782, Loss: 0.1148, Recon: 0.0193, VQ: 0.0955, Per
plexity: 50.69
Epoch 120/500, Batch 700/782, Loss: 0.1092, Recon: 0.0196, VQ: 0.0895, Per
plexity: 50.01

=== Epoch 120/500 Summary ===
Train Loss: 0.1127 | Val Loss: 0.1118
Train Recon: 0.0199 | Val Recon: 0.0208
Train VQ: 0.0928 | Val VQ: 0.0910
Train Perplexity: 50.83 | Val Perplexity: 50.68
No improvement. Patience: 8/20
Best Loss: 0.1117 (Epoch 112)

Epoch 121/500, Batch 100/782, Loss: 0.1049, Recon: 0.0195, VQ: 0.0854, Perplexity: 50.37
Epoch 121/500, Batch 200/782, Loss: 0.1029, Recon: 0.0192, VQ: 0.0837, Perplexity: 49.43
Epoch 121/500, Batch 300/782, Loss: 0.1157, Recon: 0.0203, VQ: 0.0953, Perplexity: 51.69
Epoch 121/500, Batch 400/782, Loss: 0.1200, Recon: 0.0220, VQ: 0.0980, Perplexity: 52.95
Epoch 121/500, Batch 500/782, Loss: 0.1017, Recon: 0.0189, VQ: 0.0828, Perplexity: 49.76
Epoch 121/500, Batch 600/782, Loss: 0.1107, Recon: 0.0198, VQ: 0.0908, Perplexity: 51.09
Epoch 121/500, Batch 700/782, Loss: 0.1049, Recon: 0.0188, VQ: 0.0861, Perplexity: 50.81

=== Epoch 121/500 Summary ===

Train Loss: 0.1124 | Val Loss: 0.1136
Train Recon: 0.0199 | Val Recon: 0.0208
Train VQ: 0.0925 | Val VQ: 0.0929
Train Perplexity: 50.88 | Val Perplexity: 51.38
No improvement. Patience: 9/20
Best Loss: 0.1117 (Epoch 112)

Epoch 122/500, Batch 100/782, Loss: 0.1172, Recon: 0.0198, VQ: 0.0974, Perplexity: 51.11
Epoch 122/500, Batch 200/782, Loss: 0.1084, Recon: 0.0206, VQ: 0.0878, Perplexity: 52.47
Epoch 122/500, Batch 300/782, Loss: 0.1236, Recon: 0.0200, VQ: 0.1037, Perplexity: 51.72
Epoch 122/500, Batch 400/782, Loss: 0.1036, Recon: 0.0183, VQ: 0.0853, Perplexity: 49.10
Epoch 122/500, Batch 500/782, Loss: 0.1345, Recon: 0.0195, VQ: 0.1150, Perplexity: 49.93
Epoch 122/500, Batch 600/782, Loss: 0.1153, Recon: 0.0205, VQ: 0.0948, Perplexity: 52.69
Epoch 122/500, Batch 700/782, Loss: 0.1163, Recon: 0.0212, VQ: 0.0951, Perplexity: 51.57

=== Epoch 122/500 Summary ===

Train Loss: 0.1126 | Val Loss: 0.1126
Train Recon: 0.0199 | Val Recon: 0.0207
Train VQ: 0.0927 | Val VQ: 0.0918
Train Perplexity: 50.96 | Val Perplexity: 51.10
No improvement. Patience: 10/20
Best Loss: 0.1117 (Epoch 112)

Epoch 123/500, Batch 100/782, Loss: 0.0977, Recon: 0.0186, VQ: 0.0791, Perplexity: 48.43
Epoch 123/500, Batch 200/782, Loss: 0.1169, Recon: 0.0213, VQ: 0.0957, Perplexity: 52.14
Epoch 123/500, Batch 300/782, Loss: 0.1198, Recon: 0.0212, VQ: 0.0986, Perplexity: 52.08
Epoch 123/500, Batch 400/782, Loss: 0.1083, Recon: 0.0193, VQ: 0.0891, Perplexity: 49.86
Epoch 123/500, Batch 500/782, Loss: 0.5886, Recon: 0.0220, VQ: 0.5666, Perplexity: 51.51
Epoch 123/500, Batch 600/782, Loss: 0.1097, Recon: 0.0200, VQ: 0.0897, Perplexity: 51.38
Epoch 123/500, Batch 700/782, Loss: 0.1027, Recon: 0.0190, VQ: 0.0837, Perplexity: 50.66

=== Epoch 123/500 Summary ===

Train Loss: 0.1123 | Val Loss: 0.1137
Train Recon: 0.0198 | Val Recon: 0.0207
Train VQ: 0.0925 | Val VQ: 0.0929
Train Perplexity: 51.02 | Val Perplexity: 51.49
No improvement. Patience: 11/20
Best Loss: 0.1117 (Epoch 112)

Epoch 124/500, Batch 100/782, Loss: 0.1084, Recon: 0.0192, VQ: 0.0892, Perplexity: 50.14
Epoch 124/500, Batch 200/782, Loss: 0.1332, Recon: 0.0232, VQ: 0.1100, Perplexity: 53.30
Epoch 124/500, Batch 300/782, Loss: 0.1078, Recon: 0.0194, VQ: 0.0884, Perplexity: 52.04
Epoch 124/500, Batch 400/782, Loss: 0.1043, Recon: 0.0198, VQ: 0.0845, Perplexity: 51.23
Epoch 124/500, Batch 500/782, Loss: 0.1118, Recon: 0.0204, VQ: 0.0914, Perplexity: 50.78
Epoch 124/500, Batch 600/782, Loss: 0.1085, Recon: 0.0193, VQ: 0.0892, Perplexity: 50.71
Epoch 124/500, Batch 700/782, Loss: 0.5037, Recon: 0.0223, VQ: 0.4813, Perplexity: 51.66

=== Epoch 124/500 Summary ===

Train Loss: 0.1122 | Val Loss: 0.1116
Train Recon: 0.0198 | Val Recon: 0.0207
Train VQ: 0.0924 | Val VQ: 0.0909
Train Perplexity: 51.08 | Val Perplexity: 51.25
No improvement. Patience: 12/20
Best Loss: 0.1117 (Epoch 112)

Epoch 125/500, Batch 100/782, Loss: 0.1076, Recon: 0.0205, VQ: 0.0871, Perplexity: 51.37
Epoch 125/500, Batch 200/782, Loss: 0.1170, Recon: 0.0215, VQ: 0.0955, Perplexity: 53.35
Epoch 125/500, Batch 300/782, Loss: 0.1253, Recon: 0.0203, VQ: 0.1051, Perplexity: 52.39
Epoch 125/500, Batch 400/782, Loss: 0.0926, Recon: 0.0166, VQ: 0.0760, Perplexity: 47.80
Epoch 125/500, Batch 500/782, Loss: 0.1027, Recon: 0.0186, VQ: 0.0841, Perplexity: 49.95
Epoch 125/500, Batch 600/782, Loss: 0.1076, Recon: 0.0205, VQ: 0.0871, Perplexity: 49.82
Epoch 125/500, Batch 700/782, Loss: 0.1072, Recon: 0.0196, VQ: 0.0876, Perplexity: 52.29

=== Epoch 125/500 Summary ===

Train Loss: 0.1122 | Val Loss: 0.1112
Train Recon: 0.0198 | Val Recon: 0.0207
Train VQ: 0.0924 | Val VQ: 0.0905
Train Perplexity: 51.12 | Val Perplexity: 50.89
✓ Best model saved! (Val Loss: 0.1112)
Best Loss: 0.1112 (Epoch 125)

Epoch 126/500, Batch 100/782, Loss: 0.1023, Recon: 0.0186, VQ: 0.0837, Perplexity: 51.74
Epoch 126/500, Batch 200/782, Loss: 0.1049, Recon: 0.0191, VQ: 0.0858, Perplexity: 51.24
Epoch 126/500, Batch 300/782, Loss: 0.1227, Recon: 0.0206, VQ: 0.1021, Per

plexity: 51.53
Epoch 126/500, Batch 400/782, Loss: 0.1142, Recon: 0.0204, VQ: 0.0938, Per
plexity: 51.43
Epoch 126/500, Batch 500/782, Loss: 0.1113, Recon: 0.0193, VQ: 0.0920, Per
plexity: 51.48
Epoch 126/500, Batch 600/782, Loss: 0.1213, Recon: 0.0232, VQ: 0.0981, Per
plexity: 52.90
Epoch 126/500, Batch 700/782, Loss: 0.1047, Recon: 0.0193, VQ: 0.0854, Per
plexity: 51.20

=== Epoch 126/500 Summary ===
Train Loss: 0.1118 | Val Loss: 0.1096
Train Recon: 0.0198 | Val Recon: 0.0207
Train VQ: 0.0920 | Val VQ: 0.0889
Train Perplexity: 51.16 | Val Perplexity: 50.88
✓ Best model saved! (Val Loss: 0.1096)
Best Loss: 0.1096 (Epoch 126)

Epoch 127/500, Batch 100/782, Loss: 0.1161, Recon: 0.0200, VQ: 0.0960, Per
plexity: 51.62
Epoch 127/500, Batch 200/782, Loss: 0.1090, Recon: 0.0193, VQ: 0.0897, Per
plexity: 50.77
Epoch 127/500, Batch 300/782, Loss: 0.1041, Recon: 0.0201, VQ: 0.0840, Per
plexity: 51.67
Epoch 127/500, Batch 400/782, Loss: 0.1115, Recon: 0.0192, VQ: 0.0923, Per
plexity: 51.43
Epoch 127/500, Batch 500/782, Loss: 0.0995, Recon: 0.0196, VQ: 0.0800, Per
plexity: 51.30
Epoch 127/500, Batch 600/782, Loss: 0.1091, Recon: 0.0199, VQ: 0.0892, Per
plexity: 51.68
Epoch 127/500, Batch 700/782, Loss: 0.1221, Recon: 0.0220, VQ: 0.1001, Per
plexity: 53.93

=== Epoch 127/500 Summary ===
Train Loss: 0.1119 | Val Loss: 0.1115
Train Recon: 0.0198 | Val Recon: 0.0207
Train VQ: 0.0921 | Val VQ: 0.0908
Train Perplexity: 51.22 | Val Perplexity: 51.51
No improvement. Patience: 1/20
Best Loss: 0.1096 (Epoch 126)

Epoch 128/500, Batch 100/782, Loss: 0.1199, Recon: 0.0207, VQ: 0.0993, Per
plexity: 51.55
Epoch 128/500, Batch 200/782, Loss: 0.1028, Recon: 0.0183, VQ: 0.0845, Per
plexity: 50.23
Epoch 128/500, Batch 300/782, Loss: 0.1231, Recon: 0.0218, VQ: 0.1013, Per
plexity: 54.04
Epoch 128/500, Batch 400/782, Loss: 0.1189, Recon: 0.0200, VQ: 0.0989, Per
plexity: 51.26
Epoch 128/500, Batch 500/782, Loss: 0.1073, Recon: 0.0203, VQ: 0.0870, Per
plexity: 51.88
Epoch 128/500, Batch 600/782, Loss: 0.1097, Recon: 0.0196, VQ: 0.0901, Per
plexity: 50.67
Epoch 128/500, Batch 700/782, Loss: 0.1167, Recon: 0.0214, VQ: 0.0953, Per
plexity: 52.13

=== Epoch 128/500 Summary ===
Train Loss: 0.1116 | Val Loss: 0.1115
Train Recon: 0.0198 | Val Recon: 0.0207
Train VQ: 0.0919 | Val VQ: 0.0908

Train Perplexity: 51.35 | Val Perplexity: 51.26

No improvement. Patience: 2/20

Best Loss: 0.1096 (Epoch 126)

Epoch 129/500, Batch 100/782, Loss: 0.1004, Recon: 0.0180, VQ: 0.0824, Perplexity: 49.56

Epoch 129/500, Batch 200/782, Loss: 0.1187, Recon: 0.0209, VQ: 0.0978, Perplexity: 52.48

Epoch 129/500, Batch 300/782, Loss: 0.1035, Recon: 0.0200, VQ: 0.0834, Perplexity: 52.02

Epoch 129/500, Batch 400/782, Loss: 0.1064, Recon: 0.0199, VQ: 0.0865, Perplexity: 49.80

Epoch 129/500, Batch 500/782, Loss: 0.1088, Recon: 0.0186, VQ: 0.0902, Perplexity: 52.16

Epoch 129/500, Batch 600/782, Loss: 0.1135, Recon: 0.0207, VQ: 0.0927, Perplexity: 52.15

Epoch 129/500, Batch 700/782, Loss: 0.1169, Recon: 0.0220, VQ: 0.0949, Perplexity: 54.24

=== Epoch 129/500 Summary ===

Train Loss: 0.1118 | Val Loss: 0.1114

Train Recon: 0.0197 | Val Recon: 0.0206

Train VQ: 0.0920 | Val VQ: 0.0908

Train Perplexity: 51.65 | Val Perplexity: 51.83

No improvement. Patience: 3/20

Best Loss: 0.1096 (Epoch 126)

Epoch 130/500, Batch 100/782, Loss: 0.1057, Recon: 0.0189, VQ: 0.0868, Perplexity: 51.25

Epoch 130/500, Batch 200/782, Loss: 0.1161, Recon: 0.0210, VQ: 0.0951, Perplexity: 53.54

Epoch 130/500, Batch 300/782, Loss: 0.1035, Recon: 0.0185, VQ: 0.0850, Perplexity: 52.01

Epoch 130/500, Batch 400/782, Loss: 0.1061, Recon: 0.0195, VQ: 0.0866, Perplexity: 52.04

Epoch 130/500, Batch 500/782, Loss: 0.1115, Recon: 0.0206, VQ: 0.0909, Perplexity: 52.50

Epoch 130/500, Batch 600/782, Loss: 0.1052, Recon: 0.0197, VQ: 0.0855, Perplexity: 51.16

Epoch 130/500, Batch 700/782, Loss: 0.1142, Recon: 0.0193, VQ: 0.0949, Perplexity: 52.70

=== Epoch 130/500 Summary ===

Train Loss: 0.1117 | Val Loss: 0.1121

Train Recon: 0.0197 | Val Recon: 0.0206

Train VQ: 0.0920 | Val VQ: 0.0915

Train Perplexity: 51.87 | Val Perplexity: 52.00

No improvement. Patience: 4/20

Best Loss: 0.1096 (Epoch 126)

Epoch 131/500, Batch 100/782, Loss: 0.0942, Recon: 0.0184, VQ: 0.0759, Perplexity: 49.80

Epoch 131/500, Batch 200/782, Loss: 0.1080, Recon: 0.0196, VQ: 0.0884, Perplexity: 52.68

Epoch 131/500, Batch 300/782, Loss: 0.1207, Recon: 0.0203, VQ: 0.1005, Perplexity: 52.75

Epoch 131/500, Batch 400/782, Loss: 0.1163, Recon: 0.0209, VQ: 0.0954, Perplexity: 52.42

Epoch 131/500, Batch 500/782, Loss: 0.1294, Recon: 0.0202, VQ: 0.1093, Perplexity: 53.43

Epoch 131/500, Batch 600/782, Loss: 0.1113, Recon: 0.0196, VQ: 0.0917, Perplexity: 51.70
Epoch 131/500, Batch 700/782, Loss: 0.1032, Recon: 0.0189, VQ: 0.0843, Perplexity: 50.62

=== Epoch 131/500 Summary ===

Train Loss: 0.1116 | Val Loss: 0.1113
Train Recon: 0.0197 | Val Recon: 0.0206
Train VQ: 0.0919 | Val VQ: 0.0907
Train Perplexity: 51.98 | Val Perplexity: 52.01
No improvement. Patience: 5/20
Best Loss: 0.1096 (Epoch 126)

Epoch 132/500, Batch 100/782, Loss: 0.1158, Recon: 0.0221, VQ: 0.0936, Perplexity: 54.14
Epoch 132/500, Batch 200/782, Loss: 0.1217, Recon: 0.0199, VQ: 0.1018, Perplexity: 53.13
Epoch 132/500, Batch 300/782, Loss: 0.1095, Recon: 0.0202, VQ: 0.0893, Perplexity: 52.93
Epoch 132/500, Batch 400/782, Loss: 0.1079, Recon: 0.0183, VQ: 0.0896, Perplexity: 50.54
Epoch 132/500, Batch 500/782, Loss: 0.1187, Recon: 0.0206, VQ: 0.0981, Perplexity: 52.78
Epoch 132/500, Batch 600/782, Loss: 0.1233, Recon: 0.0217, VQ: 0.1016, Perplexity: 54.23
Epoch 132/500, Batch 700/782, Loss: 0.1037, Recon: 0.0190, VQ: 0.0847, Perplexity: 51.11

=== Epoch 132/500 Summary ===

Train Loss: 0.1113 | Val Loss: 0.1103
Train Recon: 0.0197 | Val Recon: 0.0206
Train VQ: 0.0916 | Val VQ: 0.0897
Train Perplexity: 52.02 | Val Perplexity: 52.02
No improvement. Patience: 6/20
Best Loss: 0.1096 (Epoch 126)

Epoch 133/500, Batch 100/782, Loss: 0.1094, Recon: 0.0196, VQ: 0.0898, Perplexity: 51.28
Epoch 133/500, Batch 200/782, Loss: 0.1091, Recon: 0.0195, VQ: 0.0896, Perplexity: 52.85
Epoch 133/500, Batch 300/782, Loss: 0.1005, Recon: 0.0187, VQ: 0.0818, Perplexity: 51.32
Epoch 133/500, Batch 400/782, Loss: 0.1052, Recon: 0.0196, VQ: 0.0856, Perplexity: 52.54
Epoch 133/500, Batch 500/782, Loss: 0.1110, Recon: 0.0197, VQ: 0.0913, Perplexity: 52.96
Epoch 133/500, Batch 600/782, Loss: 0.1086, Recon: 0.0200, VQ: 0.0886, Perplexity: 51.07
Epoch 133/500, Batch 700/782, Loss: 0.1074, Recon: 0.0191, VQ: 0.0883, Perplexity: 50.66

=== Epoch 133/500 Summary ===

Train Loss: 0.1110 | Val Loss: 0.1118
Train Recon: 0.0196 | Val Recon: 0.0206
Train VQ: 0.0913 | Val VQ: 0.0912
Train Perplexity: 52.06 | Val Perplexity: 52.15
No improvement. Patience: 7/20
Best Loss: 0.1096 (Epoch 126)

Epoch 134/500, Batch 100/782, Loss: 0.1160, Recon: 0.0206, VQ: 0.0954, Per

plexity: 51.34
Epoch 134/500, Batch 200/782, Loss: 0.0930, Recon: 0.0176, VQ: 0.0754, Perplexity: 50.35
Epoch 134/500, Batch 300/782, Loss: 0.1091, Recon: 0.0191, VQ: 0.0900, Perplexity: 51.86
Epoch 134/500, Batch 400/782, Loss: 0.1176, Recon: 0.0205, VQ: 0.0971, Perplexity: 53.36
Epoch 134/500, Batch 500/782, Loss: 0.1227, Recon: 0.0209, VQ: 0.1018, Perplexity: 53.24
Epoch 134/500, Batch 600/782, Loss: 0.1001, Recon: 0.0188, VQ: 0.0813, Perplexity: 52.36
Epoch 134/500, Batch 700/782, Loss: 0.1039, Recon: 0.0197, VQ: 0.0843, Perplexity: 51.70

=== Epoch 134/500 Summary ===

Train Loss: 0.1113 | Val Loss: 0.1105
Train Recon: 0.0196 | Val Recon: 0.0206
Train VQ: 0.0916 | Val VQ: 0.0899
Train Perplexity: 52.09 | Val Perplexity: 52.10
No improvement. Patience: 8/20
Best Loss: 0.1096 (Epoch 126)

Epoch 135/500, Batch 100/782, Loss: 0.0979, Recon: 0.0188, VQ: 0.0790, Perplexity: 51.20
Epoch 135/500, Batch 200/782, Loss: 0.1121, Recon: 0.0200, VQ: 0.0922, Perplexity: 52.26
Epoch 135/500, Batch 300/782, Loss: 0.1074, Recon: 0.0190, VQ: 0.0884, Perplexity: 52.06
Epoch 135/500, Batch 400/782, Loss: 0.1035, Recon: 0.0189, VQ: 0.0845, Perplexity: 51.11
Epoch 135/500, Batch 500/782, Loss: 0.1114, Recon: 0.0201, VQ: 0.0913, Perplexity: 53.20
Epoch 135/500, Batch 600/782, Loss: 0.1138, Recon: 0.0206, VQ: 0.0932, Perplexity: 53.79
Epoch 135/500, Batch 700/782, Loss: 0.1219, Recon: 0.0193, VQ: 0.1026, Perplexity: 51.98

=== Epoch 135/500 Summary ===

Train Loss: 0.1111 | Val Loss: 0.1111
Train Recon: 0.0196 | Val Recon: 0.0206
Train VQ: 0.0914 | Val VQ: 0.0905
Train Perplexity: 52.19 | Val Perplexity: 52.63
No improvement. Patience: 9/20
Best Loss: 0.1096 (Epoch 126)

Epoch 136/500, Batch 100/782, Loss: 0.1543, Recon: 0.0197, VQ: 0.1346, Perplexity: 52.13
Epoch 136/500, Batch 200/782, Loss: 0.1061, Recon: 0.0199, VQ: 0.0862, Perplexity: 51.58
Epoch 136/500, Batch 300/782, Loss: 0.1079, Recon: 0.0202, VQ: 0.0877, Perplexity: 51.88
Epoch 136/500, Batch 400/782, Loss: 0.1095, Recon: 0.0199, VQ: 0.0897, Perplexity: 53.43
Epoch 136/500, Batch 500/782, Loss: 0.1069, Recon: 0.0200, VQ: 0.0870, Perplexity: 52.90
Epoch 136/500, Batch 600/782, Loss: 0.0995, Recon: 0.0195, VQ: 0.0800, Perplexity: 52.68
Epoch 136/500, Batch 700/782, Loss: 0.1196, Recon: 0.0197, VQ: 0.0999, Perplexity: 52.78

=== Epoch 136/500 Summary ===

Train Loss: 0.1109 | Val Loss: 0.1107

Train Recon: 0.0196 | Val Recon: 0.0206

Train VQ: 0.0913 | Val VQ: 0.0900

Train Perplexity: 52.24 | Val Perplexity: 52.56

No improvement. Patience: 10/20

Best Loss: 0.1096 (Epoch 126)

Epoch 137/500, Batch 100/782, Loss: 0.1024, Recon: 0.0181, VQ: 0.0843, Perplexity: 51.84

Epoch 137/500, Batch 200/782, Loss: 0.0998, Recon: 0.0178, VQ: 0.0820, Perplexity: 51.15

Epoch 137/500, Batch 300/782, Loss: 0.1084, Recon: 0.0194, VQ: 0.0890, Perplexity: 51.29

Epoch 137/500, Batch 400/782, Loss: 0.1012, Recon: 0.0190, VQ: 0.0822, Perplexity: 51.85

Epoch 137/500, Batch 500/782, Loss: 0.1061, Recon: 0.0196, VQ: 0.0865, Perplexity: 52.06

Epoch 137/500, Batch 600/782, Loss: 0.1086, Recon: 0.0200, VQ: 0.0886, Perplexity: 52.82

Epoch 137/500, Batch 700/782, Loss: 0.1073, Recon: 0.0201, VQ: 0.0872, Perplexity: 53.85

=== Epoch 137/500 Summary ===

Train Loss: 0.1107 | Val Loss: 0.1095

Train Recon: 0.0196 | Val Recon: 0.0206

Train VQ: 0.0911 | Val VQ: 0.0889

Train Perplexity: 52.32 | Val Perplexity: 52.14

✓ Best model saved! (Val Loss: 0.1095)

Best Loss: 0.1095 (Epoch 137)

Epoch 138/500, Batch 100/782, Loss: 0.1177, Recon: 0.0201, VQ: 0.0976, Perplexity: 52.35

Epoch 138/500, Batch 200/782, Loss: 0.1037, Recon: 0.0187, VQ: 0.0850, Perplexity: 52.07

Epoch 138/500, Batch 300/782, Loss: 0.1111, Recon: 0.0195, VQ: 0.0916, Perplexity: 51.31

Epoch 138/500, Batch 400/782, Loss: 0.1149, Recon: 0.0194, VQ: 0.0956, Perplexity: 50.96

Epoch 138/500, Batch 500/782, Loss: 0.1111, Recon: 0.0199, VQ: 0.0912, Perplexity: 54.08

Epoch 138/500, Batch 600/782, Loss: 0.1073, Recon: 0.0196, VQ: 0.0877, Perplexity: 54.08

Epoch 138/500, Batch 700/782, Loss: 0.1063, Recon: 0.0197, VQ: 0.0866, Perplexity: 52.91

=== Epoch 138/500 Summary ===

Train Loss: 0.1104 | Val Loss: 0.1104

Train Recon: 0.0196 | Val Recon: 0.0205

Train VQ: 0.0908 | Val VQ: 0.0899

Train Perplexity: 52.48 | Val Perplexity: 52.47

No improvement. Patience: 1/20

Best Loss: 0.1095 (Epoch 137)

Epoch 139/500, Batch 100/782, Loss: 0.1117, Recon: 0.0195, VQ: 0.0922, Perplexity: 53.76

Epoch 139/500, Batch 200/782, Loss: 0.1076, Recon: 0.0203, VQ: 0.0873, Perplexity: 52.00

Epoch 139/500, Batch 300/782, Loss: 0.1093, Recon: 0.0196, VQ: 0.0897, Perplexity: 51.18

Epoch 139/500, Batch 400/782, Loss: 0.1098, Recon: 0.0194, VQ: 0.0904, Perplexity: 53.72
Epoch 139/500, Batch 500/782, Loss: 0.1125, Recon: 0.0205, VQ: 0.0920, Perplexity: 52.51
Epoch 139/500, Batch 600/782, Loss: 0.1129, Recon: 0.0200, VQ: 0.0929, Perplexity: 53.47
Epoch 139/500, Batch 700/782, Loss: 0.1005, Recon: 0.0187, VQ: 0.0818, Perplexity: 52.27

=== Epoch 139/500 Summary ===

Train Loss: 0.1105 | Val Loss: 0.1109
Train Recon: 0.0196 | Val Recon: 0.0205
Train VQ: 0.0910 | Val VQ: 0.0904
Train Perplexity: 52.48 | Val Perplexity: 52.54
No improvement. Patience: 2/20
Best Loss: 0.1095 (Epoch 137)

Epoch 140/500, Batch 100/782, Loss: 0.0947, Recon: 0.0177, VQ: 0.0770, Perplexity: 51.18
Epoch 140/500, Batch 200/782, Loss: 0.2175, Recon: 0.0219, VQ: 0.1955, Perplexity: 54.22
Epoch 140/500, Batch 300/782, Loss: 0.1169, Recon: 0.0208, VQ: 0.0961, Perplexity: 54.15
Epoch 140/500, Batch 400/782, Loss: 0.1052, Recon: 0.0189, VQ: 0.0863, Perplexity: 54.09
Epoch 140/500, Batch 500/782, Loss: 0.1171, Recon: 0.0215, VQ: 0.0956, Perplexity: 54.80
Epoch 140/500, Batch 600/782, Loss: 0.0918, Recon: 0.0173, VQ: 0.0745, Perplexity: 49.58
Epoch 140/500, Batch 700/782, Loss: 0.1116, Recon: 0.0193, VQ: 0.0923, Perplexity: 52.13

=== Epoch 140/500 Summary ===

Train Loss: 0.1104 | Val Loss: 0.1103
Train Recon: 0.0195 | Val Recon: 0.0205
Train VQ: 0.0909 | Val VQ: 0.0898
Train Perplexity: 52.58 | Val Perplexity: 52.60
No improvement. Patience: 3/20
Best Loss: 0.1095 (Epoch 137)

Epoch 141/500, Batch 100/782, Loss: 0.1052, Recon: 0.0193, VQ: 0.0859, Perplexity: 52.90
Epoch 141/500, Batch 200/782, Loss: 0.1159, Recon: 0.0198, VQ: 0.0960, Perplexity: 51.66
Epoch 141/500, Batch 300/782, Loss: 0.1006, Recon: 0.0178, VQ: 0.0828, Perplexity: 50.48
Epoch 141/500, Batch 400/782, Loss: 0.1117, Recon: 0.0198, VQ: 0.0919, Perplexity: 51.69
Epoch 141/500, Batch 500/782, Loss: 0.1040, Recon: 0.0185, VQ: 0.0855, Perplexity: 50.98
Epoch 141/500, Batch 600/782, Loss: 0.1017, Recon: 0.0194, VQ: 0.0823, Perplexity: 52.72
Epoch 141/500, Batch 700/782, Loss: 0.2595, Recon: 0.0232, VQ: 0.2363, Perplexity: 55.74

=== Epoch 141/500 Summary ===

Train Loss: 0.1102 | Val Loss: 0.1108
Train Recon: 0.0195 | Val Recon: 0.0205
Train VQ: 0.0907 | Val VQ: 0.0903
Train Perplexity: 52.59 | Val Perplexity: 52.81

No improvement. Patience: 4/20
Best Loss: 0.1095 (Epoch 137)

Epoch 142/500, Batch 100/782, Loss: 0.1013, Recon: 0.0181, VQ: 0.0833, Perplexity: 52.78
Epoch 142/500, Batch 200/782, Loss: 0.1123, Recon: 0.0200, VQ: 0.0923, Perplexity: 53.47
Epoch 142/500, Batch 300/782, Loss: 0.1154, Recon: 0.0216, VQ: 0.0938, Perplexity: 54.34
Epoch 142/500, Batch 400/782, Loss: 0.1083, Recon: 0.0184, VQ: 0.0899, Perplexity: 53.05
Epoch 142/500, Batch 500/782, Loss: 0.1033, Recon: 0.0182, VQ: 0.0850, Perplexity: 52.02
Epoch 142/500, Batch 600/782, Loss: 0.1036, Recon: 0.0178, VQ: 0.0858, Perplexity: 52.72
Epoch 142/500, Batch 700/782, Loss: 0.1086, Recon: 0.0206, VQ: 0.0880, Perplexity: 54.06

=== Epoch 142/500 Summary ===
Train Loss: 0.1102 | Val Loss: 0.1111
Train Recon: 0.0195 | Val Recon: 0.0205
Train VQ: 0.0907 | Val VQ: 0.0906
Train Perplexity: 52.73 | Val Perplexity: 53.26
No improvement. Patience: 5/20
Best Loss: 0.1095 (Epoch 137)

Epoch 143/500, Batch 100/782, Loss: 0.1295, Recon: 0.0213, VQ: 0.1082, Perplexity: 54.71
Epoch 143/500, Batch 200/782, Loss: 0.1158, Recon: 0.0208, VQ: 0.0949, Perplexity: 54.38
Epoch 143/500, Batch 300/782, Loss: 0.1081, Recon: 0.0197, VQ: 0.0885, Perplexity: 52.79
Epoch 143/500, Batch 400/782, Loss: 0.1168, Recon: 0.0207, VQ: 0.0961, Perplexity: 52.86
Epoch 143/500, Batch 500/782, Loss: 0.1179, Recon: 0.0204, VQ: 0.0975, Perplexity: 53.99
Epoch 143/500, Batch 600/782, Loss: 0.1161, Recon: 0.0207, VQ: 0.0954, Perplexity: 53.62
Epoch 143/500, Batch 700/782, Loss: 0.1083, Recon: 0.0198, VQ: 0.0886, Perplexity: 53.37

=== Epoch 143/500 Summary ===
Train Loss: 0.1100 | Val Loss: 0.1095
Train Recon: 0.0195 | Val Recon: 0.0204
Train VQ: 0.0905 | Val VQ: 0.0891
Train Perplexity: 53.05 | Val Perplexity: 52.87
No improvement. Patience: 6/20
Best Loss: 0.1095 (Epoch 137)

Epoch 144/500, Batch 100/782, Loss: 0.1061, Recon: 0.0189, VQ: 0.0872, Perplexity: 53.07
Epoch 144/500, Batch 200/782, Loss: 0.1154, Recon: 0.0206, VQ: 0.0948, Perplexity: 54.19
Epoch 144/500, Batch 300/782, Loss: 0.1022, Recon: 0.0184, VQ: 0.0839, Perplexity: 51.81
Epoch 144/500, Batch 400/782, Loss: 0.1134, Recon: 0.0219, VQ: 0.0915, Perplexity: 53.69
Epoch 144/500, Batch 500/782, Loss: 0.1007, Recon: 0.0194, VQ: 0.0813, Perplexity: 53.10
Epoch 144/500, Batch 600/782, Loss: 0.1083, Recon: 0.0193, VQ: 0.0890, Per

plexity: 54.34

Epoch 144/500, Batch 700/782, Loss: 0.1076, Recon: 0.0204, VQ: 0.0871, Perplexity: 53.33

=== Epoch 144/500 Summary ===

Train Loss: 0.1099 | Val Loss: 0.1105

Train Recon: 0.0195 | Val Recon: 0.0206

Train VQ: 0.0904 | Val VQ: 0.0899

Train Perplexity: 53.33 | Val Perplexity: 53.21

No improvement. Patience: 7/20

Best Loss: 0.1095 (Epoch 137)

Epoch 145/500, Batch 100/782, Loss: 0.1014, Recon: 0.0184, VQ: 0.0830, Perplexity: 52.69

Epoch 145/500, Batch 200/782, Loss: 0.0997, Recon: 0.0185, VQ: 0.0812, Perplexity: 52.70

Epoch 145/500, Batch 300/782, Loss: 0.0966, Recon: 0.0179, VQ: 0.0786, Perplexity: 49.60

Epoch 145/500, Batch 400/782, Loss: 0.1035, Recon: 0.0186, VQ: 0.0850, Perplexity: 50.77

Epoch 145/500, Batch 500/782, Loss: 0.1121, Recon: 0.0199, VQ: 0.0922, Perplexity: 52.46

Epoch 145/500, Batch 600/782, Loss: 0.1089, Recon: 0.0191, VQ: 0.0898, Perplexity: 54.16

Epoch 145/500, Batch 700/782, Loss: 0.1037, Recon: 0.0196, VQ: 0.0842, Perplexity: 52.51

=== Epoch 145/500 Summary ===

Train Loss: 0.1096 | Val Loss: 0.1105

Train Recon: 0.0194 | Val Recon: 0.0204

Train VQ: 0.0902 | Val VQ: 0.0900

Train Perplexity: 53.44 | Val Perplexity: 53.69

No improvement. Patience: 8/20

Best Loss: 0.1095 (Epoch 137)

Epoch 146/500, Batch 100/782, Loss: 0.0965, Recon: 0.0183, VQ: 0.0782, Perplexity: 52.39

Epoch 146/500, Batch 200/782, Loss: 0.1052, Recon: 0.0187, VQ: 0.0865, Perplexity: 52.73

Epoch 146/500, Batch 300/782, Loss: 0.1129, Recon: 0.0206, VQ: 0.0923, Perplexity: 53.98

Epoch 146/500, Batch 400/782, Loss: 0.1011, Recon: 0.0187, VQ: 0.0824, Perplexity: 54.13

Epoch 146/500, Batch 500/782, Loss: 0.1021, Recon: 0.0197, VQ: 0.0825, Perplexity: 52.30

Epoch 146/500, Batch 600/782, Loss: 0.1102, Recon: 0.0193, VQ: 0.0909, Perplexity: 53.28

Epoch 146/500, Batch 700/782, Loss: 0.1074, Recon: 0.0197, VQ: 0.0877, Perplexity: 53.22

=== Epoch 146/500 Summary ===

Train Loss: 0.1095 | Val Loss: 0.1103

Train Recon: 0.0194 | Val Recon: 0.0205

Train VQ: 0.0901 | Val VQ: 0.0898

Train Perplexity: 53.48 | Val Perplexity: 53.43

No improvement. Patience: 9/20

Best Loss: 0.1095 (Epoch 137)

Epoch 147/500, Batch 100/782, Loss: 0.1043, Recon: 0.0193, VQ: 0.0850, Perplexity: 53.67

Epoch 147/500, Batch 200/782, Loss: 0.1036, Recon: 0.0191, VQ: 0.0845, Perplexity: 52.89
Epoch 147/500, Batch 300/782, Loss: 0.1057, Recon: 0.0195, VQ: 0.0862, Perplexity: 54.03
Epoch 147/500, Batch 400/782, Loss: 0.1017, Recon: 0.0188, VQ: 0.0828, Perplexity: 53.09
Epoch 147/500, Batch 500/782, Loss: 0.0993, Recon: 0.0192, VQ: 0.0801, Perplexity: 52.75
Epoch 147/500, Batch 600/782, Loss: 0.1085, Recon: 0.0196, VQ: 0.0889, Perplexity: 52.78
Epoch 147/500, Batch 700/782, Loss: 0.1019, Recon: 0.0192, VQ: 0.0827, Perplexity: 52.61

=== Epoch 147/500 Summary ===

Train Loss: 0.1095 | Val Loss: 0.1099
Train Recon: 0.0194 | Val Recon: 0.0204
Train VQ: 0.0901 | Val VQ: 0.0894
Train Perplexity: 53.59 | Val Perplexity: 53.60
No improvement. Patience: 10/20
Best Loss: 0.1095 (Epoch 137)

Epoch 148/500, Batch 100/782, Loss: 0.1197, Recon: 0.0193, VQ: 0.1004, Perplexity: 54.89
Epoch 148/500, Batch 200/782, Loss: 0.1040, Recon: 0.0183, VQ: 0.0857, Perplexity: 52.14
Epoch 148/500, Batch 300/782, Loss: 0.1079, Recon: 0.0205, VQ: 0.0874, Perplexity: 53.62
Epoch 148/500, Batch 400/782, Loss: 0.1112, Recon: 0.0198, VQ: 0.0914, Perplexity: 55.45
Epoch 148/500, Batch 500/782, Loss: 0.1188, Recon: 0.0204, VQ: 0.0984, Perplexity: 56.50
Epoch 148/500, Batch 600/782, Loss: 0.1215, Recon: 0.0192, VQ: 0.1024, Perplexity: 54.64
Epoch 148/500, Batch 700/782, Loss: 0.1050, Recon: 0.0187, VQ: 0.0863, Perplexity: 53.49

=== Epoch 148/500 Summary ===

Train Loss: 0.1094 | Val Loss: 0.1101
Train Recon: 0.0194 | Val Recon: 0.0205
Train VQ: 0.0900 | Val VQ: 0.0897
Train Perplexity: 53.60 | Val Perplexity: 53.50
No improvement. Patience: 11/20
Best Loss: 0.1095 (Epoch 137)

Epoch 149/500, Batch 100/782, Loss: 0.1008, Recon: 0.0191, VQ: 0.0817, Perplexity: 54.25
Epoch 149/500, Batch 200/782, Loss: 0.1211, Recon: 0.0205, VQ: 0.1006, Perplexity: 56.19
Epoch 149/500, Batch 300/782, Loss: 0.1039, Recon: 0.0189, VQ: 0.0850, Perplexity: 51.94
Epoch 149/500, Batch 400/782, Loss: 0.1044, Recon: 0.0178, VQ: 0.0867, Perplexity: 52.92
Epoch 149/500, Batch 500/782, Loss: 0.1194, Recon: 0.0210, VQ: 0.0984, Perplexity: 56.47
Epoch 149/500, Batch 600/782, Loss: 0.1064, Recon: 0.0204, VQ: 0.0861, Perplexity: 54.97
Epoch 149/500, Batch 700/782, Loss: 0.0977, Recon: 0.0186, VQ: 0.0791, Perplexity: 51.60

=== Epoch 149/500 Summary ===

Train Loss: 0.1092 | Val Loss: 0.1090
Train Recon: 0.0194 | Val Recon: 0.0204
Train VQ: 0.0899 | Val VQ: 0.0886
Train Perplexity: 53.64 | Val Perplexity: 53.66
✓ Best model saved! (Val Loss: 0.1090)
Best Loss: 0.1090 (Epoch 149)

Epoch 150/500, Batch 100/782, Loss: 0.0993, Recon: 0.0185, VQ: 0.0809, Perplexity: 52.91
Epoch 150/500, Batch 200/782, Loss: 0.1221, Recon: 0.0216, VQ: 0.1005, Perplexity: 56.31
Epoch 150/500, Batch 300/782, Loss: 0.1150, Recon: 0.0204, VQ: 0.0946, Perplexity: 55.35
Epoch 150/500, Batch 400/782, Loss: 0.1027, Recon: 0.0194, VQ: 0.0834, Perplexity: 52.24
Epoch 150/500, Batch 500/782, Loss: 0.1061, Recon: 0.0192, VQ: 0.0869, Perplexity: 54.31
Epoch 150/500, Batch 600/782, Loss: 0.1171, Recon: 0.0201, VQ: 0.0969, Perplexity: 55.84
Epoch 150/500, Batch 700/782, Loss: 0.1101, Recon: 0.0194, VQ: 0.0907, Perplexity: 54.47

=== Epoch 150/500 Summary ===
Train Loss: 0.1091 | Val Loss: 0.1104
Train Recon: 0.0194 | Val Recon: 0.0204
Train VQ: 0.0897 | Val VQ: 0.0900
Train Perplexity: 53.68 | Val Perplexity: 53.85
No improvement. Patience: 1/20
Best Loss: 0.1090 (Epoch 149)

Epoch 151/500, Batch 100/782, Loss: 0.1082, Recon: 0.0186, VQ: 0.0896, Perplexity: 54.47
Epoch 151/500, Batch 200/782, Loss: 0.1055, Recon: 0.0200, VQ: 0.0855, Perplexity: 55.16
Epoch 151/500, Batch 300/782, Loss: 0.1069, Recon: 0.0186, VQ: 0.0883, Perplexity: 52.50
Epoch 151/500, Batch 400/782, Loss: 0.1187, Recon: 0.0213, VQ: 0.0975, Perplexity: 56.34
Epoch 151/500, Batch 500/782, Loss: 0.1231, Recon: 0.0209, VQ: 0.1022, Perplexity: 54.14
Epoch 151/500, Batch 600/782, Loss: 0.0979, Recon: 0.0178, VQ: 0.0801, Perplexity: 51.55
Epoch 151/500, Batch 700/782, Loss: 0.1209, Recon: 0.0201, VQ: 0.1008, Perplexity: 54.13

=== Epoch 151/500 Summary ===
Train Loss: 0.1091 | Val Loss: 0.1087
Train Recon: 0.0194 | Val Recon: 0.0205
Train VQ: 0.0898 | Val VQ: 0.0882
Train Perplexity: 53.75 | Val Perplexity: 53.72
✓ Best model saved! (Val Loss: 0.1087)
Best Loss: 0.1087 (Epoch 151)

Epoch 152/500, Batch 100/782, Loss: 0.1153, Recon: 0.0203, VQ: 0.0951, Perplexity: 54.09
Epoch 152/500, Batch 200/782, Loss: 0.1156, Recon: 0.0209, VQ: 0.0947, Perplexity: 56.09
Epoch 152/500, Batch 300/782, Loss: 0.1009, Recon: 0.0179, VQ: 0.0830, Perplexity: 51.68
Epoch 152/500, Batch 400/782, Loss: 0.1192, Recon: 0.0193, VQ: 0.0999, Per

plexity: 55.54
Epoch 152/500, Batch 500/782, Loss: 0.1175, Recon: 0.0210, VQ: 0.0965, Perplexity: 55.36
Epoch 152/500, Batch 600/782, Loss: 0.1017, Recon: 0.0180, VQ: 0.0837, Perplexity: 53.02
Epoch 152/500, Batch 700/782, Loss: 0.1045, Recon: 0.0188, VQ: 0.0857, Perplexity: 53.15

=== Epoch 152/500 Summary ===

Train Loss: 0.1090 | Val Loss: 0.1092
Train Recon: 0.0194 | Val Recon: 0.0204
Train VQ: 0.0897 | Val VQ: 0.0888
Train Perplexity: 53.81 | Val Perplexity: 53.64
No improvement. Patience: 1/20
Best Loss: 0.1087 (Epoch 151)

Epoch 153/500, Batch 100/782, Loss: 0.0977, Recon: 0.0180, VQ: 0.0797, Perplexity: 52.62
Epoch 153/500, Batch 200/782, Loss: 0.1068, Recon: 0.0202, VQ: 0.0866, Perplexity: 53.83
Epoch 153/500, Batch 300/782, Loss: 0.1348, Recon: 0.0218, VQ: 0.1130, Perplexity: 56.55
Epoch 153/500, Batch 400/782, Loss: 0.1034, Recon: 0.0190, VQ: 0.0845, Perplexity: 52.87
Epoch 153/500, Batch 500/782, Loss: 0.0963, Recon: 0.0177, VQ: 0.0786, Perplexity: 50.96
Epoch 153/500, Batch 600/782, Loss: 0.0944, Recon: 0.0177, VQ: 0.0767, Perplexity: 51.34
Epoch 153/500, Batch 700/782, Loss: 0.1170, Recon: 0.0202, VQ: 0.0969, Perplexity: 55.48

=== Epoch 153/500 Summary ===

Train Loss: 0.1091 | Val Loss: 0.1094
Train Recon: 0.0193 | Val Recon: 0.0204
Train VQ: 0.0897 | Val VQ: 0.0890
Train Perplexity: 53.86 | Val Perplexity: 53.87
No improvement. Patience: 2/20
Best Loss: 0.1087 (Epoch 151)

Epoch 154/500, Batch 100/782, Loss: 0.1150, Recon: 0.0204, VQ: 0.0946, Perplexity: 54.89
Epoch 154/500, Batch 200/782, Loss: 0.0975, Recon: 0.0187, VQ: 0.0788, Perplexity: 53.23
Epoch 154/500, Batch 300/782, Loss: 0.1107, Recon: 0.0196, VQ: 0.0911, Perplexity: 53.92
Epoch 154/500, Batch 400/782, Loss: 0.1100, Recon: 0.0193, VQ: 0.0907, Perplexity: 54.54
Epoch 154/500, Batch 500/782, Loss: 0.1084, Recon: 0.0180, VQ: 0.0904, Perplexity: 51.70
Epoch 154/500, Batch 600/782, Loss: 0.1218, Recon: 0.0212, VQ: 0.1006, Perplexity: 57.33
Epoch 154/500, Batch 700/782, Loss: 0.1077, Recon: 0.0197, VQ: 0.0880, Perplexity: 54.12

=== Epoch 154/500 Summary ===

Train Loss: 0.1090 | Val Loss: 0.1083
Train Recon: 0.0193 | Val Recon: 0.0204
Train VQ: 0.0897 | Val VQ: 0.0879
Train Perplexity: 53.93 | Val Perplexity: 53.73
✓ Best model saved! (Val Loss: 0.1083)

Best Loss: 0.1083 (Epoch 154)

Epoch 155/500, Batch 100/782, Loss: 0.1388, Recon: 0.0199, VQ: 0.1189, Perplexity: 55.42
Epoch 155/500, Batch 200/782, Loss: 0.1038, Recon: 0.0193, VQ: 0.0844, Perplexity: 53.84
Epoch 155/500, Batch 300/782, Loss: 0.1092, Recon: 0.0197, VQ: 0.0895, Perplexity: 55.91
Epoch 155/500, Batch 400/782, Loss: 0.0988, Recon: 0.0184, VQ: 0.0804, Perplexity: 52.74
Epoch 155/500, Batch 500/782, Loss: 0.1166, Recon: 0.0191, VQ: 0.0975, Perplexity: 54.54
Epoch 155/500, Batch 600/782, Loss: 0.1212, Recon: 0.0197, VQ: 0.1015, Perplexity: 54.17
Epoch 155/500, Batch 700/782, Loss: 0.1283, Recon: 0.0196, VQ: 0.1087, Perplexity: 53.22

=== Epoch 155/500 Summary ===

Train Loss: 0.1088 | Val Loss: 0.1085
Train Recon: 0.0193 | Val Recon: 0.0204
Train VQ: 0.0895 | Val VQ: 0.0881
Train Perplexity: 54.03 | Val Perplexity: 53.74
No improvement. Patience: 1/20
Best Loss: 0.1083 (Epoch 154)

Epoch 156/500, Batch 100/782, Loss: 0.1052, Recon: 0.0199, VQ: 0.0852, Perplexity: 54.36
Epoch 156/500, Batch 200/782, Loss: 0.1007, Recon: 0.0182, VQ: 0.0825, Perplexity: 53.43
Epoch 156/500, Batch 300/782, Loss: 0.1090, Recon: 0.0195, VQ: 0.0895, Perplexity: 55.80
Epoch 156/500, Batch 400/782, Loss: 0.1060, Recon: 0.0194, VQ: 0.0866, Perplexity: 53.58
Epoch 156/500, Batch 500/782, Loss: 0.1036, Recon: 0.0191, VQ: 0.0845, Perplexity: 52.90
Epoch 156/500, Batch 600/782, Loss: 0.1019, Recon: 0.0186, VQ: 0.0833, Perplexity: 54.30
Epoch 156/500, Batch 700/782, Loss: 0.1035, Recon: 0.0191, VQ: 0.0844, Perplexity: 54.24

=== Epoch 156/500 Summary ===

Train Loss: 0.1089 | Val Loss: 0.1097
Train Recon: 0.0193 | Val Recon: 0.0204
Train VQ: 0.0895 | Val VQ: 0.0893
Train Perplexity: 54.09 | Val Perplexity: 54.18
No improvement. Patience: 2/20
Best Loss: 0.1083 (Epoch 154)

Epoch 157/500, Batch 100/782, Loss: 0.1141, Recon: 0.0202, VQ: 0.0939, Perplexity: 56.32
Epoch 157/500, Batch 200/782, Loss: 0.1122, Recon: 0.0211, VQ: 0.0911, Perplexity: 55.58
Epoch 157/500, Batch 300/782, Loss: 0.0973, Recon: 0.0180, VQ: 0.0793, Perplexity: 52.59
Epoch 157/500, Batch 400/782, Loss: 0.1322, Recon: 0.0208, VQ: 0.1113, Perplexity: 56.20
Epoch 157/500, Batch 500/782, Loss: 0.1106, Recon: 0.0199, VQ: 0.0907, Perplexity: 55.89
Epoch 157/500, Batch 600/782, Loss: 0.1082, Recon: 0.0202, VQ: 0.0880, Perplexity: 55.08

Epoch 157/500, Batch 700/782, Loss: 0.1052, Recon: 0.0193, VQ: 0.0859, Perplexity: 52.45

=== Epoch 157/500 Summary ===

Train Loss: 0.1086 | Val Loss: 0.1080

Train Recon: 0.0193 | Val Recon: 0.0203

Train VQ: 0.0893 | Val VQ: 0.0877

Train Perplexity: 54.18 | Val Perplexity: 53.84

✓ Best model saved! (Val Loss: 0.1080)

Best Loss: 0.1080 (Epoch 157)

Epoch 158/500, Batch 100/782, Loss: 0.1100, Recon: 0.0199, VQ: 0.0901, Perplexity: 55.82

Epoch 158/500, Batch 200/782, Loss: 0.1001, Recon: 0.0183, VQ: 0.0818, Perplexity: 53.32

Epoch 158/500, Batch 300/782, Loss: 0.0912, Recon: 0.0173, VQ: 0.0739, Perplexity: 52.34

Epoch 158/500, Batch 400/782, Loss: 0.1076, Recon: 0.0198, VQ: 0.0878, Perplexity: 53.78

Epoch 158/500, Batch 500/782, Loss: 0.1136, Recon: 0.0197, VQ: 0.0939, Perplexity: 55.98

Epoch 158/500, Batch 600/782, Loss: 0.0988, Recon: 0.0185, VQ: 0.0804, Perplexity: 52.33

Epoch 158/500, Batch 700/782, Loss: 0.1155, Recon: 0.0202, VQ: 0.0953, Perplexity: 54.62

=== Epoch 158/500 Summary ===

Train Loss: 0.1085 | Val Loss: 0.1089

Train Recon: 0.0193 | Val Recon: 0.0204

Train VQ: 0.0892 | Val VQ: 0.0885

Train Perplexity: 54.21 | Val Perplexity: 54.21

No improvement. Patience: 1/20

Best Loss: 0.1080 (Epoch 157)

Epoch 159/500, Batch 100/782, Loss: 0.1245, Recon: 0.0210, VQ: 0.1035, Perplexity: 57.37

Epoch 159/500, Batch 200/782, Loss: 0.1051, Recon: 0.0202, VQ: 0.0849, Perplexity: 55.12

Epoch 159/500, Batch 300/782, Loss: 0.1139, Recon: 0.0190, VQ: 0.0950, Perplexity: 54.09

Epoch 159/500, Batch 400/782, Loss: 0.1029, Recon: 0.0192, VQ: 0.0836, Perplexity: 53.63

Epoch 159/500, Batch 500/782, Loss: 0.1044, Recon: 0.0187, VQ: 0.0857, Perplexity: 53.97

Epoch 159/500, Batch 600/782, Loss: 0.0940, Recon: 0.0174, VQ: 0.0766, Perplexity: 52.43

Epoch 159/500, Batch 700/782, Loss: 0.0960, Recon: 0.0182, VQ: 0.0778, Perplexity: 54.81

=== Epoch 159/500 Summary ===

Train Loss: 0.1084 | Val Loss: 0.1083

Train Recon: 0.0193 | Val Recon: 0.0204

Train VQ: 0.0891 | Val VQ: 0.0879

Train Perplexity: 54.26 | Val Perplexity: 54.12

No improvement. Patience: 2/20

Best Loss: 0.1080 (Epoch 157)

Epoch 160/500, Batch 100/782, Loss: 0.0935, Recon: 0.0174, VQ: 0.0761, Perplexity: 52.59

Epoch 160/500, Batch 200/782, Loss: 0.1059, Recon: 0.0187, VQ: 0.0871, Per

plexity: 52.61
Epoch 160/500, Batch 300/782, Loss: 0.1327, Recon: 0.0210, VQ: 0.1117, Per
plexity: 55.26
Epoch 160/500, Batch 400/782, Loss: 0.1092, Recon: 0.0198, VQ: 0.0894, Per
plexity: 56.10
Epoch 160/500, Batch 500/782, Loss: 0.1049, Recon: 0.0197, VQ: 0.0852, Per
plexity: 53.87
Epoch 160/500, Batch 600/782, Loss: 0.1082, Recon: 0.0196, VQ: 0.0886, Per
plexity: 54.93
Epoch 160/500, Batch 700/782, Loss: 0.1021, Recon: 0.0192, VQ: 0.0829, Per
plexity: 53.99

=== Epoch 160/500 Summary ===
Train Loss: 0.1082 | Val Loss: 0.1094
Train Recon: 0.0193 | Val Recon: 0.0203
Train VQ: 0.0890 | Val VQ: 0.0891
Train Perplexity: 54.30 | Val Perplexity: 54.19
No improvement. Patience: 3/20
Best Loss: 0.1080 (Epoch 157)

Epoch 161/500, Batch 100/782, Loss: 0.1030, Recon: 0.0188, VQ: 0.0842, Per
plexity: 55.44
Epoch 161/500, Batch 200/782, Loss: 0.0926, Recon: 0.0171, VQ: 0.0755, Per
plexity: 52.02
Epoch 161/500, Batch 300/782, Loss: 0.1189, Recon: 0.0195, VQ: 0.0994, Per
plexity: 56.27
Epoch 161/500, Batch 400/782, Loss: 0.1070, Recon: 0.0196, VQ: 0.0875, Per
plexity: 54.56
Epoch 161/500, Batch 500/782, Loss: 0.1130, Recon: 0.0191, VQ: 0.0939, Per
plexity: 52.29
Epoch 161/500, Batch 600/782, Loss: 0.1028, Recon: 0.0182, VQ: 0.0846, Per
plexity: 52.82
Epoch 161/500, Batch 700/782, Loss: 0.1131, Recon: 0.0210, VQ: 0.0921, Per
plexity: 55.44

=== Epoch 161/500 Summary ===
Train Loss: 0.1082 | Val Loss: 0.1092
Train Recon: 0.0193 | Val Recon: 0.0204
Train VQ: 0.0890 | Val VQ: 0.0888
Train Perplexity: 54.29 | Val Perplexity: 54.46
No improvement. Patience: 4/20
Best Loss: 0.1080 (Epoch 157)

Epoch 162/500, Batch 100/782, Loss: 0.0954, Recon: 0.0188, VQ: 0.0766, Per
plexity: 53.16
Epoch 162/500, Batch 200/782, Loss: 0.1098, Recon: 0.0195, VQ: 0.0903, Per
plexity: 53.88
Epoch 162/500, Batch 300/782, Loss: 0.1135, Recon: 0.0195, VQ: 0.0940, Per
plexity: 54.25
Epoch 162/500, Batch 400/782, Loss: 0.0998, Recon: 0.0185, VQ: 0.0813, Per
plexity: 53.19
Epoch 162/500, Batch 500/782, Loss: 0.1188, Recon: 0.0184, VQ: 0.1004, Per
plexity: 52.73
Epoch 162/500, Batch 600/782, Loss: 0.1082, Recon: 0.0194, VQ: 0.0888, Per
plexity: 54.99
Epoch 162/500, Batch 700/782, Loss: 0.0943, Recon: 0.0174, VQ: 0.0769, Per
plexity: 50.83

=== Epoch 162/500 Summary ===
Train Loss: 0.1083 | Val Loss: 0.1091

Train Recon: 0.0192 | Val Recon: 0.0204
Train VQ: 0.0891 | Val VQ: 0.0887
Train Perplexity: 54.36 | Val Perplexity: 54.58
No improvement. Patience: 5/20
Best Loss: 0.1080 (Epoch 157)

Epoch 163/500, Batch 100/782, Loss: 0.1147, Recon: 0.0204, VQ: 0.0943, Perplexity: 55.78
Epoch 163/500, Batch 200/782, Loss: 0.1192, Recon: 0.0185, VQ: 0.1007, Perplexity: 54.19
Epoch 163/500, Batch 300/782, Loss: 0.1011, Recon: 0.0193, VQ: 0.0818, Perplexity: 52.96
Epoch 163/500, Batch 400/782, Loss: 0.1095, Recon: 0.0192, VQ: 0.0902, Perplexity: 55.57
Epoch 163/500, Batch 500/782, Loss: 0.0919, Recon: 0.0170, VQ: 0.0749, Perplexity: 52.96
Epoch 163/500, Batch 600/782, Loss: 0.1047, Recon: 0.0191, VQ: 0.0856, Perplexity: 55.06
Epoch 163/500, Batch 700/782, Loss: 0.1084, Recon: 0.0196, VQ: 0.0888, Perplexity: 56.22

=== Epoch 163/500 Summary ===
Train Loss: 0.1082 | Val Loss: 0.1082
Train Recon: 0.0192 | Val Recon: 0.0203
Train VQ: 0.0890 | Val VQ: 0.0879
Train Perplexity: 54.37 | Val Perplexity: 54.54
No improvement. Patience: 6/20
Best Loss: 0.1080 (Epoch 157)

Epoch 164/500, Batch 100/782, Loss: 0.1034, Recon: 0.0191, VQ: 0.0843, Perplexity: 54.43
Epoch 164/500, Batch 200/782, Loss: 0.0972, Recon: 0.0182, VQ: 0.0790, Perplexity: 53.65
Epoch 164/500, Batch 300/782, Loss: 0.1069, Recon: 0.0187, VQ: 0.0882, Perplexity: 53.85
Epoch 164/500, Batch 400/782, Loss: 0.1031, Recon: 0.0196, VQ: 0.0835, Perplexity: 55.78
Epoch 164/500, Batch 500/782, Loss: 0.1201, Recon: 0.0193, VQ: 0.1007, Perplexity: 51.62
Epoch 164/500, Batch 600/782, Loss: 0.1045, Recon: 0.0190, VQ: 0.0855, Perplexity: 54.90
Epoch 164/500, Batch 700/782, Loss: 0.1047, Recon: 0.0197, VQ: 0.0850, Perplexity: 55.43

=== Epoch 164/500 Summary ===
Train Loss: 0.1081 | Val Loss: 0.1087
Train Recon: 0.0192 | Val Recon: 0.0204
Train VQ: 0.0889 | Val VQ: 0.0883
Train Perplexity: 54.44 | Val Perplexity: 54.63
No improvement. Patience: 7/20
Best Loss: 0.1080 (Epoch 157)

Epoch 165/500, Batch 100/782, Loss: 0.1144, Recon: 0.0200, VQ: 0.0944, Perplexity: 55.97
Epoch 165/500, Batch 200/782, Loss: 0.1015, Recon: 0.0195, VQ: 0.0820, Perplexity: 55.90
Epoch 165/500, Batch 300/782, Loss: 0.0978, Recon: 0.0171, VQ: 0.0807, Perplexity: 53.37
Epoch 165/500, Batch 400/782, Loss: 0.1219, Recon: 0.0214, VQ: 0.1005, Perplexity: 56.06

Epoch 165/500, Batch 500/782, Loss: 0.0980, Recon: 0.0177, VQ: 0.0803, Perplexity: 52.56
Epoch 165/500, Batch 600/782, Loss: 0.1196, Recon: 0.0201, VQ: 0.0994, Perplexity: 56.24
Epoch 165/500, Batch 700/782, Loss: 0.1141, Recon: 0.0218, VQ: 0.0922, Perplexity: 56.47

=== Epoch 165/500 Summary ===

Train Loss: 0.1081 | Val Loss: 0.1098
Train Recon: 0.0192 | Val Recon: 0.0204
Train VQ: 0.0889 | Val VQ: 0.0893
Train Perplexity: 54.45 | Val Perplexity: 54.76
No improvement. Patience: 8/20
Best Loss: 0.1080 (Epoch 157)

Epoch 166/500, Batch 100/782, Loss: 0.0957, Recon: 0.0180, VQ: 0.0777, Perplexity: 53.12
Epoch 166/500, Batch 200/782, Loss: 0.1224, Recon: 0.0197, VQ: 0.1028, Perplexity: 55.82
Epoch 166/500, Batch 300/782, Loss: 0.1086, Recon: 0.0191, VQ: 0.0895, Perplexity: 54.50
Epoch 166/500, Batch 400/782, Loss: 0.1000, Recon: 0.0185, VQ: 0.0815, Perplexity: 55.06
Epoch 166/500, Batch 500/782, Loss: 0.0946, Recon: 0.0179, VQ: 0.0767, Perplexity: 52.69
Epoch 166/500, Batch 600/782, Loss: 0.1017, Recon: 0.0182, VQ: 0.0834, Perplexity: 53.30
Epoch 166/500, Batch 700/782, Loss: 0.1253, Recon: 0.0221, VQ: 0.1032, Perplexity: 55.77

=== Epoch 166/500 Summary ===

Train Loss: 0.1081 | Val Loss: 0.1076
Train Recon: 0.0192 | Val Recon: 0.0203
Train VQ: 0.0889 | Val VQ: 0.0873
Train Perplexity: 54.51 | Val Perplexity: 54.80
✓ Best model saved! (Val Loss: 0.1076)
Best Loss: 0.1076 (Epoch 166)

Epoch 167/500, Batch 100/782, Loss: 0.0974, Recon: 0.0186, VQ: 0.0788, Perplexity: 53.32
Epoch 167/500, Batch 200/782, Loss: 0.1044, Recon: 0.0184, VQ: 0.0859, Perplexity: 53.00
Epoch 167/500, Batch 300/782, Loss: 0.1119, Recon: 0.0190, VQ: 0.0929, Perplexity: 55.81
Epoch 167/500, Batch 400/782, Loss: 0.1056, Recon: 0.0191, VQ: 0.0865, Perplexity: 53.03
Epoch 167/500, Batch 500/782, Loss: 0.1014, Recon: 0.0184, VQ: 0.0830, Perplexity: 54.01
Epoch 167/500, Batch 600/782, Loss: 0.0988, Recon: 0.0183, VQ: 0.0805, Perplexity: 54.13
Epoch 167/500, Batch 700/782, Loss: 0.0975, Recon: 0.0179, VQ: 0.0797, Perplexity: 53.69

=== Epoch 167/500 Summary ===

Train Loss: 0.1081 | Val Loss: 0.1070
Train Recon: 0.0192 | Val Recon: 0.0203
Train VQ: 0.0889 | Val VQ: 0.0866
Train Perplexity: 54.60 | Val Perplexity: 54.43
✓ Best model saved! (Val Loss: 0.1070)
Best Loss: 0.1070 (Epoch 167)

Epoch 168/500, Batch 100/782, Loss: 0.1077, Recon: 0.0194, VQ: 0.0883, Perplexity: 55.93
Epoch 168/500, Batch 200/782, Loss: 0.1028, Recon: 0.0199, VQ: 0.0829, Perplexity: 54.96
Epoch 168/500, Batch 300/782, Loss: 0.1092, Recon: 0.0181, VQ: 0.0912, Perplexity: 52.04
Epoch 168/500, Batch 400/782, Loss: 0.0952, Recon: 0.0179, VQ: 0.0773, Perplexity: 54.78
Epoch 168/500, Batch 500/782, Loss: 0.1116, Recon: 0.0200, VQ: 0.0917, Perplexity: 56.63
Epoch 168/500, Batch 600/782, Loss: 0.1076, Recon: 0.0194, VQ: 0.0882, Perplexity: 55.09
Epoch 168/500, Batch 700/782, Loss: 0.1031, Recon: 0.0190, VQ: 0.0841, Perplexity: 54.21

=== Epoch 168/500 Summary ===

Train Loss: 0.1079 | Val Loss: 0.1088
Train Recon: 0.0192 | Val Recon: 0.0203
Train VQ: 0.0887 | Val VQ: 0.0885
Train Perplexity: 54.67 | Val Perplexity: 54.86
No improvement. Patience: 1/20
Best Loss: 0.1070 (Epoch 167)

Epoch 169/500, Batch 100/782, Loss: 0.0922, Recon: 0.0173, VQ: 0.0749, Perplexity: 53.05
Epoch 169/500, Batch 200/782, Loss: 0.1047, Recon: 0.0194, VQ: 0.0853, Perplexity: 54.86
Epoch 169/500, Batch 300/782, Loss: 0.1149, Recon: 0.0211, VQ: 0.0938, Perplexity: 54.80
Epoch 169/500, Batch 400/782, Loss: 0.1155, Recon: 0.0207, VQ: 0.0949, Perplexity: 55.25
Epoch 169/500, Batch 500/782, Loss: 0.1120, Recon: 0.0179, VQ: 0.0942, Perplexity: 53.57
Epoch 169/500, Batch 600/782, Loss: 0.1104, Recon: 0.0194, VQ: 0.0910, Perplexity: 54.39
Epoch 169/500, Batch 700/782, Loss: 0.0993, Recon: 0.0187, VQ: 0.0806, Perplexity: 53.71

=== Epoch 169/500 Summary ===

Train Loss: 0.1078 | Val Loss: 0.1087
Train Recon: 0.0192 | Val Recon: 0.0203
Train VQ: 0.0886 | Val VQ: 0.0884
Train Perplexity: 54.69 | Val Perplexity: 55.05
No improvement. Patience: 2/20
Best Loss: 0.1070 (Epoch 167)

Epoch 170/500, Batch 100/782, Loss: 0.0998, Recon: 0.0185, VQ: 0.0813, Perplexity: 54.00
Epoch 170/500, Batch 200/782, Loss: 0.1070, Recon: 0.0197, VQ: 0.0873, Perplexity: 54.64
Epoch 170/500, Batch 300/782, Loss: 0.0961, Recon: 0.0175, VQ: 0.0786, Perplexity: 53.03
Epoch 170/500, Batch 400/782, Loss: 0.0903, Recon: 0.0174, VQ: 0.0729, Perplexity: 52.17
Epoch 170/500, Batch 500/782, Loss: 0.1039, Recon: 0.0184, VQ: 0.0855, Perplexity: 54.33
Epoch 170/500, Batch 600/782, Loss: 0.1002, Recon: 0.0184, VQ: 0.0818, Perplexity: 55.24
Epoch 170/500, Batch 700/782, Loss: 0.1055, Recon: 0.0197, VQ: 0.0858, Per

plexity: 56.08

=== Epoch 170/500 Summary ===

Train Loss: 0.1079 | Val Loss: 0.1090

Train Recon: 0.0192 | Val Recon: 0.0204

Train VQ: 0.0887 | Val VQ: 0.0886

Train Perplexity: 54.85 | Val Perplexity: 54.90

No improvement. Patience: 3/20

Best Loss: 0.1070 (Epoch 167)

Epoch 171/500, Batch 100/782, Loss: 0.1067, Recon: 0.0178, VQ: 0.0888, Perplexity: 54.84

Epoch 171/500, Batch 200/782, Loss: 0.1047, Recon: 0.0193, VQ: 0.0854, Perplexity: 57.39

Epoch 171/500, Batch 300/782, Loss: 0.1130, Recon: 0.0205, VQ: 0.0925, Perplexity: 56.34

Epoch 171/500, Batch 400/782, Loss: 0.1103, Recon: 0.0197, VQ: 0.0905, Perplexity: 56.30

Epoch 171/500, Batch 500/782, Loss: 0.0993, Recon: 0.0174, VQ: 0.0820, Perplexity: 53.72

Epoch 171/500, Batch 600/782, Loss: 0.1100, Recon: 0.0205, VQ: 0.0895, Perplexity: 57.73

Epoch 171/500, Batch 700/782, Loss: 0.1646, Recon: 0.0198, VQ: 0.1448, Perplexity: 54.34

=== Epoch 171/500 Summary ===

Train Loss: 0.1074 | Val Loss: 0.1075

Train Recon: 0.0191 | Val Recon: 0.0203

Train VQ: 0.0883 | Val VQ: 0.0872

Train Perplexity: 54.90 | Val Perplexity: 54.81

No improvement. Patience: 4/20

Best Loss: 0.1070 (Epoch 167)

Epoch 172/500, Batch 100/782, Loss: 0.1379, Recon: 0.0193, VQ: 0.1186, Perplexity: 54.27

Epoch 172/500, Batch 200/782, Loss: 0.1148, Recon: 0.0208, VQ: 0.0940, Perplexity: 57.11

Epoch 172/500, Batch 300/782, Loss: 0.1062, Recon: 0.0192, VQ: 0.0871, Perplexity: 55.29

Epoch 172/500, Batch 400/782, Loss: 0.1057, Recon: 0.0197, VQ: 0.0860, Perplexity: 55.87

Epoch 172/500, Batch 500/782, Loss: 0.1040, Recon: 0.0181, VQ: 0.0859, Perplexity: 53.94

Epoch 172/500, Batch 600/782, Loss: 0.1017, Recon: 0.0181, VQ: 0.0836, Perplexity: 53.95

Epoch 172/500, Batch 700/782, Loss: 0.1170, Recon: 0.0209, VQ: 0.0961, Perplexity: 55.32

=== Epoch 172/500 Summary ===

Train Loss: 0.1075 | Val Loss: 0.1085

Train Recon: 0.0191 | Val Recon: 0.0203

Train VQ: 0.0884 | Val VQ: 0.0881

Train Perplexity: 54.98 | Val Perplexity: 55.35

No improvement. Patience: 5/20

Best Loss: 0.1070 (Epoch 167)

Epoch 173/500, Batch 100/782, Loss: 0.1034, Recon: 0.0196, VQ: 0.0837, Perplexity: 55.47

Epoch 173/500, Batch 200/782, Loss: 0.1646, Recon: 0.0200, VQ: 0.1445, Perplexity: 55.63

Epoch 173/500, Batch 300/782, Loss: 0.1056, Recon: 0.0195, VQ: 0.0860, Perplexity: 56.41
Epoch 173/500, Batch 400/782, Loss: 0.1062, Recon: 0.0194, VQ: 0.0869, Perplexity: 56.40
Epoch 173/500, Batch 500/782, Loss: 0.1066, Recon: 0.0202, VQ: 0.0864, Perplexity: 55.09
Epoch 173/500, Batch 600/782, Loss: 0.1087, Recon: 0.0203, VQ: 0.0884, Perplexity: 55.91
Epoch 173/500, Batch 700/782, Loss: 0.0912, Recon: 0.0170, VQ: 0.0741, Perplexity: 51.68

=== Epoch 173/500 Summary ===

Train Loss: 0.1076 | Val Loss: 0.1086
Train Recon: 0.0191 | Val Recon: 0.0203
Train VQ: 0.0885 | Val VQ: 0.0883
Train Perplexity: 55.10 | Val Perplexity: 55.17
No improvement. Patience: 6/20
Best Loss: 0.1070 (Epoch 167)

Epoch 174/500, Batch 100/782, Loss: 0.1010, Recon: 0.0196, VQ: 0.0813, Perplexity: 55.72
Epoch 174/500, Batch 200/782, Loss: 0.0990, Recon: 0.0181, VQ: 0.0810, Perplexity: 52.60
Epoch 174/500, Batch 300/782, Loss: 0.1015, Recon: 0.0178, VQ: 0.0837, Perplexity: 55.42
Epoch 174/500, Batch 400/782, Loss: 0.1102, Recon: 0.0192, VQ: 0.0910, Perplexity: 56.83
Epoch 174/500, Batch 500/782, Loss: 0.1058, Recon: 0.0181, VQ: 0.0876, Perplexity: 55.12
Epoch 174/500, Batch 600/782, Loss: 0.1107, Recon: 0.0187, VQ: 0.0920, Perplexity: 53.86
Epoch 174/500, Batch 700/782, Loss: 0.0942, Recon: 0.0181, VQ: 0.0761, Perplexity: 52.66

=== Epoch 174/500 Summary ===

Train Loss: 0.1074 | Val Loss: 0.1073
Train Recon: 0.0191 | Val Recon: 0.0203
Train VQ: 0.0883 | Val VQ: 0.0870
Train Perplexity: 55.11 | Val Perplexity: 55.18
No improvement. Patience: 7/20
Best Loss: 0.1070 (Epoch 167)

Epoch 175/500, Batch 100/782, Loss: 0.1172, Recon: 0.0195, VQ: 0.0977, Perplexity: 55.16
Epoch 175/500, Batch 200/782, Loss: 0.1145, Recon: 0.0195, VQ: 0.0950, Perplexity: 55.18
Epoch 175/500, Batch 300/782, Loss: 0.1009, Recon: 0.0190, VQ: 0.0819, Perplexity: 54.20
Epoch 175/500, Batch 400/782, Loss: 0.1054, Recon: 0.0194, VQ: 0.0860, Perplexity: 55.50
Epoch 175/500, Batch 500/782, Loss: 0.1015, Recon: 0.0180, VQ: 0.0835, Perplexity: 54.01
Epoch 175/500, Batch 600/782, Loss: 0.0992, Recon: 0.0191, VQ: 0.0801, Perplexity: 54.56
Epoch 175/500, Batch 700/782, Loss: 0.0949, Recon: 0.0173, VQ: 0.0776, Perplexity: 52.60

=== Epoch 175/500 Summary ===

Train Loss: 0.1073 | Val Loss: 0.1088
Train Recon: 0.0191 | Val Recon: 0.0203

Train VQ: 0.0882 | Val VQ: 0.0884
Train Perplexity: 55.23 | Val Perplexity: 55.24
No improvement. Patience: 8/20
Best Loss: 0.1070 (Epoch 167)

Epoch 176/500, Batch 100/782, Loss: 0.1172, Recon: 0.0203, VQ: 0.0969, Perplexity: 57.16
Epoch 176/500, Batch 200/782, Loss: 0.1088, Recon: 0.0184, VQ: 0.0904, Perplexity: 55.72
Epoch 176/500, Batch 300/782, Loss: 0.1089, Recon: 0.0203, VQ: 0.0886, Perplexity: 55.85
Epoch 176/500, Batch 400/782, Loss: 0.0896, Recon: 0.0163, VQ: 0.0733, Perplexity: 50.40
Epoch 176/500, Batch 500/782, Loss: 0.1003, Recon: 0.0184, VQ: 0.0819, Perplexity: 54.58
Epoch 176/500, Batch 600/782, Loss: 0.1071, Recon: 0.0197, VQ: 0.0874, Perplexity: 56.14
Epoch 176/500, Batch 700/782, Loss: 0.1120, Recon: 0.0185, VQ: 0.0935, Perplexity: 54.34

=== Epoch 176/500 Summary ===
Train Loss: 0.1073 | Val Loss: 0.1078
Train Recon: 0.0191 | Val Recon: 0.0203
Train VQ: 0.0882 | Val VQ: 0.0876
Train Perplexity: 55.33 | Val Perplexity: 55.30
No improvement. Patience: 9/20
Best Loss: 0.1070 (Epoch 167)

Epoch 177/500, Batch 100/782, Loss: 0.0916, Recon: 0.0170, VQ: 0.0746, Perplexity: 52.79
Epoch 177/500, Batch 200/782, Loss: 0.0953, Recon: 0.0175, VQ: 0.0779, Perplexity: 53.70
Epoch 177/500, Batch 300/782, Loss: 0.1097, Recon: 0.0186, VQ: 0.0911, Perplexity: 54.37
Epoch 177/500, Batch 400/782, Loss: 0.1233, Recon: 0.0196, VQ: 0.1036, Perplexity: 54.30
Epoch 177/500, Batch 500/782, Loss: 0.1071, Recon: 0.0190, VQ: 0.0881, Perplexity: 55.26
Epoch 177/500, Batch 600/782, Loss: 0.1104, Recon: 0.0200, VQ: 0.0904, Perplexity: 56.49
Epoch 177/500, Batch 700/782, Loss: 0.1119, Recon: 0.0206, VQ: 0.0913, Perplexity: 57.22

=== Epoch 177/500 Summary ===
Train Loss: 0.1075 | Val Loss: 0.1075
Train Recon: 0.0191 | Val Recon: 0.0203
Train VQ: 0.0884 | Val VQ: 0.0871
Train Perplexity: 55.37 | Val Perplexity: 55.20
No improvement. Patience: 10/20
Best Loss: 0.1070 (Epoch 167)

Epoch 178/500, Batch 100/782, Loss: 0.1122, Recon: 0.0199, VQ: 0.0924, Perplexity: 57.06
Epoch 178/500, Batch 200/782, Loss: 0.1132, Recon: 0.0195, VQ: 0.0937, Perplexity: 56.37
Epoch 178/500, Batch 300/782, Loss: 0.1169, Recon: 0.0185, VQ: 0.0985, Perplexity: 52.52
Epoch 178/500, Batch 400/782, Loss: 0.1046, Recon: 0.0180, VQ: 0.0866, Perplexity: 55.14
Epoch 178/500, Batch 500/782, Loss: 0.1086, Recon: 0.0197, VQ: 0.0889, Per

plexity: 56.34
Epoch 178/500, Batch 600/782, Loss: 0.1153, Recon: 0.0199, VQ: 0.0953, Per
plexity: 56.08
Epoch 178/500, Batch 700/782, Loss: 0.1044, Recon: 0.0196, VQ: 0.0848, Per
plexity: 56.86

=== Epoch 178/500 Summary ===
Train Loss: 0.1074 | Val Loss: 0.1070
Train Recon: 0.0191 | Val Recon: 0.0203
Train VQ: 0.0883 | Val VQ: 0.0867
Train Perplexity: 55.46 | Val Perplexity: 55.18
No improvement. Patience: 11/20
Best Loss: 0.1070 (Epoch 167)

Epoch 179/500, Batch 100/782, Loss: 0.1228, Recon: 0.0216, VQ: 0.1012, Per
plexity: 57.30
Epoch 179/500, Batch 200/782, Loss: 0.0972, Recon: 0.0184, VQ: 0.0788, Per
plexity: 52.52
Epoch 179/500, Batch 300/782, Loss: 0.1129, Recon: 0.0192, VQ: 0.0938, Per
plexity: 56.72
Epoch 179/500, Batch 400/782, Loss: 0.1141, Recon: 0.0207, VQ: 0.0934, Per
plexity: 56.85
Epoch 179/500, Batch 500/782, Loss: 0.1011, Recon: 0.0185, VQ: 0.0827, Per
plexity: 55.82
Epoch 179/500, Batch 600/782, Loss: 0.1011, Recon: 0.0188, VQ: 0.0823, Per
plexity: 54.17
Epoch 179/500, Batch 700/782, Loss: 0.0950, Recon: 0.0178, VQ: 0.0771, Per
plexity: 52.98

=== Epoch 179/500 Summary ===
Train Loss: 0.1075 | Val Loss: 0.1084
Train Recon: 0.0191 | Val Recon: 0.0203
Train VQ: 0.0884 | Val VQ: 0.0881
Train Perplexity: 55.47 | Val Perplexity: 55.65
No improvement. Patience: 12/20
Best Loss: 0.1070 (Epoch 167)

Epoch 180/500, Batch 100/782, Loss: 0.1060, Recon: 0.0190, VQ: 0.0870, Per
plexity: 54.67
Epoch 180/500, Batch 200/782, Loss: 0.0968, Recon: 0.0181, VQ: 0.0787, Per
plexity: 55.50
Epoch 180/500, Batch 300/782, Loss: 0.1030, Recon: 0.0183, VQ: 0.0847, Per
plexity: 56.27
Epoch 180/500, Batch 400/782, Loss: 0.1023, Recon: 0.0190, VQ: 0.0833, Per
plexity: 55.40
Epoch 180/500, Batch 500/782, Loss: 0.1127, Recon: 0.0209, VQ: 0.0918, Per
plexity: 57.94
Epoch 180/500, Batch 600/782, Loss: 0.1033, Recon: 0.0188, VQ: 0.0845, Per
plexity: 54.83
Epoch 180/500, Batch 700/782, Loss: 0.0973, Recon: 0.0184, VQ: 0.0789, Per
plexity: 54.57

=== Epoch 180/500 Summary ===
Train Loss: 0.1079 | Val Loss: 0.1102
Train Recon: 0.0191 | Val Recon: 0.0204
Train VQ: 0.0889 | Val VQ: 0.0898
Train Perplexity: 55.54 | Val Perplexity: 55.97
No improvement. Patience: 13/20
Best Loss: 0.1070 (Epoch 167)

Epoch 181/500, Batch 100/782, Loss: 0.0995, Recon: 0.0179, VQ: 0.0816, Perplexity: 55.16
Epoch 181/500, Batch 200/782, Loss: 0.1093, Recon: 0.0198, VQ: 0.0895, Perplexity: 55.78
Epoch 181/500, Batch 300/782, Loss: 0.1075, Recon: 0.0193, VQ: 0.0881, Perplexity: 56.05
Epoch 181/500, Batch 400/782, Loss: 0.1199, Recon: 0.0205, VQ: 0.0994, Perplexity: 57.70
Epoch 181/500, Batch 500/782, Loss: 0.1160, Recon: 0.0198, VQ: 0.0961, Perplexity: 56.11
Epoch 181/500, Batch 600/782, Loss: 0.0977, Recon: 0.0178, VQ: 0.0800, Perplexity: 55.92
Epoch 181/500, Batch 700/782, Loss: 0.1097, Recon: 0.0198, VQ: 0.0899, Perplexity: 55.80

=== Epoch 181/500 Summary ===

Train Loss: 0.1078 | Val Loss: 0.1076
Train Recon: 0.0191 | Val Recon: 0.0203
Train VQ: 0.0887 | Val VQ: 0.0873
Train Perplexity: 55.55 | Val Perplexity: 55.34
No improvement. Patience: 14/20
Best Loss: 0.1070 (Epoch 167)

Epoch 182/500, Batch 100/782, Loss: 0.1103, Recon: 0.0201, VQ: 0.0902, Perplexity: 57.13
Epoch 182/500, Batch 200/782, Loss: 0.1095, Recon: 0.0194, VQ: 0.0901, Perplexity: 56.66
Epoch 182/500, Batch 300/782, Loss: 0.0982, Recon: 0.0185, VQ: 0.0797, Perplexity: 56.57
Epoch 182/500, Batch 400/782, Loss: 0.1096, Recon: 0.0198, VQ: 0.0898, Perplexity: 56.18
Epoch 182/500, Batch 500/782, Loss: 0.0942, Recon: 0.0172, VQ: 0.0770, Perplexity: 53.70
Epoch 182/500, Batch 600/782, Loss: 0.1045, Recon: 0.0182, VQ: 0.0863, Perplexity: 54.37
Epoch 182/500, Batch 700/782, Loss: 0.1172, Recon: 0.0191, VQ: 0.0981, Perplexity: 54.28

=== Epoch 182/500 Summary ===

Train Loss: 0.1076 | Val Loss: 0.1079
Train Recon: 0.0190 | Val Recon: 0.0203
Train VQ: 0.0886 | Val VQ: 0.0876
Train Perplexity: 55.58 | Val Perplexity: 55.66
No improvement. Patience: 15/20
Best Loss: 0.1070 (Epoch 167)

Epoch 183/500, Batch 100/782, Loss: 0.1242, Recon: 0.0205, VQ: 0.1037, Perplexity: 55.36
Epoch 183/500, Batch 200/782, Loss: 0.1232, Recon: 0.0188, VQ: 0.1044, Perplexity: 55.47
Epoch 183/500, Batch 300/782, Loss: 0.1274, Recon: 0.0187, VQ: 0.1087, Perplexity: 54.12
Epoch 183/500, Batch 400/782, Loss: 0.1148, Recon: 0.0199, VQ: 0.0949, Perplexity: 57.25
Epoch 183/500, Batch 500/782, Loss: 0.1055, Recon: 0.0188, VQ: 0.0867, Perplexity: 55.59
Epoch 183/500, Batch 600/782, Loss: 0.1215, Recon: 0.0206, VQ: 0.1008, Perplexity: 56.59
Epoch 183/500, Batch 700/782, Loss: 0.1086, Recon: 0.0194, VQ: 0.0892, Perplexity: 56.43

=== Epoch 183/500 Summary ===

Train Loss: 0.1075 | Val Loss: 0.1090
Train Recon: 0.0190 | Val Recon: 0.0203
Train VQ: 0.0885 | Val VQ: 0.0887
Train Perplexity: 55.68 | Val Perplexity: 55.52
No improvement. Patience: 16/20
Best Loss: 0.1070 (Epoch 167)

Epoch 184/500, Batch 100/782, Loss: 0.1299, Recon: 0.0202, VQ: 0.1097, Perplexity: 56.45
Epoch 184/500, Batch 200/782, Loss: 0.1372, Recon: 0.0219, VQ: 0.1153, Perplexity: 60.50
Epoch 184/500, Batch 300/782, Loss: 0.0950, Recon: 0.0176, VQ: 0.0774, Perplexity: 52.94
Epoch 184/500, Batch 400/782, Loss: 0.1036, Recon: 0.0194, VQ: 0.0842, Perplexity: 56.94
Epoch 184/500, Batch 500/782, Loss: 0.1080, Recon: 0.0199, VQ: 0.0881, Perplexity: 55.85
Epoch 184/500, Batch 600/782, Loss: 0.1228, Recon: 0.0204, VQ: 0.1024, Perplexity: 57.12
Epoch 184/500, Batch 700/782, Loss: 0.1055, Recon: 0.0195, VQ: 0.0860, Perplexity: 55.89

=== Epoch 184/500 Summary ===

Train Loss: 0.1075 | Val Loss: 0.1086
Train Recon: 0.0190 | Val Recon: 0.0203
Train VQ: 0.0885 | Val VQ: 0.0883
Train Perplexity: 55.67 | Val Perplexity: 55.65
No improvement. Patience: 17/20
Best Loss: 0.1070 (Epoch 167)

Epoch 185/500, Batch 100/782, Loss: 0.1081, Recon: 0.0179, VQ: 0.0901, Perplexity: 53.65
Epoch 185/500, Batch 200/782, Loss: 0.1052, Recon: 0.0193, VQ: 0.0859, Perplexity: 55.41
Epoch 185/500, Batch 300/782, Loss: 0.1016, Recon: 0.0186, VQ: 0.0830, Perplexity: 54.81
Epoch 185/500, Batch 400/782, Loss: 0.1068, Recon: 0.0177, VQ: 0.0891, Perplexity: 55.53
Epoch 185/500, Batch 500/782, Loss: 0.1130, Recon: 0.0204, VQ: 0.0926, Perplexity: 56.19
Epoch 185/500, Batch 600/782, Loss: 0.1120, Recon: 0.0203, VQ: 0.0916, Perplexity: 56.48
Epoch 185/500, Batch 700/782, Loss: 0.1272, Recon: 0.0195, VQ: 0.1077, Perplexity: 56.08

=== Epoch 185/500 Summary ===

Train Loss: 0.1078 | Val Loss: 0.1065
Train Recon: 0.0190 | Val Recon: 0.0204
Train VQ: 0.0888 | Val VQ: 0.0861
Train Perplexity: 55.75 | Val Perplexity: 55.53
✓ Best model saved! (Val Loss: 0.1065)
Best Loss: 0.1065 (Epoch 185)

Epoch 186/500, Batch 100/782, Loss: 0.1020, Recon: 0.0184, VQ: 0.0836, Perplexity: 53.96
Epoch 186/500, Batch 200/782, Loss: 0.0987, Recon: 0.0182, VQ: 0.0805, Perplexity: 53.78
Epoch 186/500, Batch 300/782, Loss: 0.1084, Recon: 0.0202, VQ: 0.0881, Per

plexity: 57.48
Epoch 186/500, Batch 400/782, Loss: 0.1087, Recon: 0.0189, VQ: 0.0899, Per
plexity: 58.39
Epoch 186/500, Batch 500/782, Loss: 0.0993, Recon: 0.0183, VQ: 0.0810, Per
plexity: 54.95
Epoch 186/500, Batch 600/782, Loss: 0.0991, Recon: 0.0179, VQ: 0.0812, Per
plexity: 53.23
Epoch 186/500, Batch 700/782, Loss: 0.1145, Recon: 0.0200, VQ: 0.0945, Per
plexity: 56.84

=== Epoch 186/500 Summary ===
Train Loss: 0.1074 | Val Loss: 0.1091
Train Recon: 0.0190 | Val Recon: 0.0203
Train VQ: 0.0884 | Val VQ: 0.0888
Train Perplexity: 55.87 | Val Perplexity: 55.31
No improvement. Patience: 1/20
Best Loss: 0.1065 (Epoch 185)

Epoch 187/500, Batch 100/782, Loss: 0.0958, Recon: 0.0181, VQ: 0.0777, Per
plexity: 53.81
Epoch 187/500, Batch 200/782, Loss: 0.0972, Recon: 0.0182, VQ: 0.0790, Per
plexity: 55.83
Epoch 187/500, Batch 300/782, Loss: 0.1080, Recon: 0.0178, VQ: 0.0901, Per
plexity: 55.12
Epoch 187/500, Batch 400/782, Loss: 0.1079, Recon: 0.0183, VQ: 0.0896, Per
plexity: 55.20
Epoch 187/500, Batch 500/782, Loss: 0.1132, Recon: 0.0197, VQ: 0.0936, Per
plexity: 55.83
Epoch 187/500, Batch 600/782, Loss: 0.1032, Recon: 0.0189, VQ: 0.0843, Per
plexity: 57.43
Epoch 187/500, Batch 700/782, Loss: 0.1057, Recon: 0.0196, VQ: 0.0860, Per
plexity: 54.97

=== Epoch 187/500 Summary ===
Train Loss: 0.1075 | Val Loss: 0.1082
Train Recon: 0.0190 | Val Recon: 0.0203
Train VQ: 0.0885 | Val VQ: 0.0879
Train Perplexity: 55.92 | Val Perplexity: 55.69
No improvement. Patience: 2/20
Best Loss: 0.1065 (Epoch 185)

Epoch 188/500, Batch 100/782, Loss: 0.1105, Recon: 0.0198, VQ: 0.0907, Per
plexity: 55.98
Epoch 188/500, Batch 200/782, Loss: 0.1007, Recon: 0.0181, VQ: 0.0827, Per
plexity: 54.12
Epoch 188/500, Batch 300/782, Loss: 0.1274, Recon: 0.0215, VQ: 0.1059, Per
plexity: 59.38
Epoch 188/500, Batch 400/782, Loss: 0.0984, Recon: 0.0171, VQ: 0.0814, Per
plexity: 55.01
Epoch 188/500, Batch 500/782, Loss: 0.1093, Recon: 0.0192, VQ: 0.0901, Per
plexity: 57.89
Epoch 188/500, Batch 600/782, Loss: 0.1005, Recon: 0.0179, VQ: 0.0826, Per
plexity: 54.92
Epoch 188/500, Batch 700/782, Loss: 0.1045, Recon: 0.0186, VQ: 0.0859, Per
plexity: 55.65

=== Epoch 188/500 Summary ===
Train Loss: 0.1075 | Val Loss: 0.1079
Train Recon: 0.0190 | Val Recon: 0.0202
Train VQ: 0.0885 | Val VQ: 0.0877

Train Perplexity: 56.08 | Val Perplexity: 56.25

No improvement. Patience: 3/20

Best Loss: 0.1065 (Epoch 185)

Epoch 189/500, Batch 100/782, Loss: 0.0930, Recon: 0.0180, VQ: 0.0751, Perplexity: 53.14

Epoch 189/500, Batch 200/782, Loss: 0.0978, Recon: 0.0179, VQ: 0.0798, Perplexity: 55.68

Epoch 189/500, Batch 300/782, Loss: 0.1191, Recon: 0.0201, VQ: 0.0991, Perplexity: 55.50

Epoch 189/500, Batch 400/782, Loss: 0.1115, Recon: 0.0201, VQ: 0.0914, Perplexity: 55.68

Epoch 189/500, Batch 500/782, Loss: 0.1130, Recon: 0.0205, VQ: 0.0925, Perplexity: 58.89

Epoch 189/500, Batch 600/782, Loss: 0.1117, Recon: 0.0198, VQ: 0.0919, Perplexity: 57.53

Epoch 189/500, Batch 700/782, Loss: 0.1085, Recon: 0.0205, VQ: 0.0880, Perplexity: 56.05

=== Epoch 189/500 Summary ===

Train Loss: 0.1076 | Val Loss: 0.1082

Train Recon: 0.0190 | Val Recon: 0.0202

Train VQ: 0.0886 | Val VQ: 0.0879

Train Perplexity: 56.28 | Val Perplexity: 56.30

No improvement. Patience: 4/20

Best Loss: 0.1065 (Epoch 185)

Epoch 190/500, Batch 100/782, Loss: 0.0954, Recon: 0.0179, VQ: 0.0775, Perplexity: 53.36

Epoch 190/500, Batch 200/782, Loss: 0.0940, Recon: 0.0181, VQ: 0.0759, Perplexity: 54.27

Epoch 190/500, Batch 300/782, Loss: 0.1112, Recon: 0.0193, VQ: 0.0919, Perplexity: 57.68

Epoch 190/500, Batch 400/782, Loss: 0.1006, Recon: 0.0185, VQ: 0.0821, Perplexity: 56.75

Epoch 190/500, Batch 500/782, Loss: 0.1050, Recon: 0.0188, VQ: 0.0862, Perplexity: 56.23

Epoch 190/500, Batch 600/782, Loss: 0.1198, Recon: 0.0212, VQ: 0.0987, Perplexity: 57.84

Epoch 190/500, Batch 700/782, Loss: 0.0925, Recon: 0.0171, VQ: 0.0754, Perplexity: 54.08

=== Epoch 190/500 Summary ===

Train Loss: 0.1074 | Val Loss: 0.1080

Train Recon: 0.0190 | Val Recon: 0.0202

Train VQ: 0.0884 | Val VQ: 0.0878

Train Perplexity: 56.41 | Val Perplexity: 55.97

No improvement. Patience: 5/20

Best Loss: 0.1065 (Epoch 185)

Epoch 191/500, Batch 100/782, Loss: 0.1038, Recon: 0.0198, VQ: 0.0841, Perplexity: 57.09

Epoch 191/500, Batch 200/782, Loss: 0.1059, Recon: 0.0188, VQ: 0.0871, Perplexity: 56.79

Epoch 191/500, Batch 300/782, Loss: 0.0990, Recon: 0.0179, VQ: 0.0811, Perplexity: 54.21

Epoch 191/500, Batch 400/782, Loss: 0.0987, Recon: 0.0180, VQ: 0.0807, Perplexity: 54.24

Epoch 191/500, Batch 500/782, Loss: 0.1198, Recon: 0.0205, VQ: 0.0993, Perplexity: 56.81

Epoch 191/500, Batch 600/782, Loss: 0.1127, Recon: 0.0193, VQ: 0.0934, Perplexity: 57.77
Epoch 191/500, Batch 700/782, Loss: 0.1051, Recon: 0.0194, VQ: 0.0857, Perplexity: 58.41

=== Epoch 191/500 Summary ===

Train Loss: 0.1072 | Val Loss: 0.1082
Train Recon: 0.0190 | Val Recon: 0.0202
Train VQ: 0.0883 | Val VQ: 0.0880
Train Perplexity: 56.50 | Val Perplexity: 56.65
No improvement. Patience: 6/20
Best Loss: 0.1065 (Epoch 185)

Epoch 192/500, Batch 100/782, Loss: 0.1059, Recon: 0.0188, VQ: 0.0871, Perplexity: 56.45
Epoch 192/500, Batch 200/782, Loss: 0.1100, Recon: 0.0196, VQ: 0.0903, Perplexity: 57.39
Epoch 192/500, Batch 300/782, Loss: 0.0966, Recon: 0.0190, VQ: 0.0776, Perplexity: 55.76
Epoch 192/500, Batch 400/782, Loss: 0.1008, Recon: 0.0193, VQ: 0.0815, Perplexity: 57.24
Epoch 192/500, Batch 500/782, Loss: 0.1123, Recon: 0.0171, VQ: 0.0952, Perplexity: 55.84
Epoch 192/500, Batch 600/782, Loss: 0.1092, Recon: 0.0186, VQ: 0.0906, Perplexity: 56.39
Epoch 192/500, Batch 700/782, Loss: 0.0940, Recon: 0.0181, VQ: 0.0758, Perplexity: 54.28

=== Epoch 192/500 Summary ===

Train Loss: 0.1072 | Val Loss: 0.1077
Train Recon: 0.0189 | Val Recon: 0.0202
Train VQ: 0.0883 | Val VQ: 0.0875
Train Perplexity: 56.63 | Val Perplexity: 56.72
No improvement. Patience: 7/20
Best Loss: 0.1065 (Epoch 185)

Epoch 193/500, Batch 100/782, Loss: 0.1475, Recon: 0.0190, VQ: 0.1285, Perplexity: 55.65
Epoch 193/500, Batch 200/782, Loss: 0.0985, Recon: 0.0180, VQ: 0.0805, Perplexity: 57.45
Epoch 193/500, Batch 300/782, Loss: 0.1015, Recon: 0.0189, VQ: 0.0826, Perplexity: 57.55
Epoch 193/500, Batch 400/782, Loss: 0.0929, Recon: 0.0174, VQ: 0.0754, Perplexity: 55.97
Epoch 193/500, Batch 500/782, Loss: 0.1051, Recon: 0.0199, VQ: 0.0852, Perplexity: 57.66
Epoch 193/500, Batch 600/782, Loss: 0.1213, Recon: 0.0213, VQ: 0.1000, Perplexity: 58.04
Epoch 193/500, Batch 700/782, Loss: 0.1378, Recon: 0.0195, VQ: 0.1183, Perplexity: 57.83

=== Epoch 193/500 Summary ===

Train Loss: 0.1072 | Val Loss: 0.1096
Train Recon: 0.0189 | Val Recon: 0.0202
Train VQ: 0.0882 | Val VQ: 0.0894
Train Perplexity: 56.74 | Val Perplexity: 57.17
No improvement. Patience: 8/20
Best Loss: 0.1065 (Epoch 185)

Epoch 194/500, Batch 100/782, Loss: 0.1228, Recon: 0.0187, VQ: 0.1041, Per

plexity: 55.59
Epoch 194/500, Batch 200/782, Loss: 0.1120, Recon: 0.0189, VQ: 0.0931, Perplexity: 57.13
Epoch 194/500, Batch 300/782, Loss: 0.1004, Recon: 0.0179, VQ: 0.0825, Perplexity: 54.88
Epoch 194/500, Batch 400/782, Loss: 0.1137, Recon: 0.0202, VQ: 0.0935, Perplexity: 59.11
Epoch 194/500, Batch 500/782, Loss: 0.0969, Recon: 0.0183, VQ: 0.0786, Perplexity: 54.61
Epoch 194/500, Batch 600/782, Loss: 0.1040, Recon: 0.0197, VQ: 0.0843, Perplexity: 56.32
Epoch 194/500, Batch 700/782, Loss: 0.1052, Recon: 0.0188, VQ: 0.0864, Perplexity: 57.75

=== Epoch 194/500 Summary ===

Train Loss: 0.1075 | Val Loss: 0.1074
Train Recon: 0.0189 | Val Recon: 0.0202
Train VQ: 0.0885 | Val VQ: 0.0872
Train Perplexity: 56.82 | Val Perplexity: 56.95
No improvement. Patience: 9/20
Best Loss: 0.1065 (Epoch 185)

Epoch 195/500, Batch 100/782, Loss: 0.1018, Recon: 0.0184, VQ: 0.0834, Perplexity: 55.04
Epoch 195/500, Batch 200/782, Loss: 0.1013, Recon: 0.0184, VQ: 0.0828, Perplexity: 55.57
Epoch 195/500, Batch 300/782, Loss: 0.1035, Recon: 0.0196, VQ: 0.0839, Perplexity: 58.96
Epoch 195/500, Batch 400/782, Loss: 0.1036, Recon: 0.0198, VQ: 0.0837, Perplexity: 56.49
Epoch 195/500, Batch 500/782, Loss: 0.1002, Recon: 0.0178, VQ: 0.0825, Perplexity: 55.11
Epoch 195/500, Batch 600/782, Loss: 0.1023, Recon: 0.0199, VQ: 0.0824, Perplexity: 57.78
Epoch 195/500, Batch 700/782, Loss: 0.1062, Recon: 0.0185, VQ: 0.0877, Perplexity: 57.27

=== Epoch 195/500 Summary ===

Train Loss: 0.1071 | Val Loss: 0.1087
Train Recon: 0.0189 | Val Recon: 0.0202
Train VQ: 0.0882 | Val VQ: 0.0885
Train Perplexity: 56.92 | Val Perplexity: 56.73
No improvement. Patience: 10/20
Best Loss: 0.1065 (Epoch 185)

Epoch 196/500, Batch 100/782, Loss: 0.0932, Recon: 0.0179, VQ: 0.0753, Perplexity: 54.13
Epoch 196/500, Batch 200/782, Loss: 0.0934, Recon: 0.0176, VQ: 0.0758, Perplexity: 55.60
Epoch 196/500, Batch 300/782, Loss: 0.0973, Recon: 0.0178, VQ: 0.0796, Perplexity: 55.90
Epoch 196/500, Batch 400/782, Loss: 0.1076, Recon: 0.0191, VQ: 0.0885, Perplexity: 56.49
Epoch 196/500, Batch 500/782, Loss: 0.1186, Recon: 0.0208, VQ: 0.0978, Perplexity: 57.77
Epoch 196/500, Batch 600/782, Loss: 0.1130, Recon: 0.0199, VQ: 0.0932, Perplexity: 58.99
Epoch 196/500, Batch 700/782, Loss: 0.1092, Recon: 0.0191, VQ: 0.0900, Perplexity: 57.30

=== Epoch 196/500 Summary ===

Train Loss: 0.1073 | Val Loss: 0.1085

Train Recon: 0.0189 | Val Recon: 0.0201

Train VQ: 0.0884 | Val VQ: 0.0884

Train Perplexity: 57.00 | Val Perplexity: 57.25

No improvement. Patience: 11/20

Best Loss: 0.1065 (Epoch 185)

Epoch 197/500, Batch 100/782, Loss: 0.1086, Recon: 0.0199, VQ: 0.0887, Perplexity: 59.26

Epoch 197/500, Batch 200/782, Loss: 0.1016, Recon: 0.0187, VQ: 0.0829, Perplexity: 55.83

Epoch 197/500, Batch 300/782, Loss: 0.1148, Recon: 0.0206, VQ: 0.0943, Perplexity: 58.12

Epoch 197/500, Batch 400/782, Loss: 0.1066, Recon: 0.0190, VQ: 0.0876, Perplexity: 57.25

Epoch 197/500, Batch 500/782, Loss: 0.0959, Recon: 0.0181, VQ: 0.0778, Perplexity: 56.35

Epoch 197/500, Batch 600/782, Loss: 0.1009, Recon: 0.0190, VQ: 0.0819, Perplexity: 57.14

Epoch 197/500, Batch 700/782, Loss: 0.1055, Recon: 0.0199, VQ: 0.0856, Perplexity: 56.13

=== Epoch 197/500 Summary ===

Train Loss: 0.1071 | Val Loss: 0.1059

Train Recon: 0.0189 | Val Recon: 0.0202

Train VQ: 0.0882 | Val VQ: 0.0857

Train Perplexity: 57.09 | Val Perplexity: 56.72

✓ Best model saved! (Val Loss: 0.1059)

Best Loss: 0.1059 (Epoch 197)

Epoch 198/500, Batch 100/782, Loss: 0.0983, Recon: 0.0190, VQ: 0.0793, Perplexity: 56.45

Epoch 198/500, Batch 200/782, Loss: 0.0970, Recon: 0.0186, VQ: 0.0784, Perplexity: 55.62

Epoch 198/500, Batch 300/782, Loss: 0.1030, Recon: 0.0187, VQ: 0.0843, Perplexity: 58.45

Epoch 198/500, Batch 400/782, Loss: 0.1123, Recon: 0.0197, VQ: 0.0926, Perplexity: 57.81

Epoch 198/500, Batch 500/782, Loss: 0.1032, Recon: 0.0192, VQ: 0.0840, Perplexity: 57.45

Epoch 198/500, Batch 600/782, Loss: 0.1141, Recon: 0.0184, VQ: 0.0957, Perplexity: 56.64

Epoch 198/500, Batch 700/782, Loss: 0.1035, Recon: 0.0190, VQ: 0.0845, Perplexity: 58.34

=== Epoch 198/500 Summary ===

Train Loss: 0.1070 | Val Loss: 0.1058

Train Recon: 0.0189 | Val Recon: 0.0202

Train VQ: 0.0881 | Val VQ: 0.0857

Train Perplexity: 57.22 | Val Perplexity: 57.19

No improvement. Patience: 1/20

Best Loss: 0.1059 (Epoch 197)

Epoch 199/500, Batch 100/782, Loss: 0.0978, Recon: 0.0181, VQ: 0.0796, Perplexity: 56.61

Epoch 199/500, Batch 200/782, Loss: 0.1371, Recon: 0.0192, VQ: 0.1179, Perplexity: 59.06

Epoch 199/500, Batch 300/782, Loss: 0.1229, Recon: 0.0203, VQ: 0.1026, Perplexity: 57.85

Epoch 199/500, Batch 400/782, Loss: 0.1034, Recon: 0.0189, VQ: 0.0846, Perplexity: 58.71
Epoch 199/500, Batch 500/782, Loss: 0.1007, Recon: 0.0202, VQ: 0.0805, Perplexity: 56.47
Epoch 199/500, Batch 600/782, Loss: 0.1030, Recon: 0.0191, VQ: 0.0839, Perplexity: 57.99
Epoch 199/500, Batch 700/782, Loss: 0.1070, Recon: 0.0182, VQ: 0.0888, Perplexity: 56.68

=== Epoch 199/500 Summary ===

Train Loss: 0.1070 | Val Loss: 0.1077
Train Recon: 0.0189 | Val Recon: 0.0202
Train VQ: 0.0881 | Val VQ: 0.0875
Train Perplexity: 57.34 | Val Perplexity: 57.08
No improvement. Patience: 2/20
Best Loss: 0.1059 (Epoch 197)

Epoch 200/500, Batch 100/782, Loss: 0.1024, Recon: 0.0190, VQ: 0.0835, Perplexity: 56.35
Epoch 200/500, Batch 200/782, Loss: 0.1064, Recon: 0.0191, VQ: 0.0872, Perplexity: 59.05
Epoch 200/500, Batch 300/782, Loss: 0.1112, Recon: 0.0197, VQ: 0.0915, Perplexity: 57.44
Epoch 200/500, Batch 400/782, Loss: 0.1092, Recon: 0.0200, VQ: 0.0892, Perplexity: 57.98
Epoch 200/500, Batch 500/782, Loss: 0.1111, Recon: 0.0199, VQ: 0.0912, Perplexity: 58.29
Epoch 200/500, Batch 600/782, Loss: 0.1032, Recon: 0.0185, VQ: 0.0847, Perplexity: 56.23
Epoch 200/500, Batch 700/782, Loss: 0.1117, Recon: 0.0190, VQ: 0.0927, Perplexity: 57.23

=== Epoch 200/500 Summary ===

Train Loss: 0.1071 | Val Loss: 0.1065
Train Recon: 0.0189 | Val Recon: 0.0201
Train VQ: 0.0882 | Val VQ: 0.0863
Train Perplexity: 57.36 | Val Perplexity: 57.22
No improvement. Patience: 3/20
Best Loss: 0.1059 (Epoch 197)

Epoch 201/500, Batch 100/782, Loss: 0.0966, Recon: 0.0181, VQ: 0.0785, Perplexity: 56.72
Epoch 201/500, Batch 200/782, Loss: 0.1052, Recon: 0.0186, VQ: 0.0866, Perplexity: 57.54
Epoch 201/500, Batch 300/782, Loss: 0.0976, Recon: 0.0183, VQ: 0.0794, Perplexity: 57.64
Epoch 201/500, Batch 400/782, Loss: 0.1005, Recon: 0.0185, VQ: 0.0820, Perplexity: 54.94
Epoch 201/500, Batch 500/782, Loss: 0.1004, Recon: 0.0175, VQ: 0.0829, Perplexity: 58.25
Epoch 201/500, Batch 600/782, Loss: 0.1056, Recon: 0.0191, VQ: 0.0865, Perplexity: 57.72
Epoch 201/500, Batch 700/782, Loss: 0.0989, Recon: 0.0180, VQ: 0.0810, Perplexity: 56.49

=== Epoch 201/500 Summary ===

Train Loss: 0.1067 | Val Loss: 0.1072
Train Recon: 0.0189 | Val Recon: 0.0201
Train VQ: 0.0878 | Val VQ: 0.0871
Train Perplexity: 57.43 | Val Perplexity: 57.06

No improvement. Patience: 4/20
Best Loss: 0.1059 (Epoch 197)

Epoch 202/500, Batch 100/782, Loss: 0.1022, Recon: 0.0182, VQ: 0.0841, Perplexity: 56.95
Epoch 202/500, Batch 200/782, Loss: 0.1034, Recon: 0.0176, VQ: 0.0857, Perplexity: 59.54
Epoch 202/500, Batch 300/782, Loss: 0.0968, Recon: 0.0182, VQ: 0.0787, Perplexity: 57.27
Epoch 202/500, Batch 400/782, Loss: 0.1010, Recon: 0.0187, VQ: 0.0823, Perplexity: 56.99
Epoch 202/500, Batch 500/782, Loss: 0.1114, Recon: 0.0190, VQ: 0.0924, Perplexity: 57.43
Epoch 202/500, Batch 600/782, Loss: 0.1131, Recon: 0.0196, VQ: 0.0935, Perplexity: 58.57
Epoch 202/500, Batch 700/782, Loss: 0.0970, Recon: 0.0178, VQ: 0.0791, Perplexity: 56.21

=== Epoch 202/500 Summary ===
Train Loss: 0.1067 | Val Loss: 0.1075
Train Recon: 0.0189 | Val Recon: 0.0202
Train VQ: 0.0879 | Val VQ: 0.0873
Train Perplexity: 57.55 | Val Perplexity: 57.26
No improvement. Patience: 5/20
Best Loss: 0.1059 (Epoch 197)

Epoch 203/500, Batch 100/782, Loss: 0.1020, Recon: 0.0188, VQ: 0.0832, Perplexity: 56.98
Epoch 203/500, Batch 200/782, Loss: 0.1150, Recon: 0.0208, VQ: 0.0942, Perplexity: 58.94
Epoch 203/500, Batch 300/782, Loss: 0.1179, Recon: 0.0200, VQ: 0.0979, Perplexity: 58.58
Epoch 203/500, Batch 400/782, Loss: 0.1068, Recon: 0.0202, VQ: 0.0866, Perplexity: 59.25
Epoch 203/500, Batch 500/782, Loss: 0.1177, Recon: 0.0195, VQ: 0.0982, Perplexity: 59.46
Epoch 203/500, Batch 600/782, Loss: 0.1005, Recon: 0.0182, VQ: 0.0823, Perplexity: 56.01
Epoch 203/500, Batch 700/782, Loss: 0.1336, Recon: 0.0205, VQ: 0.1131, Perplexity: 58.36

=== Epoch 203/500 Summary ===
Train Loss: 0.1069 | Val Loss: 0.1065
Train Recon: 0.0189 | Val Recon: 0.0201
Train VQ: 0.0880 | Val VQ: 0.0863
Train Perplexity: 57.60 | Val Perplexity: 57.58
No improvement. Patience: 6/20
Best Loss: 0.1059 (Epoch 197)

Epoch 204/500, Batch 100/782, Loss: 0.1032, Recon: 0.0181, VQ: 0.0852, Perplexity: 55.76
Epoch 204/500, Batch 200/782, Loss: 0.0999, Recon: 0.0178, VQ: 0.0821, Perplexity: 57.20
Epoch 204/500, Batch 300/782, Loss: 0.0988, Recon: 0.0178, VQ: 0.0810, Perplexity: 54.73
Epoch 204/500, Batch 400/782, Loss: 0.1031, Recon: 0.0191, VQ: 0.0839, Perplexity: 57.82
Epoch 204/500, Batch 500/782, Loss: 0.1143, Recon: 0.0189, VQ: 0.0954, Perplexity: 59.08
Epoch 204/500, Batch 600/782, Loss: 0.1023, Recon: 0.0190, VQ: 0.0834, Per

plexity: 58.87

Epoch 204/500, Batch 700/782, Loss: 0.1034, Recon: 0.0189, VQ: 0.0845, Perplexity: 58.04

=== Epoch 204/500 Summary ===

Train Loss: 0.1065 | Val Loss: 0.1066

Train Recon: 0.0188 | Val Recon: 0.0201

Train VQ: 0.0877 | Val VQ: 0.0865

Train Perplexity: 57.69 | Val Perplexity: 57.35

No improvement. Patience: 7/20

Best Loss: 0.1059 (Epoch 197)

Epoch 205/500, Batch 100/782, Loss: 0.1091, Recon: 0.0190, VQ: 0.0901, Perplexity: 58.28

Epoch 205/500, Batch 200/782, Loss: 0.1058, Recon: 0.0188, VQ: 0.0870, Perplexity: 58.28

Epoch 205/500, Batch 300/782, Loss: 0.0949, Recon: 0.0178, VQ: 0.0771, Perplexity: 55.14

Epoch 205/500, Batch 400/782, Loss: 0.1066, Recon: 0.0194, VQ: 0.0871, Perplexity: 57.84

Epoch 205/500, Batch 500/782, Loss: 0.1051, Recon: 0.0188, VQ: 0.0864, Perplexity: 57.77

Epoch 205/500, Batch 600/782, Loss: 0.1085, Recon: 0.0183, VQ: 0.0903, Perplexity: 56.89

Epoch 205/500, Batch 700/782, Loss: 0.1098, Recon: 0.0191, VQ: 0.0907, Perplexity: 59.18

=== Epoch 205/500 Summary ===

Train Loss: 0.1067 | Val Loss: 0.1085

Train Recon: 0.0188 | Val Recon: 0.0200

Train VQ: 0.0879 | Val VQ: 0.0884

Train Perplexity: 57.69 | Val Perplexity: 57.81

No improvement. Patience: 8/20

Best Loss: 0.1059 (Epoch 197)

Epoch 206/500, Batch 100/782, Loss: 0.1163, Recon: 0.0197, VQ: 0.0967, Perplexity: 58.36

Epoch 206/500, Batch 200/782, Loss: 0.0940, Recon: 0.0170, VQ: 0.0770, Perplexity: 54.51

Epoch 206/500, Batch 300/782, Loss: 0.1304, Recon: 0.0198, VQ: 0.1106, Perplexity: 57.25

Epoch 206/500, Batch 400/782, Loss: 0.1020, Recon: 0.0195, VQ: 0.0825, Perplexity: 58.74

Epoch 206/500, Batch 500/782, Loss: 0.1175, Recon: 0.0209, VQ: 0.0967, Perplexity: 61.42

Epoch 206/500, Batch 600/782, Loss: 0.1015, Recon: 0.0185, VQ: 0.0830, Perplexity: 56.35

Epoch 206/500, Batch 700/782, Loss: 0.1017, Recon: 0.0188, VQ: 0.0829, Perplexity: 57.57

=== Epoch 206/500 Summary ===

Train Loss: 0.1068 | Val Loss: 0.1072

Train Recon: 0.0188 | Val Recon: 0.0200

Train VQ: 0.0880 | Val VQ: 0.0872

Train Perplexity: 57.78 | Val Perplexity: 57.94

No improvement. Patience: 9/20

Best Loss: 0.1059 (Epoch 197)

Epoch 207/500, Batch 100/782, Loss: 0.0951, Recon: 0.0169, VQ: 0.0783, Perplexity: 55.88

Epoch 207/500, Batch 200/782, Loss: 0.1094, Recon: 0.0193, VQ: 0.0901, Perplexity: 59.45
Epoch 207/500, Batch 300/782, Loss: 0.1039, Recon: 0.0191, VQ: 0.0848, Perplexity: 56.79
Epoch 207/500, Batch 400/782, Loss: 0.1212, Recon: 0.0214, VQ: 0.0998, Perplexity: 60.75
Epoch 207/500, Batch 500/782, Loss: 0.0909, Recon: 0.0167, VQ: 0.0741, Perplexity: 54.97
Epoch 207/500, Batch 600/782, Loss: 0.1067, Recon: 0.0183, VQ: 0.0884, Perplexity: 58.16
Epoch 207/500, Batch 700/782, Loss: 0.1133, Recon: 0.0202, VQ: 0.0930, Perplexity: 59.30

=== Epoch 207/500 Summary ===

Train Loss: 0.1068 | Val Loss: 0.1072
Train Recon: 0.0188 | Val Recon: 0.0201
Train VQ: 0.0880 | Val VQ: 0.0871
Train Perplexity: 57.79 | Val Perplexity: 57.65
No improvement. Patience: 10/20
Best Loss: 0.1059 (Epoch 197)

Epoch 208/500, Batch 100/782, Loss: 0.2203, Recon: 0.0203, VQ: 0.2000, Perplexity: 59.00
Epoch 208/500, Batch 200/782, Loss: 0.1045, Recon: 0.0183, VQ: 0.0863, Perplexity: 57.64
Epoch 208/500, Batch 300/782, Loss: 0.1055, Recon: 0.0184, VQ: 0.0871, Perplexity: 56.69
Epoch 208/500, Batch 400/782, Loss: 0.1112, Recon: 0.0204, VQ: 0.0909, Perplexity: 59.70
Epoch 208/500, Batch 500/782, Loss: 0.1055, Recon: 0.0204, VQ: 0.0850, Perplexity: 59.44
Epoch 208/500, Batch 600/782, Loss: 0.1012, Recon: 0.0176, VQ: 0.0836, Perplexity: 54.18
Epoch 208/500, Batch 700/782, Loss: 0.1014, Recon: 0.0183, VQ: 0.0831, Perplexity: 55.78

=== Epoch 208/500 Summary ===

Train Loss: 0.1070 | Val Loss: 0.1077
Train Recon: 0.0188 | Val Recon: 0.0200
Train VQ: 0.0882 | Val VQ: 0.0876
Train Perplexity: 57.81 | Val Perplexity: 57.88
No improvement. Patience: 11/20
Best Loss: 0.1059 (Epoch 197)

Epoch 209/500, Batch 100/782, Loss: 0.1017, Recon: 0.0182, VQ: 0.0834, Perplexity: 56.33
Epoch 209/500, Batch 200/782, Loss: 0.0992, Recon: 0.0183, VQ: 0.0809, Perplexity: 56.66
Epoch 209/500, Batch 300/782, Loss: 0.1031, Recon: 0.0194, VQ: 0.0837, Perplexity: 57.85
Epoch 209/500, Batch 400/782, Loss: 0.0964, Recon: 0.0180, VQ: 0.0784, Perplexity: 56.45
Epoch 209/500, Batch 500/782, Loss: 0.1036, Recon: 0.0197, VQ: 0.0840, Perplexity: 56.13
Epoch 209/500, Batch 600/782, Loss: 0.1102, Recon: 0.0199, VQ: 0.0903, Perplexity: 57.99
Epoch 209/500, Batch 700/782, Loss: 0.0968, Recon: 0.0182, VQ: 0.0787, Perplexity: 58.00

=== Epoch 209/500 Summary ===

Train Loss: 0.1070 | Val Loss: 0.1080
Train Recon: 0.0188 | Val Recon: 0.0201
Train VQ: 0.0882 | Val VQ: 0.0880
Train Perplexity: 57.85 | Val Perplexity: 57.61
No improvement. Patience: 12/20
Best Loss: 0.1059 (Epoch 197)

Epoch 210/500, Batch 100/782, Loss: 0.1079, Recon: 0.0197, VQ: 0.0882, Perplexity: 60.73
Epoch 210/500, Batch 200/782, Loss: 0.1034, Recon: 0.0175, VQ: 0.0859, Perplexity: 55.66
Epoch 210/500, Batch 300/782, Loss: 0.1149, Recon: 0.0188, VQ: 0.0961, Perplexity: 58.70
Epoch 210/500, Batch 400/782, Loss: 0.0936, Recon: 0.0176, VQ: 0.0760, Perplexity: 55.61
Epoch 210/500, Batch 500/782, Loss: 0.1014, Recon: 0.0180, VQ: 0.0833, Perplexity: 55.48
Epoch 210/500, Batch 600/782, Loss: 0.1019, Recon: 0.0178, VQ: 0.0841, Perplexity: 56.89
Epoch 210/500, Batch 700/782, Loss: 0.1031, Recon: 0.0180, VQ: 0.0851, Perplexity: 58.09

=== Epoch 210/500 Summary ===
Train Loss: 0.1067 | Val Loss: 0.1082
Train Recon: 0.0188 | Val Recon: 0.0200
Train VQ: 0.0879 | Val VQ: 0.0881
Train Perplexity: 57.89 | Val Perplexity: 57.78
No improvement. Patience: 13/20
Best Loss: 0.1059 (Epoch 197)

Epoch 211/500, Batch 100/782, Loss: 0.0938, Recon: 0.0170, VQ: 0.0768, Perplexity: 57.36
Epoch 211/500, Batch 200/782, Loss: 0.1030, Recon: 0.0186, VQ: 0.0844, Perplexity: 57.34
Epoch 211/500, Batch 300/782, Loss: 0.1363, Recon: 0.0175, VQ: 0.1187, Perplexity: 55.91
Epoch 211/500, Batch 400/782, Loss: 0.1066, Recon: 0.0195, VQ: 0.0870, Perplexity: 58.40
Epoch 211/500, Batch 500/782, Loss: 0.1065, Recon: 0.0200, VQ: 0.0865, Perplexity: 58.38
Epoch 211/500, Batch 600/782, Loss: 0.1014, Recon: 0.0197, VQ: 0.0816, Perplexity: 56.73
Epoch 211/500, Batch 700/782, Loss: 0.1045, Recon: 0.0191, VQ: 0.0854, Perplexity: 56.80

=== Epoch 211/500 Summary ===
Train Loss: 0.1070 | Val Loss: 0.1064
Train Recon: 0.0188 | Val Recon: 0.0201
Train VQ: 0.0883 | Val VQ: 0.0863
Train Perplexity: 57.91 | Val Perplexity: 57.67
No improvement. Patience: 14/20
Best Loss: 0.1059 (Epoch 197)

Epoch 212/500, Batch 100/782, Loss: 0.1090, Recon: 0.0177, VQ: 0.0913, Perplexity: 56.44
Epoch 212/500, Batch 200/782, Loss: 0.0982, Recon: 0.0185, VQ: 0.0797, Perplexity: 58.62
Epoch 212/500, Batch 300/782, Loss: 0.2193, Recon: 0.0200, VQ: 0.1993, Perplexity: 58.89
Epoch 212/500, Batch 400/782, Loss: 0.1095, Recon: 0.0183, VQ: 0.0912, Per

plexity: 55.60
Epoch 212/500, Batch 500/782, Loss: 0.0857, Recon: 0.0167, VQ: 0.0690, Per
plexity: 54.42
Epoch 212/500, Batch 600/782, Loss: 0.0983, Recon: 0.0187, VQ: 0.0795, Per
plexity: 56.60
Epoch 212/500, Batch 700/782, Loss: 0.1000, Recon: 0.0187, VQ: 0.0813, Per
plexity: 58.50

=== Epoch 212/500 Summary ===

Train Loss: 0.1068 | Val Loss: 0.1070
Train Recon: 0.0188 | Val Recon: 0.0200
Train VQ: 0.0881 | Val VQ: 0.0870
Train Perplexity: 57.96 | Val Perplexity: 57.46
No improvement. Patience: 15/20
Best Loss: 0.1059 (Epoch 197)

Epoch 213/500, Batch 100/782, Loss: 0.0956, Recon: 0.0194, VQ: 0.0762, Per
plexity: 57.91
Epoch 213/500, Batch 200/782, Loss: 0.1047, Recon: 0.0187, VQ: 0.0860, Per
plexity: 58.41
Epoch 213/500, Batch 300/782, Loss: 0.1067, Recon: 0.0204, VQ: 0.0864, Per
plexity: 59.10
Epoch 213/500, Batch 400/782, Loss: 0.1083, Recon: 0.0202, VQ: 0.0881, Per
plexity: 59.79
Epoch 213/500, Batch 500/782, Loss: 0.1023, Recon: 0.0183, VQ: 0.0840, Per
plexity: 58.59
Epoch 213/500, Batch 600/782, Loss: 0.1059, Recon: 0.0189, VQ: 0.0870, Per
plexity: 60.04
Epoch 213/500, Batch 700/782, Loss: 0.1070, Recon: 0.0185, VQ: 0.0885, Per
plexity: 57.42

=== Epoch 213/500 Summary ===

Train Loss: 0.1066 | Val Loss: 0.1075
Train Recon: 0.0187 | Val Recon: 0.0201
Train VQ: 0.0878 | Val VQ: 0.0875
Train Perplexity: 57.96 | Val Perplexity: 58.17
No improvement. Patience: 16/20
Best Loss: 0.1059 (Epoch 197)

Epoch 214/500, Batch 100/782, Loss: 0.1024, Recon: 0.0196, VQ: 0.0827, Per
plexity: 58.96
Epoch 214/500, Batch 200/782, Loss: 0.0983, Recon: 0.0175, VQ: 0.0808, Per
plexity: 56.74
Epoch 214/500, Batch 300/782, Loss: 0.1044, Recon: 0.0193, VQ: 0.0851, Per
plexity: 59.25
Epoch 214/500, Batch 400/782, Loss: 0.1106, Recon: 0.0187, VQ: 0.0920, Per
plexity: 56.49
Epoch 214/500, Batch 500/782, Loss: 0.1194, Recon: 0.0207, VQ: 0.0987, Per
plexity: 61.14
Epoch 214/500, Batch 600/782, Loss: 0.1047, Recon: 0.0177, VQ: 0.0870, Per
plexity: 57.67
Epoch 214/500, Batch 700/782, Loss: 0.0919, Recon: 0.0174, VQ: 0.0745, Per
plexity: 54.94

=== Epoch 214/500 Summary ===

Train Loss: 0.1069 | Val Loss: 0.1074
Train Recon: 0.0187 | Val Recon: 0.0200
Train VQ: 0.0881 | Val VQ: 0.0874
Train Perplexity: 58.02 | Val Perplexity: 57.97
No improvement. Patience: 17/20

Best Loss: 0.1059 (Epoch 197)

Epoch 215/500, Batch 100/782, Loss: 0.1161, Recon: 0.0199, VQ: 0.0962, Perplexity: 58.28
Epoch 215/500, Batch 200/782, Loss: 0.1110, Recon: 0.0196, VQ: 0.0913, Perplexity: 56.99
Epoch 215/500, Batch 300/782, Loss: 0.1106, Recon: 0.0193, VQ: 0.0913, Perplexity: 57.62
Epoch 215/500, Batch 400/782, Loss: 0.0924, Recon: 0.0174, VQ: 0.0750, Perplexity: 56.93
Epoch 215/500, Batch 500/782, Loss: 0.1075, Recon: 0.0190, VQ: 0.0885, Perplexity: 58.43
Epoch 215/500, Batch 600/782, Loss: 0.1087, Recon: 0.0194, VQ: 0.0893, Perplexity: 59.50
Epoch 215/500, Batch 700/782, Loss: 0.1067, Recon: 0.0192, VQ: 0.0876, Perplexity: 58.72

=== Epoch 215/500 Summary ===

Train Loss: 0.1068 | Val Loss: 0.1066
Train Recon: 0.0187 | Val Recon: 0.0200
Train VQ: 0.0881 | Val VQ: 0.0865
Train Perplexity: 58.07 | Val Perplexity: 57.82
No improvement. Patience: 18/20
Best Loss: 0.1059 (Epoch 197)

Epoch 216/500, Batch 100/782, Loss: 0.0984, Recon: 0.0169, VQ: 0.0815, Perplexity: 57.13
Epoch 216/500, Batch 200/782, Loss: 0.0948, Recon: 0.0182, VQ: 0.0766, Perplexity: 54.91
Epoch 216/500, Batch 300/782, Loss: 0.0934, Recon: 0.0168, VQ: 0.0766, Perplexity: 54.81
Epoch 216/500, Batch 400/782, Loss: 0.1001, Recon: 0.0182, VQ: 0.0820, Perplexity: 58.75
Epoch 216/500, Batch 500/782, Loss: 0.1076, Recon: 0.0190, VQ: 0.0885, Perplexity: 58.98
Epoch 216/500, Batch 600/782, Loss: 0.1089, Recon: 0.0204, VQ: 0.0885, Perplexity: 59.08
Epoch 216/500, Batch 700/782, Loss: 0.1172, Recon: 0.0187, VQ: 0.0985, Perplexity: 58.74

=== Epoch 216/500 Summary ===

Train Loss: 0.1066 | Val Loss: 0.1071
Train Recon: 0.0187 | Val Recon: 0.0200
Train VQ: 0.0879 | Val VQ: 0.0871
Train Perplexity: 58.16 | Val Perplexity: 57.92
No improvement. Patience: 19/20
Best Loss: 0.1059 (Epoch 197)

Epoch 217/500, Batch 100/782, Loss: 0.0943, Recon: 0.0178, VQ: 0.0765, Perplexity: 56.70
Epoch 217/500, Batch 200/782, Loss: 0.0960, Recon: 0.0176, VQ: 0.0785, Perplexity: 56.91
Epoch 217/500, Batch 300/782, Loss: 0.1123, Recon: 0.0184, VQ: 0.0939, Perplexity: 59.34
Epoch 217/500, Batch 400/782, Loss: 0.1039, Recon: 0.0190, VQ: 0.0849, Perplexity: 58.90
Epoch 217/500, Batch 500/782, Loss: 0.1106, Recon: 0.0202, VQ: 0.0903, Perplexity: 59.97
Epoch 217/500, Batch 600/782, Loss: 0.1018, Recon: 0.0191, VQ: 0.0827, Perplexity: 60.86

Epoch 217/500, Batch 700/782, Loss: 0.1071, Recon: 0.0193, VQ: 0.0879, Perplexity: 59.38

=== Epoch 217/500 Summary ===

Train Loss: 0.1064 | Val Loss: 0.1050

Train Recon: 0.0187 | Val Recon: 0.0200

Train VQ: 0.0877 | Val VQ: 0.0850

Train Perplexity: 58.27 | Val Perplexity: 58.26

✓ Best model saved! (Val Loss: 0.1050)

Best Loss: 0.1050 (Epoch 217)

Epoch 218/500, Batch 100/782, Loss: 0.0988, Recon: 0.0179, VQ: 0.0809, Perplexity: 56.83

Epoch 218/500, Batch 200/782, Loss: 0.0929, Recon: 0.0168, VQ: 0.0761, Perplexity: 57.13

Epoch 218/500, Batch 300/782, Loss: 0.1101, Recon: 0.0203, VQ: 0.0898, Perplexity: 59.94

Epoch 218/500, Batch 400/782, Loss: 0.1048, Recon: 0.0188, VQ: 0.0861, Perplexity: 56.27

Epoch 218/500, Batch 500/782, Loss: 0.1140, Recon: 0.0193, VQ: 0.0947, Perplexity: 60.63

Epoch 218/500, Batch 600/782, Loss: 0.1205, Recon: 0.0198, VQ: 0.1007, Perplexity: 59.42

Epoch 218/500, Batch 700/782, Loss: 0.0992, Recon: 0.0186, VQ: 0.0806, Perplexity: 57.55

=== Epoch 218/500 Summary ===

Train Loss: 0.1060 | Val Loss: 0.1054

Train Recon: 0.0187 | Val Recon: 0.0200

Train VQ: 0.0873 | Val VQ: 0.0854

Train Perplexity: 58.42 | Val Perplexity: 58.32

No improvement. Patience: 1/20

Best Loss: 0.1050 (Epoch 217)

Epoch 219/500, Batch 100/782, Loss: 0.1010, Recon: 0.0184, VQ: 0.0826, Perplexity: 59.36

Epoch 219/500, Batch 200/782, Loss: 0.1079, Recon: 0.0185, VQ: 0.0894, Perplexity: 58.63

Epoch 219/500, Batch 300/782, Loss: 0.1009, Recon: 0.0186, VQ: 0.0822, Perplexity: 57.53

Epoch 219/500, Batch 400/782, Loss: 0.1137, Recon: 0.0199, VQ: 0.0938, Perplexity: 58.95

Epoch 219/500, Batch 500/782, Loss: 0.1033, Recon: 0.0186, VQ: 0.0847, Perplexity: 57.92

Epoch 219/500, Batch 600/782, Loss: 0.1118, Recon: 0.0199, VQ: 0.0919, Perplexity: 59.97

Epoch 219/500, Batch 700/782, Loss: 0.2598, Recon: 0.0193, VQ: 0.2406, Perplexity: 59.29

=== Epoch 219/500 Summary ===

Train Loss: 0.1056 | Val Loss: 0.1061

Train Recon: 0.0187 | Val Recon: 0.0200

Train VQ: 0.0869 | Val VQ: 0.0861

Train Perplexity: 58.57 | Val Perplexity: 58.43

No improvement. Patience: 2/20

Best Loss: 0.1050 (Epoch 217)

Epoch 220/500, Batch 100/782, Loss: 0.0877, Recon: 0.0166, VQ: 0.0711, Perplexity: 54.47

Epoch 220/500, Batch 200/782, Loss: 0.0934, Recon: 0.0171, VQ: 0.0763, Per

plexity: 56.77
Epoch 220/500, Batch 300/782, Loss: 0.1047, Recon: 0.0189, VQ: 0.0858, Per
plexity: 60.07
Epoch 220/500, Batch 400/782, Loss: 0.1016, Recon: 0.0182, VQ: 0.0835, Per
plexity: 58.93
Epoch 220/500, Batch 500/782, Loss: 0.1033, Recon: 0.0185, VQ: 0.0848, Per
plexity: 58.07
Epoch 220/500, Batch 600/782, Loss: 0.0913, Recon: 0.0175, VQ: 0.0738, Per
plexity: 56.91
Epoch 220/500, Batch 700/782, Loss: 0.1211, Recon: 0.0221, VQ: 0.0990, Per
plexity: 62.16

=== Epoch 220/500 Summary ===

Train Loss: 0.1049 | Val Loss: 0.1058
Train Recon: 0.0187 | Val Recon: 0.0200
Train VQ: 0.0862 | Val VQ: 0.0858
Train Perplexity: 58.65 | Val Perplexity: 58.72
No improvement. Patience: 3/20
Best Loss: 0.1050 (Epoch 217)

Epoch 221/500, Batch 100/782, Loss: 0.0897, Recon: 0.0174, VQ: 0.0723, Per
plexity: 57.66
Epoch 221/500, Batch 200/782, Loss: 0.1153, Recon: 0.0202, VQ: 0.0952, Per
plexity: 60.47
Epoch 221/500, Batch 300/782, Loss: 0.1067, Recon: 0.0182, VQ: 0.0885, Per
plexity: 58.89
Epoch 221/500, Batch 400/782, Loss: 0.0949, Recon: 0.0172, VQ: 0.0777, Per
plexity: 56.18
Epoch 221/500, Batch 500/782, Loss: 0.1027, Recon: 0.0178, VQ: 0.0849, Per
plexity: 59.60
Epoch 221/500, Batch 600/782, Loss: 0.1468, Recon: 0.0191, VQ: 0.1277, Per
plexity: 59.07
Epoch 221/500, Batch 700/782, Loss: 0.0996, Recon: 0.0176, VQ: 0.0819, Per
plexity: 57.13

=== Epoch 221/500 Summary ===

Train Loss: 0.1049 | Val Loss: 0.1059
Train Recon: 0.0187 | Val Recon: 0.0200
Train VQ: 0.0862 | Val VQ: 0.0859
Train Perplexity: 58.69 | Val Perplexity: 58.82
No improvement. Patience: 4/20
Best Loss: 0.1050 (Epoch 217)

Epoch 222/500, Batch 100/782, Loss: 0.1105, Recon: 0.0197, VQ: 0.0908, Per
plexity: 61.44
Epoch 222/500, Batch 200/782, Loss: 0.0969, Recon: 0.0182, VQ: 0.0787, Per
plexity: 58.70
Epoch 222/500, Batch 300/782, Loss: 0.1172, Recon: 0.0192, VQ: 0.0980, Per
plexity: 60.29
Epoch 222/500, Batch 400/782, Loss: 0.1082, Recon: 0.0176, VQ: 0.0906, Per
plexity: 57.94
Epoch 222/500, Batch 500/782, Loss: 0.1001, Recon: 0.0188, VQ: 0.0814, Per
plexity: 58.47
Epoch 222/500, Batch 600/782, Loss: 0.1042, Recon: 0.0189, VQ: 0.0853, Per
plexity: 59.91
Epoch 222/500, Batch 700/782, Loss: 0.0917, Recon: 0.0175, VQ: 0.0742, Per
plexity: 56.84

=== Epoch 222/500 Summary ===

Train Loss: 0.1047 | Val Loss: 0.1055

Train Recon: 0.0186 | Val Recon: 0.0200
Train VQ: 0.0861 | Val VQ: 0.0855
Train Perplexity: 58.73 | Val Perplexity: 58.78
No improvement. Patience: 5/20
Best Loss: 0.1050 (Epoch 217)

Epoch 223/500, Batch 100/782, Loss: 0.1287, Recon: 0.0204, VQ: 0.1083, Perplexity: 59.42
Epoch 223/500, Batch 200/782, Loss: 0.1036, Recon: 0.0182, VQ: 0.0855, Perplexity: 58.50
Epoch 223/500, Batch 300/782, Loss: 0.1122, Recon: 0.0189, VQ: 0.0933, Perplexity: 58.46
Epoch 223/500, Batch 400/782, Loss: 0.1011, Recon: 0.0189, VQ: 0.0822, Perplexity: 58.37
Epoch 223/500, Batch 500/782, Loss: 0.1060, Recon: 0.0187, VQ: 0.0873, Perplexity: 59.67
Epoch 223/500, Batch 600/782, Loss: 0.1237, Recon: 0.0195, VQ: 0.1042, Perplexity: 59.33
Epoch 223/500, Batch 700/782, Loss: 0.1092, Recon: 0.0197, VQ: 0.0895, Perplexity: 59.18

=== Epoch 223/500 Summary ===
Train Loss: 0.1047 | Val Loss: 0.1043
Train Recon: 0.0186 | Val Recon: 0.0200
Train VQ: 0.0860 | Val VQ: 0.0843
Train Perplexity: 58.80 | Val Perplexity: 58.56
✓ Best model saved! (Val Loss: 0.1043)
Best Loss: 0.1043 (Epoch 223)

Epoch 224/500, Batch 100/782, Loss: 0.0993, Recon: 0.0180, VQ: 0.0813, Perplexity: 57.40
Epoch 224/500, Batch 200/782, Loss: 0.1038, Recon: 0.0195, VQ: 0.0843, Perplexity: 59.04
Epoch 224/500, Batch 300/782, Loss: 0.1269, Recon: 0.0202, VQ: 0.1067, Perplexity: 61.07
Epoch 224/500, Batch 400/782, Loss: 0.0984, Recon: 0.0174, VQ: 0.0810, Perplexity: 58.98
Epoch 224/500, Batch 500/782, Loss: 0.1027, Recon: 0.0184, VQ: 0.0843, Perplexity: 59.40
Epoch 224/500, Batch 600/782, Loss: 0.0920, Recon: 0.0174, VQ: 0.0746, Perplexity: 57.62
Epoch 224/500, Batch 700/782, Loss: 0.1059, Recon: 0.0192, VQ: 0.0868, Perplexity: 57.23

=== Epoch 224/500 Summary ===
Train Loss: 0.1044 | Val Loss: 0.1039
Train Recon: 0.0186 | Val Recon: 0.0200
Train VQ: 0.0858 | Val VQ: 0.0839
Train Perplexity: 58.88 | Val Perplexity: 58.54
✓ Best model saved! (Val Loss: 0.1039)
Best Loss: 0.1039 (Epoch 224)

Epoch 225/500, Batch 100/782, Loss: 0.1091, Recon: 0.0188, VQ: 0.0904, Perplexity: 59.55
Epoch 225/500, Batch 200/782, Loss: 0.0966, Recon: 0.0174, VQ: 0.0791, Perplexity: 58.03
Epoch 225/500, Batch 300/782, Loss: 0.0946, Recon: 0.0171, VQ: 0.0775, Perplexity: 57.80
Epoch 225/500, Batch 400/782, Loss: 0.1652, Recon: 0.0208, VQ: 0.1444, Perplexity: 58.14

Epoch 225/500, Batch 500/782, Loss: 0.0914, Recon: 0.0173, VQ: 0.0740, Perplexity: 56.50
Epoch 225/500, Batch 600/782, Loss: 0.1078, Recon: 0.0185, VQ: 0.0893, Perplexity: 59.97
Epoch 225/500, Batch 700/782, Loss: 0.1059, Recon: 0.0187, VQ: 0.0872, Perplexity: 59.46

=== Epoch 225/500 Summary ===

Train Loss: 0.1043 | Val Loss: 0.1057
Train Recon: 0.0186 | Val Recon: 0.0200
Train VQ: 0.0857 | Val VQ: 0.0857
Train Perplexity: 59.01 | Val Perplexity: 58.95
No improvement. Patience: 1/20
Best Loss: 0.1039 (Epoch 224)

Epoch 226/500, Batch 100/782, Loss: 0.1003, Recon: 0.0184, VQ: 0.0819, Perplexity: 58.97
Epoch 226/500, Batch 200/782, Loss: 0.0858, Recon: 0.0166, VQ: 0.0692, Perplexity: 56.38
Epoch 226/500, Batch 300/782, Loss: 0.0949, Recon: 0.0176, VQ: 0.0773, Perplexity: 57.25
Epoch 226/500, Batch 400/782, Loss: 0.0937, Recon: 0.0175, VQ: 0.0762, Perplexity: 59.38
Epoch 226/500, Batch 500/782, Loss: 0.0968, Recon: 0.0187, VQ: 0.0781, Perplexity: 58.16
Epoch 226/500, Batch 600/782, Loss: 0.1105, Recon: 0.0185, VQ: 0.0920, Perplexity: 58.58
Epoch 226/500, Batch 700/782, Loss: 0.0937, Recon: 0.0178, VQ: 0.0759, Perplexity: 57.82

=== Epoch 226/500 Summary ===

Train Loss: 0.1043 | Val Loss: 0.1039
Train Recon: 0.0186 | Val Recon: 0.0200
Train VQ: 0.0857 | Val VQ: 0.0840
Train Perplexity: 59.33 | Val Perplexity: 59.26
No improvement. Patience: 2/20
Best Loss: 0.1039 (Epoch 224)

Epoch 227/500, Batch 100/782, Loss: 0.1008, Recon: 0.0184, VQ: 0.0824, Perplexity: 58.70
Epoch 227/500, Batch 200/782, Loss: 0.1029, Recon: 0.0188, VQ: 0.0840, Perplexity: 59.15
Epoch 227/500, Batch 300/782, Loss: 0.1077, Recon: 0.0187, VQ: 0.0890, Perplexity: 61.00
Epoch 227/500, Batch 400/782, Loss: 0.1082, Recon: 0.0196, VQ: 0.0886, Perplexity: 61.44
Epoch 227/500, Batch 500/782, Loss: 0.1073, Recon: 0.0211, VQ: 0.0863, Perplexity: 60.19
Epoch 227/500, Batch 600/782, Loss: 0.1019, Recon: 0.0178, VQ: 0.0842, Perplexity: 58.71
Epoch 227/500, Batch 700/782, Loss: 0.0938, Recon: 0.0182, VQ: 0.0755, Perplexity: 58.96

=== Epoch 227/500 Summary ===

Train Loss: 0.1038 | Val Loss: 0.1049
Train Recon: 0.0186 | Val Recon: 0.0200
Train VQ: 0.0853 | Val VQ: 0.0849
Train Perplexity: 59.57 | Val Perplexity: 59.76
No improvement. Patience: 3/20
Best Loss: 0.1039 (Epoch 224)

Epoch 228/500, Batch 100/782, Loss: 0.0989, Recon: 0.0180, VQ: 0.0809, Perplexity: 57.86
Epoch 228/500, Batch 200/782, Loss: 0.1143, Recon: 0.0195, VQ: 0.0948, Perplexity: 62.67
Epoch 228/500, Batch 300/782, Loss: 0.0868, Recon: 0.0162, VQ: 0.0706, Perplexity: 57.78
Epoch 228/500, Batch 400/782, Loss: 0.0913, Recon: 0.0174, VQ: 0.0739, Perplexity: 58.04
Epoch 228/500, Batch 500/782, Loss: 0.0958, Recon: 0.0182, VQ: 0.0776, Perplexity: 59.43
Epoch 228/500, Batch 600/782, Loss: 0.0963, Recon: 0.0182, VQ: 0.0781, Perplexity: 58.92
Epoch 228/500, Batch 700/782, Loss: 0.0951, Recon: 0.0179, VQ: 0.0772, Perplexity: 59.62

=== Epoch 228/500 Summary ===

Train Loss: 0.1036 | Val Loss: 0.1041
Train Recon: 0.0186 | Val Recon: 0.0199
Train VQ: 0.0851 | Val VQ: 0.0842
Train Perplexity: 59.75 | Val Perplexity: 59.61
No improvement. Patience: 4/20
Best Loss: 0.1039 (Epoch 224)

Epoch 229/500, Batch 100/782, Loss: 0.1168, Recon: 0.0191, VQ: 0.0977, Perplexity: 60.54
Epoch 229/500, Batch 200/782, Loss: 0.0982, Recon: 0.0193, VQ: 0.0789, Perplexity: 58.63
Epoch 229/500, Batch 300/782, Loss: 0.1018, Recon: 0.0188, VQ: 0.0830, Perplexity: 61.77
Epoch 229/500, Batch 400/782, Loss: 0.1110, Recon: 0.0187, VQ: 0.0923, Perplexity: 60.02
Epoch 229/500, Batch 500/782, Loss: 0.0963, Recon: 0.0175, VQ: 0.0788, Perplexity: 58.80
Epoch 229/500, Batch 600/782, Loss: 0.1009, Recon: 0.0172, VQ: 0.0838, Perplexity: 58.49
Epoch 229/500, Batch 700/782, Loss: 0.0969, Recon: 0.0173, VQ: 0.0797, Perplexity: 58.27

=== Epoch 229/500 Summary ===

Train Loss: 0.1036 | Val Loss: 0.1037
Train Recon: 0.0186 | Val Recon: 0.0199
Train VQ: 0.0850 | Val VQ: 0.0837
Train Perplexity: 59.79 | Val Perplexity: 59.72
✓ Best model saved! (Val Loss: 0.1037)
Best Loss: 0.1037 (Epoch 229)

Epoch 230/500, Batch 100/782, Loss: 0.1088, Recon: 0.0205, VQ: 0.0883, Perplexity: 61.69
Epoch 230/500, Batch 200/782, Loss: 0.1057, Recon: 0.0185, VQ: 0.0871, Perplexity: 57.61
Epoch 230/500, Batch 300/782, Loss: 0.0933, Recon: 0.0179, VQ: 0.0754, Perplexity: 57.38
Epoch 230/500, Batch 400/782, Loss: 0.1086, Recon: 0.0201, VQ: 0.0885, Perplexity: 61.67
Epoch 230/500, Batch 500/782, Loss: 0.1049, Recon: 0.0184, VQ: 0.0866, Perplexity: 58.89
Epoch 230/500, Batch 600/782, Loss: 0.0975, Recon: 0.0178, VQ: 0.0796, Perplexity: 58.90
Epoch 230/500, Batch 700/782, Loss: 0.1078, Recon: 0.0204, VQ: 0.0873, Per

plexity: 60.96

=== Epoch 230/500 Summary ===

Train Loss: 0.1036 | Val Loss: 0.1032
Train Recon: 0.0186 | Val Recon: 0.0199
Train VQ: 0.0851 | Val VQ: 0.0833
Train Perplexity: 59.82 | Val Perplexity: 59.96
✓ Best model saved! (Val Loss: 0.1032)
Best Loss: 0.1032 (Epoch 230)

Epoch 231/500, Batch 100/782, Loss: 0.1029, Recon: 0.0185, VQ: 0.0845, Perplexity: 60.75
Epoch 231/500, Batch 200/782, Loss: 0.0961, Recon: 0.0181, VQ: 0.0780, Perplexity: 58.99
Epoch 231/500, Batch 300/782, Loss: 0.0914, Recon: 0.0183, VQ: 0.0731, Perplexity: 58.71
Epoch 231/500, Batch 400/782, Loss: 0.1093, Recon: 0.0181, VQ: 0.0912, Perplexity: 58.55
Epoch 231/500, Batch 500/782, Loss: 0.0930, Recon: 0.0176, VQ: 0.0753, Perplexity: 58.69
Epoch 231/500, Batch 600/782, Loss: 0.1180, Recon: 0.0204, VQ: 0.0977, Perplexity: 63.17
Epoch 231/500, Batch 700/782, Loss: 0.0998, Recon: 0.0181, VQ: 0.0817, Perplexity: 58.21

=== Epoch 231/500 Summary ===

Train Loss: 0.1033 | Val Loss: 0.1039
Train Recon: 0.0185 | Val Recon: 0.0200
Train VQ: 0.0847 | Val VQ: 0.0840
Train Perplexity: 59.89 | Val Perplexity: 60.12
No improvement. Patience: 1/20
Best Loss: 0.1032 (Epoch 230)

Epoch 232/500, Batch 100/782, Loss: 0.0954, Recon: 0.0172, VQ: 0.0782, Perplexity: 59.47
Epoch 232/500, Batch 200/782, Loss: 0.0979, Recon: 0.0172, VQ: 0.0807, Perplexity: 58.99
Epoch 232/500, Batch 300/782, Loss: 0.0953, Recon: 0.0181, VQ: 0.0773, Perplexity: 58.19
Epoch 232/500, Batch 400/782, Loss: 0.1037, Recon: 0.0190, VQ: 0.0847, Perplexity: 60.29
Epoch 232/500, Batch 500/782, Loss: 0.0900, Recon: 0.0168, VQ: 0.0732, Perplexity: 58.76
Epoch 232/500, Batch 600/782, Loss: 0.1033, Recon: 0.0196, VQ: 0.0837, Perplexity: 60.22
Epoch 232/500, Batch 700/782, Loss: 0.3481, Recon: 0.0201, VQ: 0.3280, Perplexity: 58.99

=== Epoch 232/500 Summary ===

Train Loss: 0.1036 | Val Loss: 0.1037
Train Recon: 0.0185 | Val Recon: 0.0199
Train VQ: 0.0851 | Val VQ: 0.0838
Train Perplexity: 59.91 | Val Perplexity: 60.13
No improvement. Patience: 2/20
Best Loss: 0.1032 (Epoch 230)

Epoch 233/500, Batch 100/782, Loss: 0.1132, Recon: 0.0169, VQ: 0.0963, Perplexity: 59.45
Epoch 233/500, Batch 200/782, Loss: 0.1085, Recon: 0.0202, VQ: 0.0884, Perplexity: 62.14

Epoch 233/500, Batch 300/782, Loss: 0.1270, Recon: 0.0189, VQ: 0.1081, Perplexity: 61.65
Epoch 233/500, Batch 400/782, Loss: 0.0986, Recon: 0.0178, VQ: 0.0808, Perplexity: 60.72
Epoch 233/500, Batch 500/782, Loss: 0.1176, Recon: 0.0200, VQ: 0.0976, Perplexity: 64.45
Epoch 233/500, Batch 600/782, Loss: 0.1471, Recon: 0.0198, VQ: 0.1273, Perplexity: 60.89
Epoch 233/500, Batch 700/782, Loss: 0.1059, Recon: 0.0190, VQ: 0.0869, Perplexity: 60.16

=== Epoch 233/500 Summary ===

Train Loss: 0.1034 | Val Loss: 0.1030
Train Recon: 0.0185 | Val Recon: 0.0200
Train VQ: 0.0849 | Val VQ: 0.0830
Train Perplexity: 59.93 | Val Perplexity: 59.76
✓ Best model saved! (Val Loss: 0.1030)
Best Loss: 0.1030 (Epoch 233)

Epoch 234/500, Batch 100/782, Loss: 0.1115, Recon: 0.0185, VQ: 0.0930, Perplexity: 57.49
Epoch 234/500, Batch 200/782, Loss: 0.1002, Recon: 0.0186, VQ: 0.0816, Perplexity: 60.25
Epoch 234/500, Batch 300/782, Loss: 0.1045, Recon: 0.0188, VQ: 0.0856, Perplexity: 61.80
Epoch 234/500, Batch 400/782, Loss: 0.1072, Recon: 0.0195, VQ: 0.0877, Perplexity: 60.51
Epoch 234/500, Batch 500/782, Loss: 0.0942, Recon: 0.0181, VQ: 0.0761, Perplexity: 59.89
Epoch 234/500, Batch 600/782, Loss: 0.0936, Recon: 0.0172, VQ: 0.0764, Perplexity: 57.69
Epoch 234/500, Batch 700/782, Loss: 0.1008, Recon: 0.0182, VQ: 0.0827, Perplexity: 61.43

=== Epoch 234/500 Summary ===

Train Loss: 0.1030 | Val Loss: 0.1040
Train Recon: 0.0185 | Val Recon: 0.0199
Train VQ: 0.0845 | Val VQ: 0.0840
Train Perplexity: 60.07 | Val Perplexity: 60.05
No improvement. Patience: 1/20
Best Loss: 0.1030 (Epoch 233)

Epoch 235/500, Batch 100/782, Loss: 0.1032, Recon: 0.0185, VQ: 0.0847, Perplexity: 59.51
Epoch 235/500, Batch 200/782, Loss: 0.1239, Recon: 0.0207, VQ: 0.1033, Perplexity: 61.50
Epoch 235/500, Batch 300/782, Loss: 0.1074, Recon: 0.0202, VQ: 0.0873, Perplexity: 61.15
Epoch 235/500, Batch 400/782, Loss: 0.0953, Recon: 0.0173, VQ: 0.0780, Perplexity: 59.52
Epoch 235/500, Batch 500/782, Loss: 0.1021, Recon: 0.0200, VQ: 0.0821, Perplexity: 62.05
Epoch 235/500, Batch 600/782, Loss: 0.1093, Recon: 0.0193, VQ: 0.0900, Perplexity: 60.71
Epoch 235/500, Batch 700/782, Loss: 0.3459, Recon: 0.0200, VQ: 0.3259, Perplexity: 61.16

=== Epoch 235/500 Summary ===

Train Loss: 0.1030 | Val Loss: 0.1041
Train Recon: 0.0185 | Val Recon: 0.0200

Train VQ: 0.0846 | Val VQ: 0.0841
Train Perplexity: 60.08 | Val Perplexity: 59.97
No improvement. Patience: 2/20
Best Loss: 0.1030 (Epoch 233)

Epoch 236/500, Batch 100/782, Loss: 0.0931, Recon: 0.0175, VQ: 0.0756, Perplexity: 58.76
Epoch 236/500, Batch 200/782, Loss: 0.1131, Recon: 0.0192, VQ: 0.0939, Perplexity: 59.52
Epoch 236/500, Batch 300/782, Loss: 0.1241, Recon: 0.0204, VQ: 0.1036, Perplexity: 61.61
Epoch 236/500, Batch 400/782, Loss: 0.0985, Recon: 0.0191, VQ: 0.0794, Perplexity: 61.90
Epoch 236/500, Batch 500/782, Loss: 0.0931, Recon: 0.0179, VQ: 0.0752, Perplexity: 58.96
Epoch 236/500, Batch 600/782, Loss: 0.0986, Recon: 0.0183, VQ: 0.0802, Perplexity: 58.09
Epoch 236/500, Batch 700/782, Loss: 0.1272, Recon: 0.0192, VQ: 0.1080, Perplexity: 62.43

=== Epoch 236/500 Summary ===
Train Loss: 0.1035 | Val Loss: 0.1037
Train Recon: 0.0185 | Val Recon: 0.0199
Train VQ: 0.0850 | Val VQ: 0.0838
Train Perplexity: 60.06 | Val Perplexity: 60.08
No improvement. Patience: 3/20
Best Loss: 0.1030 (Epoch 233)

Epoch 237/500, Batch 100/782, Loss: 0.1045, Recon: 0.0191, VQ: 0.0854, Perplexity: 61.37
Epoch 237/500, Batch 200/782, Loss: 0.1035, Recon: 0.0186, VQ: 0.0849, Perplexity: 60.65
Epoch 237/500, Batch 300/782, Loss: 0.1078, Recon: 0.0200, VQ: 0.0878, Perplexity: 59.09
Epoch 237/500, Batch 400/782, Loss: 0.1132, Recon: 0.0208, VQ: 0.0923, Perplexity: 62.26
Epoch 237/500, Batch 500/782, Loss: 0.0990, Recon: 0.0179, VQ: 0.0811, Perplexity: 59.48
Epoch 237/500, Batch 600/782, Loss: 0.1036, Recon: 0.0187, VQ: 0.0848, Perplexity: 61.18
Epoch 237/500, Batch 700/782, Loss: 0.1032, Recon: 0.0193, VQ: 0.0840, Perplexity: 61.19

=== Epoch 237/500 Summary ===
Train Loss: 0.1031 | Val Loss: 0.1045
Train Recon: 0.0185 | Val Recon: 0.0199
Train VQ: 0.0846 | Val VQ: 0.0846
Train Perplexity: 60.10 | Val Perplexity: 60.16
No improvement. Patience: 4/20
Best Loss: 0.1030 (Epoch 233)

Epoch 238/500, Batch 100/782, Loss: 0.0875, Recon: 0.0164, VQ: 0.0711, Perplexity: 57.65
Epoch 238/500, Batch 200/782, Loss: 0.1039, Recon: 0.0191, VQ: 0.0848, Perplexity: 57.99
Epoch 238/500, Batch 300/782, Loss: 0.1411, Recon: 0.0203, VQ: 0.1208, Perplexity: 60.00
Epoch 238/500, Batch 400/782, Loss: 0.1058, Recon: 0.0198, VQ: 0.0860, Perplexity: 61.64
Epoch 238/500, Batch 500/782, Loss: 0.1078, Recon: 0.0198, VQ: 0.0880, Per

plexity: 62.54
Epoch 238/500, Batch 600/782, Loss: 0.0983, Recon: 0.0179, VQ: 0.0804, Per
plexity: 60.13
Epoch 238/500, Batch 700/782, Loss: 0.0972, Recon: 0.0187, VQ: 0.0785, Per
plexity: 60.31

=== Epoch 238/500 Summary ===
Train Loss: 0.1031 | Val Loss: 0.1032
Train Recon: 0.0185 | Val Recon: 0.0200
Train VQ: 0.0846 | Val VQ: 0.0832
Train Perplexity: 60.16 | Val Perplexity: 59.77
No improvement. Patience: 5/20
Best Loss: 0.1030 (Epoch 233)

Epoch 239/500, Batch 100/782, Loss: 0.1105, Recon: 0.0204, VQ: 0.0901, Per
plexity: 62.59
Epoch 239/500, Batch 200/782, Loss: 0.1058, Recon: 0.0170, VQ: 0.0889, Per
plexity: 58.68
Epoch 239/500, Batch 300/782, Loss: 0.1098, Recon: 0.0191, VQ: 0.0907, Per
plexity: 62.20
Epoch 239/500, Batch 400/782, Loss: 0.1000, Recon: 0.0178, VQ: 0.0822, Per
plexity: 59.10
Epoch 239/500, Batch 500/782, Loss: 0.1262, Recon: 0.0186, VQ: 0.1076, Per
plexity: 61.26
Epoch 239/500, Batch 600/782, Loss: 0.0982, Recon: 0.0179, VQ: 0.0803, Per
plexity: 59.22
Epoch 239/500, Batch 700/782, Loss: 0.1029, Recon: 0.0189, VQ: 0.0840, Per
plexity: 61.66

=== Epoch 239/500 Summary ===
Train Loss: 0.1030 | Val Loss: 0.1038
Train Recon: 0.0185 | Val Recon: 0.0199
Train VQ: 0.0846 | Val VQ: 0.0838
Train Perplexity: 60.14 | Val Perplexity: 60.28
No improvement. Patience: 6/20
Best Loss: 0.1030 (Epoch 233)

Epoch 240/500, Batch 100/782, Loss: 0.1045, Recon: 0.0182, VQ: 0.0863, Per
plexity: 61.06
Epoch 240/500, Batch 200/782, Loss: 0.1056, Recon: 0.0188, VQ: 0.0867, Per
plexity: 60.98
Epoch 240/500, Batch 300/782, Loss: 0.0994, Recon: 0.0179, VQ: 0.0815, Per
plexity: 57.64
Epoch 240/500, Batch 400/782, Loss: 0.1096, Recon: 0.0184, VQ: 0.0912, Per
plexity: 61.30
Epoch 240/500, Batch 500/782, Loss: 0.0956, Recon: 0.0177, VQ: 0.0779, Per
plexity: 60.07
Epoch 240/500, Batch 600/782, Loss: 0.0902, Recon: 0.0169, VQ: 0.0732, Per
plexity: 58.63
Epoch 240/500, Batch 700/782, Loss: 0.1064, Recon: 0.0179, VQ: 0.0885, Per
plexity: 58.77

=== Epoch 240/500 Summary ===
Train Loss: 0.1031 | Val Loss: 0.1047
Train Recon: 0.0185 | Val Recon: 0.0200
Train VQ: 0.0847 | Val VQ: 0.0847
Train Perplexity: 60.19 | Val Perplexity: 60.63
No improvement. Patience: 7/20
Best Loss: 0.1030 (Epoch 233)

Epoch 241/500, Batch 100/782, Loss: 0.0956, Recon: 0.0174, VQ: 0.0782, Perplexity: 59.82
Epoch 241/500, Batch 200/782, Loss: 0.1073, Recon: 0.0190, VQ: 0.0883, Perplexity: 60.24
Epoch 241/500, Batch 300/782, Loss: 0.1046, Recon: 0.0185, VQ: 0.0861, Perplexity: 62.12
Epoch 241/500, Batch 400/782, Loss: 0.1193, Recon: 0.0198, VQ: 0.0995, Perplexity: 62.76
Epoch 241/500, Batch 500/782, Loss: 0.1019, Recon: 0.0184, VQ: 0.0836, Perplexity: 61.40
Epoch 241/500, Batch 600/782, Loss: 0.1003, Recon: 0.0178, VQ: 0.0825, Perplexity: 60.50
Epoch 241/500, Batch 700/782, Loss: 0.0991, Recon: 0.0188, VQ: 0.0803, Perplexity: 61.65

=== Epoch 241/500 Summary ===

Train Loss: 0.1031 | Val Loss: 0.1038
Train Recon: 0.0184 | Val Recon: 0.0200
Train VQ: 0.0846 | Val VQ: 0.0839
Train Perplexity: 60.25 | Val Perplexity: 59.69
No improvement. Patience: 8/20
Best Loss: 0.1030 (Epoch 233)

Epoch 242/500, Batch 100/782, Loss: 0.0971, Recon: 0.0179, VQ: 0.0792, Perplexity: 57.47
Epoch 242/500, Batch 200/782, Loss: 0.1074, Recon: 0.0189, VQ: 0.0884, Perplexity: 61.37
Epoch 242/500, Batch 300/782, Loss: 0.0980, Recon: 0.0182, VQ: 0.0798, Perplexity: 61.58
Epoch 242/500, Batch 400/782, Loss: 0.1121, Recon: 0.0195, VQ: 0.0926, Perplexity: 60.68
Epoch 242/500, Batch 500/782, Loss: 0.0941, Recon: 0.0180, VQ: 0.0761, Perplexity: 59.54
Epoch 242/500, Batch 600/782, Loss: 0.0889, Recon: 0.0155, VQ: 0.0734, Perplexity: 56.85
Epoch 242/500, Batch 700/782, Loss: 0.1043, Recon: 0.0192, VQ: 0.0851, Perplexity: 62.22

=== Epoch 242/500 Summary ===

Train Loss: 0.1030 | Val Loss: 0.1037
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0845 | Val VQ: 0.0837
Train Perplexity: 60.21 | Val Perplexity: 60.40
No improvement. Patience: 9/20
Best Loss: 0.1030 (Epoch 233)

Epoch 243/500, Batch 100/782, Loss: 0.1061, Recon: 0.0184, VQ: 0.0877, Perplexity: 62.25
Epoch 243/500, Batch 200/782, Loss: 0.1050, Recon: 0.0184, VQ: 0.0866, Perplexity: 58.20
Epoch 243/500, Batch 300/782, Loss: 0.1006, Recon: 0.0185, VQ: 0.0821, Perplexity: 60.61
Epoch 243/500, Batch 400/782, Loss: 0.0997, Recon: 0.0186, VQ: 0.0811, Perplexity: 60.24
Epoch 243/500, Batch 500/782, Loss: 0.1026, Recon: 0.0188, VQ: 0.0838, Perplexity: 61.21
Epoch 243/500, Batch 600/782, Loss: 0.1121, Recon: 0.0201, VQ: 0.0920, Perplexity: 61.71
Epoch 243/500, Batch 700/782, Loss: 0.0978, Recon: 0.0179, VQ: 0.0799, Perplexity: 59.43

=== Epoch 243/500 Summary ===

Train Loss: 0.1029 | Val Loss: 0.1052
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0845 | Val VQ: 0.0852
Train Perplexity: 60.23 | Val Perplexity: 60.56
No improvement. Patience: 10/20
Best Loss: 0.1030 (Epoch 233)

Epoch 244/500, Batch 100/782, Loss: 0.1117, Recon: 0.0204, VQ: 0.0912, Perplexity: 62.91
Epoch 244/500, Batch 200/782, Loss: 0.0951, Recon: 0.0172, VQ: 0.0779, Perplexity: 58.00
Epoch 244/500, Batch 300/782, Loss: 0.1097, Recon: 0.0194, VQ: 0.0903, Perplexity: 62.09
Epoch 244/500, Batch 400/782, Loss: 0.4254, Recon: 0.0216, VQ: 0.4038, Perplexity: 62.96
Epoch 244/500, Batch 500/782, Loss: 0.1321, Recon: 0.0191, VQ: 0.1130, Perplexity: 59.67
Epoch 244/500, Batch 600/782, Loss: 0.1019, Recon: 0.0176, VQ: 0.0842, Perplexity: 61.49
Epoch 244/500, Batch 700/782, Loss: 0.1072, Recon: 0.0196, VQ: 0.0876, Perplexity: 62.04

=== Epoch 244/500 Summary ===

Train Loss: 0.1031 | Val Loss: 0.1042
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0847 | Val VQ: 0.0843
Train Perplexity: 60.29 | Val Perplexity: 60.36
No improvement. Patience: 11/20
Best Loss: 0.1030 (Epoch 233)

Epoch 245/500, Batch 100/782, Loss: 0.1004, Recon: 0.0172, VQ: 0.0832, Perplexity: 58.15
Epoch 245/500, Batch 200/782, Loss: 0.1035, Recon: 0.0196, VQ: 0.0838, Perplexity: 61.99
Epoch 245/500, Batch 300/782, Loss: 0.1042, Recon: 0.0190, VQ: 0.0851, Perplexity: 61.10
Epoch 245/500, Batch 400/782, Loss: 0.1052, Recon: 0.0182, VQ: 0.0870, Perplexity: 62.46
Epoch 245/500, Batch 500/782, Loss: 0.1014, Recon: 0.0190, VQ: 0.0824, Perplexity: 60.14
Epoch 245/500, Batch 600/782, Loss: 0.0938, Recon: 0.0175, VQ: 0.0763, Perplexity: 58.63
Epoch 245/500, Batch 700/782, Loss: 0.1067, Recon: 0.0187, VQ: 0.0880, Perplexity: 61.90

=== Epoch 245/500 Summary ===

Train Loss: 0.1030 | Val Loss: 0.1037
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0846 | Val VQ: 0.0838
Train Perplexity: 60.31 | Val Perplexity: 60.45
No improvement. Patience: 12/20
Best Loss: 0.1030 (Epoch 233)

Epoch 246/500, Batch 100/782, Loss: 0.0939, Recon: 0.0171, VQ: 0.0768, Perplexity: 59.27
Epoch 246/500, Batch 200/782, Loss: 0.0988, Recon: 0.0177, VQ: 0.0811, Perplexity: 58.55
Epoch 246/500, Batch 300/782, Loss: 0.0949, Recon: 0.0179, VQ: 0.0770, Perplexity: 58.55

plexity: 60.40
Epoch 246/500, Batch 400/782, Loss: 0.0875, Recon: 0.0170, VQ: 0.0705, Per
plexity: 58.67
Epoch 246/500, Batch 500/782, Loss: 0.0979, Recon: 0.0185, VQ: 0.0794, Per
plexity: 61.02
Epoch 246/500, Batch 600/782, Loss: 0.1104, Recon: 0.0182, VQ: 0.0921, Per
plexity: 60.37
Epoch 246/500, Batch 700/782, Loss: 0.0820, Recon: 0.0155, VQ: 0.0665, Per
plexity: 56.26

=== Epoch 246/500 Summary ===
Train Loss: 0.1031 | Val Loss: 0.1047
Train Recon: 0.0184 | Val Recon: 0.0200
Train VQ: 0.0847 | Val VQ: 0.0847
Train Perplexity: 60.33 | Val Perplexity: 60.58
No improvement. Patience: 13/20
Best Loss: 0.1030 (Epoch 233)

Epoch 247/500, Batch 100/782, Loss: 0.1087, Recon: 0.0193, VQ: 0.0894, Per
plexity: 59.63
Epoch 247/500, Batch 200/782, Loss: 0.0983, Recon: 0.0181, VQ: 0.0802, Per
plexity: 60.16
Epoch 247/500, Batch 300/782, Loss: 0.1007, Recon: 0.0180, VQ: 0.0826, Per
plexity: 59.20
Epoch 247/500, Batch 400/782, Loss: 0.1055, Recon: 0.0193, VQ: 0.0862, Per
plexity: 61.54
Epoch 247/500, Batch 500/782, Loss: 0.0972, Recon: 0.0168, VQ: 0.0805, Per
plexity: 59.17
Epoch 247/500, Batch 600/782, Loss: 0.0983, Recon: 0.0184, VQ: 0.0799, Per
plexity: 62.29
Epoch 247/500, Batch 700/782, Loss: 0.0970, Recon: 0.0176, VQ: 0.0794, Per
plexity: 60.08

=== Epoch 247/500 Summary ===
Train Loss: 0.1029 | Val Loss: 0.1031
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0844 | Val VQ: 0.0832
Train Perplexity: 60.35 | Val Perplexity: 60.23
No improvement. Patience: 14/20
Best Loss: 0.1030 (Epoch 233)

Epoch 248/500, Batch 100/782, Loss: 0.0995, Recon: 0.0171, VQ: 0.0825, Per
plexity: 57.89
Epoch 248/500, Batch 200/782, Loss: 0.1037, Recon: 0.0180, VQ: 0.0857, Per
plexity: 60.46
Epoch 248/500, Batch 300/782, Loss: 0.0894, Recon: 0.0170, VQ: 0.0724, Per
plexity: 57.97
Epoch 248/500, Batch 400/782, Loss: 0.0898, Recon: 0.0172, VQ: 0.0725, Per
plexity: 59.07
Epoch 248/500, Batch 500/782, Loss: 0.1034, Recon: 0.0187, VQ: 0.0847, Per
plexity: 59.07
Epoch 248/500, Batch 600/782, Loss: 0.1068, Recon: 0.0186, VQ: 0.0882, Per
plexity: 61.00
Epoch 248/500, Batch 700/782, Loss: 0.0986, Recon: 0.0185, VQ: 0.0801, Per
plexity: 59.68

=== Epoch 248/500 Summary ===
Train Loss: 0.1029 | Val Loss: 0.1038
Train Recon: 0.0184 | Val Recon: 0.0200
Train VQ: 0.0845 | Val VQ: 0.0838

Train Perplexity: 60.38 | Val Perplexity: 60.35
No improvement. Patience: 15/20
Best Loss: 0.1030 (Epoch 233)

Epoch 249/500, Batch 100/782, Loss: 0.1080, Recon: 0.0191, VQ: 0.0889, Perplexity: 61.65
Epoch 249/500, Batch 200/782, Loss: 0.0976, Recon: 0.0183, VQ: 0.0793, Perplexity: 60.83
Epoch 249/500, Batch 300/782, Loss: 0.0968, Recon: 0.0184, VQ: 0.0784, Perplexity: 60.34
Epoch 249/500, Batch 400/782, Loss: 0.0959, Recon: 0.0173, VQ: 0.0786, Perplexity: 61.11
Epoch 249/500, Batch 500/782, Loss: 0.0854, Recon: 0.0162, VQ: 0.0692, Perplexity: 56.97
Epoch 249/500, Batch 600/782, Loss: 0.0921, Recon: 0.0182, VQ: 0.0738, Perplexity: 60.95
Epoch 249/500, Batch 700/782, Loss: 0.0981, Recon: 0.0175, VQ: 0.0806, Perplexity: 61.01

=== Epoch 249/500 Summary ===
Train Loss: 0.1030 | Val Loss: 0.1028
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0846 | Val VQ: 0.0828
Train Perplexity: 60.42 | Val Perplexity: 60.30
✓ Best model saved! (Val Loss: 0.1028)
Best Loss: 0.1028 (Epoch 249)

Epoch 250/500, Batch 100/782, Loss: 0.1115, Recon: 0.0181, VQ: 0.0934, Perplexity: 60.40
Epoch 250/500, Batch 200/782, Loss: 0.0908, Recon: 0.0176, VQ: 0.0732, Perplexity: 58.44
Epoch 250/500, Batch 300/782, Loss: 0.0933, Recon: 0.0171, VQ: 0.0763, Perplexity: 58.05
Epoch 250/500, Batch 400/782, Loss: 0.1146, Recon: 0.0200, VQ: 0.0946, Perplexity: 62.88
Epoch 250/500, Batch 500/782, Loss: 0.1034, Recon: 0.0185, VQ: 0.0849, Perplexity: 59.76
Epoch 250/500, Batch 600/782, Loss: 0.0955, Recon: 0.0174, VQ: 0.0781, Perplexity: 61.42
Epoch 250/500, Batch 700/782, Loss: 0.1002, Recon: 0.0181, VQ: 0.0822, Perplexity: 60.53

=== Epoch 250/500 Summary ===
Train Loss: 0.1029 | Val Loss: 0.1041
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0845 | Val VQ: 0.0842
Train Perplexity: 60.42 | Val Perplexity: 60.49
No improvement. Patience: 1/20
Best Loss: 0.1028 (Epoch 249)

Epoch 251/500, Batch 100/782, Loss: 0.0866, Recon: 0.0165, VQ: 0.0701, Perplexity: 57.62
Epoch 251/500, Batch 200/782, Loss: 0.0910, Recon: 0.0172, VQ: 0.0738, Perplexity: 59.19
Epoch 251/500, Batch 300/782, Loss: 0.0980, Recon: 0.0182, VQ: 0.0798, Perplexity: 59.74
Epoch 251/500, Batch 400/782, Loss: 0.0992, Recon: 0.0180, VQ: 0.0813, Perplexity: 62.57
Epoch 251/500, Batch 500/782, Loss: 0.1074, Recon: 0.0192, VQ: 0.0883, Perplexity: 61.93

Epoch 251/500, Batch 600/782, Loss: 0.1116, Recon: 0.0207, VQ: 0.0909, Perplexity: 63.15
Epoch 251/500, Batch 700/782, Loss: 0.1068, Recon: 0.0194, VQ: 0.0875, Perplexity: 62.25

=== Epoch 251/500 Summary ===

Train Loss: 0.1028 | Val Loss: 0.1033
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0844 | Val VQ: 0.0834
Train Perplexity: 60.49 | Val Perplexity: 60.57
No improvement. Patience: 2/20
Best Loss: 0.1028 (Epoch 249)

Epoch 252/500, Batch 100/782, Loss: 0.0913, Recon: 0.0168, VQ: 0.0744, Perplexity: 59.52
Epoch 252/500, Batch 200/782, Loss: 0.0985, Recon: 0.0175, VQ: 0.0809, Perplexity: 58.26
Epoch 252/500, Batch 300/782, Loss: 0.0973, Recon: 0.0187, VQ: 0.0787, Perplexity: 60.97
Epoch 252/500, Batch 400/782, Loss: 0.1174, Recon: 0.0184, VQ: 0.0990, Perplexity: 61.56
Epoch 252/500, Batch 500/782, Loss: 0.1156, Recon: 0.0200, VQ: 0.0957, Perplexity: 63.22
Epoch 252/500, Batch 600/782, Loss: 0.1142, Recon: 0.0204, VQ: 0.0937, Perplexity: 61.53
Epoch 252/500, Batch 700/782, Loss: 0.1086, Recon: 0.0197, VQ: 0.0889, Perplexity: 62.70

=== Epoch 252/500 Summary ===

Train Loss: 0.1026 | Val Loss: 0.1031
Train Recon: 0.0184 | Val Recon: 0.0200
Train VQ: 0.0842 | Val VQ: 0.0832
Train Perplexity: 60.47 | Val Perplexity: 60.36
No improvement. Patience: 3/20
Best Loss: 0.1028 (Epoch 249)

Epoch 253/500, Batch 100/782, Loss: 0.1004, Recon: 0.0188, VQ: 0.0816, Perplexity: 61.07
Epoch 253/500, Batch 200/782, Loss: 0.0970, Recon: 0.0182, VQ: 0.0788, Perplexity: 59.35
Epoch 253/500, Batch 300/782, Loss: 0.1220, Recon: 0.0199, VQ: 0.1021, Perplexity: 61.61
Epoch 253/500, Batch 400/782, Loss: 0.0883, Recon: 0.0166, VQ: 0.0718, Perplexity: 59.06
Epoch 253/500, Batch 500/782, Loss: 0.0983, Recon: 0.0183, VQ: 0.0800, Perplexity: 60.22
Epoch 253/500, Batch 600/782, Loss: 0.0916, Recon: 0.0170, VQ: 0.0746, Perplexity: 58.15
Epoch 253/500, Batch 700/782, Loss: 0.1015, Recon: 0.0189, VQ: 0.0826, Perplexity: 61.49

=== Epoch 253/500 Summary ===

Train Loss: 0.1027 | Val Loss: 0.1037
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0843 | Val VQ: 0.0837
Train Perplexity: 60.56 | Val Perplexity: 60.54
No improvement. Patience: 4/20
Best Loss: 0.1028 (Epoch 249)

Epoch 254/500, Batch 100/782, Loss: 0.1057, Recon: 0.0195, VQ: 0.0862, Per

plexity: 61.03
Epoch 254/500, Batch 200/782, Loss: 0.0940, Recon: 0.0180, VQ: 0.0760, Perplexity: 59.75
Epoch 254/500, Batch 300/782, Loss: 0.0853, Recon: 0.0158, VQ: 0.0696, Perplexity: 56.18
Epoch 254/500, Batch 400/782, Loss: 0.0906, Recon: 0.0170, VQ: 0.0736, Perplexity: 59.22
Epoch 254/500, Batch 500/782, Loss: 0.1066, Recon: 0.0185, VQ: 0.0880, Perplexity: 61.36
Epoch 254/500, Batch 600/782, Loss: 0.0925, Recon: 0.0177, VQ: 0.0748, Perplexity: 60.03
Epoch 254/500, Batch 700/782, Loss: 0.1018, Recon: 0.0189, VQ: 0.0829, Perplexity: 62.39

=== Epoch 254/500 Summary ===

Train Loss: 0.1024 | Val Loss: 0.1029
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0840 | Val VQ: 0.0830
Train Perplexity: 60.58 | Val Perplexity: 60.37
No improvement. Patience: 5/20
Best Loss: 0.1028 (Epoch 249)

Epoch 255/500, Batch 100/782, Loss: 0.1212, Recon: 0.0192, VQ: 0.1020, Perplexity: 62.16
Epoch 255/500, Batch 200/782, Loss: 0.1161, Recon: 0.0187, VQ: 0.0974, Perplexity: 60.34
Epoch 255/500, Batch 300/782, Loss: 0.0888, Recon: 0.0168, VQ: 0.0720, Perplexity: 59.08
Epoch 255/500, Batch 400/782, Loss: 0.0901, Recon: 0.0173, VQ: 0.0728, Perplexity: 57.93
Epoch 255/500, Batch 500/782, Loss: 0.0942, Recon: 0.0173, VQ: 0.0769, Perplexity: 59.83
Epoch 255/500, Batch 600/782, Loss: 0.0996, Recon: 0.0186, VQ: 0.0809, Perplexity: 60.74
Epoch 255/500, Batch 700/782, Loss: 0.1017, Recon: 0.0182, VQ: 0.0835, Perplexity: 60.29

=== Epoch 255/500 Summary ===

Train Loss: 0.1024 | Val Loss: 0.1024
Train Recon: 0.0184 | Val Recon: 0.0199
Train VQ: 0.0841 | Val VQ: 0.0824
Train Perplexity: 60.62 | Val Perplexity: 60.56
✓ Best model saved! (Val Loss: 0.1024)
Best Loss: 0.1024 (Epoch 255)

Epoch 256/500, Batch 100/782, Loss: 0.0959, Recon: 0.0180, VQ: 0.0779, Perplexity: 59.60
Epoch 256/500, Batch 200/782, Loss: 0.1027, Recon: 0.0190, VQ: 0.0837, Perplexity: 62.28
Epoch 256/500, Batch 300/782, Loss: 0.1180, Recon: 0.0191, VQ: 0.0989, Perplexity: 62.48
Epoch 256/500, Batch 400/782, Loss: 0.1082, Recon: 0.0191, VQ: 0.0891, Perplexity: 62.84
Epoch 256/500, Batch 500/782, Loss: 0.1445, Recon: 0.0196, VQ: 0.1248, Perplexity: 62.07
Epoch 256/500, Batch 600/782, Loss: 0.0972, Recon: 0.0182, VQ: 0.0790, Perplexity: 60.75
Epoch 256/500, Batch 700/782, Loss: 0.1014, Recon: 0.0188, VQ: 0.0826, Perplexity: 62.35

=== Epoch 256/500 Summary ===

Train Loss: 0.1024 | Val Loss: 0.1028

Train Recon: 0.0184 | Val Recon: 0.0199

Train VQ: 0.0840 | Val VQ: 0.0829

Train Perplexity: 60.74 | Val Perplexity: 60.29

No improvement. Patience: 1/20

Best Loss: 0.1024 (Epoch 255)

Epoch 257/500, Batch 100/782, Loss: 0.0939, Recon: 0.0175, VQ: 0.0764, Perplexity: 59.24

Epoch 257/500, Batch 200/782, Loss: 0.0972, Recon: 0.0186, VQ: 0.0786, Perplexity: 62.68

Epoch 257/500, Batch 300/782, Loss: 0.1077, Recon: 0.0183, VQ: 0.0894, Perplexity: 60.94

Epoch 257/500, Batch 400/782, Loss: 0.1131, Recon: 0.0197, VQ: 0.0934, Perplexity: 61.60

Epoch 257/500, Batch 500/782, Loss: 0.1006, Recon: 0.0195, VQ: 0.0811, Perplexity: 62.07

Epoch 257/500, Batch 600/782, Loss: 0.0882, Recon: 0.0168, VQ: 0.0714, Perplexity: 58.87

Epoch 257/500, Batch 700/782, Loss: 0.0967, Recon: 0.0180, VQ: 0.0787, Perplexity: 61.50

=== Epoch 257/500 Summary ===

Train Loss: 0.1023 | Val Loss: 0.1016

Train Recon: 0.0183 | Val Recon: 0.0199

Train VQ: 0.0839 | Val VQ: 0.0816

Train Perplexity: 60.86 | Val Perplexity: 60.73

✓ Best model saved! (Val Loss: 0.1016)

Best Loss: 0.1016 (Epoch 257)

Epoch 258/500, Batch 100/782, Loss: 0.1065, Recon: 0.0194, VQ: 0.0871, Perplexity: 62.15

Epoch 258/500, Batch 200/782, Loss: 0.0989, Recon: 0.0173, VQ: 0.0816, Perplexity: 60.22

Epoch 258/500, Batch 300/782, Loss: 0.1014, Recon: 0.0175, VQ: 0.0839, Perplexity: 59.63

Epoch 258/500, Batch 400/782, Loss: 0.1064, Recon: 0.0182, VQ: 0.0883, Perplexity: 60.16

Epoch 258/500, Batch 500/782, Loss: 0.0998, Recon: 0.0185, VQ: 0.0813, Perplexity: 62.23

Epoch 258/500, Batch 600/782, Loss: 0.1015, Recon: 0.0194, VQ: 0.0821, Perplexity: 63.01

Epoch 258/500, Batch 700/782, Loss: 0.1106, Recon: 0.0190, VQ: 0.0916, Perplexity: 58.36

=== Epoch 258/500 Summary ===

Train Loss: 0.1022 | Val Loss: 0.1038

Train Recon: 0.0183 | Val Recon: 0.0199

Train VQ: 0.0839 | Val VQ: 0.0838

Train Perplexity: 61.01 | Val Perplexity: 61.04

No improvement. Patience: 1/20

Best Loss: 0.1016 (Epoch 257)

Epoch 259/500, Batch 100/782, Loss: 0.0894, Recon: 0.0172, VQ: 0.0722, Perplexity: 59.65

Epoch 259/500, Batch 200/782, Loss: 0.1027, Recon: 0.0187, VQ: 0.0840, Perplexity: 62.82

Epoch 259/500, Batch 300/782, Loss: 0.1026, Recon: 0.0190, VQ: 0.0836, Perplexity: 62.53

Epoch 259/500, Batch 400/782, Loss: 0.0940, Recon: 0.0175, VQ: 0.0764, Perplexity: 60.30
Epoch 259/500, Batch 500/782, Loss: 0.1104, Recon: 0.0193, VQ: 0.0911, Perplexity: 63.40
Epoch 259/500, Batch 600/782, Loss: 0.1095, Recon: 0.0197, VQ: 0.0898, Perplexity: 62.18
Epoch 259/500, Batch 700/782, Loss: 0.1799, Recon: 0.0204, VQ: 0.1596, Perplexity: 62.90

=== Epoch 259/500 Summary ===

Train Loss: 0.1023 | Val Loss: 0.1037
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0840 | Val VQ: 0.0838
Train Perplexity: 61.05 | Val Perplexity: 61.41
No improvement. Patience: 2/20
Best Loss: 0.1016 (Epoch 257)

Epoch 260/500, Batch 100/782, Loss: 0.1032, Recon: 0.0191, VQ: 0.0841, Perplexity: 62.91
Epoch 260/500, Batch 200/782, Loss: 0.0971, Recon: 0.0180, VQ: 0.0791, Perplexity: 59.37
Epoch 260/500, Batch 300/782, Loss: 0.0906, Recon: 0.0170, VQ: 0.0735, Perplexity: 61.42
Epoch 260/500, Batch 400/782, Loss: 0.1058, Recon: 0.0190, VQ: 0.0868, Perplexity: 61.77
Epoch 260/500, Batch 500/782, Loss: 0.0924, Recon: 0.0173, VQ: 0.0751, Perplexity: 60.24
Epoch 260/500, Batch 600/782, Loss: 0.0909, Recon: 0.0168, VQ: 0.0741, Perplexity: 57.53
Epoch 260/500, Batch 700/782, Loss: 0.1111, Recon: 0.0210, VQ: 0.0901, Perplexity: 64.14

=== Epoch 260/500 Summary ===

Train Loss: 0.1021 | Val Loss: 0.1023
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0838 | Val VQ: 0.0824
Train Perplexity: 61.20 | Val Perplexity: 60.77
No improvement. Patience: 3/20
Best Loss: 0.1016 (Epoch 257)

Epoch 261/500, Batch 100/782, Loss: 0.1105, Recon: 0.0203, VQ: 0.0902, Perplexity: 64.42
Epoch 261/500, Batch 200/782, Loss: 0.1026, Recon: 0.0187, VQ: 0.0839, Perplexity: 61.31
Epoch 261/500, Batch 300/782, Loss: 0.1087, Recon: 0.0192, VQ: 0.0895, Perplexity: 62.70
Epoch 261/500, Batch 400/782, Loss: 0.1262, Recon: 0.0198, VQ: 0.1063, Perplexity: 64.33
Epoch 261/500, Batch 500/782, Loss: 0.1013, Recon: 0.0176, VQ: 0.0837, Perplexity: 60.44
Epoch 261/500, Batch 600/782, Loss: 0.1053, Recon: 0.0191, VQ: 0.0862, Perplexity: 61.14
Epoch 261/500, Batch 700/782, Loss: 0.0924, Recon: 0.0180, VQ: 0.0744, Perplexity: 59.28

=== Epoch 261/500 Summary ===

Train Loss: 0.1021 | Val Loss: 0.1047
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0838 | Val VQ: 0.0847
Train Perplexity: 61.18 | Val Perplexity: 61.36

No improvement. Patience: 4/20
Best Loss: 0.1016 (Epoch 257)

Epoch 262/500, Batch 100/782, Loss: 0.1283, Recon: 0.0193, VQ: 0.1089, Perplexity: 62.42
Epoch 262/500, Batch 200/782, Loss: 0.0931, Recon: 0.0170, VQ: 0.0762, Perplexity: 59.32
Epoch 262/500, Batch 300/782, Loss: 0.0948, Recon: 0.0170, VQ: 0.0778, Perplexity: 61.58
Epoch 262/500, Batch 400/782, Loss: 0.1078, Recon: 0.0186, VQ: 0.0892, Perplexity: 62.89
Epoch 262/500, Batch 500/782, Loss: 0.1026, Recon: 0.0197, VQ: 0.0830, Perplexity: 62.37
Epoch 262/500, Batch 600/782, Loss: 0.1063, Recon: 0.0187, VQ: 0.0877, Perplexity: 61.90
Epoch 262/500, Batch 700/782, Loss: 0.1039, Recon: 0.0201, VQ: 0.0838, Perplexity: 62.15

=== Epoch 262/500 Summary ===
Train Loss: 0.1018 | Val Loss: 0.1043
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0835 | Val VQ: 0.0844
Train Perplexity: 61.33 | Val Perplexity: 61.52
No improvement. Patience: 5/20
Best Loss: 0.1016 (Epoch 257)

Epoch 263/500, Batch 100/782, Loss: 0.1036, Recon: 0.0185, VQ: 0.0850, Perplexity: 59.48
Epoch 263/500, Batch 200/782, Loss: 0.0943, Recon: 0.0180, VQ: 0.0763, Perplexity: 60.82
Epoch 263/500, Batch 300/782, Loss: 0.1075, Recon: 0.0184, VQ: 0.0891, Perplexity: 61.61
Epoch 263/500, Batch 400/782, Loss: 0.1035, Recon: 0.0185, VQ: 0.0850, Perplexity: 60.33
Epoch 263/500, Batch 500/782, Loss: 0.0917, Recon: 0.0171, VQ: 0.0746, Perplexity: 59.96
Epoch 263/500, Batch 600/782, Loss: 0.1062, Recon: 0.0199, VQ: 0.0863, Perplexity: 62.36
Epoch 263/500, Batch 700/782, Loss: 0.1226, Recon: 0.0198, VQ: 0.1029, Perplexity: 61.70

=== Epoch 263/500 Summary ===
Train Loss: 0.1019 | Val Loss: 0.1029
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0836 | Val VQ: 0.0830
Train Perplexity: 61.36 | Val Perplexity: 61.50
No improvement. Patience: 6/20
Best Loss: 0.1016 (Epoch 257)

Epoch 264/500, Batch 100/782, Loss: 0.0968, Recon: 0.0179, VQ: 0.0789, Perplexity: 59.96
Epoch 264/500, Batch 200/782, Loss: 0.1000, Recon: 0.0174, VQ: 0.0825, Perplexity: 62.39
Epoch 264/500, Batch 300/782, Loss: 0.0978, Recon: 0.0182, VQ: 0.0797, Perplexity: 61.56
Epoch 264/500, Batch 400/782, Loss: 0.0920, Recon: 0.0174, VQ: 0.0746, Perplexity: 60.93
Epoch 264/500, Batch 500/782, Loss: 0.1030, Recon: 0.0182, VQ: 0.0849, Perplexity: 61.34
Epoch 264/500, Batch 600/782, Loss: 0.1001, Recon: 0.0185, VQ: 0.0816, Per

plexity: 61.98

Epoch 264/500, Batch 700/782, Loss: 0.0943, Recon: 0.0167, VQ: 0.0776, Perplexity: 60.46

=== Epoch 264/500 Summary ===

Train Loss: 0.1021 | Val Loss: 0.1021

Train Recon: 0.0183 | Val Recon: 0.0199

Train VQ: 0.0838 | Val VQ: 0.0823

Train Perplexity: 61.35 | Val Perplexity: 61.26

No improvement. Patience: 7/20

Best Loss: 0.1016 (Epoch 257)

Epoch 265/500, Batch 100/782, Loss: 0.0951, Recon: 0.0175, VQ: 0.0776, Perplexity: 59.34

Epoch 265/500, Batch 200/782, Loss: 0.1051, Recon: 0.0194, VQ: 0.0857, Perplexity: 64.04

Epoch 265/500, Batch 300/782, Loss: 0.0948, Recon: 0.0185, VQ: 0.0763, Perplexity: 62.07

Epoch 265/500, Batch 400/782, Loss: 0.1048, Recon: 0.0199, VQ: 0.0849, Perplexity: 63.36

Epoch 265/500, Batch 500/782, Loss: 0.0931, Recon: 0.0175, VQ: 0.0756, Perplexity: 61.13

Epoch 265/500, Batch 600/782, Loss: 0.0962, Recon: 0.0185, VQ: 0.0778, Perplexity: 61.04

Epoch 265/500, Batch 700/782, Loss: 0.1033, Recon: 0.0191, VQ: 0.0842, Perplexity: 61.05

=== Epoch 265/500 Summary ===

Train Loss: 0.1017 | Val Loss: 0.1030

Train Recon: 0.0183 | Val Recon: 0.0199

Train VQ: 0.0834 | Val VQ: 0.0830

Train Perplexity: 61.40 | Val Perplexity: 61.04

No improvement. Patience: 8/20

Best Loss: 0.1016 (Epoch 257)

Epoch 266/500, Batch 100/782, Loss: 0.0937, Recon: 0.0171, VQ: 0.0766, Perplexity: 61.73

Epoch 266/500, Batch 200/782, Loss: 0.0860, Recon: 0.0164, VQ: 0.0696, Perplexity: 60.30

Epoch 266/500, Batch 300/782, Loss: 0.0983, Recon: 0.0181, VQ: 0.0803, Perplexity: 59.34

Epoch 266/500, Batch 400/782, Loss: 0.1160, Recon: 0.0208, VQ: 0.0951, Perplexity: 64.72

Epoch 266/500, Batch 500/782, Loss: 0.1145, Recon: 0.0198, VQ: 0.0947, Perplexity: 62.02

Epoch 266/500, Batch 600/782, Loss: 0.0919, Recon: 0.0174, VQ: 0.0745, Perplexity: 61.29

Epoch 266/500, Batch 700/782, Loss: 0.1034, Recon: 0.0185, VQ: 0.0849, Perplexity: 61.62

=== Epoch 266/500 Summary ===

Train Loss: 0.1016 | Val Loss: 0.1026

Train Recon: 0.0183 | Val Recon: 0.0199

Train VQ: 0.0834 | Val VQ: 0.0827

Train Perplexity: 61.51 | Val Perplexity: 61.53

No improvement. Patience: 9/20

Best Loss: 0.1016 (Epoch 257)

Epoch 267/500, Batch 100/782, Loss: 0.0859, Recon: 0.0162, VQ: 0.0696, Perplexity: 59.25

Epoch 267/500, Batch 200/782, Loss: 0.1047, Recon: 0.0200, VQ: 0.0847, Perplexity: 63.30
Epoch 267/500, Batch 300/782, Loss: 0.1044, Recon: 0.0194, VQ: 0.0850, Perplexity: 64.14
Epoch 267/500, Batch 400/782, Loss: 0.1075, Recon: 0.0200, VQ: 0.0875, Perplexity: 62.57
Epoch 267/500, Batch 500/782, Loss: 0.1118, Recon: 0.0191, VQ: 0.0928, Perplexity: 63.23
Epoch 267/500, Batch 600/782, Loss: 0.0992, Recon: 0.0182, VQ: 0.0810, Perplexity: 63.23
Epoch 267/500, Batch 700/782, Loss: 0.1071, Recon: 0.0202, VQ: 0.0869, Perplexity: 63.07

=== Epoch 267/500 Summary ===

Train Loss: 0.1015 | Val Loss: 0.1030
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0833 | Val VQ: 0.0832
Train Perplexity: 61.57 | Val Perplexity: 61.77
No improvement. Patience: 10/20
Best Loss: 0.1016 (Epoch 257)

Epoch 268/500, Batch 100/782, Loss: 0.1051, Recon: 0.0198, VQ: 0.0853, Perplexity: 63.76
Epoch 268/500, Batch 200/782, Loss: 0.1173, Recon: 0.0180, VQ: 0.0993, Perplexity: 62.01
Epoch 268/500, Batch 300/782, Loss: 0.1110, Recon: 0.0185, VQ: 0.0925, Perplexity: 62.32
Epoch 268/500, Batch 400/782, Loss: 0.1166, Recon: 0.0202, VQ: 0.0964, Perplexity: 62.07
Epoch 268/500, Batch 500/782, Loss: 0.0889, Recon: 0.0168, VQ: 0.0721, Perplexity: 59.88
Epoch 268/500, Batch 600/782, Loss: 0.1010, Recon: 0.0186, VQ: 0.0824, Perplexity: 62.46
Epoch 268/500, Batch 700/782, Loss: 0.1157, Recon: 0.0195, VQ: 0.0963, Perplexity: 62.79

=== Epoch 268/500 Summary ===

Train Loss: 0.1015 | Val Loss: 0.1020
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0832 | Val VQ: 0.0821
Train Perplexity: 61.66 | Val Perplexity: 61.20
No improvement. Patience: 11/20
Best Loss: 0.1016 (Epoch 257)

Epoch 269/500, Batch 100/782, Loss: 0.1094, Recon: 0.0193, VQ: 0.0901, Perplexity: 62.36
Epoch 269/500, Batch 200/782, Loss: 0.1000, Recon: 0.0188, VQ: 0.0812, Perplexity: 61.79
Epoch 269/500, Batch 300/782, Loss: 0.1095, Recon: 0.0184, VQ: 0.0911, Perplexity: 62.54
Epoch 269/500, Batch 400/782, Loss: 0.0943, Recon: 0.0177, VQ: 0.0766, Perplexity: 59.78
Epoch 269/500, Batch 500/782, Loss: 0.1058, Recon: 0.0178, VQ: 0.0880, Perplexity: 60.03
Epoch 269/500, Batch 600/782, Loss: 0.1109, Recon: 0.0190, VQ: 0.0919, Perplexity: 64.03
Epoch 269/500, Batch 700/782, Loss: 0.0964, Recon: 0.0176, VQ: 0.0788, Perplexity: 61.08

=== Epoch 269/500 Summary ===

Train Loss: 0.1014 | Val Loss: 0.1023
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0832 | Val VQ: 0.0824
Train Perplexity: 61.72 | Val Perplexity: 61.27
No improvement. Patience: 12/20
Best Loss: 0.1016 (Epoch 257)

Epoch 270/500, Batch 100/782, Loss: 0.1059, Recon: 0.0196, VQ: 0.0863, Perplexity: 64.04
Epoch 270/500, Batch 200/782, Loss: 0.0958, Recon: 0.0160, VQ: 0.0799, Perplexity: 58.97
Epoch 270/500, Batch 300/782, Loss: 0.0993, Recon: 0.0189, VQ: 0.0805, Perplexity: 62.41
Epoch 270/500, Batch 400/782, Loss: 0.1047, Recon: 0.0183, VQ: 0.0864, Perplexity: 61.29
Epoch 270/500, Batch 500/782, Loss: 0.1031, Recon: 0.0186, VQ: 0.0846, Perplexity: 62.86
Epoch 270/500, Batch 600/782, Loss: 0.1040, Recon: 0.0196, VQ: 0.0844, Perplexity: 64.21
Epoch 270/500, Batch 700/782, Loss: 0.1011, Recon: 0.0196, VQ: 0.0815, Perplexity: 61.97

=== Epoch 270/500 Summary ===
Train Loss: 0.1013 | Val Loss: 0.1029
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0831 | Val VQ: 0.0831
Train Perplexity: 61.75 | Val Perplexity: 62.02
No improvement. Patience: 13/20
Best Loss: 0.1016 (Epoch 257)

Epoch 271/500, Batch 100/782, Loss: 0.1109, Recon: 0.0198, VQ: 0.0910, Perplexity: 65.03
Epoch 271/500, Batch 200/782, Loss: 0.0987, Recon: 0.0182, VQ: 0.0805, Perplexity: 62.98
Epoch 271/500, Batch 300/782, Loss: 0.0970, Recon: 0.0170, VQ: 0.0800, Perplexity: 61.36
Epoch 271/500, Batch 400/782, Loss: 0.0978, Recon: 0.0176, VQ: 0.0802, Perplexity: 60.51
Epoch 271/500, Batch 500/782, Loss: 0.0961, Recon: 0.0182, VQ: 0.0780, Perplexity: 63.02
Epoch 271/500, Batch 600/782, Loss: 0.0989, Recon: 0.0188, VQ: 0.0801, Perplexity: 61.68
Epoch 271/500, Batch 700/782, Loss: 0.1003, Recon: 0.0180, VQ: 0.0824, Perplexity: 62.43

=== Epoch 271/500 Summary ===
Train Loss: 0.1014 | Val Loss: 0.1021
Train Recon: 0.0183 | Val Recon: 0.0198
Train VQ: 0.0831 | Val VQ: 0.0822
Train Perplexity: 61.80 | Val Perplexity: 61.77
No improvement. Patience: 14/20
Best Loss: 0.1016 (Epoch 257)

Epoch 272/500, Batch 100/782, Loss: 0.0985, Recon: 0.0178, VQ: 0.0807, Perplexity: 61.36
Epoch 272/500, Batch 200/782, Loss: 0.0928, Recon: 0.0173, VQ: 0.0754, Perplexity: 61.71
Epoch 272/500, Batch 300/782, Loss: 0.0956, Recon: 0.0187, VQ: 0.0769, Perplexity: 61.62
Epoch 272/500, Batch 400/782, Loss: 0.1006, Recon: 0.0183, VQ: 0.0824, Per

plexity: 62.89
Epoch 272/500, Batch 500/782, Loss: 0.1032, Recon: 0.0185, VQ: 0.0847, Per
plexity: 63.58
Epoch 272/500, Batch 600/782, Loss: 0.0963, Recon: 0.0181, VQ: 0.0782, Per
plexity: 60.45
Epoch 272/500, Batch 700/782, Loss: 0.0980, Recon: 0.0181, VQ: 0.0799, Per
plexity: 61.81

=== Epoch 272/500 Summary ===

Train Loss: 0.1013 | Val Loss: 0.1014
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0831 | Val VQ: 0.0815
Train Perplexity: 61.89 | Val Perplexity: 61.85
✓ Best model saved! (Val Loss: 0.1014)
Best Loss: 0.1014 (Epoch 272)

Epoch 273/500, Batch 100/782, Loss: 0.0947, Recon: 0.0185, VQ: 0.0762, Per
plexity: 61.15
Epoch 273/500, Batch 200/782, Loss: 0.1104, Recon: 0.0194, VQ: 0.0910, Per
plexity: 63.06
Epoch 273/500, Batch 300/782, Loss: 0.0937, Recon: 0.0176, VQ: 0.0761, Per
plexity: 61.69
Epoch 273/500, Batch 400/782, Loss: 0.1063, Recon: 0.0192, VQ: 0.0870, Per
plexity: 64.41
Epoch 273/500, Batch 500/782, Loss: 0.0894, Recon: 0.0171, VQ: 0.0723, Per
plexity: 59.11
Epoch 273/500, Batch 600/782, Loss: 0.0997, Recon: 0.0184, VQ: 0.0813, Per
plexity: 62.67
Epoch 273/500, Batch 700/782, Loss: 0.1012, Recon: 0.0177, VQ: 0.0836, Per
plexity: 61.43

=== Epoch 273/500 Summary ===

Train Loss: 0.1013 | Val Loss: 0.1013
Train Recon: 0.0183 | Val Recon: 0.0199
Train VQ: 0.0830 | Val VQ: 0.0814
Train Perplexity: 61.92 | Val Perplexity: 61.69
✓ Best model saved! (Val Loss: 0.1013)
Best Loss: 0.1013 (Epoch 273)

Epoch 274/500, Batch 100/782, Loss: 0.0985, Recon: 0.0189, VQ: 0.0796, Per
plexity: 61.78
Epoch 274/500, Batch 200/782, Loss: 0.0929, Recon: 0.0180, VQ: 0.0749, Per
plexity: 60.45
Epoch 274/500, Batch 300/782, Loss: 0.1268, Recon: 0.0198, VQ: 0.1071, Per
plexity: 65.37
Epoch 274/500, Batch 400/782, Loss: 0.1059, Recon: 0.0191, VQ: 0.0868, Per
plexity: 62.81
Epoch 274/500, Batch 500/782, Loss: 0.1087, Recon: 0.0200, VQ: 0.0888, Per
plexity: 65.16
Epoch 274/500, Batch 600/782, Loss: 0.1115, Recon: 0.0188, VQ: 0.0927, Per
plexity: 62.43
Epoch 274/500, Batch 700/782, Loss: 0.0887, Recon: 0.0171, VQ: 0.0716, Per
plexity: 61.54

=== Epoch 274/500 Summary ===

Train Loss: 0.1009 | Val Loss: 0.1028
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0826 | Val VQ: 0.0829
Train Perplexity: 61.99 | Val Perplexity: 62.02
No improvement. Patience: 1/20

Best Loss: 0.1013 (Epoch 273)

Epoch 275/500, Batch 100/782, Loss: 0.1005, Recon: 0.0179, VQ: 0.0826, Perplexity: 63.29
Epoch 275/500, Batch 200/782, Loss: 0.1114, Recon: 0.0207, VQ: 0.0907, Perplexity: 64.98
Epoch 275/500, Batch 300/782, Loss: 0.1024, Recon: 0.0190, VQ: 0.0834, Perplexity: 64.89
Epoch 275/500, Batch 400/782, Loss: 0.1112, Recon: 0.0187, VQ: 0.0925, Perplexity: 62.25
Epoch 275/500, Batch 500/782, Loss: 0.1047, Recon: 0.0185, VQ: 0.0862, Perplexity: 63.15
Epoch 275/500, Batch 600/782, Loss: 0.1070, Recon: 0.0181, VQ: 0.0888, Perplexity: 62.20
Epoch 275/500, Batch 700/782, Loss: 0.0951, Recon: 0.0173, VQ: 0.0778, Perplexity: 59.94

=== Epoch 275/500 Summary ===

Train Loss: 0.1010 | Val Loss: 0.1020
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0827 | Val VQ: 0.0821
Train Perplexity: 62.00 | Val Perplexity: 61.73
No improvement. Patience: 2/20
Best Loss: 0.1013 (Epoch 273)

Epoch 276/500, Batch 100/782, Loss: 0.0940, Recon: 0.0176, VQ: 0.0764, Perplexity: 61.50
Epoch 276/500, Batch 200/782, Loss: 0.1755, Recon: 0.0186, VQ: 0.1569, Perplexity: 63.54
Epoch 276/500, Batch 300/782, Loss: 0.0834, Recon: 0.0160, VQ: 0.0673, Perplexity: 59.40
Epoch 276/500, Batch 400/782, Loss: 0.0892, Recon: 0.0175, VQ: 0.0717, Perplexity: 60.76
Epoch 276/500, Batch 500/782, Loss: 0.1019, Recon: 0.0186, VQ: 0.0833, Perplexity: 62.69
Epoch 276/500, Batch 600/782, Loss: 0.0983, Recon: 0.0186, VQ: 0.0797, Perplexity: 63.30
Epoch 276/500, Batch 700/782, Loss: 0.1050, Recon: 0.0196, VQ: 0.0854, Perplexity: 63.91

=== Epoch 276/500 Summary ===

Train Loss: 0.1007 | Val Loss: 0.1012
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0825 | Val VQ: 0.0814
Train Perplexity: 62.06 | Val Perplexity: 61.98
✓ Best model saved! (Val Loss: 0.1012)
Best Loss: 0.1012 (Epoch 276)

Epoch 277/500, Batch 100/782, Loss: 0.0961, Recon: 0.0171, VQ: 0.0789, Perplexity: 61.65
Epoch 277/500, Batch 200/782, Loss: 0.1002, Recon: 0.0187, VQ: 0.0816, Perplexity: 62.59
Epoch 277/500, Batch 300/782, Loss: 0.1304, Recon: 0.0205, VQ: 0.1099, Perplexity: 65.07
Epoch 277/500, Batch 400/782, Loss: 0.1412, Recon: 0.0176, VQ: 0.1236, Perplexity: 60.36
Epoch 277/500, Batch 500/782, Loss: 0.0919, Recon: 0.0174, VQ: 0.0746, Perplexity: 61.40
Epoch 277/500, Batch 600/782, Loss: 0.0998, Recon: 0.0189, VQ: 0.0809, Perplexity: 62.21

Epoch 277/500, Batch 700/782, Loss: 0.0957, Recon: 0.0170, VQ: 0.0787, Perplexity: 62.14

=== Epoch 277/500 Summary ===

Train Loss: 0.1010 | Val Loss: 0.1010
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0827 | Val VQ: 0.0812
Train Perplexity: 62.13 | Val Perplexity: 62.09
✓ Best model saved! (Val Loss: 0.1010)
Best Loss: 0.1010 (Epoch 277)

Epoch 278/500, Batch 100/782, Loss: 0.0909, Recon: 0.0178, VQ: 0.0731, Perplexity: 60.40
Epoch 278/500, Batch 200/782, Loss: 0.1018, Recon: 0.0187, VQ: 0.0830, Perplexity: 64.31
Epoch 278/500, Batch 300/782, Loss: 0.0957, Recon: 0.0175, VQ: 0.0782, Perplexity: 61.08
Epoch 278/500, Batch 400/782, Loss: 0.1038, Recon: 0.0184, VQ: 0.0855, Perplexity: 62.74
Epoch 278/500, Batch 500/782, Loss: 0.0858, Recon: 0.0166, VQ: 0.0692, Perplexity: 59.50
Epoch 278/500, Batch 600/782, Loss: 0.0964, Recon: 0.0169, VQ: 0.0796, Perplexity: 61.12
Epoch 278/500, Batch 700/782, Loss: 0.0885, Recon: 0.0181, VQ: 0.0704, Perplexity: 60.01

=== Epoch 278/500 Summary ===

Train Loss: 0.1006 | Val Loss: 0.1020
Train Recon: 0.0182 | Val Recon: 0.0199
Train VQ: 0.0824 | Val VQ: 0.0821
Train Perplexity: 62.13 | Val Perplexity: 62.34
No improvement. Patience: 1/20
Best Loss: 0.1010 (Epoch 277)

Epoch 279/500, Batch 100/782, Loss: 0.1126, Recon: 0.0178, VQ: 0.0948, Perplexity: 63.47
Epoch 279/500, Batch 200/782, Loss: 0.1003, Recon: 0.0192, VQ: 0.0812, Perplexity: 63.31
Epoch 279/500, Batch 300/782, Loss: 0.1024, Recon: 0.0189, VQ: 0.0835, Perplexity: 63.77
Epoch 279/500, Batch 400/782, Loss: 0.1008, Recon: 0.0174, VQ: 0.0834, Perplexity: 61.52
Epoch 279/500, Batch 500/782, Loss: 0.0880, Recon: 0.0171, VQ: 0.0709, Perplexity: 61.18
Epoch 279/500, Batch 600/782, Loss: 0.0925, Recon: 0.0178, VQ: 0.0747, Perplexity: 63.77
Epoch 279/500, Batch 700/782, Loss: 0.0986, Recon: 0.0180, VQ: 0.0805, Perplexity: 61.38

=== Epoch 279/500 Summary ===

Train Loss: 0.1007 | Val Loss: 0.1014
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0825 | Val VQ: 0.0815
Train Perplexity: 62.30 | Val Perplexity: 62.62
No improvement. Patience: 2/20
Best Loss: 0.1010 (Epoch 277)

Epoch 280/500, Batch 100/782, Loss: 0.0993, Recon: 0.0195, VQ: 0.0798, Perplexity: 64.41
Epoch 280/500, Batch 200/782, Loss: 0.1033, Recon: 0.0190, VQ: 0.0842, Per

plexity: 62.66
Epoch 280/500, Batch 300/782, Loss: 0.0996, Recon: 0.0182, VQ: 0.0814, Per
plexity: 62.41
Epoch 280/500, Batch 400/782, Loss: 0.0981, Recon: 0.0177, VQ: 0.0804, Per
plexity: 62.27
Epoch 280/500, Batch 500/782, Loss: 0.1062, Recon: 0.0182, VQ: 0.0880, Per
plexity: 60.15
Epoch 280/500, Batch 600/782, Loss: 0.1006, Recon: 0.0186, VQ: 0.0820, Per
plexity: 63.51
Epoch 280/500, Batch 700/782, Loss: 0.0957, Recon: 0.0182, VQ: 0.0775, Per
plexity: 62.46

=== Epoch 280/500 Summary ===
Train Loss: 0.1008 | Val Loss: 0.1010
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0826 | Val VQ: 0.0812
Train Perplexity: 62.56 | Val Perplexity: 62.52
No improvement. Patience: 3/20
Best Loss: 0.1010 (Epoch 277)

Epoch 281/500, Batch 100/782, Loss: 0.0991, Recon: 0.0175, VQ: 0.0816, Per
plexity: 61.57
Epoch 281/500, Batch 200/782, Loss: 0.0916, Recon: 0.0172, VQ: 0.0745, Per
plexity: 61.89
Epoch 281/500, Batch 300/782, Loss: 0.0932, Recon: 0.0181, VQ: 0.0751, Per
plexity: 62.38
Epoch 281/500, Batch 400/782, Loss: 0.1010, Recon: 0.0193, VQ: 0.0817, Per
plexity: 63.72
Epoch 281/500, Batch 500/782, Loss: 0.1020, Recon: 0.0181, VQ: 0.0839, Per
plexity: 62.99
Epoch 281/500, Batch 600/782, Loss: 0.0999, Recon: 0.0186, VQ: 0.0813, Per
plexity: 63.61
Epoch 281/500, Batch 700/782, Loss: 0.1004, Recon: 0.0188, VQ: 0.0816, Per
plexity: 64.29

=== Epoch 281/500 Summary ===
Train Loss: 0.1008 | Val Loss: 0.1011
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0826 | Val VQ: 0.0814
Train Perplexity: 62.85 | Val Perplexity: 62.82
No improvement. Patience: 4/20
Best Loss: 0.1010 (Epoch 277)

Epoch 282/500, Batch 100/782, Loss: 0.1003, Recon: 0.0170, VQ: 0.0833, Per
plexity: 62.24
Epoch 282/500, Batch 200/782, Loss: 0.1049, Recon: 0.0185, VQ: 0.0864, Per
plexity: 64.16
Epoch 282/500, Batch 300/782, Loss: 0.0958, Recon: 0.0172, VQ: 0.0785, Per
plexity: 63.72
Epoch 282/500, Batch 400/782, Loss: 0.0980, Recon: 0.0192, VQ: 0.0788, Per
plexity: 61.88
Epoch 282/500, Batch 500/782, Loss: 0.0942, Recon: 0.0180, VQ: 0.0762, Per
plexity: 61.13
Epoch 282/500, Batch 600/782, Loss: 0.0864, Recon: 0.0164, VQ: 0.0701, Per
plexity: 59.84
Epoch 282/500, Batch 700/782, Loss: 0.1106, Recon: 0.0197, VQ: 0.0909, Per
plexity: 66.23

=== Epoch 282/500 Summary ===
Train Loss: 0.1005 | Val Loss: 0.1013

Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0823 | Val VQ: 0.0815
Train Perplexity: 62.94 | Val Perplexity: 63.00
No improvement. Patience: 5/20
Best Loss: 0.1010 (Epoch 277)

Epoch 283/500, Batch 100/782, Loss: 0.1779, Recon: 0.0193, VQ: 0.1586, Perplexity: 64.35
Epoch 283/500, Batch 200/782, Loss: 0.1020, Recon: 0.0178, VQ: 0.0843, Perplexity: 63.08
Epoch 283/500, Batch 300/782, Loss: 0.1028, Recon: 0.0187, VQ: 0.0841, Perplexity: 63.19
Epoch 283/500, Batch 400/782, Loss: 0.0937, Recon: 0.0183, VQ: 0.0754, Perplexity: 62.94
Epoch 283/500, Batch 500/782, Loss: 0.0994, Recon: 0.0176, VQ: 0.0818, Perplexity: 63.86
Epoch 283/500, Batch 600/782, Loss: 0.1066, Recon: 0.0192, VQ: 0.0874, Perplexity: 63.96
Epoch 283/500, Batch 700/782, Loss: 0.1031, Recon: 0.0186, VQ: 0.0845, Perplexity: 65.42

=== Epoch 283/500 Summary ===
Train Loss: 0.1006 | Val Loss: 0.1005
Train Recon: 0.0182 | Val Recon: 0.0197
Train VQ: 0.0824 | Val VQ: 0.0808
Train Perplexity: 63.00 | Val Perplexity: 62.84
✓ Best model saved! (Val Loss: 0.1005)
Best Loss: 0.1005 (Epoch 283)

Epoch 284/500, Batch 100/782, Loss: 0.1053, Recon: 0.0187, VQ: 0.0867, Perplexity: 61.85
Epoch 284/500, Batch 200/782, Loss: 0.0836, Recon: 0.0167, VQ: 0.0669, Perplexity: 58.48
Epoch 284/500, Batch 300/782, Loss: 0.0971, Recon: 0.0176, VQ: 0.0794, Perplexity: 63.24
Epoch 284/500, Batch 400/782, Loss: 0.0995, Recon: 0.0178, VQ: 0.0817, Perplexity: 62.80
Epoch 284/500, Batch 500/782, Loss: 0.1021, Recon: 0.0190, VQ: 0.0831, Perplexity: 64.83
Epoch 284/500, Batch 600/782, Loss: 0.0922, Recon: 0.0182, VQ: 0.0740, Perplexity: 63.18
Epoch 284/500, Batch 700/782, Loss: 0.1093, Recon: 0.0189, VQ: 0.0904, Perplexity: 66.06

=== Epoch 284/500 Summary ===
Train Loss: 0.1004 | Val Loss: 0.1007
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0822 | Val VQ: 0.0809
Train Perplexity: 63.10 | Val Perplexity: 63.05
No improvement. Patience: 1/20
Best Loss: 0.1005 (Epoch 283)

Epoch 285/500, Batch 100/782, Loss: 0.0974, Recon: 0.0186, VQ: 0.0789, Perplexity: 64.60
Epoch 285/500, Batch 200/782, Loss: 0.0912, Recon: 0.0168, VQ: 0.0744, Perplexity: 63.08
Epoch 285/500, Batch 300/782, Loss: 0.1072, Recon: 0.0201, VQ: 0.0872, Perplexity: 65.28
Epoch 285/500, Batch 400/782, Loss: 0.0961, Recon: 0.0178, VQ: 0.0783, Perplexity: 63.30

Epoch 285/500, Batch 500/782, Loss: 0.1002, Recon: 0.0186, VQ: 0.0815, Perplexity: 64.73
Epoch 285/500, Batch 600/782, Loss: 0.0969, Recon: 0.0182, VQ: 0.0787, Perplexity: 63.97
Epoch 285/500, Batch 700/782, Loss: 0.0947, Recon: 0.0176, VQ: 0.0771, Perplexity: 61.98

=== Epoch 285/500 Summary ===

Train Loss: 0.1003 | Val Loss: 0.1022
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0821 | Val VQ: 0.0825
Train Perplexity: 63.11 | Val Perplexity: 63.34
No improvement. Patience: 2/20
Best Loss: 0.1005 (Epoch 283)

Epoch 286/500, Batch 100/782, Loss: 0.1046, Recon: 0.0184, VQ: 0.0862, Perplexity: 61.95
Epoch 286/500, Batch 200/782, Loss: 0.0893, Recon: 0.0169, VQ: 0.0724, Perplexity: 62.85
Epoch 286/500, Batch 300/782, Loss: 0.0942, Recon: 0.0171, VQ: 0.0772, Perplexity: 61.39
Epoch 286/500, Batch 400/782, Loss: 0.0975, Recon: 0.0174, VQ: 0.0801, Perplexity: 59.82
Epoch 286/500, Batch 500/782, Loss: 0.0977, Recon: 0.0192, VQ: 0.0786, Perplexity: 64.75
Epoch 286/500, Batch 600/782, Loss: 0.0942, Recon: 0.0176, VQ: 0.0766, Perplexity: 61.27
Epoch 286/500, Batch 700/782, Loss: 0.1045, Recon: 0.0183, VQ: 0.0862, Perplexity: 63.33

=== Epoch 286/500 Summary ===

Train Loss: 0.1004 | Val Loss: 0.1006
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0822 | Val VQ: 0.0808
Train Perplexity: 63.18 | Val Perplexity: 63.24
No improvement. Patience: 3/20
Best Loss: 0.1005 (Epoch 283)

Epoch 287/500, Batch 100/782, Loss: 0.2424, Recon: 0.0199, VQ: 0.2226, Perplexity: 62.56
Epoch 287/500, Batch 200/782, Loss: 0.0841, Recon: 0.0163, VQ: 0.0678, Perplexity: 58.16
Epoch 287/500, Batch 300/782, Loss: 0.0853, Recon: 0.0175, VQ: 0.0678, Perplexity: 60.24
Epoch 287/500, Batch 400/782, Loss: 0.0998, Recon: 0.0185, VQ: 0.0812, Perplexity: 65.63
Epoch 287/500, Batch 500/782, Loss: 0.0916, Recon: 0.0174, VQ: 0.0742, Perplexity: 63.46
Epoch 287/500, Batch 600/782, Loss: 0.1010, Recon: 0.0185, VQ: 0.0825, Perplexity: 66.02
Epoch 287/500, Batch 700/782, Loss: 0.0982, Recon: 0.0189, VQ: 0.0793, Perplexity: 64.29

=== Epoch 287/500 Summary ===

Train Loss: 0.1001 | Val Loss: 0.1003
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0819 | Val VQ: 0.0805
Train Perplexity: 63.24 | Val Perplexity: 62.86
✓ Best model saved! (Val Loss: 0.1003)
Best Loss: 0.1003 (Epoch 287)

Epoch 288/500, Batch 100/782, Loss: 0.1019, Recon: 0.0191, VQ: 0.0828, Perplexity: 64.14
Epoch 288/500, Batch 200/782, Loss: 0.1248, Recon: 0.0182, VQ: 0.1066, Perplexity: 61.74
Epoch 288/500, Batch 300/782, Loss: 0.0935, Recon: 0.0175, VQ: 0.0760, Perplexity: 61.61
Epoch 288/500, Batch 400/782, Loss: 0.1129, Recon: 0.0197, VQ: 0.0932, Perplexity: 64.84
Epoch 288/500, Batch 500/782, Loss: 0.0951, Recon: 0.0180, VQ: 0.0771, Perplexity: 62.23
Epoch 288/500, Batch 600/782, Loss: 0.1000, Recon: 0.0192, VQ: 0.0808, Perplexity: 65.35
Epoch 288/500, Batch 700/782, Loss: 0.1107, Recon: 0.0194, VQ: 0.0913, Perplexity: 65.79

=== Epoch 288/500 Summary ===

Train Loss: 0.1003 | Val Loss: 0.0996
Train Recon: 0.0182 | Val Recon: 0.0198
Train VQ: 0.0821 | Val VQ: 0.0798
Train Perplexity: 63.28 | Val Perplexity: 62.80
✓ Best model saved! (Val Loss: 0.0996)
Best Loss: 0.0996 (Epoch 288)

Epoch 289/500, Batch 100/782, Loss: 0.0909, Recon: 0.0169, VQ: 0.0741, Perplexity: 61.77
Epoch 289/500, Batch 200/782, Loss: 0.0934, Recon: 0.0183, VQ: 0.0751, Perplexity: 61.62
Epoch 289/500, Batch 300/782, Loss: 0.0964, Recon: 0.0176, VQ: 0.0788, Perplexity: 63.50
Epoch 289/500, Batch 400/782, Loss: 0.1060, Recon: 0.0198, VQ: 0.0862, Perplexity: 65.73
Epoch 289/500, Batch 500/782, Loss: 0.1076, Recon: 0.0194, VQ: 0.0881, Perplexity: 64.58
Epoch 289/500, Batch 600/782, Loss: 0.0938, Recon: 0.0172, VQ: 0.0765, Perplexity: 60.76
Epoch 289/500, Batch 700/782, Loss: 0.0936, Recon: 0.0173, VQ: 0.0763, Perplexity: 61.29

=== Epoch 289/500 Summary ===

Train Loss: 0.1001 | Val Loss: 0.1005
Train Recon: 0.0182 | Val Recon: 0.0197
Train VQ: 0.0819 | Val VQ: 0.0808
Train Perplexity: 63.31 | Val Perplexity: 63.09
No improvement. Patience: 1/20
Best Loss: 0.0996 (Epoch 288)

Epoch 290/500, Batch 100/782, Loss: 0.1583, Recon: 0.0179, VQ: 0.1403, Perplexity: 63.77
Epoch 290/500, Batch 200/782, Loss: 0.0992, Recon: 0.0177, VQ: 0.0815, Perplexity: 63.53
Epoch 290/500, Batch 300/782, Loss: 0.0983, Recon: 0.0181, VQ: 0.0803, Perplexity: 63.17
Epoch 290/500, Batch 400/782, Loss: 0.1133, Recon: 0.0198, VQ: 0.0936, Perplexity: 65.29
Epoch 290/500, Batch 500/782, Loss: 0.1178, Recon: 0.0197, VQ: 0.0980, Perplexity: 65.00
Epoch 290/500, Batch 600/782, Loss: 0.0986, Recon: 0.0177, VQ: 0.0808, Perplexity: 62.39
Epoch 290/500, Batch 700/782, Loss: 0.1009, Recon: 0.0187, VQ: 0.0822, Per

plexity: 63.04

=== Epoch 290/500 Summary ===

Train Loss: 0.1003 | Val Loss: 0.1012

Train Recon: 0.0182 | Val Recon: 0.0198

Train VQ: 0.0821 | Val VQ: 0.0814

Train Perplexity: 63.31 | Val Perplexity: 63.26

No improvement. Patience: 2/20

Best Loss: 0.0996 (Epoch 288)

Epoch 291/500, Batch 100/782, Loss: 0.1127, Recon: 0.0200, VQ: 0.0928, Perplexity: 65.48

Epoch 291/500, Batch 200/782, Loss: 0.1240, Recon: 0.0192, VQ: 0.1048, Perplexity: 65.33

Epoch 291/500, Batch 300/782, Loss: 0.0825, Recon: 0.0163, VQ: 0.0662, Perplexity: 60.12

Epoch 291/500, Batch 400/782, Loss: 0.0901, Recon: 0.0176, VQ: 0.0725, Perplexity: 62.99

Epoch 291/500, Batch 500/782, Loss: 0.0922, Recon: 0.0180, VQ: 0.0742, Perplexity: 63.75

Epoch 291/500, Batch 600/782, Loss: 0.0905, Recon: 0.0173, VQ: 0.0732, Perplexity: 63.00

Epoch 291/500, Batch 700/782, Loss: 0.0878, Recon: 0.0177, VQ: 0.0701, Perplexity: 61.93

=== Epoch 291/500 Summary ===

Train Loss: 0.1002 | Val Loss: 0.1005

Train Recon: 0.0181 | Val Recon: 0.0197

Train VQ: 0.0821 | Val VQ: 0.0807

Train Perplexity: 63.33 | Val Perplexity: 63.35

No improvement. Patience: 3/20

Best Loss: 0.0996 (Epoch 288)

Epoch 292/500, Batch 100/782, Loss: 0.0995, Recon: 0.0192, VQ: 0.0803, Perplexity: 65.28

Epoch 292/500, Batch 200/782, Loss: 0.0927, Recon: 0.0183, VQ: 0.0744, Perplexity: 64.78

Epoch 292/500, Batch 300/782, Loss: 0.1008, Recon: 0.0166, VQ: 0.0842, Perplexity: 62.54

Epoch 292/500, Batch 400/782, Loss: 0.1094, Recon: 0.0197, VQ: 0.0897, Perplexity: 66.17

Epoch 292/500, Batch 500/782, Loss: 0.1135, Recon: 0.0177, VQ: 0.0958, Perplexity: 62.32

Epoch 292/500, Batch 600/782, Loss: 0.0950, Recon: 0.0181, VQ: 0.0769, Perplexity: 63.61

Epoch 292/500, Batch 700/782, Loss: 0.0940, Recon: 0.0168, VQ: 0.0772, Perplexity: 60.52

=== Epoch 292/500 Summary ===

Train Loss: 0.1001 | Val Loss: 0.1014

Train Recon: 0.0182 | Val Recon: 0.0197

Train VQ: 0.0819 | Val VQ: 0.0817

Train Perplexity: 63.40 | Val Perplexity: 63.56

No improvement. Patience: 4/20

Best Loss: 0.0996 (Epoch 288)

Epoch 293/500, Batch 100/782, Loss: 0.0971, Recon: 0.0182, VQ: 0.0789, Perplexity: 64.42

Epoch 293/500, Batch 200/782, Loss: 0.0965, Recon: 0.0186, VQ: 0.0778, Perplexity: 64.46

Epoch 293/500, Batch 300/782, Loss: 0.1026, Recon: 0.0197, VQ: 0.0829, Perplexity: 64.46
Epoch 293/500, Batch 400/782, Loss: 0.1108, Recon: 0.0193, VQ: 0.0915, Perplexity: 65.56
Epoch 293/500, Batch 500/782, Loss: 0.1023, Recon: 0.0200, VQ: 0.0823, Perplexity: 65.36
Epoch 293/500, Batch 600/782, Loss: 0.0928, Recon: 0.0179, VQ: 0.0749, Perplexity: 62.55
Epoch 293/500, Batch 700/782, Loss: 0.0980, Recon: 0.0185, VQ: 0.0795, Perplexity: 64.05

=== Epoch 293/500 Summary ===

Train Loss: 0.1001 | Val Loss: 0.1010
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0820 | Val VQ: 0.0812
Train Perplexity: 63.43 | Val Perplexity: 63.64
No improvement. Patience: 5/20
Best Loss: 0.0996 (Epoch 288)

Epoch 294/500, Batch 100/782, Loss: 0.1088, Recon: 0.0183, VQ: 0.0905, Perplexity: 65.08
Epoch 294/500, Batch 200/782, Loss: 0.0999, Recon: 0.0173, VQ: 0.0827, Perplexity: 61.44
Epoch 294/500, Batch 300/782, Loss: 0.1121, Recon: 0.0184, VQ: 0.0936, Perplexity: 64.68
Epoch 294/500, Batch 400/782, Loss: 0.0857, Recon: 0.0168, VQ: 0.0689, Perplexity: 60.20
Epoch 294/500, Batch 500/782, Loss: 0.0984, Recon: 0.0190, VQ: 0.0793, Perplexity: 64.08
Epoch 294/500, Batch 600/782, Loss: 0.1154, Recon: 0.0204, VQ: 0.0951, Perplexity: 65.63
Epoch 294/500, Batch 700/782, Loss: 0.1116, Recon: 0.0199, VQ: 0.0917, Perplexity: 63.41

=== Epoch 294/500 Summary ===

Train Loss: 0.1002 | Val Loss: 0.1025
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0821 | Val VQ: 0.0828
Train Perplexity: 63.52 | Val Perplexity: 63.37
No improvement. Patience: 6/20
Best Loss: 0.0996 (Epoch 288)

Epoch 295/500, Batch 100/782, Loss: 0.1012, Recon: 0.0186, VQ: 0.0826, Perplexity: 63.99
Epoch 295/500, Batch 200/782, Loss: 0.0918, Recon: 0.0168, VQ: 0.0750, Perplexity: 59.45
Epoch 295/500, Batch 300/782, Loss: 0.0957, Recon: 0.0193, VQ: 0.0764, Perplexity: 63.60
Epoch 295/500, Batch 400/782, Loss: 0.0929, Recon: 0.0174, VQ: 0.0755, Perplexity: 61.48
Epoch 295/500, Batch 500/782, Loss: 0.0814, Recon: 0.0162, VQ: 0.0652, Perplexity: 59.58
Epoch 295/500, Batch 600/782, Loss: 0.1023, Recon: 0.0186, VQ: 0.0837, Perplexity: 63.94
Epoch 295/500, Batch 700/782, Loss: 0.0970, Recon: 0.0178, VQ: 0.0792, Perplexity: 63.31

=== Epoch 295/500 Summary ===

Train Loss: 0.1003 | Val Loss: 0.1008
Train Recon: 0.0181 | Val Recon: 0.0197

Train VQ: 0.0821 | Val VQ: 0.0811
Train Perplexity: 63.54 | Val Perplexity: 63.77
No improvement. Patience: 7/20
Best Loss: 0.0996 (Epoch 288)

Epoch 296/500, Batch 100/782, Loss: 0.0978, Recon: 0.0188, VQ: 0.0791, Perplexity: 61.88
Epoch 296/500, Batch 200/782, Loss: 0.1001, Recon: 0.0178, VQ: 0.0823, Perplexity: 62.34
Epoch 296/500, Batch 300/782, Loss: 0.1026, Recon: 0.0174, VQ: 0.0853, Perplexity: 64.14
Epoch 296/500, Batch 400/782, Loss: 0.1033, Recon: 0.0195, VQ: 0.0837, Perplexity: 64.12
Epoch 296/500, Batch 500/782, Loss: 0.1013, Recon: 0.0194, VQ: 0.0820, Perplexity: 67.39
Epoch 296/500, Batch 600/782, Loss: 0.0998, Recon: 0.0193, VQ: 0.0805, Perplexity: 62.73
Epoch 296/500, Batch 700/782, Loss: 0.0954, Recon: 0.0160, VQ: 0.0794, Perplexity: 61.83

=== Epoch 296/500 Summary ===
Train Loss: 0.1001 | Val Loss: 0.1013
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0820 | Val VQ: 0.0816
Train Perplexity: 63.58 | Val Perplexity: 63.60
No improvement. Patience: 8/20
Best Loss: 0.0996 (Epoch 288)

Epoch 297/500, Batch 100/782, Loss: 0.1066, Recon: 0.0200, VQ: 0.0866, Perplexity: 65.87
Epoch 297/500, Batch 200/782, Loss: 0.0923, Recon: 0.0176, VQ: 0.0747, Perplexity: 63.24
Epoch 297/500, Batch 300/782, Loss: 0.1109, Recon: 0.0197, VQ: 0.0912, Perplexity: 66.18
Epoch 297/500, Batch 400/782, Loss: 0.1003, Recon: 0.0182, VQ: 0.0821, Perplexity: 63.28
Epoch 297/500, Batch 500/782, Loss: 0.0937, Recon: 0.0176, VQ: 0.0761, Perplexity: 62.19
Epoch 297/500, Batch 600/782, Loss: 0.0973, Recon: 0.0180, VQ: 0.0793, Perplexity: 63.77
Epoch 297/500, Batch 700/782, Loss: 0.1048, Recon: 0.0191, VQ: 0.0857, Perplexity: 66.15

=== Epoch 297/500 Summary ===
Train Loss: 0.1001 | Val Loss: 0.1006
Train Recon: 0.0181 | Val Recon: 0.0198
Train VQ: 0.0819 | Val VQ: 0.0809
Train Perplexity: 63.65 | Val Perplexity: 63.10
No improvement. Patience: 9/20
Best Loss: 0.0996 (Epoch 288)

Epoch 298/500, Batch 100/782, Loss: 0.1013, Recon: 0.0184, VQ: 0.0828, Perplexity: 64.57
Epoch 298/500, Batch 200/782, Loss: 0.1018, Recon: 0.0191, VQ: 0.0827, Perplexity: 65.31
Epoch 298/500, Batch 300/782, Loss: 0.1011, Recon: 0.0190, VQ: 0.0821, Perplexity: 65.75
Epoch 298/500, Batch 400/782, Loss: 0.0971, Recon: 0.0181, VQ: 0.0790, Perplexity: 62.87
Epoch 298/500, Batch 500/782, Loss: 0.0911, Recon: 0.0168, VQ: 0.0742, Per

plexity: 61.67
Epoch 298/500, Batch 600/782, Loss: 0.0937, Recon: 0.0172, VQ: 0.0765, Per
plexity: 61.83
Epoch 298/500, Batch 700/782, Loss: 0.0965, Recon: 0.0186, VQ: 0.0779, Per
plexity: 62.14

=== Epoch 298/500 Summary ===
Train Loss: 0.1001 | Val Loss: 0.1012
Train Recon: 0.0181 | Val Recon: 0.0198
Train VQ: 0.0820 | Val VQ: 0.0814
Train Perplexity: 63.68 | Val Perplexity: 63.72
No improvement. Patience: 10/20
Best Loss: 0.0996 (Epoch 288)

Epoch 299/500, Batch 100/782, Loss: 0.0949, Recon: 0.0174, VQ: 0.0774, Per
plexity: 62.78
Epoch 299/500, Batch 200/782, Loss: 0.0975, Recon: 0.0189, VQ: 0.0786, Per
plexity: 65.86
Epoch 299/500, Batch 300/782, Loss: 0.1041, Recon: 0.0191, VQ: 0.0850, Per
plexity: 65.55
Epoch 299/500, Batch 400/782, Loss: 0.0923, Recon: 0.0169, VQ: 0.0753, Per
plexity: 62.72
Epoch 299/500, Batch 500/782, Loss: 0.0853, Recon: 0.0171, VQ: 0.0682, Per
plexity: 60.39
Epoch 299/500, Batch 600/782, Loss: 0.1025, Recon: 0.0178, VQ: 0.0847, Per
plexity: 63.59
Epoch 299/500, Batch 700/782, Loss: 0.0924, Recon: 0.0175, VQ: 0.0749, Per
plexity: 61.39

=== Epoch 299/500 Summary ===
Train Loss: 0.0999 | Val Loss: 0.1005
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0819 | Val VQ: 0.0808
Train Perplexity: 63.70 | Val Perplexity: 63.74
No improvement. Patience: 11/20
Best Loss: 0.0996 (Epoch 288)

Epoch 300/500, Batch 100/782, Loss: 0.0818, Recon: 0.0157, VQ: 0.0661, Per
plexity: 60.56
Epoch 300/500, Batch 200/782, Loss: 0.0879, Recon: 0.0170, VQ: 0.0709, Per
plexity: 62.69
Epoch 300/500, Batch 300/782, Loss: 0.1135, Recon: 0.0173, VQ: 0.0962, Per
plexity: 61.69
Epoch 300/500, Batch 400/782, Loss: 0.1002, Recon: 0.0188, VQ: 0.0814, Per
plexity: 65.51
Epoch 300/500, Batch 500/782, Loss: 0.1023, Recon: 0.0187, VQ: 0.0836, Per
plexity: 63.86
Epoch 300/500, Batch 600/782, Loss: 0.0893, Recon: 0.0167, VQ: 0.0726, Per
plexity: 63.85
Epoch 300/500, Batch 700/782, Loss: 0.0949, Recon: 0.0178, VQ: 0.0771, Per
plexity: 64.63

=== Epoch 300/500 Summary ===
Train Loss: 0.1000 | Val Loss: 0.1022
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0819 | Val VQ: 0.0825
Train Perplexity: 63.78 | Val Perplexity: 64.04
No improvement. Patience: 12/20
Best Loss: 0.0996 (Epoch 288)

Epoch 301/500, Batch 100/782, Loss: 0.1088, Recon: 0.0191, VQ: 0.0897, Perplexity: 64.93
Epoch 301/500, Batch 200/782, Loss: 0.0976, Recon: 0.0185, VQ: 0.0791, Perplexity: 64.82
Epoch 301/500, Batch 300/782, Loss: 0.0892, Recon: 0.0173, VQ: 0.0720, Perplexity: 61.61
Epoch 301/500, Batch 400/782, Loss: 0.1055, Recon: 0.0196, VQ: 0.0859, Perplexity: 64.67
Epoch 301/500, Batch 500/782, Loss: 0.0924, Recon: 0.0180, VQ: 0.0744, Perplexity: 63.39
Epoch 301/500, Batch 600/782, Loss: 0.1041, Recon: 0.0195, VQ: 0.0846, Perplexity: 64.78
Epoch 301/500, Batch 700/782, Loss: 0.0949, Recon: 0.0174, VQ: 0.0774, Perplexity: 63.48

=== Epoch 301/500 Summary ===

Train Loss: 0.0999 | Val Loss: 0.1017
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0818 | Val VQ: 0.0819
Train Perplexity: 63.88 | Val Perplexity: 64.21
No improvement. Patience: 13/20
Best Loss: 0.0996 (Epoch 288)

Epoch 302/500, Batch 100/782, Loss: 0.1126, Recon: 0.0174, VQ: 0.0952, Perplexity: 61.15
Epoch 302/500, Batch 200/782, Loss: 0.0956, Recon: 0.0182, VQ: 0.0773, Perplexity: 63.75
Epoch 302/500, Batch 300/782, Loss: 0.0982, Recon: 0.0188, VQ: 0.0795, Perplexity: 62.24
Epoch 302/500, Batch 400/782, Loss: 0.1004, Recon: 0.0189, VQ: 0.0814, Perplexity: 66.77
Epoch 302/500, Batch 500/782, Loss: 0.0798, Recon: 0.0159, VQ: 0.0638, Perplexity: 59.42
Epoch 302/500, Batch 600/782, Loss: 0.0918, Recon: 0.0170, VQ: 0.0748, Perplexity: 62.60
Epoch 302/500, Batch 700/782, Loss: 0.0876, Recon: 0.0168, VQ: 0.0708, Perplexity: 59.57

=== Epoch 302/500 Summary ===

Train Loss: 0.1002 | Val Loss: 0.1012
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0821 | Val VQ: 0.0815
Train Perplexity: 63.95 | Val Perplexity: 64.08
No improvement. Patience: 14/20
Best Loss: 0.0996 (Epoch 288)

Epoch 303/500, Batch 100/782, Loss: 0.1004, Recon: 0.0172, VQ: 0.0831, Perplexity: 64.02
Epoch 303/500, Batch 200/782, Loss: 0.0981, Recon: 0.0188, VQ: 0.0793, Perplexity: 65.49
Epoch 303/500, Batch 300/782, Loss: 0.0952, Recon: 0.0167, VQ: 0.0785, Perplexity: 61.04
Epoch 303/500, Batch 400/782, Loss: 0.0999, Recon: 0.0184, VQ: 0.0815, Perplexity: 64.91
Epoch 303/500, Batch 500/782, Loss: 0.0898, Recon: 0.0171, VQ: 0.0727, Perplexity: 62.49
Epoch 303/500, Batch 600/782, Loss: 0.2316, Recon: 0.0179, VQ: 0.2136, Perplexity: 64.79
Epoch 303/500, Batch 700/782, Loss: 0.0993, Recon: 0.0182, VQ: 0.0812, Perplexity: 64.73

=== Epoch 303/500 Summary ===

Train Loss: 0.1000 | Val Loss: 0.1016
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0820 | Val VQ: 0.0819
Train Perplexity: 63.97 | Val Perplexity: 64.04
No improvement. Patience: 15/20
Best Loss: 0.0996 (Epoch 288)

Epoch 304/500, Batch 100/782, Loss: 0.0928, Recon: 0.0176, VQ: 0.0752, Perplexity: 64.10
Epoch 304/500, Batch 200/782, Loss: 0.0987, Recon: 0.0187, VQ: 0.0800, Perplexity: 63.95
Epoch 304/500, Batch 300/782, Loss: 0.2738, Recon: 0.0195, VQ: 0.2543, Perplexity: 64.98
Epoch 304/500, Batch 400/782, Loss: 0.1131, Recon: 0.0201, VQ: 0.0930, Perplexity: 67.44
Epoch 304/500, Batch 500/782, Loss: 0.0960, Recon: 0.0172, VQ: 0.0788, Perplexity: 63.67
Epoch 304/500, Batch 600/782, Loss: 0.0948, Recon: 0.0175, VQ: 0.0773, Perplexity: 63.43
Epoch 304/500, Batch 700/782, Loss: 0.1024, Recon: 0.0178, VQ: 0.0846, Perplexity: 64.56

=== Epoch 304/500 Summary ===

Train Loss: 0.1002 | Val Loss: 0.1013
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0821 | Val VQ: 0.0816
Train Perplexity: 64.01 | Val Perplexity: 64.11
No improvement. Patience: 16/20
Best Loss: 0.0996 (Epoch 288)

Epoch 305/500, Batch 100/782, Loss: 0.1063, Recon: 0.0182, VQ: 0.0881, Perplexity: 65.96
Epoch 305/500, Batch 200/782, Loss: 0.0973, Recon: 0.0184, VQ: 0.0790, Perplexity: 64.29
Epoch 305/500, Batch 300/782, Loss: 0.0990, Recon: 0.0183, VQ: 0.0807, Perplexity: 64.77
Epoch 305/500, Batch 400/782, Loss: 0.1204, Recon: 0.0192, VQ: 0.1012, Perplexity: 64.13
Epoch 305/500, Batch 500/782, Loss: 0.1037, Recon: 0.0198, VQ: 0.0839, Perplexity: 66.59
Epoch 305/500, Batch 600/782, Loss: 0.0946, Recon: 0.0170, VQ: 0.0776, Perplexity: 63.19
Epoch 305/500, Batch 700/782, Loss: 0.1016, Recon: 0.0180, VQ: 0.0836, Perplexity: 64.25

=== Epoch 305/500 Summary ===

Train Loss: 0.0999 | Val Loss: 0.1007
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0819 | Val VQ: 0.0810
Train Perplexity: 64.08 | Val Perplexity: 64.15
No improvement. Patience: 17/20
Best Loss: 0.0996 (Epoch 288)

Epoch 306/500, Batch 100/782, Loss: 0.0954, Recon: 0.0182, VQ: 0.0772, Perplexity: 62.78
Epoch 306/500, Batch 200/782, Loss: 0.1046, Recon: 0.0195, VQ: 0.0851, Perplexity: 64.77
Epoch 306/500, Batch 300/782, Loss: 0.1134, Recon: 0.0201, VQ: 0.0934, Per

plexity: 66.74
Epoch 306/500, Batch 400/782, Loss: 0.0922, Recon: 0.0169, VQ: 0.0753, Per
plexity: 63.00
Epoch 306/500, Batch 500/782, Loss: 0.1032, Recon: 0.0187, VQ: 0.0845, Per
plexity: 65.54
Epoch 306/500, Batch 600/782, Loss: 0.1118, Recon: 0.0184, VQ: 0.0935, Per
plexity: 64.90
Epoch 306/500, Batch 700/782, Loss: 0.0931, Recon: 0.0169, VQ: 0.0762, Per
plexity: 64.53

=== Epoch 306/500 Summary ===
Train Loss: 0.1001 | Val Loss: 0.1008
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0820 | Val VQ: 0.0811
Train Perplexity: 64.09 | Val Perplexity: 64.02
No improvement. Patience: 18/20
Best Loss: 0.0996 (Epoch 288)

Epoch 307/500, Batch 100/782, Loss: 0.1090, Recon: 0.0197, VQ: 0.0893, Per
plexity: 64.50
Epoch 307/500, Batch 200/782, Loss: 0.0937, Recon: 0.0163, VQ: 0.0774, Per
plexity: 62.73
Epoch 307/500, Batch 300/782, Loss: 0.1030, Recon: 0.0192, VQ: 0.0838, Per
plexity: 64.93
Epoch 307/500, Batch 400/782, Loss: 0.0985, Recon: 0.0185, VQ: 0.0800, Per
plexity: 63.66
Epoch 307/500, Batch 500/782, Loss: 0.1199, Recon: 0.0180, VQ: 0.1019, Per
plexity: 63.84
Epoch 307/500, Batch 600/782, Loss: 0.1068, Recon: 0.0178, VQ: 0.0889, Per
plexity: 63.88
Epoch 307/500, Batch 700/782, Loss: 0.1040, Recon: 0.0182, VQ: 0.0859, Per
plexity: 64.55

=== Epoch 307/500 Summary ===
Train Loss: 0.1001 | Val Loss: 0.1009
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0821 | Val VQ: 0.0812
Train Perplexity: 64.06 | Val Perplexity: 63.98
No improvement. Patience: 19/20
Best Loss: 0.0996 (Epoch 288)

Epoch 308/500, Batch 100/782, Loss: 0.1041, Recon: 0.0186, VQ: 0.0855, Per
plexity: 65.68
Epoch 308/500, Batch 200/782, Loss: 0.0878, Recon: 0.0162, VQ: 0.0716, Per
plexity: 62.91
Epoch 308/500, Batch 300/782, Loss: 0.0940, Recon: 0.0175, VQ: 0.0765, Per
plexity: 64.76
Epoch 308/500, Batch 400/782, Loss: 0.1139, Recon: 0.0205, VQ: 0.0934, Per
plexity: 66.98
Epoch 308/500, Batch 500/782, Loss: 0.0925, Recon: 0.0170, VQ: 0.0755, Per
plexity: 61.38
Epoch 308/500, Batch 600/782, Loss: 0.1073, Recon: 0.0170, VQ: 0.0902, Per
plexity: 62.96
Epoch 308/500, Batch 700/782, Loss: 0.0964, Recon: 0.0186, VQ: 0.0778, Per
plexity: 65.53

=== Epoch 308/500 Summary ===
Train Loss: 0.0999 | Val Loss: 0.1007
Train Recon: 0.0181 | Val Recon: 0.0197
Train VQ: 0.0819 | Val VQ: 0.0811

Train Perplexity: 64.19 | Val Perplexity: 63.95

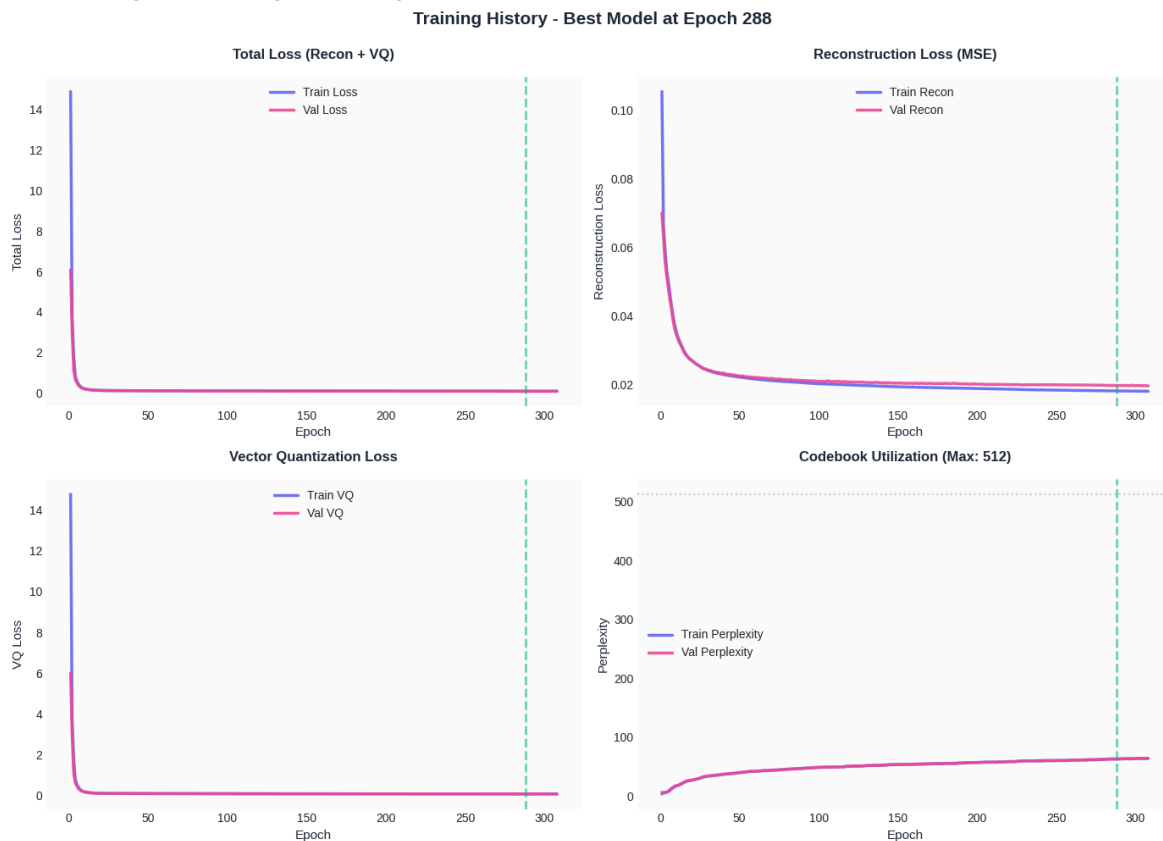
No improvement. Patience: 20/20

Best Loss: 0.0996 (Epoch 288)

⚠ Early stopping triggered after 308 epochs

Best model was at epoch 288 with loss 0.0996

Plotting training history...



Loading best model from epoch 288...

```
/usr/local/lib/python3.11/dist-packages/torch/nn/parallel/_functions.py:7
0: UserWarning: Was asked to gather along dimension 0, but all input tensors were scalars; will instead unsqueeze and return a vector.
warnings.warn(
```

=== Generating Modern Visualizations ===

1. Creating DNA-style sequences...

```
/tmp/ipykernel_19/3665887241.py:451: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.
plt.tight_layout()
```

DNA-Style Codeword Sequences



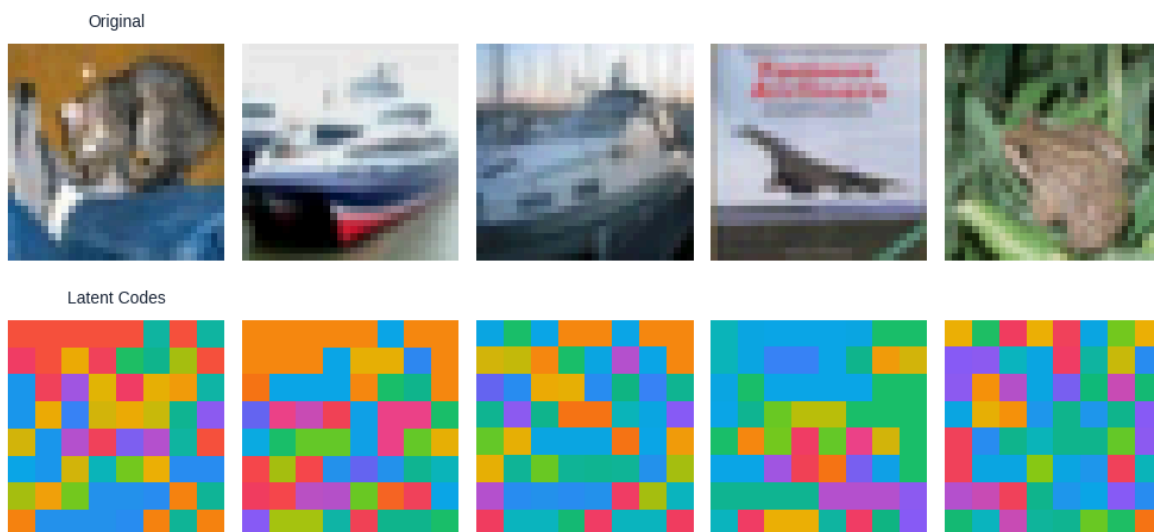
2. Creating 2D genome map...



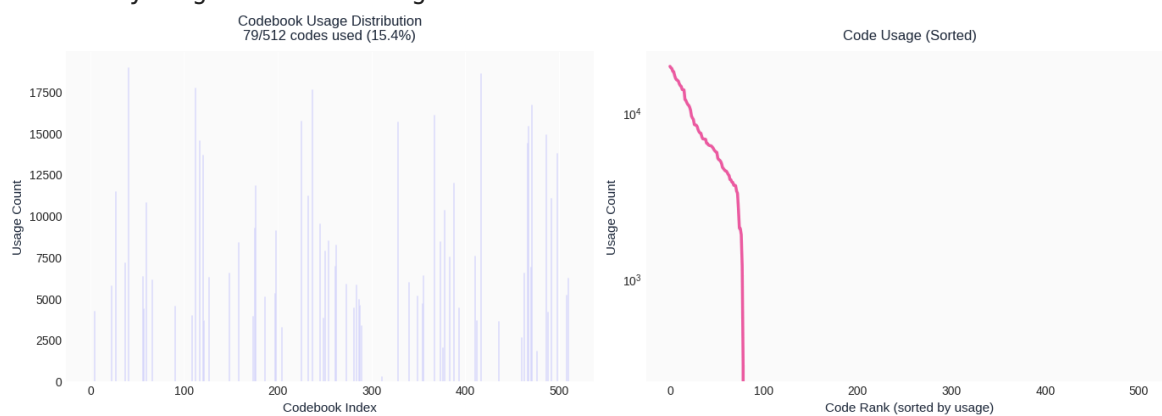
3. Creating ring plot...



4. Visualizing latent codes...

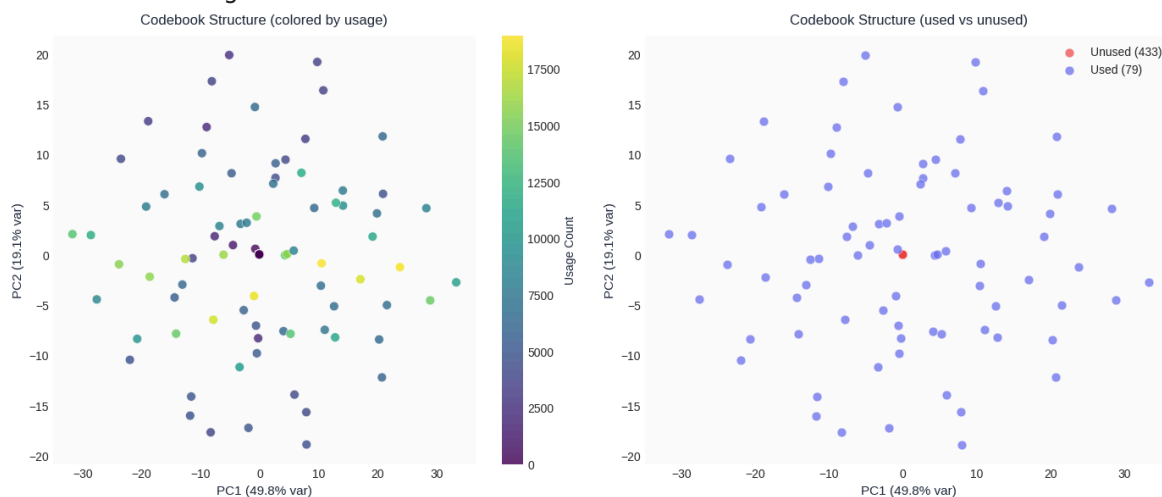


5. Analyzing codebook usage...



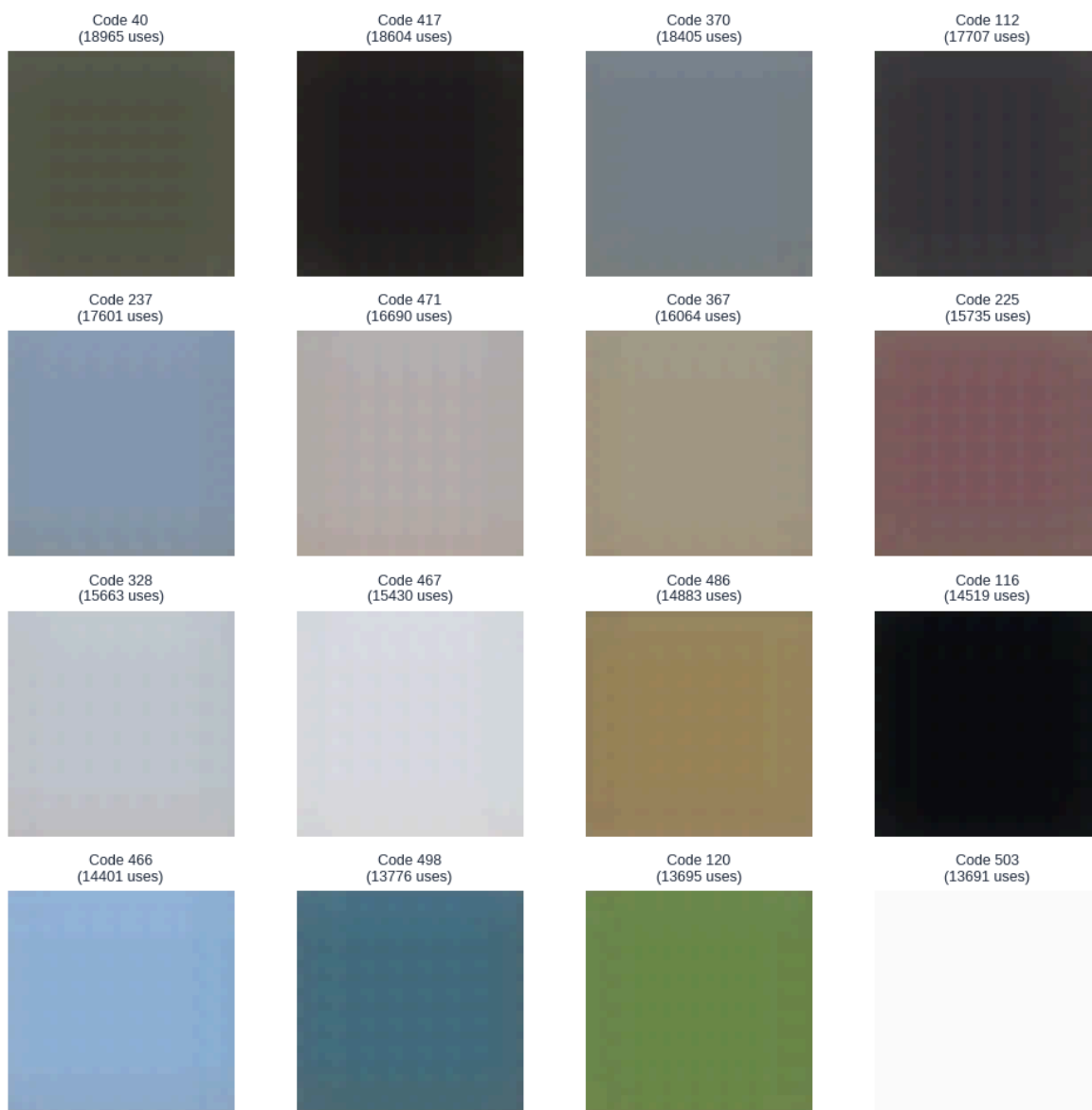
- Active codes: 79/512 (15.4%)
- Most used code: 18965 times
- Least used active code: 300 times

6. Visualizing codebook structure...



7. Visualizing individual codebook entries...

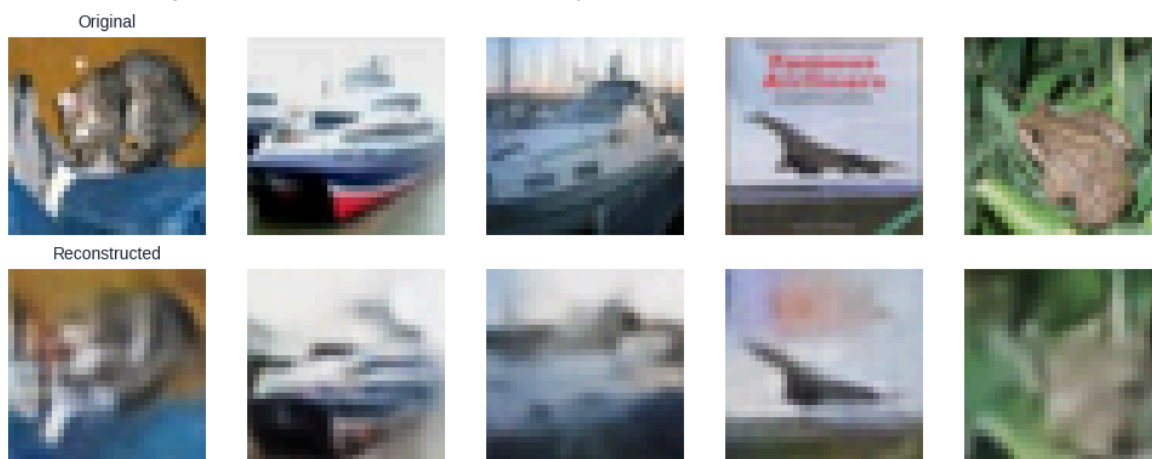
Top 16 Most Used Codebook Entries



8. Visualizing codebook interpolation...



9. Creating basic reconstruction comparison...



=== All Visualizations Complete ===

Generated files:

0. vqvae_training_history.png – Modern training curves
1. vqvae_reconstruction.png – Original vs Reconstructed
2. vqvae_dna_sequences.png – DNA-style linear sequences
3. vqvae_genome_map.png – 2D genome map (no colorbar)
4. vqvae_circular_genome.png – Modern ring plot
5. vqvae_latent_codes.png – Discrete latent representations
6. vqvae_codebook_usage.png – Usage statistics
7. vqvae_codebook_structure.png – PCA visualization
8. vqvae_top_codes.png – Most frequently used codes
9. vqvae_interpolation.png – Smooth interpolation

All visualizations use modern, soft color palette!

Ring plot inner radius: 0.8 (adjustable in config)

In []: